

Igniting the Mathematical Spark

A Tour of Mathematics for Middle School Students

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Preface: Don't Rush to Memorize Formulas

This book is a tour of mathematics for middle school students. It does not follow the textbook's order to check what you "ought to know." Instead, it starts with real, intriguing questions and takes you along to see how mathematicians pose problems, invent concepts, look for evidence, and connect their ideas.

As you read, you can temporarily set aside standard answers and lists of formulas—but please keep two things: your curiosity and the habit of asking "why?" For every key conclusion, we offer an intuitive explanation, a necessary proof, or an experiment you can check for yourself.

The reflection questions at the end of each chapter have three levels: observation, proof, and challenge. We give hints, but no direct answers. Getting stuck is normal. Getting stuck is precisely where mathematics begins.

Part I

From Middle School Algebra—Symbols, Equations, and Functions

Chapter 1 Numbers and Symbols: Turning Specific Problems into Portable Models

1.1 A Hundred-Yuan Shopping List

The day before school starts, your mother hands you a hundred-yuan note. “Go and buy the stationery you’ll need this term. Notebooks cost six yuan each and gel pens cost four yuan each. Choose the quantities yourself, and try to spend exactly one hundred yuan.”

It sounds like a small errand, but standing before the shelves in the stationery shop, you discover a proper problem: how many notebooks and pens should you buy to spend exactly one hundred yuan? Your mother might even add another condition: “You need both notebooks and pens. You can’t buy just one kind.”

Start with a guess. Ten notebooks cost sixty yuan, leaving forty yuan for exactly ten pens. There is one answer: 10 notebooks and 10 pens. Are there other ways? 16 notebooks cost ninety-six yuan; add one pen and the total is exactly one hundred. There seems to be more than one answer. That raises a more ambitious question:

How many ways are there? Can you guarantee you have missed none and counted none twice?

This question can leave you stuck. Unlike “How much do 10 notebooks and 10 pens cost?”, it cannot be answered with a single calculation. You have to find *all* the answers and explain why none are missing. Calling out a number on instinct is not enough. You say there are 8 ways; your classmate says 9. Who is right? How do you decide?

That is where this chapter begins. We will start by patiently trying arithmetic and see how the work gets out of hand. Then we will introduce our main characters—*algebraic symbols*—and turn “each individual problem” into “a whole class of problems.” Finally, you will have a small model that solves more than this shop’s problem. You can carry it, unchanged, into taxi fares, mobile phone plans, mixing drinks, dividing New Year gift money, and many other situations that seem unrelated.

1.2 The Arithmetic Approach: When Trying Every Case Gets Out of Hand

Anyone can think of the simplest method: try the possibilities one by one.

You must buy at least 1 notebook and at most 16 (17 would cost more than one hundred yuan). For each number of notebooks, work out the money left, then check whether it is divisible by four with a quotient of at least 1. We obtain this table:

Notebooks x	Money left (yuan)	Gel pens y
2	$100 - 12 = 88$	22
4	$100 - 24 = 76$	19
6	$100 - 36 = 64$	16
8	$100 - 48 = 52$	13
10	$100 - 60 = 40$	10
12	$100 - 72 = 28$	7
14	$100 - 84 = 16$	4
16	$100 - 96 = 4$	1

After careful checking, there are 8 ways in total. Notice one detail: an odd number of notebooks never works. The remaining money would be an amount such as 94 or 82 yuan, which leaves a remainder when divided by 4. You cannot buy a whole number of pens. We discovered this by counting the money case by case.

The problem seems solved. But consider the cost of this method: 16 subtractions, 16 divisions, and constant attention to whether the money left can be spent exactly. If your mother changes the budget to one hundred and fifty or two hundred yuan, you must redo the table. If the shop adds another item—stickers at two yuan each, say—and the condition becomes “buy all three kinds and spend exactly one hundred yuan,” enumeration no longer fits in a single short table. It takes dozens or even hundreds of rows: the possibilities with 1 sticker, then with 2 stickers, and so on. Doing this by hand, an error is only a matter of time.

Arithmetic fails here not because its calculations are wrong, but because *every new problem requires a fresh start*. Solving the hundred-yuan problem leaves us no reusable result for a hundred and twenty yuan; counting combinations of two items gives us no ready method for three. We need more than faster trial and error. We need a way to write down the process of trying itself.

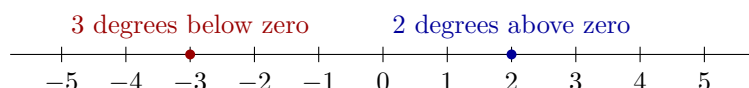
One-Minute Experiment

Take a sheet of paper and repeat the enumeration above with a budget of 120 yuan. Notebooks still cost 6 yuan, pens still cost 4 yuan, and you must buy at least one of each. Count how many divisions you perform. Before reading on, ask yourself: is there a pattern repeated in those divisions? If you had to describe the same pattern to a robot that knew no arithmetic, how would you write it?

1.3 A Quick Roll Call: The Numbers We Already Know

Before bringing algebraic symbols onstage, let's greet some old friends. From this chapter onward, they will often appear together in the same expression.

The *natural numbers* $0, 1, 2, 3, \dots$ are the counting numbers. Counting notebooks or sharing sweets belongs to their territory. The *integers* add the negative numbers $-1, -2, -3, \dots$, which describe quantities in the “opposite direction”: three degrees below zero, a debt of five yuan, the second basement floor in a lift. Place the integers on a straight line and you get our familiar number line:



Fractions, together with their close relatives decimals and percentages, deal with parts of a whole. Half a pizza is $\frac{1}{2}$; a six-yuan notebook sold at seventy-five percent of its price costs $6 \times 0.75 = 4.5$ yuan; 60% of a class joins the spring outing. *Ratios* describe the relative sizes of quantities. A map scale of $1 : 10000$ says that 1 centimetre on the map represents 100 metres on the ground. Sharing 12 chocolates between two people in the ratio $2 : 1$ means dividing them into 3 equal portions, with one person taking two portions and the other taking one.

Each kind of number has its own character. Division can take you outside the natural numbers: 7 divided by 2 has no natural-number answer. The same is true of the integers. The world of fractions is much broader: adding, subtracting, multiplying, or dividing two fractions (with a nonzero divisor) always gives another fraction. Percentages are just fractions with denominator 100. Prices described as seventy-five, eighty, or ninety-five percent of the original all tell you what percentage of the original price you must pay.

Why this roll call? Because whenever we write an expression, we must first decide which numbers its quantities are allowed to take. Numbers of stationery items must be natural numbers; temperatures can be integers or fractions; discounts are percentages. *The world of numbers a problem lives in often determines whether it has a solution and how many solutions it has.* If buying “3.5 notebooks” were allowed, the hundred-yuan problem would suddenly have endlessly many possibilities—but the shop does not sell half a notebook. We will use this point again shortly.

1.4 Naming What We Don’t Know: Symbols and Algebraic Expressions

Return to the hundred-yuan list. Each step in our enumeration did the same thing: suppose we buy some number of notebooks, calculate, and then adjust. Long ago, mathematicians realized that instead of repeatedly writing “suppose we buy some number,” we could give this as-yet-unknown number a temporary name: a symbol, usually written as a letter.

Call the number of notebooks x and the number of gel pens y . The money spent is

$$6x + 4y,$$

and the requirement to spend exactly one hundred yuan becomes

$$6x + 4y = 100.$$

Pause to look at this line. It accomplishes our first major step: *that entire table and all those divisions have been compressed into one line*. The line makes no particular guess about how many notebooks to buy. It speaks about every possible purchase at once. Has the budget become a hundred and twenty yuan? No need to rethink the problem; replace the right-hand side with 120: $6x + 4y = 120$. Have the unit prices changed? Replace 6 and 4. The expression's skeleton stays exactly the same.

An expression such as $6x + 4y$, made from numbers, symbols representing numbers, and operation symbols, is called an *algebraic expression*.

Terminology Card: Algebraic Expression

Intuitive explanation: The skeleton left when you replace the changing parts of an arithmetic problem with symbols.

Formal definition: An expression joining numbers and symbols that represent numbers through operations such as addition, subtraction, multiplication, division, and powers.

An example: With notebooks at 6 yuan each and pens at 4 yuan each, the total cost of x notebooks and y pens is the algebraic expression $6x + 4y$. Set $x = 10$ and $y = 10$, and it becomes the ordinary arithmetic calculation $6 \times 10 + 4 \times 10 = 100$.

A symbol's first role here is to represent an *unknown*: a particular number whose value we do not yet know and must uncover using the conditions. In $6x + 4y = 100$, x and y are two unknowns. The shop's restriction that " x, y are both positive integers"—remember how the world of numbers matters—will help narrow the answers to a finite set.

Do not underestimate the step of giving something a name. Mathematicians worked toward it for over a thousand years. Early mathematical texts had no symbolic notation; a problem occupied a whole paragraph of everyday language: "There is a quantity. Take three times that quantity and add two. The result is fifty. What is the quantity?" Before calculating, the reader had to decipher the prose. Mathematicians in different places invented their own abbreviations and marks, could not read one another's "codes," and circulated many arguments about which notation was more convenient. Only as letter notation gradually became standardized could the same problem be compressed into the seven symbols $3x + 2 = 50$. The triumph of notation was not about saving ink. Once all the conditions live in an expression, reasoning can operate directly on that expression without relying on a mental ledger. Every transformation, solution, and generalization later in this chapter benefits from that triumph.

Writing conditions with symbols has another immediate benefit: we can *reason* directly with expressions instead of repeatedly counting money. Consider why an odd number of notebooks

is impossible. The expression $6x + 4y$ is always even: $6x$ is even, $4y$ is even, and their sum is even. Since 100 is also even, parity alone gives no contradiction. But the remainder when $6x$ is divided by 4 depends on x . For odd x , $6x$ takes values such as 6, 18, 30, each a multiple of 4 plus 2. Adding $4y$ leaves remainder 2, whereas 100 leaves remainder 0: a contradiction. Thus x must be even. A short argument replaces the eight crossed-out attempts in the table.

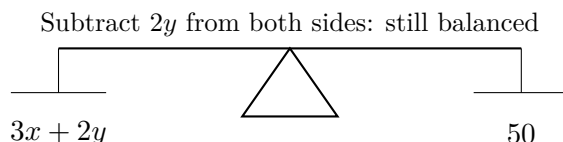
1.5 The Rules of the Balance: Transforming Equations and Finding a Unique Solution

Writing $6x + 4y = 100$ is only the beginning. We must also know how to solve it. First, let's get the terminology straight:

Definition 1.1 (Equation). An equality containing unknowns is called an *equation*. Values of the unknowns that make its two sides equal are called *solutions*; finding them is called *solving the equation*.

All the techniques of solving equations rest on an old and simple picture: a balance.

An equality is like a balanced scale. The left pan holds $6x + 4y$, the right pan holds 100, and the beam is level. We can perform the same operation on both sides without upsetting the balance: add equal weights, remove equal weights, double both loads, or halve them. The balance is preserved throughout.



Theorem 1.2 (The Balance Principle). *If two expressions are equal, they remain equal when we add or subtract the same number or expression on both sides, multiply both sides by the same number, or divide both sides by the same nonzero number.*

This principle sounds unremarkable, but it is the constitution of the world of equations. Divide both sides of $6x + 4y = 100$ by 2 to obtain the cleaner $3x + 2y = 50$. Then “remove” $2y$ from both sides by subtracting it, giving $3x = 50 - 2y$. Divide by 3: $x = \frac{50 - 2y}{3}$. Each step merely rearranges the balance while keeping it level. By the end, x has almost been singled out: choose y and you can immediately calculate x . Our table repeatedly uses this expression for $y = 22, 19, 16, \dots$

The classic setting for the balance principle is a linear equation in one unknown: an equation with just one unknown, appearing only to the first power. There is a particularly clean result:

Theorem 1.3 (The Solution of a Linear Equation in One Unknown). *Let $a \neq 0$, and let b, c be any known numbers. Then the equation $ax + b = c$ has exactly one solution,*

$$x = \frac{c - b}{a}.$$

Proof. There are two steps: first find a solution, then show there cannot be a second one.

First, use the balance principle to isolate x . Subtract b from both sides to get $ax = c - b$. Since $a \neq 0$, divide both sides by a , giving $x = \frac{c-b}{a}$. Substitute this value into the original equation to check it: $a \cdot \frac{c-b}{a} + b = (c-b) + b = c$. The equality holds, so this is a solution.

Second, prove uniqueness. Suppose x_1 and x_2 both solve the equation, so $ax_1 + b = c$ and $ax_2 + b = c$. Subtract the equalities, removing equal quantities from the two pans, to obtain $a(x_1 - x_2) = 0$. Since $a \neq 0$, and a product is zero only if at least one factor is zero, we have $x_1 - x_2 = 0$, hence $x_1 = x_2$. The supposed two solutions are the same. This completes the proof. $\square$

Notice a small but essential detail: before dividing by a , we must establish $a \neq 0$. The rule allowing division by the same number has a condition: that number cannot be zero. Division by zero is undefined, so the balance rule has a clear boundary here. This is no mere quibble. When $a = 0$, the equation becomes $0 \cdot x + b = c$. If $b = c$, every x works; if $b \neq c$, there is no solution. Neither case has exactly one solution. *A theorem's conditions are as much a part of it as its conclusion.*

Here is a numerical example of the theorem's power. The stationery shop offers two promotions. Plan A takes twenty-five yuan off purchases of at least one hundred yuan; Plan B charges seventy-five percent of the original price. At what purchase amount do they offer equal savings? Let the original price be x yuan, assuming $x \geq 100$ so Plan A applies. You pay $x - 25$ yuan under A and $0.75x$ yuan under B. Equal savings means $x - 25 = 0.75x$. Subtract $0.75x$ to get $0.25x = 25$, then divide by 0.25 : $x = 100$. They tie at exactly one hundred yuan; beyond that, B is cheaper because its discount grows with the total, while A's reduction stays fixed. Percentages, linear equations, and the balance principle have worked together on an everyday problem.

1.6 The World Is Not Always Exact: Inequalities

If your mother relaxes her instruction to “buy what you like within one hundred yuan, but don't overspend,” an equality no longer captures it. The amount $6x + 4y$ need not equal a fixed number; it must satisfy

$$6x + 4y \leq 100.$$

This is an *inequality*. Most real-world constraints look like this: stay within a budget, finish before a deadline, keep below a weight limit, or score at least a passing grade. Equalities describe an exact fit; inequalities describe an acceptable range. The latter territory is much larger.

Inequalities can also be transformed, but their rules come with an extra warning. Adding or subtracting the same number on both sides preserves the direction. Multiplying or dividing by a *positive* number preserves it too. But multiplying or dividing by a *negative* number requires reversing the inequality sign.

Why? One look at the number line explains it. $2 < 3$ means 2 lies to the left of 3. Multiplying both sides by -1 gives -2 and -3 , reflected across 0. Now -2 lies to the *right* of -3 , so $-2 > -3$.

Multiplication by a negative number mirrors the number line, exchanging left and right.

Counterexample Alert

Alert one: “Multiplying both sides of an inequality by the same number preserves its direction.” Wrong. Try multiplying by -1 . Starting with $5 < 7$, it would be absurd to conclude $-5 < -7$; the correct result is $-5 > -7$. Only a *positive* multiplier preserves direction. A negative multiplier reverses it, and multiplying by zero turns both sides into 0, making the strict inequality collapse.

Alert two: “An algebraic symbol represents a hidden unknown and becomes useless once you solve for it.” Also wrong. In $6x + 4y = 100$, x, y can take 8 different pairs of values; the symbols represent all the candidates. In the next section, you will see symbols stand for any natural number. Thinking of a symbol only as one concealed number misses much of its power.

Inequalities and equalities often work together. Return to the three-item problem: notebooks at 6 yuan, pens at 4 yuan, stickers at 2 yuan, buying all three kinds and spending exactly one hundred yuan. If we buy z stickers, the first two quantities must satisfy $6x + 4y = 100 - 2z$, with $x \geq 1$ and $y \geq 1$. Those inequalities limit z : $100 - 2z$ must cover at least one notebook and one pen, so $100 - 2z \geq 10$, giving $z \leq 45$; also $z \geq 1$. Thus z ranges only from 1 to 45. For each z , solve a two-unknown problem we already understand. Inequalities have trimmed a daunting three-unknown enumeration into a bounded task we can handle in batches.

1.7 Another Job for Symbols: Representing a Whole Class of Numbers

So far, symbols have stood for particular numbers that are not yet known. But they have a second, perhaps more important job: speaking about *all* numbers at once.

Consider an old pattern. What is the sum of the natural numbers from 1 to 100? Brute force requires 99 additions. Legend says that the young Gauss answered 5050 within seconds in class. The story’s details may have been embellished, but its underlying idea is sound, and symbols express it in a single sentence.

Theorem 1.4 (The Sum of Consecutive Natural Numbers). *For every positive integer n ,*

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Write the sum twice, once forward and once backward, with the terms aligned:

$$\begin{array}{ccccccc} 1 & + & 2 & + \cdots + & (n-1) & + & n \\ n & + & (n-1) & + \cdots + & 2 & + & 1 \end{array}$$

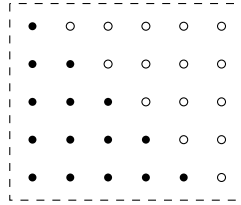
Each vertical pair sums to $n+1$: first $1+n$, then $2+(n-1)$, and so on, ending with $n+1$. There are n pairs, so the two rows together total $n(n+1)$. They are two copies of the same sum, hence

$$2(1 + 2 + \cdots + n) = n(n+1),$$

and dividing both sides by 2 using the balance principle gives the result. This completes the proof. $\square$

Substitute $n = 100$: $\frac{100 \times 101}{2} = 5050$, just as in the legend. But the theorem says much more than how to add from 1 to 100. It works when n is 7, 1000, or 1000000. *One symbol, n , combines infinitely many addition problems into one.* That is a symbol's second job: representing any member of an entire class of numbers, rather than one unknown.

The formula also has a geometric face. Picture $1 + 2 + \cdots + n$ as an arrangement of dots: 1 in the first row, 2 in the second, and so on to n in the last, forming a triangle. Take two identical dot triangles and turn one over. Together they form a rectangle with n rows and $n + 1$ columns:



Filled dots: $1 + 2 + 3 + 4 + 5$; open dots: the flipped copy; together: 5×6

In the proof, we wrote the sum twice and paired the terms. In the picture, we took two copies and made a rectangle. Algebra states a fact and geometry states it again. This habit of expressing the same idea in two ways will return throughout the book.

Terminology Card: Variables, Parameters, and Constants

Intuitive explanation: Three roles in a play. Variables are the actors who move; parameters are scenery fixed for one performance but replaceable for another; constants are props that never change.

Formal definition: A quantity that can take different values within a problem is a variable. A quantity held fixed for a given problem but allowed to change between related problems is a parameter. A number that does not change is a constant.

An example: In $6x + 4y = 100$, x, y are variables and 6, 4, 100 are constants. Generalize to $px + qy = m$, with prices p, q and budget m . Then x, y remain variables, while p, q, m are parameters. Changing the shop or the budget changes a set of parameters while leaving the model intact.

Distinguishing these three roles is a key step from solving one problem to building a model. Writing $6x + 4y = 100$ addresses your mother's particular list. Writing $px + qy = m$ addresses the same kind of problem for *all* prices and *all* budgets. Parameters let a model travel farther.

1.8 A Different Perspective: Models Can Travel

Looking back, this chapter has done one thing: translate a specific problem into a model made of symbols, equalities, and inequalities. The same model repeatedly appears in very different settings. This is what makes algebra so rewarding.

A taxi fare: the initial charge is 10 yuan, followed by 2 yuan per kilometre. For a journey of x kilometres, with $x \geq 3$ and the first 3 kilometres covered by the initial charge, the fare is $2(x - 3) + 10$, or $2x + 4$ yuan. With only 30 yuan, how far can you go? Solve $2x + 4 \leq 30$ to get $x \leq 13$. A mobile phone plan: a monthly fee of 19 yuan includes 10 GB of data, with additional data charged at 3 yuan per GB. With a monthly spending limit of 50 yuan, how much data can you use? Let the extra amount be x GB. Then $19 + 3x \leq 50$, giving $x \leq \frac{31}{3}$, or at most 10 whole extra GB. Mixing a drink: concentrate and water are mixed in the ratio 1 : 5. How much concentrate makes 1.2 litres? If the concentrate is x litres, then $x + 5x = 1.2$, so $x = 0.2$ litres.

See the common thread? Strip away the details of fares, data, and drinks, and each involves a quantity of the form $ax + b$ with an equality or inequality. The balance rules you learned in the stationery shop work just as well in the taxi, the phone plan, and the kitchen, without changing a word.

Here is the sentence this chapter most wants you to take away: *algebra is not about calculating one problem faster; it is about solving many similar problems at once*. Enumeration grows with the size of the problem: double the budget and the table doubles. The effort of building a model is spent once; each new problem becomes a routine matter of changing parameters and solving an equation. Arithmetic asks for this problem's answer. Algebra asks what the answers to a whole class of problems look like. Moving from one case to a class is exactly what "portable" means in this chapter's title.

Connections

This ability to translate is one of the modern world's fundamental technologies. Navigation software turns a road network into constraints between variables and solves an enormous "how do I arrive fastest?" problem. Online shopping systems turn stock levels, coupons, and discount rules into hundreds or thousands of expressions like $6x + 4y \leq 100$, automatically deciding whether an order can go through and what to charge. Every variable you declare in a program is an electronic incarnation of this chapter's algebraic symbols. Each later chapter adds another language to this translator: Chapter 2 adds change, Chapters 3 and 4 add shape, and later come randomness and infinity.

We should also state a model's boundaries honestly. Compressing a taxi journey into $2x + 4$ ignores traffic and detours. Comparing discounts through $x - 25$ and $0.75x$ assumes you really will spend at least one hundred yuan. *A model translates reality, and every translation involves choices*. A good model captures the main structure and omits irrelevant details. Judging those choices is a shared task for modelers and mathematicians. Chapters 2 and 18 will revisit it.

Keep Climbing

Every thread in this chapter leads to a larger world. Equations whose unknowns must be integers, such as the stationery-shop version of $6x + 4y = 100$, are called *Diophantine equations*. They are a classic hunting ground of number theory, which Chapter 13 will visit. Using inequalities to define a feasible region and finding the best plan within it is the domain of *linear programming*, used daily in operations research and economics to allocate real goods and money. Even the rule that matching operations preserve balance is distilled into the axioms of groups and fields in university *abstract algebra*. Mathematicians ask: what other worlds have balances of their own? Keywords: Diophantine equations, linear programming, axiomatization.

Mathematical Toolbox

Keep the new tools from this chapter:

- **Introduce symbols:** Name an unknown number or an arbitrary member of a class of numbers. This is the first step in modeling.
- **Algebraic expressions:** Isolate what changes and retain a reusable skeleton, such as $6x + 4y$.
- **The balance principle:** Adding, subtracting, multiplying, or dividing equally on both sides preserves equality, provided the divisor is nonzero. This underlies equation solving.
- **Transforming inequalities:** Reverse the sign when multiplying or dividing by a negative number. Equalities describe exact matches; inequalities describe ranges. They often work together.
- **Distinguish roles:** Variables vary, parameters change between problems, and constants stay fixed. Parameters make models portable.
- **Ask which numbers are allowed:** Must unknowns be natural numbers, integers, or unrestricted numbers? This determines how many solutions there are, or whether any exist.

Reflection Questions

Observation

1. Look again at the hundred-yuan solutions: $(x, y) = (2, 22), (4, 19), (6, 16), \dots$. Moving to the next purchase increases x by 2 and decreases y by 3. Why precisely “add 2, subtract 3”? (Hint: changing the purchase must leave $6x + 4y$ unchanged. Increase x and calculate how y must respond.)
2. Plot several pairs (x, y) satisfying $6x + 4y \leq 100$, with $x \geq 1$ and $y \geq 1$. Put x horizontally and

y vertically, so each purchase becomes a point. What do you notice about their arrangement? (Hint: first plot the points satisfying $6x + 4y = 100$, then see which side the other points lie on.)

Proof

1. Prove that the sum of any two consecutive integers is odd. (Hint: call the smaller integer n , express the other using n , and add.)
2. Prove $1 + 3 + 5 + \cdots + (2n - 1) = n^2$: the sum of the first n odd numbers is a perfect square. (Hint: imitate the proof that writes the sum twice and pairs terms, or draw dots and add an L-shaped layer around the corner at each step.)

Challenge

1. The three-item problem is $6x + 4y + 2z = 100$, with x, y, z positive integers. Design a counting procedure that misses nothing and counts nothing twice, then find the total number of purchases. (Hint: fix z first and return to the two-unknown case solved in this chapter. What conditions must the right-hand side of $6x + 4y = 100 - 2z$ satisfy for solutions to exist?)
2. Use 40 metres of fencing to enclose a rectangular vegetable plot against a wall. The wall forms one side, so only three sides need fencing. Both dimensions must be whole numbers of metres. What arrangement maximizes the area? (Hint: let the side parallel to the wall be x metres and each other side be y metres. Use the fencing length to relate x, y , express the area using x alone, and try values in a table. How might the answer change without a wall?)

The Door to the Next Chapter

The final challenge hides a question this chapter's tools cannot fully answer. With the fencing length fixed, x and y are tied together. Every change in x forces y to change, and the area S changes too. We found a largest area by trying values in a table, but something still feels uncertain. What about nonintegers? Could $x = 13.5$ metres be better? We tried a few points and claimed to see the whole picture.

This chapter has focused on individual numbers. But the real subject here is *change itself*: how S rises and falls continuously as x varies. Mathematicians have a new tool for this kind of question: draw the relationship on coordinate paper. The horizontal direction represents x and the vertical direction S . Each possible arrangement becomes a point, and together the points form a portrait of the relationship. The portrait shows the highest point, the uphill approach, and the downhill direction. We no longer need to guess or worry about missing intermediate values such as 13.5.

This portrait is the graph of a *function*. Translating the same relationship from an expression into a picture raises a new set of questions. Why is the graph a parabola? What does its steepness tell us? What does the intersection of two graphs mean? In Chapter 2, we move from solving an equation to studying the shape formed by all its answers.

Chapter 2 Equations and Functions: From Finding One Answer to Studying Them All

2.1 A Taxi Meter and a Cage of Chickens and Rabbits

Rain starts falling at dusk, and you hail a taxi. As you settle in, a number appears on the meter: 11. The car moves, and you watch the little red screen. At first it ignores you. After a certain distance, it starts clicking upward at intervals, adding 2 yuan each time. At your destination it stops at 21 yuan.

The driver explains the simple rule: the initial fare is 11 yuan for 3 kilometres; beyond that, each kilometre adds 2 yuan. Suddenly you wonder: how far did you just travel?

Even a primary school student could attempt this question. But change it a little: what if the meter shows 35 yuan, or 100 yuan? What if the company changes its rule to 13 yuan for the first 2 kilometres and 3 yuan per kilometre afterward? Every change requires a fresh calculation. Is there a method that handles all such problems at once?

Now consider an old problem that has appeared in Chinese arithmetic books for over a thousand years. A cage contains chickens and rabbits. Count from above and there are 35 heads; count from below and there are 94 legs. How many chickens and rabbits are there? No price or charging rule is given, only two tangled pieces of information. Before reading on, make an intuitive guess: are there more chickens or rabbits? Roughly how many of each?

The meter and the cage seem unrelated: one concerns money, the other animals. By the end of this chapter, you will see them as two faces of the same mathematical family. Learning to recognize that family takes us beyond finding a single answer into a broader landscape: studying the patterns of all answers.

2.2 Guessing Until Nightfall: The Failure of Trial and Error

The natural response to the cage problem is to guess. Suppose all 35 animals are chickens. They have $35 \times 2 = 70$ legs, 24 short of 94, so we need some rabbits. Try 20 chickens and 15 rabbits: $20 \times 2 + 15 \times 4 = 100$ legs, 6 too many. Adjust to 23 chickens and 12 rabbits: $23 \times 2 + 12 \times 4 = 94$. A hit! There are 23 chickens and 12 rabbits.

Getting the answer feels good. But recall the process honestly: each guess took two or three multiplications and an addition, and a wrong guess meant starting again. Would you still want

to guess with 3500 heads and 9400 legs? More fundamentally, guessing depends on luck and patience rather than a reliable method. You cannot promise someone that following these steps will certainly find the answer. You can only say that enough attempts will probably hit it.

The taxi problem is similar. Try 5 kilometres: $11 + 2 \times (5 - 3) = 15$ yuan, too little. Try 10: $11 + 2 \times (10 - 3) = 25$ yuan, too much. Try 8: $11 + 2 \times (8 - 3) = 21$ yuan, just right. Three attempts, each like searching for a key in the dark.

Chapter 1 showed the power of symbols: a symbol can represent any number, allowing one equality to summarize infinitely many specific problems. Our first major step here is to turn guessing into translation. Translate the problem into equalities, then let the equalities tell us the answer.

2.3 Translating Problems into Equalities

Start with the taxi. What we do not know is the distance in kilometres, so name it x . Naming an unknown is itself an invention: an invisible quantity becomes a symbol we can manipulate on paper.

The charging rule has two parts. The first 3 kilometres cost a fixed 11 yuan; the extra distance, $(x - 3)$ kilometres, costs 2 yuan per kilometre. Thus “the fare is 21 yuan” translates into

$$11 + 2(x - 3) = 21.$$

An equality containing an unknown is an **equation**. Its two sides resemble the pans of a balance. Operations that preserve the balance—adding or subtracting the same number, or multiplying or dividing by the same nonzero number on both sides—preserve the equality. Using that principle,

$$2(x - 3) = 10, \quad x - 3 = 5, \quad x = 8.$$

The journey was 8 kilometres. This time we deduced it. If the fare were 35 yuan, we would replace 21 by 35 and follow the same steps. If the charging rule changed, we would change the equation’s numbers. An equation compresses infinitely many fare questions into one line.

Does the cage problem have just one unknown? It has two: the number of chickens and the number of rabbits. Name them x and y . The two given conditions translate into two equations:

$$\begin{cases} x + y = 35, \\ 2x + 4y = 94. \end{cases}$$

Equations bound together form a **system of equations**. A simple idea solves it: use one equation to replace an unknown. From the first, $y = 35 - x$. Substitute into the second:

$$2x + 4(35 - x) = 94, \quad 140 - 2x = 94, \quad x = 23,$$

so $y = 35 - 23 = 12$. This matches our guess, but now we know the answer is right and unique, because every reasoning step can be checked.

An elegant old solution imagines a whistle blowing and every animal raising two legs. There are now $94 - 35 \times 2 = 24$ legs on the ground. The chickens are sitting down; only rabbits remain standing, each on two legs. Thus there are $24 \div 2 = 12$ rabbits. Delightful ingenuity—but notice that it does the same work as elimination. Subtract twice $x + y = 35$ from $2x + 4y = 94$, and you remove exactly the two-legs-per-animal contribution. Ingenuity solves one problem; equations solve a class. Ingenuity is the scenery and equations are the road. Mathematics favors roads without rejecting scenery.

In these equations, unknowns appear only to the first power, with no squares or cubes. Hence we call them linear equations in one unknown or systems of linear equations in two unknowns. Are all problems so obliging?

Suppose your family wants to enclose a rectangular vegetable garden with 20 metres of fencing and an area of exactly 24 square metres. What are its dimensions? Let the length be x metres. The width is $(10 - x)$ metres, since length plus width equals half the perimeter. The area condition becomes

$$x(10 - x) = 24, \quad \text{which rearranges to} \quad x^2 - 10x + 24 = 0.$$

Now the unknown is squared! Simply balancing and isolating terms is not enough: you cannot combine x^2 and x as like terms and divide. We are stuck, which is precisely the situation mathematicians find most interesting.

2.4 Completing the Square: A Universal Key to Quadratics

For $x^2 - 10x + 24 = 0$, first try factoring. Can two numbers have product 24 and sum 10? A few attempts find 4 and 6. Thus

$$x^2 - 10x + 24 = (x - 4)(x - 6) = 0.$$

A product is zero only if at least one factor is zero, so $x = 4$ or $x = 6$. Length 6 metres and width 4 metres, or vice versa, gives area $6 \times 4 = 24$. The check succeeds.

But this approach has a weakness: we must happen to find the right numbers. What about $x^2 - 10x + 23 = 0$? No two integers have product 23 and sum 10. Does that mean no solution? Not necessarily; perhaps the answers are nonintegers. We need a method independent of guessing.

That method is **completing the square**. Its central idea comes from a familiar identity, $(x + a)^2 = x^2 + 2ax + a^2$. Read it backward: adding a^2 to $x^2 + 2ax$ completes a perfect square. In $x^2 - 10x + 23 = 0$, the initial terms $x^2 - 10x$ give $2a = -10$, so $a = -5$ and the required square is 25. Add 2 to both sides, turning 23 into 25:

$$x^2 - 10x + 25 = 2, \quad (x - 5)^2 = 2, \quad x - 5 = \pm\sqrt{2}.$$

Thus $x = 5 + \sqrt{2}$ or $x = 5 - \sqrt{2}$, approximately 6.41 and 3.59. Noninteger answers are within reach too. The value of completing the square is its equal treatment of every quadratic equation. Apply the procedure to the most general quadratic and we obtain this chapter's central theorem.

Theorem 2.1 (The Quadratic Formula). *Let $a \neq 0$. When $b^2 - 4ac \geq 0$, all real solutions of $ax^2 + bx + c = 0$ are*

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

When $b^2 - 4ac = 0$, the two solutions coincide. When $b^2 - 4ac < 0$, there are no real solutions.

Proof. Since $a \neq 0$, divide both sides by a :

$$x^2 + \frac{b}{a}x + \frac{c}{a} = 0.$$

The coefficient of the linear term is $\frac{b}{a}$, so completing the square requires the square of half that coefficient, $\left(\frac{b}{2a}\right)^2$. Add the completing term and move $\frac{c}{a}$ to the right:

$$x^2 + \frac{b}{a}x + \left(\frac{b}{2a}\right)^2 = \left(\frac{b}{2a}\right)^2 - \frac{c}{a} = \frac{b^2}{4a^2} - \frac{4ac}{4a^2} = \frac{b^2 - 4ac}{4a^2}.$$

The left-hand side is a perfect square:

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}.$$

A real square cannot be negative, so the right side must be nonnegative. This gives the condition $b^2 - 4ac \geq 0$. When it holds, take square roots, remembering both signs:

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a},$$

then move $\frac{b}{2a}$ to the right and combine over a common denominator to obtain the result. $\square$

This formula deserves respect. Centuries ago, mathematicians argued, wagered, and dueled over equation-solving formulas—actual duels did occur, in the story of cubics that Chapter 5 will tell. Today the formula rests quietly on your page. The expression $b^2 - 4ac$ has its own name, the **discriminant**. Like a diagnostic reading, its sign tells you whether the equation has real solutions before you calculate them.

Example 2.2. Solve $2x^2 - 7x + 3 = 0$ with the quadratic formula. Here $a = 2$, $b = -7$, $c = 3$, and the discriminant is $49 - 24 = 25 > 0$. Thus

$$x = \frac{7 \pm 5}{4}, \quad x_1 = 3, \quad x_2 = \frac{1}{2}.$$

Check in the original equation: $2 \times 9 - 7 \times 3 + 3 = 0$ and $2 \times \frac{1}{4} - 7 \times \frac{1}{2} + 3 = 0$. Both hold.

2.5 From Finding One Answer to Studying Them All

So far, we have repeatedly found the number or numbers satisfying a given condition. That is the equation-solving perspective.

Return to the taxi. A fare of 21 yuan corresponds to 8 kilometres; 35 yuan corresponds to 15 kilometres. We could keep solving forever, gaining only one point each time. Can we present the whole relationship between distance and fare at once?

Yes. For distance $x \geq 3$, the fare is $11 + 2(x - 3)$, or $2x + 5$. This expression is no longer one answer but a **rule**. Give it any allowed distance and it returns a fare: 4 gives 13, 10 gives 25, 100 gives 205. The rule pairs infinitely many inputs with outputs like a machine that never stops.

Such a rule has a mathematical name, perhaps the most important name in all of secondary-school mathematics.

Terminology Card: Function

Intuitive explanation: An input-output machine. Put a number in and a number comes out. The same input always gives the same output.

Formal definition: Consider quantities x and y . If each value of x in a specified domain corresponds to exactly one determined value of y , then y is a function of x , written $y = f(x)$. The quantity x is the independent variable, and f denotes the correspondence rule.

An example: $f(x) = 2x + 5$ is the meter's rule for $x \geq 3$: $f(4) = 13$ and $f(10) = 25$.

At first, $f(x)$ can feel awkward. Are f and x being multiplied? No. Read it as “the value of f at x ”: the result of putting x through the rule f . Previously we could only refer vaguely to a relationship. Now we can name each machine: f might be the meter, g a temperature conversion, and h a phone plan. Then we can compare and combine them.

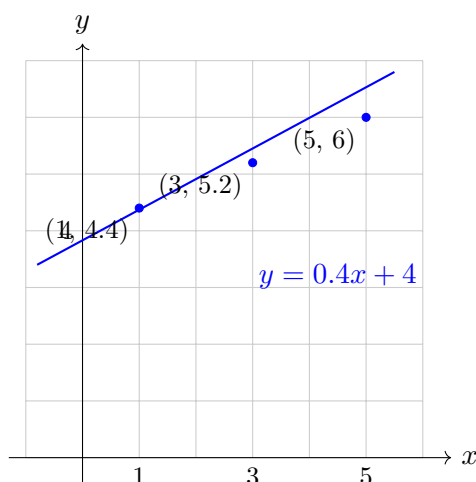
A function machine has three portraits: a formula such as $f(x) = 2x + 5$, a table of inputs and outputs, and a graph plotting each pair as a point. Formulas are precise, tables concrete, and graphs visual. They are three faces of one rule, and we will move repeatedly between them.

Notice the definition's phrase “exactly one.” A machine that sometimes outputs 5 and sometimes 7 for the same input is not a function; it is a faulty machine. Different inputs giving the same output are perfectly acceptable. A machine that tells each person today's day of the week gives everyone the same answer.

Moving from equations to functions is a major shift in viewpoint. An equation asks which x makes $2x + 5 = 21$ true, focusing on one value. A function asks what the rule $f(x) = 2x + 5$ looks like, whether it changes faster or slower, and whether it has a maximum or minimum. Its gaze spans all values. One is a detective search for a particular suspect; the other studies a city's map. In the second half of this chapter, we learn to read maps.

2.6 Graphs: The Tracks Left by Rules

A machine's input and output form a number pair, such as $(4, 13)$ or $(10, 25)$. An ordered pair is exactly a point on coordinate paper. Plot every point produced by a function and you obtain its **graph**. Think of it as the rule's tracks, like footprints in snow revealing where it goes.

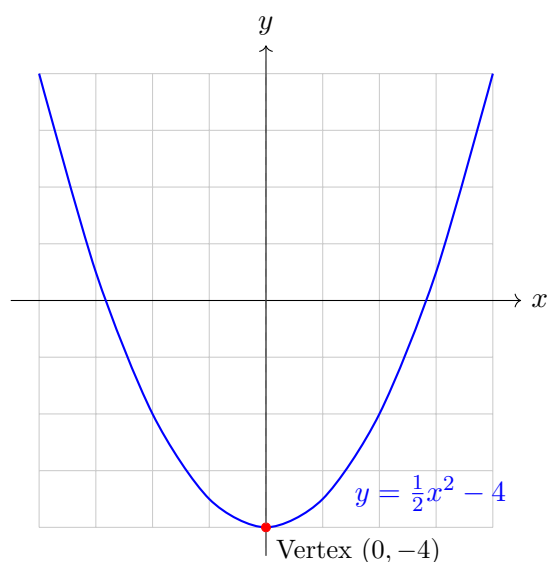


The figure shows the line $y = 0.4x + 4$. Why is its track straight? Look at the rule: each increase of 1 in x adds 0.4 to y ; an increase of 5 adds 2. The rate of change is constant, so the track cannot bend. Functions of the form $y = kx + b$, with fixed k, b and $k \neq 0$, are called **linear functions**; their graphs are straight lines.

Try a different rule: $y = x^2$. Plot a few points:

x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9

Notice the symmetry: $(-2, 4)$ and $(2, 4)$ have the same height. Connect the points. To fit the page, we draw the scaled and shifted version $y = \frac{1}{2}x^2 - 4$:



This curve is a **parabola**. The table explains why it is not straight. From $x = 0$ to 1, y moves only a little; from $x = 2$ to 3, it jumps much farther. The rate of change increases, so the track bends. A ball thrown into the air and a fountain's jet follow this curve, which is why it looks familiar.

Another relationship worth meeting is **inverse proportion**. A class pays 300 yuan to hire a vehicle for an outing. The fare per person depends on the number attending, x : $y = \frac{300}{x}$. With 10 people, each pays 30 yuan; with 30, each pays 10; with 60, each pays 5. More people means a lower price, but the drop slows. Going from 10 to 20 people reduces each fare by 15 yuan; going from 50 to 60 reduces it by only 1 yuan. The graph approaches the axes ever more closely without touching them; it is one branch of a hyperbola. Speed and time for a fixed distance, and length and width for a fixed rectangle area, are also inversely proportional.

Lines, parabolas, and hyperbolas: three tracks, three temperaments. A line changes steadily; a parabola's rate changes; a positive reciprocal curve drops steeply and then flattens. Why draw a graph when we already know the rule? Because our eyes and brains process shapes faster than numbers. We can see whether a curve rises or falls, is steep or gentle, and has high or low points. Extracting the same information from a formula takes work. A graph gives a rule a visual examination.

2.7 Slope and Intercept: Reading a Function's Character

In a line's equation $y = kx + b$, what do k and b control?

Start with b . When $x = 0$, $y = b$, so b is the height where the line meets the y -axis, called the **intercept**. In a taxi example, an intercept resembles a fixed charge, the part of the bill present before distance-dependent charges are added.

Now k . Each increase of 1 in x increases y by k , so k measures the line's **steepness**. It is the **slope**. Positive slope means uphill from left to right; negative slope means downhill. Greater absolute slope means a steeper line.

Terminology Card: Slope

Intuitive explanation: A line's gradient: the vertical rise for each horizontal unit.

Formal definition: For any two points (x_1, y_1) and (x_2, y_2) on a nonvertical line, with $x_1 \neq x_2$, its slope is

$$k = \frac{y_2 - y_1}{x_2 - x_1}.$$

An example: Through $(1, 4)$ and $(5, 12)$, the slope is $k = \frac{12 - 4}{5 - 1} = 2$: move 1 right and rise 2.

An important fact hides in the definition: the ratio is **independent** of the two points chosen. Why should different points give the same slope? Because a linear function changes at a constant rate: $y_2 - y_1 = kx_2 + b - (kx_1 + b) = k(x_2 - x_1)$. Division gives exactly k , with b canceled. Conversely, if a function's slope varies with position, its graph cannot be a line. That is why a parabola is gentle near its bottom and steeper farther up.

Slope has another important name: **rate of change**. A distance-fare line of slope 2 means 2 yuan per kilometre. The slope of a time-distance line is speed; the slope of a time-savings line is the saving rate. Understanding slope means understanding how quickly something changes.

Slope also carries units, an easily overlooked but useful fact. Fare against distance has

units of yuan per kilometre; distance against time has units of metres per second. The units themselves explain what the rate measures. Negative slopes matter too. Altitude against time has positive slope while climbing and negative slope while descending; body temperature rises with a developing fever and falls as the fever subsides. The sign tells us whether the quantity increases or decreases.

Counterexample Alert

“ y increases as x increases, so they are directly proportional.” False. Consider $y = x + 1$. As x goes from 1 to 2, y goes from 2 to 3, so it increases. But the ratio of y to x changes from 2 to 1.5, and at $x = 0$, y is not 0. Direct proportion requires a line through the origin, with $b = 0$. A related trap is “the graph rises, so it is a straight line.” The parabola $y = x^2$ also rises throughout $x > 0$, yet it curves. Rising describes direction, not shape.

The inverse-proportion function $y = \frac{300}{x}$ has a different character. It has no single uniform slope: near the origin it is extremely steep, while farther away it is almost flat. The rate of change itself changes. Acknowledging this was a deep breath in mathematical history, and Chapter 10 will face it directly.

2.8 A Different Perspective: Let Graphs Speak for Equations

Once we can read graphs, equations look different.

The equation $2x + 5 = 21$ asks for the horizontal coordinate where $y = 2x + 5$ reaches height 21. The system

$$\begin{cases} y = 2x + 5, \\ y = -x + 23 \end{cases}$$

asks where two lines intersect. Draw them: two nonparallel lines meet at exactly one point. That is the geometric reason a two-variable linear system typically has a unique solution. The equation $x^2 - 10x + 24 = 0$ asks where the parabola $y = x^2 - 10x + 24$ meets the ground, or x -axis. A positive discriminant gives two crossings; zero gives one touch before the curve rises again; a negative discriminant leaves this upward-opening parabola suspended above the ground. No real solution means no contact.

Connections

In Chapter 1, symbols compressed arithmetic problems into algebraic expressions. Here graphs translate expressions back into shapes. Equations, functions, and graphs are three languages for the same underlying relationship. An equation interrogates it for particular values of x ; a function describes the whole correspondence; a graph maps that correspondence. Chapter 4 will connect this translation system fully by turning geometric shapes directly into algebraic equations. But why trust the points, lines, and angles in a picture? Why does an apparent intersection count as a real one? Chapter 3 will answer through proof.

Changing languages has practical power. A list of monthly electricity bills for the past year may reveal little at first glance. Plot them and trends, anomalies, and cycles emerge. Temperature

curves in weather forecasts, fluctuations in stock displays, and growth curves in health reports all illustrate the same idea: rules leave tracks, and tracks reveal rules.

The reverse works too: translate a geometric problem into equations. A park fountain sends water along a parabola. The designer wants it to return to the pool 4 metres away and reach a maximum height of 2 metres. These shape requirements become equations whose solutions determine precise nozzle parameters. Shapes set requirements and equations supply answers, as engineers do every day. Chapter 4 practises this translation systematically.

One-Minute Experiment

Experiment one: draw your own parabola. Plot 9 points of $y = x^2$ for x from -4 to 4 on squared paper and connect them with a smooth curve. Then (1) fold along the y -axis to check whether the halves coincide, and (2) use a ruler to compare the curve's slope near $(1, 1)$ and $(3, 9)$, feeling how slope changes with position.

Experiment two: measure an everyday slope. Find some stairs and measure one step's height and depth, perhaps 15 and 28 centimetres. Calculate the gradient: $15 \div 28 \approx 0.54$. Make the same measurements on an accessible ramp and compare. You have done the same work as an engineer designing a ramp or a mathematician defining slope.

Mathematical Toolbox

Your new tools from this chapter:

- **Setting up equations:** Name unknowns and translate conditions into equalities. For several unknowns, use a system and eliminate by substitution.
- **Completing the square and the quadratic formula:** A general procedure for quadratic equations. The discriminant $b^2 - 4ac$ first reports whether real solutions exist.
- **Function notation $f(x)$:** Treat a correspondence rule as an object you can name and study.
- **Graphs:** Plot points to turn rules into visible tracks. Lines, parabolas, and hyperbolas are three common examples.
- **Slope and intercept:** Two parameters of a linear function, controlling steepness and starting position. Slope is rate of change.

Keep Climbing

Farther along, these ideas await: **growth rates and derivatives**, with Chapter 10 defining instantaneous slope at each point of a parabola; **composition and inverse functions**, connecting machines in series or running one backward; **exponential and logarithmic functions**, including relationships whose growth rate itself grows, such as bacteria populations and savings; and **fixed points**, inputs a machine returns unchanged. University mathematics extends functions to astonishing settings. We can speak of distances between functions, and collections of functions can themselves form spaces.

Reflection Questions

Observation

1. Draw $y = 2x + 5$ and $y = -x + 23$ on the same axes, locate their intersection, and solve the system. How do the intersection coordinates relate to the solution, and why? *Hint: the intersection lies on both lines, so its coordinates satisfy both rules.*
2. For $y = \frac{12}{x}$, how much does y change as x goes from 1 to 2? What about from 6 to 7? Why are the changes so different for equal input intervals? *Hint: calculate the actual y values and compare the changes, rather than starting with rates.*

Proof

1. Prove that two positive numbers with sum 10 have product at most 25, with equality only when both are 5. *Hint: write them as x and $10 - x$, express their product as a quadratic, and complete the square. Recall the garden problem.*
2. Let $f(x) = kx + b$ with $k > 0$. Prove that $x_1 < x_2$ implies $f(x_1) < f(x_2)$, a property called strict monotonic increase. *Hint: calculate $f(x_2) - f(x_1)$ and use $x_2 - x_1 > 0$ and $k > 0$ to determine its sign.*

Challenge

1. Let x_1, x_2 be the two solutions of $ax^2 + bx + c = 0$. Without evaluating them individually, prove $x_1 + x_2 = -\frac{b}{a}$ and $x_1x_2 = \frac{c}{a}$. *Hint: add and multiply the two expressions in the quadratic formula, watching how $\sqrt{\quad}$ disappears. Chapter 5 gives these relationships a name.*
2. A city's taxi rule charges 11 yuan within 3 kilometres, 2 yuan per kilometre from 3 to 10 kilometres, and 3 yuan per kilometre beyond 10. Write the fare y as a piecewise expression in distance x and draw its graph. Is this machine a function? What happens at $x = 3$ and $x = 10$? *Hint: check that each input has one output. The graph has no break, but its slope changes suddenly; such points will later have a special name.*

The Door to the Next Chapter

This chapter has built a habit: turn relationships into graphs and view equations through intersections. Graphs have become our trusted eyes.

But recall one small detail. To draw the parabola, you plotted 9 points and connected them with a smooth curve. Why was that justified? You saw nine points roughly following a curved track and joined them. Perhaps the curve secretly bends between two points in a way you missed. Perhaps lines that seem to intersect at $(6, 17)$ measure as $(5.98, 17.03)$. Eyes and rulers reveal clues, but never guarantee a conclusion. That is the boundary between measurement and proof.

Next we return to mathematics older than functions: points, lines, angles, triangles, and circles. We will see how the ancient Greeks began with the impossibility of drawing a perfect straight line and built humanity's first great structure of proof. We will also ask why the same conclusion, such as the famous theorem about a right triangle's sides, deserves a dozen different proofs. Graphs let us see an answer; proofs let us be sure. Now let us meet the second.

Chapter 3 Elementary Geometry: From Measurement and Construction to Proof

In the first two chapters, symbols and equations compressed many concrete problems into models we could carry from one situation to another. Most belonged to the world of quantities: prices, speeds, and times. In this chapter we enter a different world—the world of shapes. You will see that humanity’s earliest geometrical knowledge arose from practical needs: surveying land, building houses, and observing the stars. Yet geometry’s most remarkable achievement was eventually to escape the limitations of rulers and ropes and become a subject whose conclusions could be guaranteed by reasoning alone. This was the first appearance of proof in the history of mathematics, and the first appearance of a theme that will echo throughout this book.

3.1 A Pyramid in the Shadows

About twenty-six hundred years ago, a traveller arrived in Egypt. Standing before an enormous pyramid, he asked a question that left the local priests bewildered: how tall is it?

Direct measurement was impossible. The pyramid’s sides were sloping walls of stone. No ladder could reach its tip, and no rope could be lowered from the top—the tools of the day could not reach a summit more than a hundred metres high. The traveller did not climb the pyramid. He asked only for a wooden pole, a length of rope, and a sunny day.

He planted the pole vertically in the ground and waited for the sunlight. The pole cast a shadow on the ground, and the pyramid cast a great shadow across the sand. He measured three quantities: the pole’s height, the length of its shadow, and the length of the pyramid’s shadow (from the edge of its base to the shadow’s tip, plus half the length of the base, because the centre of the base lies directly below the summit). Then a simple proportion gave him the pyramid’s height, reportedly with very little error.

The traveller was Thales of ancient Greece. Legend says that the priests were astonished, but do not rush to admire him just yet. A real mathematical question lies inside this story, and it is where our chapter begins.

3.2 What Justifies It, and What Does Not

Why was Thales’s method correct? His intuition was probably this: at the same moment, sunlight falls at the same angle, so the small triangle formed by the pole and its shadow and the

large triangle formed by the pyramid and its shadow are enlarged versions of the same shape. When shapes are the same, corresponding sides have the same ratio, so

$$\frac{\text{pyramid height}}{\text{pole height}} = \frac{\text{pyramid shadow}}{\text{pole shadow}}.$$

Knowing three of these four quantities lets us calculate the fourth. For instance, if the pole is 2 metres tall, its shadow is 3 metres long, and the pyramid’s shadow (measured from its centre) is 180 metres long, then the pyramid is $2 \times 180 \div 3 = 120$ metres tall.

This reasoning sounds natural. But pause and ask one more question, as a mathematician would: why does “the same shape” imply “proportional corresponding sides”? Measure a hundred triangles and it will seem true every time. Yet a gulf separates “the measurements agree” from “it must be true.”

Consider an easy trap. Draw a triangle, measure its three interior angles with a protractor, and add them. What do you get? Probably not exactly 180° : perhaps 179° or 181° . Draw another, measure again, and you are still a little off. You might say that the protractor is inaccurate and that the sum should theoretically be 180° . Good—but where did that “theoretically” come from? You have not measured every triangle in the world, or even a hundred million. Even if a hundred million all gave something close to 180° , what about the next one? A triangle with sides a light-year long? One too small to see?

This is the division of labour between measurement and proof. Measurement is a tremendous achievement: it helps us discover patterns and make conjectures, and engineers use it to build real bridges. But measurements always carry errors and can check only finitely many examples. Geometry is full of conclusions about infinitely many cases. “The interior angles of every triangle sum to 180° ” refers to every triangle, whatever its size or proportions. A ruler cannot handle “every”; reasoning can.

This also explains why the ancient Egyptians could survey land and build pyramids without leaving behind a concept of the theorem, whereas the ancient Greeks turned geometry into a science of proof. Measurement tells us approximately what is true; proof tells us why it must be true. Our goal in this chapter is to travel from “approximately” to “necessarily.”

3.3 From “They Look the Same” to Definitions

To replace measurement with reasoning, we must first turn vague everyday words into precise definitions. You may think you understand “point,” “line,” and “angle,” but try explaining them clearly to someone who has never seen them. It is harder than it sounds.

The ancient Greeks took a somewhat surprising approach: they admitted that the most basic concepts could not truly be defined, only stipulated. A point is a position without size, a line has length without width, and a straight line can extend indefinitely in both directions. These statements tell us less about what a point “is” than about the rules it obeys. For example, we agree that exactly one straight line passes through two distinct points, and that a line segment is the shortest path between two points. Such agreements are called axioms. They are the starting

points of reasoning, just as the moves of the horse and elephant must be agreed upon before a game of Chinese chess.

With points and lines, we can create our first real object of study: join three line segments end to end to enclose a figure, and we have a triangle. Triangles are the atoms of geometry. Every polygon can be divided into triangles, so understanding triangles comes close to understanding all of plane geometry.

Next come two crucial concepts, both obtained by unpacking the everyday word “same.”

Definition 3.1 (Congruence). Two figures are **congruent** if translations, rotations, and reflections can make them coincide exactly. Congruent triangles have equal corresponding sides and equal corresponding angles. We write $\triangle ABC \cong \triangle DEF$.

Definition 3.2 (Similarity). Two figures are **similar** if they have the same shape, though their sizes may differ. Two triangles are similar if and only if their corresponding angles are equal; their corresponding side lengths then share a common ratio, called the similarity ratio.

Congruence means “coinciding exactly”; similarity means “different scalings of the same shape.” Thales used similarity to measure the pyramid: the pole triangle and pyramid triangle are similar, and their similarity ratio is the ratio of the shadow lengths.

Terminology Card: Congruence

Intuition: figures cut from the same mould, fitting perfectly on top of one another after shifting, turning, or flipping. Formal definition: two figures that coincide exactly after translations, rotations, and reflections. Example: two identical triangular pieces of paper made with a photocopier are congruent.

Terminology Card: Similarity

Intuition: a photograph and its enlargement have the same shape but different sizes. Formal definition: two figures whose corresponding angles are equal and whose corresponding sides share a common ratio. Example: a triangular plot on a map is similar to the real triangular plot; the similarity ratio is the map’s scale.

Definitions tell us what “the same” means, but not how to recognize it. We often cannot place two triangles on top of each other—you cannot move a pyramid. Fortunately, mathematicians discovered that checking triangle congruence does not require comparing all six quantities (three sides and three angles). The right three are enough:

- Three corresponding sides are equal (side–side–side).
- Two corresponding sides and their included angle are equal (side–angle–side).
- Two corresponding angles and their included side are equal (angle–side–angle).

These criteria contain claims that can be proved. For side–angle–side, for example, the intuition is that fixing the lengths of two sides and their included angle determines the third side

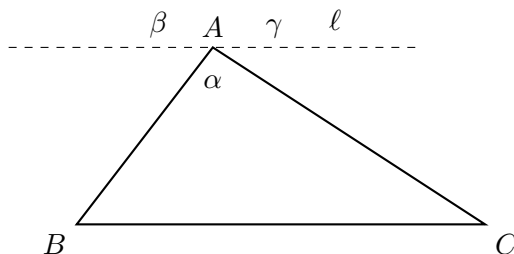
uniquely; the triangle has no room to change shape. Similarity has its own economical criterion: two equal corresponding angles suffice. Since the interior angle sum of a triangle is fixed (as we will prove shortly), fixing two angles fixes the third, and the shape is locked in place.

Counterexample Alert

“If two corresponding sides and the angle opposite one of them are equal, the triangles are congruent.” This sounds much like the earlier criteria, but it is false. Fix side AB , side AC , and angle B (notice that B is not the angle included between AB and AC). Point C must lie on the ray from B making the prescribed angle with BA , while also satisfying the prescribed length of AC . In other words, it is an intersection of that ray with an arc centred at A and having the given radius. The arc may meet the ray at **two** different points, producing one broad triangle and one narrow triangle that satisfy the same conditions but are not congruent. This trap reminds us that every word in a criterion matters: replace “included angle” with “opposite angle” and the conclusion collapses. It is also a natural place for proof by cases to enter: when the same conditions can produce two figures, each possibility must be checked separately.

Before continuing, let us supply the proof we owe. It is also our first important example of a geometrical proof technique: adding an auxiliary line.

Theorem 3.3 (The Interior Angle Sum of a Triangle). *The three interior angles of any triangle sum to 180° .*



Proof. Let the triangle be $\triangle ABC$. Through vertex A , draw a line ℓ parallel to side BC (the dashed line in the figure). This line was not in the original figure: we **added** it to help our reasoning, so it is called an auxiliary line.

Side AB crosses both parallel lines ℓ and BC , so the angle β formed by AB and ℓ to the left of A equals angle B (alternate interior angles are equal). Similarly, the angle γ formed by AC and ℓ to the right of A equals angle C . At A , however, β , angle A itself (labelled α), and γ fit together into a straight angle, or 180° . Therefore

$$\angle A + \angle B + \angle C = 180^\circ.$$

Notice that this argument measures no angle. It works for every triangle, whatever its side lengths. $\square$

The move worth savouring is “draw a parallel line.” It was not originally there; we invited it in to help. Once it arrives, angles B and C are carried over beside A . The three angles stand

shoulder to shoulder along a straight line, and the conclusion emerges. Auxiliary lines are among the most creative parts of geometrical proof: there is no fixed recipe, only an understanding of the figure and a little luck. You will see more of their performances later.

3.4 One Theorem, Two Reasons

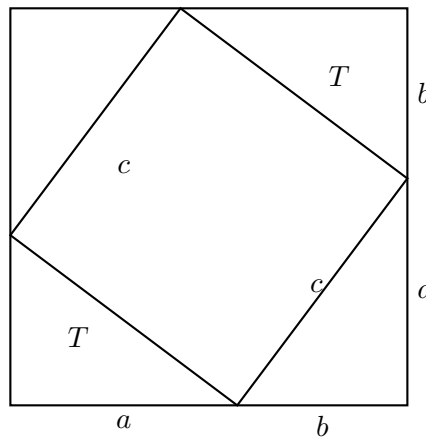
Now for this chapter's leading character. Right triangles are a special family: one angle is exactly 90° . The two sides enclosing the right angle are called the legs; the longest side, opposite the right angle, is the hypotenuse. They satisfy what may be the world's most famous theorem.

Theorem 3.4 (The Pythagorean Theorem). *The sum of the squares of the legs of a right triangle equals the square of its hypotenuse. If the legs have lengths a and b , and the hypotenuse has length c , then*

$$a^2 + b^2 = c^2.$$

Try a few numbers. 3, 4, 5: $9 + 16 = 25$, correct. 5, 12, 13: $25 + 144 = 169$, correct. The ancient Egyptians used 3, 4, 5 to make right angles with ropes: mark twelve equally spaced knots on a rope and form a triangle with sides of 3, 4, and 5 intervals, and its largest angle is a right angle. But the same issue returns: checking ten thousand triples is not a proof. Here are two entirely different proofs—one conclusion, two different reasons. That itself is an important mathematical fact: a theorem belongs to its conclusion, not exclusively to any particular proof; another proof gives another way to understand it.

Proof 1: Rearranging Areas.



Proof. Take four congruent right triangles with legs a and b and hypotenuse c . Arrange them inside a large square of side $a + b$, one triangle in each corner with its hypotenuse facing inward. The four hypotenuses enclose a central quadrilateral.

First confirm that the central region really is a square. All four sides have length c , so we need only check one angle. At a joining point along the bottom side of the large square, an acute angle α of one triangle, an acute angle β of another, and an angle of the central quadrilateral form a straight angle of 180° . But $\alpha + \beta = 90^\circ$, since they are the two acute angles of the same

right triangle. The central angle is therefore 90° . A rhombus with four equal sides and one right angle is a square, with area c^2 .

Now calculate the area of the same large square in two ways. Directly it is $(a + b)^2$. By pieces it is the four triangles plus the central square. Each triangle has area $\frac{1}{2}ab$, so

$$(a + b)^2 = 4 \times \frac{1}{2}ab + c^2 = 2ab + c^2.$$

Expanding the left side gives $a^2 + 2ab + b^2 = 2ab + c^2$. Subtract $2ab$ from both sides to obtain $a^2 + b^2 = c^2$. $\square$

What is the heart of this proof? **Area does not change when pieces are cut apart and rearranged.** The figure changes shape, but its area stays fixed: area is an invariant of cutting and rearrangement. Finding a quantity that remains unchanged amid change is often the key to a proof. This idea will return repeatedly.

Proof 2: Using Similarity.

Proof. Let $\triangle ABC$ be a right triangle with its right angle at C and hypotenuse AB . Draw the perpendicular from C to AB , meeting it at D —another auxiliary line. It splits the original triangle into two smaller triangles, $\triangle ACD$ and $\triangle CBD$.

Compare $\triangle ACD$ with the original $\triangle ABC$: they share angle A , and each has a right angle (ADC and ACB). Two corresponding angles are equal, so the triangles are similar. Likewise, $\triangle CBD$ is similar to $\triangle ABC$. Write down the ratios of corresponding sides. Set $AB = c$, $AC = b$, $BC = a$, and write AD as x and BD as y , noting that $x + y = c$. From $\triangle ACD \sim \triangle ABC$ (hypotenuse corresponding to hypotenuse, and leg AC to leg AD),

$$\frac{AC}{AB} = \frac{AD}{AC}, \quad \text{that is, } b^2 = cx.$$

Similarly, $\triangle CBD \sim \triangle ABC$ gives $a^2 = cy$. Add the two equations:

$$a^2 + b^2 = cx + cy = c(x + y) = c^2.$$

Here we used the basic property that corresponding sides of similar triangles are proportional. Its rigorous proof needs more preparation, so in this chapter we accept it as a tool, just as we agree on the rules before playing a game. $\square$

Compare the flavour of these two reasons. The first makes the theorem a story about area: a^2 , b^2 , and c^2 are no longer abstract squares of numbers but three tangible square regions that we could cut out and weigh. The theorem says that the two smaller squares can fit together to make the large square on the hypotenuse. The second makes it a story about shape: a right triangle contains two smaller versions of itself, and that self-similar structure forces the equality. Neither proof measures a length, and both work for every right triangle.

Incidentally, there may be hundreds of proofs of this theorem: some use trapezoid areas, others circles, and still others pure algebraic completion of squares or another arrangement of the hypotenuse diagram (see the exercises). Mathematicians still occasionally publish new proofs, because each new reason may illuminate a new aspect of the theorem and prove useful elsewhere.

3.5 Invariants, Cases, and Circles

Our two proofs of the Pythagorean theorem used two general weapons: invariants and auxiliary lines. This section puts a third on the table and then sends them all to work on circles.

Proof by cases means something simple: when a statement's conditions allow several different possibilities, separate them completely and solve each one. In the earlier side-side-angle counterexample, an arc and a ray may intersect at zero, one, or two points. These cases yield nonexistence, uniqueness, or two solutions; without separating them, we cannot describe the outcome clearly. Another example is deciding whether three segments can form a triangle. We need only the following fact, which follows directly from “a segment is the shortest path between two points.”

Corollary 3.5 (The Triangle Inequality). *The sum of any two sides of a triangle is greater than the third side. Conversely, if the sum of the two shorter of three segments exceeds the longest, they can form a triangle.*

Notice that this corollary already embodies the idea of cases: instead of checking all three pairs, we need check only the crucial case of whether the two shortest sides together exceed the longest.

An **invariant** reverses the question: while the figure changes, what stays fixed? Area stays fixed under cutting and rearrangement; angles and shape stay fixed under enlargement and reduction (the whole point of similarity). Circles give an even more beautiful example.

Define a circle as all points at a fixed distance (the radius r) from a fixed point (the centre), and consider its circumference. Measure a few circles with string—a coin, a cup rim, a round table—and you will find that circumference divided by diameter is always about 3.14. Again, measurement suggests a conjecture. Why must it hold? Because any two circles are similar: enlarge a circle about any point and it remains a circle. Ratios of corresponding lengths in similar figures are fixed, so the ratio of circumference to diameter is the same for every circle. This number is π . It is irrational, with a decimal expansion that never ends; 3.14 and 3.14159 are only approximations. Ancient mathematicians found an ingenious way to approach it: inscribe regular polygons inside a circle. As the number of sides grows, their perimeters approach the circumference. Bounds from regular 96-gons place π between 3.1410 and 3.1427—an achievement of Archimedes' era, and an early victory of approximating the infinite with the finite. We will return to this picture when discussing limits in Chapter 11.

Terminology Card: Invariant

Intuition: while everything else changes, it refuses to change, making it a useful yardstick and witness. Formal definition: a quantity or property preserved by a specified operation or transformation. Example: area is unchanged when paper is cut and rearranged, so calculating the same area in two arrangements can prove the Pythagorean theorem.

Area itself deserves a definition. Defining it rigorously is not easy (integration in Chapter 12 will truly address it), but at the middle-school level we can agree that a square of side 1 has

area 1, and that area is additive and unchanged by cutting and rearrangement. It follows that a rectangle has area length times width; a parallelogram has area base times height (move a right triangle from one side to the other to make a rectangle); a triangle has half the product of base and height (two congruent triangles form a parallelogram); and a circle has area πr^2 (cut it into many small sectors and alternate them to approximate a rectangle of length half the circumference, πr , and width r —another preview of approximation). Behind every formula stands a rearrangement, not a measurement.

Mathematical Toolbox

The new tools in this chapter:

- **Axioms and definitions:** reasoning begins with rules to obey; distinguish congruence and similarity within the everyday idea of sameness.
- **Congruence and similarity criteria:** the right two or three conditions determine the shape without overlaying the figures.
- **Auxiliary lines:** invite extra lines to help; parallel lines carry angles, and perpendicular lines create right angles and similarity.
- **Proof by cases:** when conditions allow several possibilities, separate them completely and handle each; overlooking cases causes the side-side-angle trap.
- **Invariants:** area under rearrangement, angles under scaling, and π shared by all circles—a fixed quantity gives us a grip on change.
- **Measurement and proof:** measurement discovers conjectures and supplies examples; proof guarantees all cases.

3.6 Another Perspective: Triangulation and the Stars

Geometry’s most spectacular applications rest on its most abstract conclusions. The ancient Greeks wanted to know the distances to the Moon and Sun. They had no rockets, only similar triangles and angles.

Their method is still used today and is called triangulation. To measure an inaccessible point C , first choose two points A and B and measure the baseline AB . Then measure the angles that the sight lines AC and BC make with that baseline. Two angles and the included side completely determine $\triangle ABC$. Although C may be far away, the triangle’s shape is fixed. Draw a scaled-down copy, or calculate using similarity and proportions, and out come AC and BC . A surveyor measuring a tower across a river, a mapping team charting a country, and astronomers observing the Moon from two stations on Earth all use the same method. Later, people even tried using Earth itself as a baseline to measure planetary distances.

Notice the premise: we trust statements such as “two angles and their included side uniquely determine a triangle” and “corresponding sides of similar triangles are proportional” to hold

equally for triangles on the ground and triangles as large as the distance to the Moon. If these were merely observations from measuring many small triangles, extending them to astronomical scales would be a huge gamble. Because they are theorems, engineers and astronomers dare to stake their work and lives on them. That is the practical value of proof: it gives knowledge a licence to operate at any scale.

Connections

Looking back: Chapter 1's symbols and proportions became similarity ratios and area formulas here; Chapter 2's equations let us solve for unknown sides in $a^2 + b^2 = c^2$. Looking ahead: throughout this chapter we drew on an unnamed plane, where points had positions but no addresses. Next chapter we will give each point an identity card bearing two numbers—its coordinates. Lines and circles will become equations; geometry problems can be calculated algebraically, and algebraic expressions can be drawn as shapes. You are about to see this chapter's theorems, especially the Pythagorean theorem, become one of the busiest formulas in the coordinate world: the distance formula.

3.7 Try It Yourself

One-Minute Experiment

Experiment 1: A Right Angle from Knots. Find a soft rope and mark 12 equally spaced knots with a marker. Form a triangle whose sides contain 3, 4, and 5 intervals, pull it taut, and compare its largest angle with the right angle of a set square. Then try 5, 12, and 13 intervals (requiring 30 intervals altogether). Think: why do you believe the taut rope produces a right angle? Is that a conclusion from measurement or from a theorem?

Experiment 2: A Paper Proof of Pythagoras. Draw a right triangle on squared paper with legs of 3 and 4 squares, then make three copies (or draw three congruent triangles). Arrange the four triangles inside a 7×7 square as in Proof 1, and count the area of the empty square in the middle. $49 - 4 \times 6 = 25$, exactly 5×5 . Repeat with different legs, such as 2 and 5 squares, and check whether the central area is $2^2 + 5^2 = 29$ squares.

Experiment 3: Approaching π . Use a compass to draw a circle of radius 5 centimetres. With a ruler, construct an inscribed regular hexagon and then a regular dodecagon, bisecting each arc. Measure the polygon perimeters, divide by the diameter of 10 centimetres, and record how the results approach 3.14. Remember: however carefully you measure, this remains measurement; exact properties of π require reasoning.

Questions to Think About

Observation

1. Draw three triangles with very different shapes: one tall and thin, one broad and flat, and one nearly equilateral. Measure and add each one's interior angles, recording each error.

Are the errors sometimes positive and sometimes negative, or always in one direction? Hint: consider the smallest divisions of your protractor and your reading habits. Where does the error come from?

2. Measure the shadow of the same vertical flagpole at different times of day. Does “pole height divided by shadow length” change? Hint: when would this ratio be the same? What changes, and what is invariant?

Proof

1. There is another arrangement of the hypotenuse diagram: put four congruent right triangles with legs a and b , where $a < b$, inside a large square of side c , with their hypotenuses along the square’s four sides and a small empty square in the middle. Hint: the small square has side $b - a$. Calculate the large square’s area in two ways.
2. Prove that the two base angles of an isosceles triangle are equal. Hint: bisect the vertex angle (or draw the altitude to the base), dividing the triangle into two pieces, and use a congruence criterion to show that the pieces are congruent.

Challenge

1. Prove that if one side of a triangle inscribed in a circle is a diameter, the angle opposite it is a right angle. Hint: join the centre to that angle’s vertex. The circle is divided into two isosceles triangles; give their four base angles two names and use the angle sum.
2. More than two thousand years ago, someone calculated Earth’s circumference using these facts: at noon on the summer solstice, sunlight reached the bottom of a well in one city, producing no shadow; at the same time in a city to the north, a pole’s shadow indicated an angular deviation of about one-fiftieth of a full turn; and the distance between the cities was known. Hint: treat sunlight as parallel lines. What figure do the two cities and Earth’s centre form? Which central angle equals the shadow’s angular deviation? What assumptions do you need, and which is only approximately true?

Keep Climbing

More than two thousand years ago, this chapter’s picture was organized systematically in a book called the *Elements*: from five axioms it derived hundreds of theorems. This axiomatic style remains a standard currency of mathematics. Further ahead you will encounter Euclid’s fifth postulate (the parallel postulate) and non-Euclidean geometry—replace it, and triangle angles no longer sum to 180° ; projective geometry, the study of perspective and shadows themselves; trigonometry, which distils similar triangles into the sine and cosine functions; analytic geometry, next chapter’s starting point, which tames curves with equations; and modern automated geometry and computer-assisted proof, where machines too can act as geometers.

The Door to the Next Chapter

This chapter taught us to prove things on a blank sheet: add auxiliary lines, find congruence, use similarity, and identify invariants. But you may have noticed a quiet inconvenience—points on the page have no names. We say “the foot of the perpendicular on the hypotenuse” or “three centimetres to the right of the centre,” relying on gestures and descriptions. What if a geometry problem has a hundred points and twenty curves? What if you want a computer with no eyes to draw figures or test conjectures? It cannot understand “the point on the left.” It understands numbers.

So a question almost childlike in its simplicity arises: can every point in the plane have a name made of numbers? If so, the distance between two points becomes an arithmetic problem, lines and circles become equations, and every theorem in this chapter gains a new language. Next chapter we will build that bridge, starting from the Pythagorean theorem and fitting the plane with coordinates.

Chapter 4 The Coordinate Bridge: Connecting Algebra and Geometry

Throughout the first three chapters, we have really inhabited two separate worlds. The algebraic world contains numbers, symbols, equations, and functions; it cares about relationships: what equals what, and what changes with what. The geometrical world contains points, lines, angles, triangles, and circles; it cares about shapes: what is congruent to what, which length is greater, and which lines are parallel. Each world is fascinating, but a river separates them: algebraic expressions do not reveal shapes, and geometrical figures cannot be calculated directly.

This chapter builds a bridge across that river. Its name is coordinates. Once it is built, you will discover that an equation can become a curve and a curve can become an equation; a geometry problem can move to the algebraic side for calculation, and an algebra problem can move to the geometrical side for a clearer view. Mathematicians have a formal name for this method: analytic geometry. Before learning that name, though, let us begin with an argument.

4.1 An Argument over Battleship

A game often called Battleship is played on squared paper. Each of two players draws a few ships on a private grid—really just groups of connected squares—and they take turns calling out positions to guess where the other player’s ships are hidden. A correct guess is a hit.

Xiaoming and Xiaohong are playing. Xiaoming calls, “Third row, fifth column!” Xiaohong looks at her paper and says, “Miss.” Xiaoming protests: “Impossible! Your ship is right there!” Only when they put their sheets side by side do they discover the problem: Xiaoming counts rows from top to bottom, while Xiaohong counts from bottom to top. His “third row” is her third row from the bottom. They are not talking about the same square at all, so of course the argument goes nowhere.

At the root of this dispute lies a question much bigger than the game: **how can we describe a position precisely?** Everyday life asks it constantly. A cinema ticket says “row 7, seat 12,” leading you to exactly one seat. A delivery rider appears as a moving dot on your phone’s map, and the system tracks his position with a set of numbers. A globe has longitude and latitude lines; Beijing lies approximately at the intersection of 116 degrees east and 40 degrees north. These methods share a feature: they pin down position with numbers instead of vague phrases like “over there” or “a little to the left.”

Battleship uses grid squares, cinema tickets use rows and seat numbers, and globes use longitude and latitude. All are relatives of this chapter's main character. We will go further: instead of merely numbering squares, we will give **every** point in the plane its own unique numerical address.

4.2 A Problem That Gets Us Stuck

Before inventing tools, let us feel why this problem is difficult.

Imagine a flagpole in your school playground. You stand somewhere nearby and want to tell a friend who cannot see you exactly where you are. First attempt: "I'm east of the flagpole." No good—a huge region lies east of it, and your friend still cannot find you. Second attempt: "I'm quite far east of the flagpole, and a little north." How far is "quite far"? How much is "a little"? Such language may suffice in daily life, but it is unacceptable in mathematics: two people's interpretations of "quite far" might differ by tens of metres.

Bring in a tool, then: measure with a tape. Third attempt: "I'm exactly 5 metres from the flagpole." Precise at last, but still insufficient. There is not just one point 5 metres from the pole, but an entire ring: every point on the circle centred at the pole with radius 5 metres fits. Your friend faces infinitely many possible positions.

"Then I'll add a condition: I'm 5 metres from the flagpole and 8 metres from the goal in the corner." Much better: draw two circles, and they have at most two intersections. But "at most two" sometimes means exactly two. You still have to add "I'm at the more northerly intersection" to finish the description. After all that, we are back to language like "more northerly."

Describing position through distances seems always to leave something unresolved. Mathematicians want a clean convention: **each position corresponds to exactly one set of numbers, and each set of numbers to exactly one position**. No extras and no omissions. This convention did not fall from the sky; people invented it step by step after wrestling long enough with problems like these. The first step begins with a line.

4.3 One Straight Line Holds All the Numbers

You have probably seen a number line, but let us reinvent it to see exactly what difficulty it resolves.

A thermometer is a vertical number line: 0 is marked on the tube, temperatures above zero go upward, those below zero downward, and each division represents one degree. Lift buttons are another relative: basement level 2 is labelled -2 , while floors 1 and 2 run upward above ground. From experiences like these, people abstracted a method. Draw a horizontal straight line and choose a point called the **origin**, representing 0. Declare rightward to be the **positive direction**. Choose a **unit length**, mark $1, 2, 3, \dots$ successively to the right of the origin, and $-1, -2, -3, \dots$ to the left. Fractions and decimals have homes too: $\frac{1}{2}$ sits halfway between 0 and 1, and -1.5 halfway between -1 and -2 .

Terminology Card: Number Line

Intuition: a straight line marked with a scale, giving each number a definite seat.

Formal definition: a straight line with a specified origin, positive direction, and unit length.

Example: the point representing -2 is two units left of the origin; the point representing 3.5 is three and a half units to its right.

The number line's main achievement is not marking a few integers, but establishing a **one-to-one correspondence**: every real number corresponds to exactly one point on the line, and every point to exactly one real number. Nothing is duplicated or omitted. A number and a point are now two descriptions of the same thing—our first little bridge, though only in one dimension.

With a number line, distances in one dimension become easy to calculate. If two points represent a and b , their distance is $|b - a|$. Why the absolute value? Distance does not care about direction. From -2 to 5 , we count 7 units rightward; from 5 to -2 , we count the same 7 units leftward. $5 - (-2) = 7$, while $(-2) - 5 = -7$. The sign depends on where we start, but the distance is the same, so the absolute value discards direction and keeps magnitude. This little formula—the absolute difference of coordinates—will soon grow into a more famous formula in the plane.

4.4 An Address for Every Point in the Plane

One dimension is settled. What about a two-dimensional plane? The answer is astonishingly simple: if one number line is not enough, use two.

A story about this idea's birth tells of the French mathematician Descartes lying ill in bed, watching a fly crawl across the ceiling. Suddenly he realized that recording its distances from two walls would identify any position on the ceiling. The tale was probably embellished by later generations and is unreliable as history, but it captures the crucial idea vividly: **two numbers suffice for a position in the plane.**

Place two number lines perpendicular to one another with their origins coinciding. One is horizontal, the **horizontal axis** (usually the x -axis), positive to the right; the other is vertical, the **vertical axis** (usually the y -axis), positive upward. This gives a **Cartesian coordinate system in the plane**. Now choose any point P . Drop a perpendicular from it to the horizontal axis, and suppose its foot corresponds to 3. Drop another to the vertical axis, with its foot corresponding to 2. We say that the **coordinates** of P are $(3, 2)$: 3 is the horizontal coordinate, and 2 the vertical coordinate. Conversely, given $(3, 2)$, find 3 on the horizontal axis and travel vertically, then find 2 on the vertical axis and travel horizontally. The two paths meet at exactly one point. Once again, points and number pairs correspond one to one.

Terminology Card: Coordinates

Intuition: a point's numerical address in the plane, consisting of an ordered pair of numbers.

Formal definition: in a Cartesian coordinate system, drop perpendiculars from a point P to the two axes. The corresponding numbers x and y , taken in order as (x, y) , are called the coordinates of P .

Example: “row 7, seat 12” and “row 12, seat 7” name different cinema seats. Likewise, $(7, 12)$ and $(12, 7)$ are different points in the plane; the order of the pair matters.

The two axes divide the plane into four regions. In the upper right both coordinates are positive; this is the first quadrant. Proceeding anticlockwise gives the second, third, and fourth quadrants. Points on the axes belong to no quadrant; they are boundary points. You need not memorize these names mechanically: a couple of drawings will make them familiar.

Notice what we have accomplished. Points are the smallest inhabitants of the geometrical world; ordered pairs are ordinary inhabitants of the algebraic world. A coordinate system puts them in exact one-to-one correspondence. This is a complete translation mechanism, not just a resemblance in a few examples. From now on, any statement about points can in principle become a statement about numbers, and conversely. The bridge's ends are aligned with the riverbanks. Now it is time to carry the first cargo across.

4.5 The Distance Formula: Pythagoras on Coordinate Paper

Our first cargo is geometry's most frequently used quantity: distance.

Chapter 3 taught us the Pythagorean theorem: the sum of the squares of a right triangle's legs equals the square of its hypotenuse. That was a purely geometrical statement about a figure. With coordinates, points become pairs of numbers. We can now ask: if A has coordinates $(1, 2)$ and B has coordinates $(4, 6)$, how far apart are A and B ?

Staring at the two pairs does not reveal the distance. The trick is to complete a right triangle: draw a horizontal line through A and a vertical line through B . They meet at C , whose coordinates are $(4, 2)$ —as high as A , as far right as B . In triangle ACB , angle C is right; the horizontal segment AC has length $4 - 1 = 3$, and the vertical segment BC has length $6 - 2 = 4$. Apply the Pythagorean theorem:

$$AB = \sqrt{AC^2 + BC^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

The distance is 5. We did just one thing: turned coordinate differences into two legs and called on Pythagoras. Replace the specific values with symbols, and the same reasoning works for any two points.

Theorem 4.1 (The Distance Formula for Two Points). *In a Cartesian coordinate system, the distance between $A(x_1, y_1)$ and $B(x_2, y_2)$ is*

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

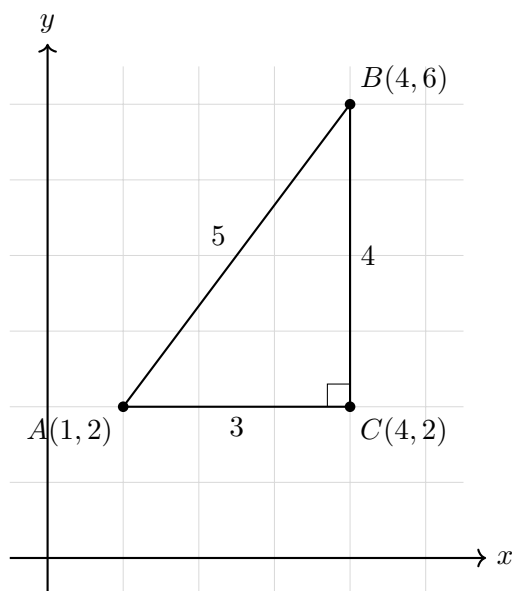
Proof. Consider two cases.

Case 1: $x_1 \neq x_2$ and $y_1 \neq y_2$. Choose $C(x_2, y_1)$, with the same vertical coordinate as A and the same horizontal coordinate as B . Then AC is horizontal and BC vertical, so they are perpendicular and ACB is a right triangle with its right angle at C . By the distance rule on the number line, $AC = |x_2 - x_1|$ and $BC = |y_2 - y_1|$. Applying the Pythagorean theorem to ACB ,

$$AB = \sqrt{AC^2 + BC^2} = \sqrt{|x_2 - x_1|^2 + |y_2 - y_1|^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2},$$

where the last step uses the fact that a number and its absolute value have the same square, so the absolute value signs can be removed.

Case 2: $x_1 = x_2$ or $y_1 = y_2$. Then A and B lie on the same vertical or horizontal line, and their distance is the absolute coordinate difference. One term in the formula is 0, so it reduces to exactly that result. Together the two cases establish the conclusion. $\square$



This formula deserves another look. On the left, AB is a geometrical object—a segment’s length. On the right is an algebraic object—arithmetic with coordinates and a square root. One formula, two languages: the first vehicle has crossed our coordinate bridge.

Counterexample Alert

“ A is at $(1, 2)$ and B at $(4, 6)$. The horizontal difference is 3 and the vertical difference 4, so the distance is $3 + 4 = 7$.” Wrong. $3 + 4$ is the length of a broken path that goes horizontally and then vertically, like a taxi following a grid of streets. The distance between two points means the straight segment, the diagonal shortcut. The Pythagorean theorem gives $\sqrt{3^2 + 4^2} = 5$, which is indeed less than 7, in agreement with the shortest-path property of a segment. Adding coordinate differences confuses road mileage with straight-line distance.

The distance formula has a useful by-product. What are the coordinates of the midpoint M of AB ? The intuitive answer is to average each pair of coordinates, and that intuition is correct:

Corollary 4.2 (The Midpoint Formula). *The midpoint M of the segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ has coordinates*

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right).$$

Think of it this way: travelling from A to B changes the horizontal coordinate by $x_2 - x_1$. Halfway along, the change is half as large, so the midpoint's horizontal coordinate is $x_1 + \frac{x_2 - x_1}{2} = \frac{x_1 + x_2}{2}$. The vertical coordinate works the same way. For a rigorous proof, substitute the coordinates of M into the distance formula and verify $AM = MB = \frac{1}{2}AB$; try it yourself. For example, the midpoint of $A(1, 2)$ and $B(4, 6)$ is $(\frac{5}{2}, 4)$, or $(2.5, 4)$, which lies exactly halfway along the segment in the figure.

4.6 An Equation Is a Shape's Identity Card

So far, we have started with figures and calculated coordinates. Now reverse the direction and ask something more playful: **if we impose a rule on points in the plane, what shape do the obedient points form?**

For instance: “the distance from the origin equals 3.” Your geometrical intuition may already shout, “A circle!” That is its definition. But pretend we do not know, and work through the coordinates. If an obedient point has coordinates (x, y) , the distance formula gives its distance from $(0, 0)$, translating our rule into algebra:

$$\sqrt{x^2 + y^2} = 3, \quad \text{squaring both sides gives} \quad x^2 + y^2 = 9.$$

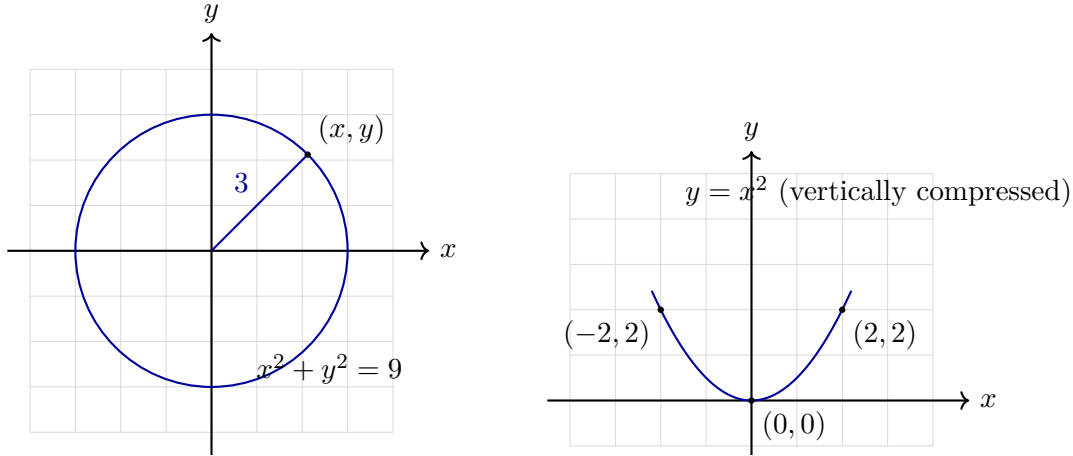
A geometrical condition—fixed distance from a point—has become an algebraic equation, $x^2 + y^2 = 9$. This equation acts like a sieve. Substitute $(3, 0)$ and get $9 + 0 = 9$: it passes and lies on the circle. Substitute $(1, 2)$ and get $1 + 4 = 5 \neq 9$: it fails and lies off the circle. Substitute $(\sqrt{5}, 2)$ and get $5 + 4 = 9$: it lies on the circle. Every one of the countless points in the plane either passes or fails, and the passing points together form exactly the circle of radius 3 centred at the origin, with no extras or omissions.

Definition 4.3 (The Equation of a Curve and Its Locus). If an equation $F(x, y) = 0$ and a curve correspond in both directions—every point on the curve has coordinates satisfying the equation, and every point whose coordinates satisfy the equation lies on the curve—then the equation is called an equation of the curve, and the curve is called the graph (or locus) of the equation.

A locus is an evocative idea: a point obeying a rule moves around the plane, and its trace is the locus. The rule written algebraically is an equation; the trace drawn geometrically is a curve. Here is a translation table:

Algebra: equation	Geometry: figure
$y = 2x - 1$	A line (slope 2, through $(0, -1)$)
$x^2 + y^2 = 9$	A circle of radius 3 centred at the origin
$y = x^2$	An upward-opening parabola with vertex at the origin

You met the line in Chapter 2: the graph of a linear function $y = kx + b$ is a straight line. Now you have a new way to say it: the locus of the linear equation in two variables $y = kx + b$ is a line. The parabola is worth checking by hand. Set $x = 0, \pm 1, \pm 2$ to obtain $y = 0, 1, 1, 4, 4$, plot the five points, and join them with a smooth curve to get that famous bowl shape.



Example 4.4. Determine whether $P(2, \sqrt{5})$ lies on the circle $x^2 + y^2 = 9$, and explain whether it lies inside or outside it.

Substitute into the left side: $2^2 + (\sqrt{5})^2 = 4 + 5 = 9$, exactly the right side, so P lies on the circle. Now consider $Q(1, 1)$: $1^2 + 1^2 = 2 < 9$, so its distance from the origin is $\sqrt{2}$, less than the radius 3, and it lies inside. In general, points with $x^2 + y^2 < 9$ lie inside, and those with a value greater than 9 lie outside. An equality identifies the curve, and two inequalities identify its two sides, all with the distance formula working behind the scenes.

One-Minute Experiment

Take coordinate paper, or draw a grid yourself, and mark $A(0, 0)$ and $B(4, 0)$. Look for points equidistant from A and B . Start by measuring: $(2, 0)$ is an obvious example; $(2, 1)$ is $\sqrt{5}$ from both, so it also qualifies. What about $(2, 3)$ and $(2, -2)$? Plot several such points and see what they form: a vertical line, precisely the perpendicular bisector of AB . Now verify it algebraically. If (x, y) is equidistant from A and B , then $\sqrt{x^2 + y^2} = \sqrt{(x - 4)^2 + y^2}$. Squaring and simplifying gives $x^2 = (x - 4)^2$; expanding gives $8x = 16$, or $x = 2$. Geometry finds the perpendicular bisector by construction; algebra produces it in two lines of simplification.

4.7 Translation 1: Solving Geometry with Algebra

Our bridge carries traffic both ways. First travel from geometry to algebra with a classic problem:

Problem (the Triangle Midsegment Theorem): the segment joining the midpoints of two sides of a triangle is parallel to the third side and half its length.

In Chapter 3, the standard approach to such a problem was to add auxiliary lines and find congruent triangles. That takes some inspiration, which does not always arrive on time.

Coordinates offer a route with a different temperament: patience is enough. Let us prove the theorem fully with coordinates.

Let the triangle's vertices be $A(x_1, y_1)$, $B(x_2, y_2)$, and $C(x_3, y_3)$. We deliberately avoid specific numbers because the theorem concerns **all** triangles; the coordinate variables represent arbitrary choices.

Step 1: translate what is given. Let M be the midpoint of AB and N the midpoint of AC . The midpoint formula gives

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right), \quad N\left(\frac{x_1 + x_3}{2}, \frac{y_1 + y_3}{2}\right).$$

Step 2: translate the “half” part of the conclusion. Calculate MN using the distance formula. The horizontal coordinate difference is

$$\frac{x_1 + x_3}{2} - \frac{x_1 + x_2}{2} = \frac{x_3 - x_2}{2},$$

and the vertical difference is similarly $\frac{y_3 - y_2}{2}$. Thus

$$MN = \sqrt{\left(\frac{x_3 - x_2}{2}\right)^2 + \left(\frac{y_3 - y_2}{2}\right)^2} = \frac{1}{2} \sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2} = \frac{1}{2} BC.$$

We have proved the half-length claim. Every step merely combines fractions, extracts a factor of $\frac{1}{2}$, or applies the distance formula; none needs a flash of inspiration.

Step 3: translate “parallel.” What does parallelism mean in coordinates? Look at direction. From M to N , the horizontal change is $\frac{x_3 - x_2}{2}$ and the vertical change $\frac{y_3 - y_2}{2}$. From B to C , they are $x_3 - x_2$ and $y_3 - y_2$. Each change in the first journey is exactly half its counterpart in the second. The proportions of horizontal and vertical steps are identical, so the directions agree and MN is parallel to BC . (If $x_3 = x_2$, both horizontal changes are 0, making both segments vertical and still parallel.) The theorem is proved.

Reflect on this method. A geometrical proof is like a detective, needing a flash of insight to discover the auxiliary line. A coordinate proof is like a mill: feed figures in to get coordinates, feed conclusions in to get equations, and grind steadily until a proof comes out. The mill is not romantic, but it never misses work. Coordinates also offer an advantage unavailable to the purely geometrical approach: **they handle whole batches**. Nearly the same routine works for many theorems, from the concurrence of three medians to diagonals bisecting one another. Algebra's spirit of solving a whole class of problems at once has crossed the bridge into geometry.

4.8 Translation 2: Solving Algebra with Geometry

Now send a vehicle in the other direction. Consider a problem that makes algebra frown:

Problem: for what value of x does

$$f(x) = \sqrt{x^2 + 9} + \sqrt{(6 - x)^2 + 16}$$

attain its minimum, and what is that minimum?

Pure algebra makes this awkward: two square roots enclose squares, completing the square does not finish the job, and squaring makes things more complicated. The algebraic tools available to a middle-school student are largely powerless. But after the previous section, a thought may occur: those radicals look remarkably like distance formulas.

The quantity $\sqrt{x^2 + 9} = \sqrt{(x - 0)^2 + (0 - 3)^2}$ measures the distance from $(x, 0)$ to $(0, 3)$. Likewise, $\sqrt{(6 - x)^2 + 16} = \sqrt{(x - 6)^2 + (0 - (-4))^2}$ measures the distance from $(x, 0)$ to $(6, -4)$. So the expression says: choose $P(x, 0)$ on the x -axis; its total distance to $A(0, 3)$ and $B(6, -4)$ is $PA + PB$. Our problem becomes:

The Translated Geometry Problem: a point P moves along the x -axis, with A above the axis and B below. Find the minimum of $PA + PB$.

Geometrically, the answer is almost immediate: a segment is the shortest path between two points. Since A and B lie on opposite sides of the x -axis, the segment joining A and B must cross the x -axis. Its intersection is the desired P . There, $PA + PB$ equals AB ; at any other position of P , $PA + PB$ is a broken path and therefore longer. Thus the minimum is

$$AB = \sqrt{(6 - 0)^2 + (-4 - 3)^2} = \sqrt{36 + 49} = \sqrt{85}.$$

$\sqrt{85}$ is approximately 9.22. To find x as well, follow AB from $(0, 3)$ to $(6, -4)$: it moves 6 units horizontally while dropping 7 vertically. Reaching the x -axis means dropping from 3 to 0, or $\frac{3}{7}$ of the total drop, so the horizontal journey is also $\frac{3}{7}$ of the total: $x = 6 \times \frac{3}{7} = \frac{18}{7}$. Substitute $x = \frac{18}{7}$ into the original expression to check that the two radicals sum to exactly $\sqrt{85}$.

Pause to appreciate what happened. We **did not calculate the algebraic expression directly**. We recognized its geometrical form and invoked a principle a three-year-old can understand: a straight path is shorter than a bent one. A forbidding algebraic minimum became an obvious geometrical fact. This is translation's second use: when one language obscures a problem, another may reveal its answer at once.

Connections

This kind of translation is no armchair game. Navigation software performs this chapter's demonstrations constantly: satellites supply latitude and longitude (coordinates), streets become lines in a plane, and the shortest route becomes a problem about distances and broken paths. A game character jumps along a parabola because a program repeatedly evaluates equations like $y = x^2$. A doctor's CT image is essentially a machine reconstructing a picture from numerical density information at points in a cross-section. Looking across the book, Chapter 2's function graphs first introduced equations becoming curves, and this chapter made that a systematic method. Chapter 7's vectors will turn translations and rotations of points into algebra too. And calculus's central activity—studying the bending and changing of curves—rests at every step on this coordinate bridge.

4.9 Setting Points in Motion: Parametric Equations

So far, equations describe static pictures: which points lie on a curve and which do not. But real-world points are almost always moving: a second hand's tip circles a clock, a thrown basketball traces an arc, and a game character runs across a screen. How can we write an equation for motion itself?

Look again at our old friend, the parabola $y = x^2$. Imagine it as the trail of a crawling ball. The ball starts at the origin, moves rightward at 1 unit per second, and changes height according to a time rule. If t is the number of seconds since departure, then

$$x = t, \quad y = t^2.$$

At $t = 0$ the ball is at $(0, 0)$; at $t = 1$, at $(1, 1)$; at $t = 2$, at $(2, 4)$; and at $t = 3$, at $(3, 9)$. At each second it has a definite position, and together the positions trace the same parabola. See the difference? $y = x^2$ describes the relationship between the coordinates, without saying when the point arrives. The pair $x = t, y = t^2$ tells us where it is at each moment: it draws the route and records a timetable. The extra variable t is a **parameter**, and the pair forms **parametric equations**.

Definition 4.5 (Parametric Equations). Suppose the coordinates (x, y) of a point moving on a curve are expressed in terms of a variable t as

$$x = f(t), \quad y = g(t),$$

and as t ranges over its allowed values, (x, y) traces exactly the curve. Then these are parametric equations of the curve, and t is the parameter.

A parameter often plays the role of time, but it need not be time: it is simply a conductor's baton—as t changes, the point moves. Here is a more physical example. Throw a ball horizontally, neglecting air resistance; it moves rightward at constant speed and accelerates downward. Simplifying the numbers rather than using precise gravitational data, write

$$x = 2t, \quad y = 5 - t^2 \quad (t \geq 0).$$

At $t = 0$ the ball is at $(0, 5)$, released at height 5; at $t = 1$, at $(2, 4)$; and at $t = 2$, at $(4, 1)$. For slightly larger t , y becomes negative and the ball has reached the ground. If we care only about the arc's shape and not the timing, we can **eliminate** the parameter. The first equation gives $t = \frac{x}{2}$; substitute into the second:

$$y = 5 - \left(\frac{x}{2}\right)^2 = 5 - \frac{x^2}{4}.$$

The parameter disappears, leaving a downward-opening parabola. Horizontal projectile motion indeed traces a parabola, in the same family as a basketball shot or a fountain's stream. Eliminating t returns us to an ordinary equation. Conversely, one curve can have many parametric descriptions, representing different ways to travel the same route. $x = t, y = t^2$ climbs

the parabola with constant horizontal speed; $x = t^3$, $y = t^6$ traces the same right-hand arc at a different pace. The locus is the stage and the parametric equations the script: one stage can host many scripts.

Why set points in motion? Because so many questions naturally involve time or progress: when a planet comes closest to the Sun, where a shell lands, or which screen pixel should hold a character in this animation frame. A static equation says which road a point is on; parametric equations say where on that road it is now. The latter is much closer to what engineers and physicists actually want to ask.

Keep Climbing

Beyond the coordinate bridge lies a vast continent. Its landmarks include **analytic geometry**, the systematic study of translating between equations and curves, whose founders are generally recognized as Descartes and Fermat, reportedly rivals over priority; **conic sections**, asking why ellipses, parabolas, and hyperbolas are called conics and how they relate to planetary orbits; **polar coordinates**, which address points using distance and direction, as on a radar screen; and **parametric equations and calculus**, asking how a timetable $x = f(t)$, $y = g(t)$ reveals velocity at each instant. It is fine if these names are unfamiliar: think of them as regions of the map yet to be unlocked.

Mathematical Toolbox

This chapter's new tools, in order of appearance:

- **The number line and one-to-one correspondence:** real numbers correspond one to one with points on a line; one-dimensional distance $= |b - a|$.
- **Cartesian coordinates in the plane:** points correspond one to one with ordered pairs (x, y) ; order matters.
- **The distance formula:** $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, essentially the Pythagorean theorem in coordinates.
- **The midpoint formula:** midpoint coordinates are the averages of the endpoint coordinates.
- **Translating equations and loci:** a point lies on a curve if and only if its coordinates satisfy the equation; a geometrical condition produces an equation, and an equation traces a curve.
- **Solving in both directions:** coordinates let us calculate geometry problems directly; algebra problems, especially those with radicals, can become distances that we recognize geometrically.
- **Parametric equations:** a third variable t records a moving point's timetable; eliminating the parameter restores an ordinary equation.

Questions to Think About

Observation

1. Set $t = -2, -1, 0, 1, 2$ in the parametric equations $x = t^2$, $y = 2t$, plot the resulting points, and join them smoothly. What shape do you see? Hint: solve the second equation for $t = \frac{y}{2}$, substitute into the first to eliminate t , and examine the equation relating x and y . How is it related to $y = x^2$?
2. On coordinate paper, shade the region satisfying $x^2 + y^2 \leq 9$, then mark the region satisfying $x + y > 0$ in a different way. What does their overlap look like? Hint: first draw the boundaries $x^2 + y^2 = 9$ and $x + y = 0$. To identify the appropriate side of a boundary, substitute a test point, such as the origin.

Proof

1. Use coordinates to prove that a quadrilateral whose diagonals bisect each other is a parallelogram. Hint: put the common midpoint at the origin, one pair of opposite vertices at (a, b) and $(-a, -b)$, and the other pair at (c, d) and $(-c, -d)$. Compare the horizontal and vertical steps along each pair of opposite sides.
2. Two fixed points are $A(-1, 0)$ and $B(1, 0)$. A moving point $P(x, y)$ satisfies $PA^2 + PB^2 = 10$. Prove that the locus of P is a circle, and identify its centre and radius. Hint: expand PA^2 and PB^2 using the distance formula (these are **squared** distances, so the radicals disappear), combine like terms, and inspect the result.

Challenge

1. Find the minimum of $\sqrt{x^2 - 2x + 2} + \sqrt{x^2 - 8x + 20}$. Hint: complete the square inside each radical and recognize two distances. This time the fixed points lie on the same side of the x -axis, so joining them directly will not work. Think of a mirror: reflect one point across the x -axis. The broken path keeps its length, and a straight line can now help.
2. A point P moves along $y = x^2$ and can be written (t, t^2) ; a fixed point A is $(2, 0)$. Express PA^2 in terms of t . Hint: apply the distance formula but do not rush to take a square root. PA^2 is a **quartic** expression in t . Middle-school completion of squares no longer reveals whether it has a minimum or where it occurs. This puzzling quartic is exactly the sort of object the next chapter will investigate.

The Door to the Next Chapter

With the bridge open, one of the first pieces of cargo across the river is a curve we have met repeatedly: the parabola.

Draw $y = x^2 - 3x + 2$ on coordinate paper, and it meets the x -axis twice. At each meeting the vertical coordinate is 0, so the horizontal coordinate must satisfy $x^2 - 3x + 2 = 0$, a quadratic equation. In other words, **intersections with the x -axis and roots of an equation are two descriptions of the same thing**: geometry sees them, and algebra calculates them. The equation $x^2 - 3x + 2 = 0$ has roots 1 and 2, so the parabola meets the x -axis exactly at $(1, 0)$ and $(2, 0)$.

A stream of new questions now crosses the bridge. Replace degree 2 with degree 3: how many times does $y = x^3 - x$ meet the x -axis? What about degrees 4 and 5? Can a polynomial curve never meet the x -axis—can an equation have no real roots at all? If so, is the problem wrong, or is our world of numbers too small and ready for expansion?

The common protagonists are expressions like $x^2 - 3x + 2$ and $x^3 - x$, assembled by adding and subtracting powers of variables: **polynomials**. They seem like simple algebraic components, but they hide roots of equations, intersections of curves, and even clues to the birth of a new number system. At our next stop, we will take polynomials apart and examine their structure.

Part II

Along Two Main Threads—Polynomials, Transformations, and Vectors

Chapter 5 Polynomials: The Structure Behind Equations

5.1 Three Measurements at a Skatepark

A new skatepark on the south side of town has a launch ramp that sends skateboarders airborne. Its designer, Zhou, has drawn the flight path: a rider leaves the ramp's edge, traces an arc through the air, and lands on a descending slope. This arc cannot be drawn casually. The landing must fall on the gentlest part of the slope, or the rider could suffer anything from a tumble to a serious accident.

Put the site into a coordinate system, using the skill from Chapter 4: take the launch point as the origin, the horizontal direction as the x -axis, and the vertical direction as the y -axis, with metres as units. Zhou knows that, ignoring air resistance, the flight path is always a quadratic curve, described by

$$y = ax^2 + bx + c$$

for three numbers a , b , and c yet to be determined. Such expressions appear everywhere: the arc of a fountain's stream, the path of a basketball shot, and the shape of a suspension bridge's main cable all belong to the same mathematical family. The difficulty is that the instruments have measured only three points on the path: the launch position $(0, 2)$, a point in the air $(1, 5)$, and the chosen safe landing point $(4, 2)$.

There are now two questions. First, does a quadratic curve pass through all three points, and if so, is it unique? Second, once the trajectory is determined, at what horizontal positions is the rider exactly 3 metres high? That means solving $ax^2 + bx + c = 3$, or finding where the curve meets a horizontal line.

Make an intuitive guess first. Three conditions determine three unknowns, so perhaps a solution exists and is probably unique. That intuition is correct, but explaining why requires this chapter's tools. You may also think solving $ax^2 + bx + c = 3$ simply means applying the quadratic formula. But what if the discriminant is negative? Is the problem wrong, or is mathematics hinting at something?

These two questions launch our chapter. We will study an expression like $ax^2 + bx + c$ as a building with internal structure, asking how its components (factors), foundations (roots), and number of storeys (degree) determine its behaviour. Then we will return to the skatepark to ask whether three points can recover a rule. Finally, we will confront the negative discriminant,

which will lead us into a whole new world of numbers.

5.2 Trying Intuition: Two Easy Traps

When asked for a curve through three points, the natural approach is to guess. Try $y = -x^2 + 4x + 2$: at $x = 0, y = 2$, correct; at $x = 1, y = 5$, correct again; at $x = 4, y = -16 + 16 + 2 = 2$, all correct! We got lucky this time. Change the points, though, and guessing becomes difficult. We cannot rely on luck every time. We need a method guaranteed to succeed that also proves uniqueness; later in the chapter we will find one.

Intuition about equations can stumble too. Many students memorize that the graph's intersections with the x -axis are the equation's roots, then infer two plausible-sounding claims:

- The equation has as many roots as the curve has encounters with the x -axis.
- If the graph does not meet the x -axis, the equation is meaningless.

The first misses a subtle possibility: the curve can touch the x -axis and bounce back without crossing it, as $y = (x - 1)^2$ does at $x = 1$. The equation then has only one root, but that root occupies two places. The second is more dangerous: the graph associated with $x^2 + 1 = 0$ indeed cannot reach the x -axis, yet by this chapter's end we will see that the equation is meaningful and prompted one of the most important expansions of the number system in mathematical history.

Counterexample Alert

“A quadratic equation either has two roots or has no real roots.” Wrong. $x^2 - 2x + 1 = 0$ has only the root $x = 1$, but it is repeated: the left side is $(x - 1)^2$, where the factor $x - 1$ appears twice. Saying only “one root” loses structural information; “one double root” is more precise. When reading a graph, remember that touching the x -axis tangentially and crossing the x -axis are fundamentally different encounters.

Both failures teach the same lesson: looking only at the answer is insufficient. We must take the expression apart and inspect its structure. Let us turn that dismantling into a formal tool.

5.3 Expressions as Buildings: Polynomials, Degree, and Roots

Take apart $3x^2 - 4x + 1$, and you find several blocks added together: $3x^2$, $-4x$, and 1 . Each is a number multiplied by a power of x (we can regard 1 as $1 \cdot x^0$). Such expressions are extremely convenient: they can be added and multiplied, and the results remain in the same family, like Lego bricks that stay within the system however you assemble them. To discuss these objects repeatedly, we give them a name.

Definition 5.1 (Polynomial). An expression of the form

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

is called a **polynomial** in x , where $a_n, a_{n-1}, \dots, a_0$ are fixed numbers called **coefficients**, and n is a nonnegative integer. If $a_n \neq 0$, its **degree** is n , and a_n is the **leading coefficient**.

Terminology Card: Degree

Intuition: the exponent of the polynomial's fastest-growing term, determining the expression's character when x is large.

Formal definition: the highest exponent of a term with nonzero coefficient.

Example: $x^3 - 7x + 6$ has degree 3; the constant 5, or $5x^0$, has degree 0.

Terminology Card: Root

Intuition: an input that makes the polynomial zero; on the graph, a position where the curve meets the x -axis.

Formal definition: if a number r satisfies $f(r) = 0$, then r is a root of the polynomial $f(x)$.

Example: 1 and 3 are roots of $x^2 - 4x + 3$, because both give 0 when substituted.

Why invent a special word, degree? Because it acts like a building's storey count, largely determining its broad features. A degree-one polynomial has a straight-line graph, with one end rising and the other falling. A quadratic has a parabola; a cubic resembles a twisted ribbon, diving at one end and emerging at the other. Degree also answers a question that troubled ancient mathematicians: how many solutions can an equation have at most?

Polynomials also deserve study because they are simple enough and versatile enough. Simple, because evaluating them requires only multiplication and addition, with no elaborate operations—exactly the expressions computers like best. Versatile, because later, in Chapters 11 and 12, we will see that even apparently unrelated functions such as $\sin x$ and e^x can be approximated as closely as we like by infinitely long polynomials. Understanding polynomials lays foundations for the whole world of functions.

Roots weld algebra to geometry: solving $f(x) = 0$ is equivalent to finding the intersections of $y = f(x)$ with the x -axis. The same object has an algebraic face and a geometrical face, and we can switch between them at will.

Now return to taking expressions apart. We can write $x^2 - 4x + 3$ as $(x - 1)(x - 3)$. Once factored, its roots almost jump out: a product is zero if and only if at least one factor is zero, so the roots are 1 and 3. Conversely, knowing the roots 1 and 3 immediately suggests the factors $x - 1$ and $x - 3$. This translation between roots and factors deserves to become a theorem.

5.4 The Factor Theorem: Translating Roots into Factors

Theorem 5.2 (The Factor Theorem). *Let $f(x)$ be a polynomial and a a number. Then a is a root of $f(x)$ if and only if $x - a$ is a factor of $f(x)$, meaning that some polynomial $q(x)$ satisfies*

$$f(x) = (x - a)q(x).$$

Proof. Check the two directions separately.

(If $x - a$ is a factor.) Suppose $f(x) = (x - a)q(x)$. Substitute $x = a$: the first factor becomes $a - a = 0$, and zero times any number is zero. Thus $f(a) = 0$, so a is a root.

(If a is a root.) Recall division with remainder from primary school: 17 divided by 5 has quotient 3 and remainder 2, so $17 = 5 \times 3 + 2$, with the remainder smaller than the divisor. Polynomials have an analogous division. Dividing $f(x)$ by the linear expression $x - a$ gives

$$f(x) = (x - a)q(x) + r,$$

where the remainder r must have degree less than the degree 1 of the divisor $x - a$, so r can only be a constant. Substitute $x = a$. The left side is $f(a) = 0$ because a is a root; the first term on the right disappears, leaving r . Hence $r = 0$, so $f(x) = (x - a)q(x)$ and $x - a$ is indeed a factor. $\square$

Example 5.3. Verify that $x = 2$ is a root of $f(x) = x^3 - 7x + 6$ and factor the polynomial completely. Substitution gives $f(2) = 8 - 14 + 6 = 0$, so it is a root and $x - 2$ is a factor. Division, or patient trial, gives

$$x^3 - 7x + 6 = (x - 2)(x^2 + 2x - 3) = (x - 2)(x - 1)(x + 3).$$

The three roots are immediately visible: 2, 1, and -3 . Once one root is found, solving a cubic has been reduced to solving a quadratic.

How exactly does the division work? Almost like long division with integers. Divide $x^3 - 7x + 6$ by $x - 2$. Start with the highest power, x^3 : what power of x multiplying $x - 2$ supplies that term? The first quotient term is x^2 . Multiply back to get $x^3 - 2x^2$ and subtract, leaving $2x^2 - 7x + 6$. For $2x^2$, the next quotient term is $2x$; multiplying back gives $2x^2 - 4x$, and subtracting leaves $-3x + 6$. The final quotient term is -3 , whose product is $-3x + 6$; subtraction clears everything, leaving remainder 0. A zero remainder is another way to say that 2 is a root. Divide using a number that is not a root and a small piece remains. That piece is exactly the function value, as the remainder theorem in Further Climbing explains.

This reduction in degree is powerful and repeatable, so an important result comes almost for free:

Corollary 5.4. *A polynomial of degree n has at most n distinct real roots.*

Proof. Every root a supplies a linear factor $x - a$ by the factor theorem, splitting the polynomial as $(x - a)q(x)$ with $q(x)$ of degree $n - 1$. Finding a root of $q(x)$ splits off another linear factor. Each step reduces the remaining degree by one. After at most n steps it reaches 0, leaving a constant with no further roots to split off. Thus there are at most n distinct roots. $\square$

Note. “At most” is essential. $x^2 + 1$ has degree two but no real roots. Degree gives an *upper bound*, not a guarantee. We will return to this gap at the chapter’s end.

5.5 Vieta's Relations: Hearing the Roots Without Factoring

Factoring is satisfying, but sometimes the coefficients make it awkward, and sometimes complete factorization is unnecessary. Can we learn something about roots without solving the equation?

For quadratics we can, in a particularly elegant way. Let the roots of $x^2 + bx + c = 0$ be α and β , possibly equal. The factor theorem gives

$$x^2 + bx + c = (x - \alpha)(x - \beta).$$

Expand the right side:

$$(x - \alpha)(x - \beta) = x^2 - (\alpha + \beta)x + \alpha\beta.$$

The two sides are the same polynomial, so corresponding coefficients must match. We obtain:

Theorem 5.5 (Vieta's Relations for Quadratics). *If α and β are the roots of $x^2 + bx + c = 0$, then*

$$\alpha + \beta = -b, \quad \alpha\beta = c.$$

For the general form $ax^2 + bx + c = 0$ with $a \neq 0$, divide both sides by a to reach the preceding case, giving $\alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$.

Example 5.6. Without solving $x^2 - 7x + 12 = 0$, we know its roots sum to 7 and multiply to 12. Which two numbers have sum 7 and product 12? 3 and 4. Now consider $x^2 + x + 1 = 0$. If it has real roots, they sum to -1 and multiply to 1. A positive product means the two real numbers have the same sign; a negative sum means both must be negative. But for two negative numbers whose product is 1, the closest possibility is -1 and -1 , whose sum is -2 —a little short of the required -1 . This shortfall is no coincidence; the chapter's ending will explain it.

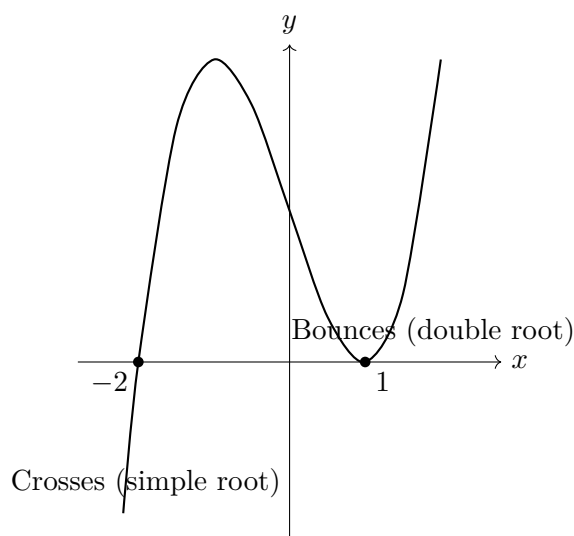
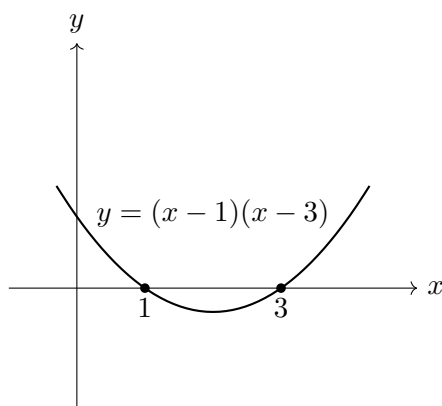
The real value of Vieta's relations is obtaining information at a distance: the coefficients are **symmetric functions** of the roots. $\alpha + \beta$ and $\alpha\beta$ share a feature: swapping α and β leaves their values unchanged. Coefficients see only symmetric information about the roots; they cannot report what they do not see, such as which root is larger. Remember the word symmetry: it is one of the keys to the next chapter. These relations are named after the sixteenth-century French mathematician Vieta. His contribution went beyond the two formulas: he began using letters systematically for known and unknown quantities. That notation made it possible to study all equations instead of solving them one at a time.

Cubic and quartic equations have Vieta relations too. For $x^3 + px^2 + qx + r = 0$, the three roots sum to $-p$, their pairwise products sum to q , and their product is $-r$. The pattern comes from the same expansion with more brackets. Check our old friend $x^3 - 7x + 6$: its roots are 1, 2, and -3 , with sum 0 (there is indeed no x^2 term), pairwise product sum $1 \times 2 + 1 \times (-3) + 2 \times (-3) = 2 - 3 - 6 = -7$ (the coefficient of x), and product -6 (the negative of the constant term 6, because a cubic's constant term is the negative product). The three coefficients store all the symmetric information about the three roots.

5.6 A Graph Is a Map of the Roots

Now translate the algebraic structure into geometry. What can we read directly from a polynomial curve? This section's answer is that the graph is factorization made visible. We will examine three features.

First, **wherever there is a root, the curve meets the x -axis**, but how it meets matters. $(x-1)(x-3)$ *crosses* decisively at 1 and 3; $(x-1)^2$ merely *kisses* the axis at 1 and bounces back. Multiplicity explains the difference: if $x-a$ appears an odd number of times, moving x past a changes the sign of the product and the curve crosses. If it appears an even number of times, the product keeps its sign and the curve touches and rebounds.



The second figure shows $y = (x+2)(x-1)^2$: it crosses at -2 and bounces at 1. The graph openly reveals the secrets of the factorization.

Second, **degree and leading coefficient determine the ends' directions**. When the absolute value of x is very large, lower-degree terms become negligible and the leading term $a_n x^n$ controls the behaviour. For even n , both ends point the same way (up if $a_n > 0$, down if $a_n < 0$); for odd n , they point in opposite directions. A by-product follows immediately: an odd-degree polynomial rises into the sky at one end and dives underground at the other, so it must cross the x -axis somewhere between them.

Corollary 5.7. *Every odd-degree polynomial has at least one real root.*

A rigorous proof needs the fact that a continuous curve cannot jump over the x -axis (the intermediate value theorem in Chapter 12 will make this precise), but the picture is persuasive: travelling from underground to the sky requires passing through ground level.

Third, **roots divide the curve into portions on which the sign stays fixed**. Want to solve $x^3 - 7x + 6 > 0$? Factor it as $(x + 3)(x - 1)(x - 2)$. The roots -3 , 1 , and 2 divide the number line into four intervals. Test one point in each to determine its sign, or construct a sign table. Each factor is positive to the right of its own root and negative to the left; the product's sign depends on whether the number of negative factors is even or odd.

Interval	$x + 3$	$x - 1$	$x - 2$	Product $f(x)$
$x < -3$	—	—	—	—
$-3 < x < 1$	+	—	—	+
$1 < x < 2$	+	+	—	—
$x > 2$	+	+	+	+

Thus $f(x) > 0$ holds for $-3 < x < 1$ or $x > 2$. Read the table row by row, with no further equations to solve. A troublesome-looking inequality is settled by a map of the roots.

5.7 A Rule from Three Points: Interpolation

Return to the skatepark. The trajectory passes through $(0, 2)$, $(1, 5)$, and $(4, 2)$; find the quadratic curve. Earlier we luckily guessed $y = -x^2 + 4x + 2$. Now we will give a reliable method and prove uniqueness along the way.

The idea is surprisingly simple: **build components that are on duty at only one point**. Construct a quadratic polynomial $\ell_1(x)$ that equals 1 at $x = 0$ and 0 at $x = 1$ and $x = 4$. The factor theorem handles the latter conditions immediately: ℓ_1 must contain $(x - 1)(x - 4)$, so $\ell_1(x) = k(x - 1)(x - 4)$. Use its value 1 at 0 to determine k : substitution gives $k(0 - 1)(0 - 4) = 4k = 1$, so $k = \frac{1}{4}$ and

$$\ell_1(x) = \frac{1}{4}(x - 1)(x - 4).$$

Similarly, construct $\ell_2(x) = -\frac{1}{3}x(x - 4)$, on duty at $x = 1$ and equal to 0 at the other two points (check that $\ell_2(1) = -\frac{1}{3} \cdot 1 \cdot (-3) = 1$), and $\ell_3(x) = \frac{1}{8}x(x - 1)$, on duty at $x = 4$.

Now, like a sound mixing desk, combine the three components according to their target heights:

$$f(x) = 2\ell_1(x) + 5\ell_2(x) + 2\ell_3(x).$$

Check: at $x = 0$, ℓ_2 and ℓ_3 are 0 while $\ell_1 = 1$, so $f(0) = 2$. Similarly, $f(1) = 5$ and $f(4) = 2$. Expanding and combining gives $f(x) = -x^2 + 4x + 2$, just as guessed, but now calculated. The same procedure works for any three points. Its name is **Lagrange interpolation**.

Note (Uniqueness). Only one quadratic polynomial passes through these three points. If both f and g satisfy the requirements, consider $f - g$. It is zero at 0, 1, and 4: a polynomial of degree at most 2 has three roots! That contradicts the two-root bound unless $f - g$ is identically zero. The bound we just proved has immediately found a use.

In general, $n+1$ points with distinct horizontal coordinates determine exactly one polynomial of degree at most n . Fewer points leave the answer nonunique; more points may not all lie on any polynomial of the chosen low degree.

There is a subtle caution here: with more points, is higher degree always better? Quite the opposite. If measurements contain errors, as real measurements always do, a high-degree curve forced through every point can oscillate wildly between them and predict badly. Engineers usually prefer a lower-degree curve passing approximately through all the points. This is fitting, a close relative of interpolation. Interpolation insists on passing exactly through the points; fitting seeks the best overall match. Knowing which to use is a basic skill in the age of data.

Connections

Interpolation is far more than a geometry game. When a phone enlarges a photograph, interpolation computes the new pixels' colours. Animators draw only keyframes, and software interpolates the frames between them. Weather stations observe only selected cities, and interpolation fills out a national temperature map. More remarkably, error-correcting codes transmit data as values of a polynomial. Even if a few points are lost along the way, interpolation can reconstruct the entire polynomial. You use this every day when listening to music or scanning a QR code.

5.8 When Solutions Run Out: The Birth of i

Now face this chapter's biggest stumbling block: $x^2 + 1 = 0$.

Over the real numbers it has no solution, since no real square is negative. Mathematicians centuries ago thought the same and consigned such equations to the wastebasket of meaninglessness. The turning point came in sixteenth-century Italy, where competitions to solve cubic equations revealed that solution formulas sometimes required square roots of negative numbers along the way. Strangely, if one *pretended* those square roots existed and pressed on using ordinary rules, the imaginary parts eventually cancelled in pairs, and the resulting answer checked perfectly in the original equation!

This was awkward: a nonexistent number had become an indispensable bridge to a correct answer. Reject it, and real roots known to exist became inaccessible; accept it, and the calculation flowed. Mathematicians argued and doubted for more than a century. Some called these numbers sophisticated quantities; others refused to write them. But their computational power was irresistible: results reached through imaginary numbers could be checked with real ones and repeatedly proved correct. After practice voted again and again, theory finally yielded. Mathematicians decided to recognize the bridge officially and grant it citizenship. Introduce a new

number i satisfying

$$i^2 = -1.$$

i is not an inhabitant of the real number line; there is no place for it there. Where does it live? The answer is beautifully reassuring. Since real numbers fill a straight line, put i *outside* that line, at $(0, 1)$ in a plane. Then a general complex number $a + bi$ is the point (a, b) : real numbers lie along the horizontal axis, and i stands one unit up the vertical axis.

Terminology Card: Complex Number

Intuition: a point in the plane whose horizontal coordinate is the real part and whose vertical coordinate is the imaginary part.

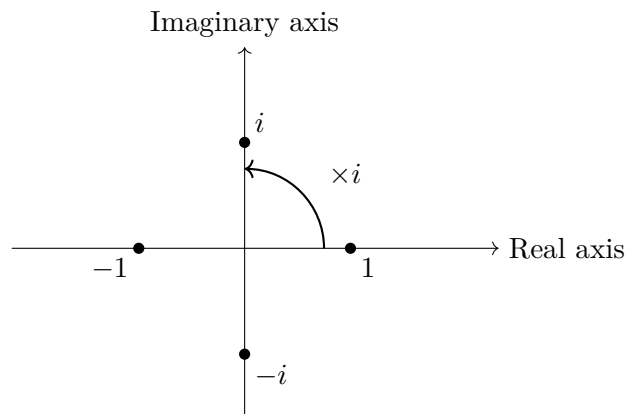
Formal definition: a number of the form $a + bi$, where a and b are real and i satisfies $i^2 = -1$.

Example: $3 - 2i$ corresponds to $(3, -2)$; the real number 5 is $5 + 0i$, and the purely imaginary number $2i$ is $0 + 2i$.

All the rules for complex arithmetic follow from one principle: calculate as in ordinary algebra, replacing i^2 with -1 whenever it appears. Addition adds corresponding components: $(1+2i)+(3-i) = 4+i$, represented geometrically by placing two arrows head to tail. Multiplication uses the distributive law:

$$(1 + 2i)(3 - i) = 3 - i + 6i - 2i^2 = 3 + 5i - 2 \times (-1) = 5 + 5i.$$

Each complex number $a+bi$ also has a mirror partner $a-bi$, called its **conjugate**. Their sum is $2a$, a real number; their product is $a^2 + b^2$, a nonnegative real number equal to the squared distance from the origin. This partnership turns division into multiplication too: multiply numerator and denominator by the denominator's conjugate, and the denominator immediately becomes real.



Complex multiplication has a geometrical interpretation: multiplying by i rotates a point 90° anticlockwise about the origin. $1 \times i = i$ turns rightward into upward; $i \times i = i^2 = -1$ turns another 90° to the left. A square root of a negative number turns out to be the square root of the action of making a half-turn. How natural!

With i , the equation $x^2 + 1 = 0$ has roots $\pm i$, and the quadratic formula gives, for $x^2 + x + 1 = 0$,

$$x = \frac{-1 \pm \sqrt{-3}}{2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i.$$

Remember the earlier shortfall? These two complex roots sum to exactly -1 and multiply to exactly 1 . Vieta's relations still hold in the complex world. More astonishing is the following fact:

Theorem 5.8 (The Fundamental Theorem of Algebra). *Every polynomial of degree n with complex coefficients, where $n \geq 1$, has at least one complex root. Repeated application of the factor theorem then splits it into a product of n linear factors: it has exactly n roots, counting repeated roots according to their multiplicities.*

The proof goes beyond this book's tools. It requires some analysis or topology; university courses in complex analysis offer particularly elegant proofs. But its message is clear: **complex numbers make degree and number of roots match perfectly**. Earlier qualifications—odd degree guarantees a real root, a quadratic may have none—disappear in the complex world. A degree- n equation always has exactly n solutions, no more and no fewer. Introducing a new kind of number brings striking order.

Looking back, number systems have expanded along a familiar road: measuring land required fractions; accounting for debts required negative numbers; measuring diagonals required irrational numbers; solving equations required complex numbers. Each time solutions run out, mathematics is knocking on a door. Behind it waits a larger number system and a tidier collection of theorems.

Keep Climbing

Further ahead you will encounter the **remainder theorem** ($f(a)$ is exactly the remainder when f is divided by $x - a$), the general **Lagrange interpolation formula**, the **conjugate root theorem** (nonreal roots of equations with real coefficients come in pairs), and **roots of unity** in the complex plane. Farther still stands a major question: why is there no general solution formula for equations of degree five and higher? Its answer gave birth to a subject that studies equations through symmetry: **group theory**.

5.9 Hands-on Time

One-Minute Experiment

Experiment 1: Predicting a Graph from Its Roots. Without using a calculator, answer these questions from $f(x) = (x+2)(x-1)(x-3)$ alone. (1) At which three positions does the curve meet the x -axis, and does it cross or bounce? (2) Where do its ends point for very large positive and negative x ? (3) Mark the intervals where $f(x) > 0$ and $f(x) < 0$ on a number line. Then plot points, sketch the curve, and check each prediction.

Experiment 2: Interpolation by Hand. Take $(0, 1)$, $(2, 3)$, and $(3, 1)$. Following the text, construct three components ℓ_1 , ℓ_2 , and ℓ_3 , each on duty at one point, and combine them by height to obtain $f(x)$. Expand and substitute all three points to check. Then choose another horizontal coordinate, such as $x = 1$, calculate its function value, plot it, and see whether the curve smoothly joins your points.

Experiment 3: Vieta the Detective. Ask a classmate to choose two integer roots of a quadratic secretly and tell you only their sum and product. Use Vieta's relations to reconstruct the equation, then solve it to check. Switch roles for three rounds and consider what it means for coefficients to see only symmetric information.

Mathematical Toolbox

This chapter's new tools:

- **Polynomials, coefficients, and degree:** a large class of expressions becomes one unified object of study.
- **The factor theorem:** roots and linear factors translate into each other; finding one root lowers the equation's degree by one.
- **Bounds on roots:** a degree- n polynomial has at most n real roots; an odd-degree polynomial has at least one.
- **Vieta's relations:** coefficients are symmetric functions of roots, revealing information without solving the equation.
- **Multiplicity and graphs:** simple roots cross the axis, repeated roots touch and bounce; the leading term determines the ends' directions.
- **Lagrange interpolation:** $n + 1$ points uniquely determine a rule of degree at most n .
- **Complex numbers:** $i^2 = -1$; in the complex world, a degree- n equation has exactly n solutions.

Questions to Think About

Observation

1. Examine sketches of $y = x^2$, $y = x^3$, $y = x^4$, and $y = x^5$. What patterns distinguish even and odd degrees for very large positive and negative x ? Hint: consider only the leading term, treating other terms as negligible small fluctuations.
2. Sketch $y = (x - 1)^3$. The point $x = 1$ is a triple root: does the curve cross or bounce there? Hint: an odd power changes sign, but growth flattens near the root. Plot $x = 0.5$, 1, and 1.5 and look closely.

Proof

1. Prove that if $x^2 + bx + c = 0$ has two real roots, they have the same sign when their product is positive and opposite signs when it is negative. Hint: use Vieta's relation $\alpha\beta = c$ directly.
2. A cubic polynomial $f(x)$ satisfies $f(1) = f(2) = f(3) = 0$ and $f(0) = 6$. Find $f(4)$. Hint: the factor theorem determines the form of f up to one unknown coefficient.

Challenge

1. Use $i^2 = -1$ to calculate $(1+i)^2$ and $(1+i)^4$. Hint: plot each result in the complex plane and observe what happens to distance from the origin and angle at each squaring. This quietly echoes the rule that multiplying by i rotates by 90° .
2. Prove that at most one polynomial of degree at most n passes through $n + 1$ points with distinct horizontal coordinates. Hint: follow the uniqueness note in the text, consider the difference of two candidates, and count its roots.

The Door to the Next Chapter

Symmetry has appeared quietly several times in this chapter. In Vieta's relations, $\alpha + \beta$ and $\alpha\beta$ remain unchanged when the two roots are exchanged; coefficients see only symmetric information about roots. And the ultimate question—why cubics and quartics have solution formulas but equations of degree five and higher do not—eventually receives this answer: **the symmetries among the roots of a quintic are too complicated to express through finitely many root extractions.**

Giving an equation's roots a symmetry check sounds extraordinary. But symmetry itself can be a precisely defined mathematical object on which we calculate. How many reflections and rotations does a square have? How do these operations combine? Does reflecting and then rotating give the same result as rotating and then reflecting? Next chapter we will turn symmetry from an adjective into a noun, from an impression into a structure, and follow it to the heart of modern algebra: groups.

Chapter 6 Transformations and Symmetry: From Moving Figures to Groups

6.1 Putting a Square Back into Its Frame

Many places still keep the New Year tradition of cutting paper decorations for windows. Fold a sheet of red paper in half, then in half again, snip a few notches with scissors, and unfold it. An astonishing pattern appears: whatever is on the left must also be on the right; whatever is above appears again when you turn the paper. The person cutting never designed these repetitions. The folds supplied them for free.

A kaleidoscope makes the effect even more immediate. A few fragments of coloured glass inside a tube lined with three mirrors create patterns as orderly as precision machinery. Turn the tube and the patterns change endlessly, yet always retain an unmistakable order. Why flowers with six or twelve petals, rather than seven or thirteen? Because the angles between the mirrors are fixed. The glass fragments are reflected repeatedly, each reflection adding another copy, and the angles determine how many copies appear. The broken glass is random; the mirrors' angles impose the order.

Behind these kinds of order lies the same mathematical question. In this chapter we will investigate it through an even simpler game.

Prepare a stiff square card and label its corners 1, 2, 3, and 4 on both sides, making sure each corner can be identified from either face. Draw a square frame of the same size on the table. How many different ways can you put the card exactly back into its frame?

Remember the rule: you may flip the card over, so consider more than rotations.

Guess first. Many people guess 4, counting rotations through 0° , 90° , 180° , and 270° . But the card can also turn over, so there are at least 8. And that is not the end of the matter. The interesting questions are just beginning.

If I perform “rotate 90° ” and “reflect across the vertical midline” in succession, does their result count as a new placement? Could there be a ninth? Further, is rotating and then reflecting the same as reflecting and then rotating?

This sounds like a tiny playground puzzle, but it has challenged serious thinkers. While studying equations in the nineteenth century, mathematicians were forced to examine successive actions carefully. From them they extracted a new mathematical concept: the group. It later became a common language throughout modern mathematics and physics. Classifying crystals,

naming elementary particles, and solving a Rubik's cube all depend on it. And it really can begin with the square card on your table.

6.2 First Guess, Then Go Wrong

Before inventing any terminology, let us see where intuition can lead us astray.

The first common guess is that order does not matter when combining a rotation and a reflection. After all, addition and multiplication commute: $3 + 5 = 5 + 3$. Why should actions be different? This guess fails decisively. Start with corner 1 in the upper left of the frame. Rotate 90° anticlockwise, and corner 1 moves to the lower left; reflect across the vertical midline, and corner 1 reaches the lower right. Reverse the order: first reflect across the vertical midline, moving corner 1 to the upper right; then rotate 90° anticlockwise, bringing corner 1 to the upper left. The two orders put corner 1 in different places. Order matters when actions are composed.

Counterexample Alert

Warning: "When two actions are performed in succession, either order gives the same result." The card experiment refutes this: rotate 90° then reflect, or reflect then rotate 90° , and corner 1 ends in different places. This is fundamentally different from multiplying numbers, where $3 \times 5 = 5 \times 3$ is unquestioned. For actions, reversing the order can completely change the outcome. You already know this in daily life: putting on shoes and then socks is quite different from putting on socks and then shoes.

A second common guess is that a square has just two symmetry axes, one horizontal and one vertical. Many students draw the two midlines and stop, forgetting that folding along either diagonal also makes the halves coincide exactly. A square has four symmetry axes, not two. Missing axes means missing half the reflected placements.

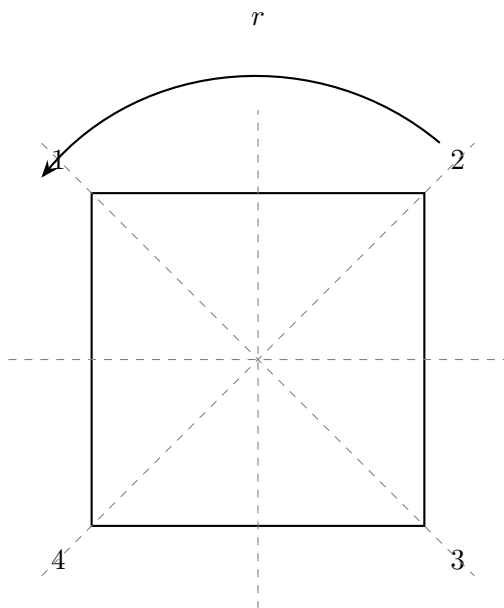
A subtler third guess is that because actions can be repeated again and again, there must be infinitely many placements. This overlooks returning to the start. Four 90° rotations leave the card exactly as if it had never moved; two reflections do the same. Actions can continue indefinitely, but their *results* repeat. We are counting final *states*, not sequences of actions, of which there are indeed infinitely many. Understanding that distinction finally pins down the problem.

Three failures have given three clues: composition depends on order, reflections come in several kinds, and returning to the start makes the total finite. To weave these clues together, we need to make actions themselves mathematical objects we can calculate with.

6.3 Making Actions into Mathematical Objects

When we say "turn the figure," we usually mean a process. The mathematician's crucial step is to regard that process as a *rule*: every point in the plane is sent to a new position.

Definition 6.1 (Transformation). A *transformation* of the plane is a rule sending each point P



Four symmetry axes (dashed) and rotation r through 90° anticlockwise

to a definite point P' , with distinct points sent to distinct points and every point reached from some point (no omissions and no collisions). The point P' is called the *image* of P under the transformation.

What difficulty does this definition resolve? It lets us examine individual points instead of staring at an entire figure. To decide whether a transformation returns a square to itself, we check where its points go—in fact, a few key points suffice. The requirements of no omissions and no collisions are not pedantry. If two points could collide at one destination, the transformation could not be undone: we could not tell where the new point came from. Without undoing, the later story of inverse transformations would fail.

The familiar transformations of middle-school geometry gather here:

- *Translation*: all points move the same distance in the same direction, as when the whole card slides three centimetres rightward.
- *Rotation*: all points turn through the same angle about a fixed point, the centre of rotation.
- *Reflection* (also called axial symmetry): all points flip across a line, the axis of symmetry, as in a mirror.
- *Scaling*: distances from a centre are all enlarged or reduced by the same factor.

The first three share a property: the distance between any two points remains unchanged. Such transformations are called *congruence transformations*, or isometries; they turn figures into congruent figures. Scaling differs: distances change while shape remains the same and size

changes, producing a *similar* figure. This chapter focuses on the first three, but scaling will return in a new form next chapter.

Definition 6.2 (Composition). Let a and b be transformations. The combined effect of first performing b and then a is another transformation, written ab and called the *composition* of a and b .

Remember this convention especially: ab means *first b , then a* . Multiplication-like notation is convenient, but read it from right to left. The convention comes from function notation: in Chapter 2, $f(g(x))$ also meant calculating g before f . If it feels unfamiliar, a couple of card experiments will help.

Composition pays off immediately. Two successive translations still amount to one translation: three centimetres rightward followed by two upward becomes a single diagonal move whose distance and direction we can calculate. What about two rotations? Turning 30° then 45° about the *same* centre is a 75° rotation about that centre—simply add the angles. With different centres, things become interesting: the result is still a rotation or a translation, but finding the new centre may take several auxiliary lines. That is why successive actions deserve study: each step is simple, but together they can surprise us.

There is also the laziest transformation: do nothing, leaving every point in place. This is the *identity transformation*, written e . Do not underestimate it: its role in the theory resembles that of 1 in multiplication.

Definition 6.3 (Symmetry Transformation). If a transformation sends every point of a figure F to a point in F , leaving the transformed figure exactly coincident with the original, it is called a *symmetry transformation* of F .

Pay attention to “exactly coincident.” Rotating a square 45° is not a symmetry transformation: its corners stick out. Translating it three centimetres rightward is not one either: the figure moves away. A symmetry transformation is a special isometry that preserves both distances and the figure as a whole.

An intriguing contrast appears here: a square has only a few symmetries, whereas a circle has infinitely many. Rotate a circle through any angle about its centre and it coincides with itself; reflect across any diameter and the same holds. Intuitively we say that a circle is more symmetrical than a square. We now have a precise measure: the more symmetry transformations a figure has, the more symmetrical it is. The square’s degree of symmetry is a finite list we can soon inventory completely.

The card problem now has precise language: we are asking how many symmetry transformations a square has, and successive actions are compositions of those symmetries.

Mathematical Toolbox

Keep these new tools handy:

- *Transformation*: translate moving a figure into a rule transporting every point, so actions can be compared precisely.
- *Composition ab* : first b , then a ; successive actions become expressions we can calculate.
- *Identity transformation e* : doing nothing provides the reference point for counting and comparison.
- *Inverse transformation*: undoing an action. The inverse of a 90° rotation is a 270° rotation; a reflection is its own inverse.
- *Symmetry transformation*: a transformation making a figure coincide with itself, an entry in our inventory of order.

Terminology Card: Inverse Transformation

Intuition: an action's undo button. *Formal definition*: if transformations a and b satisfy $ab = e$ and $ba = e$, then b is the inverse of a , written a^{-1} . *Example*: the inverse of a 90° anticlockwise rotation is a 90° clockwise rotation, or 270° anticlockwise. A reflection in any axis is its own inverse, because reflecting twice restores the original position.

6.4 Eight, and No More

Return to the card. We can now give the complete answer and *prove* it.

Write r for a 90° anticlockwise rotation about the square's centre, and f for reflection in the vertical midline. Then $r^2 = rr$ rotates 180° , r^3 rotates 270° , and $r^4 = e$. We seem able to create more and more actions, but the following theorem says they all belong to these eight.

Theorem 6.4 (The Symmetries of a Square). *A square has exactly 8 symmetry transformations:*

$$e, \quad r, \quad r^2, \quad r^3, \quad f, \quad rf, \quad r^2f, \quad r^3f,$$

namely four rotations, including the identity, and four reflections.

Proof. First show there are at most 8. Let T be any symmetry of the square. A symmetry preserves distances, and the centre, where the diagonals meet, is the unique point equidistant from the four vertices, so T must send the centre to itself. Thus the four vertices can only be sent to vertices. Corner 1 has at most 4 possible images. Once that image is fixed, consider its neighbour, corner 2. Since T preserves distance, the image of corner 2 must be adjacent to the image of corner 1, giving at most 2 choices. Finally, fixing the images of corners 1 and 2 uniquely determines the whole square's position: by side-side-side congruence, there is only one placement of a square having these adjacent vertices as a side, leaving the other vertices nowhere else to go. Thus T has at most $4 \times 2 = 8$ possibilities.

Now show that all 8 are achievable. The four rotations e, r, r^2, r^3 are clearly symmetries. The transformation f reflects in the vertical midline. The transformation rf first reflects in that midline and then rotates 90° ; checking the vertices directly shows that it equals reflection in a diagonal. Similarly, r^2f and r^3f equal reflections in the horizontal midline and the other diagonal. The square has exactly four symmetry axes, so these are four symmetries, distinct from one another and from the four rotations. Altogether there are exactly 8, no more and no fewer. $\square$

Corollary 6.5. *Every symmetry of the square has an inverse, and that inverse is among the same eight transformations.*

Proof. The inverse of r is r^3 , since $rr^3 = r^4 = e$. The inverse of r^2 is itself. Each reflection is its own inverse, since reflecting twice in the same axis returns everything to its original position. The inverse of e is e . All eight members have inverses, and none of those inverses leaves the collection. $\square$

Note. The proof's claim that side-side-side congruence fixes the square's placement expresses a more general fact: a plane isometry is completely determined by the images of three noncollinear points. The square has four vertices, but watching two adjacent ones suffices because its centre is already fixed, constraining the third point. That is why the little multiplication 4×2 controls the entire figure.

Remember this counting method: to count actions, we need not perform each one. We need only track the *images of key points*. The destinations of two adjacent vertices determine the whole square. The idea of fixing a whole object with a small amount of information will recur.

Having counted the eight transformations, take the step that best captures this chapter's spirit: calculate *all* their compositions and put them in a table. The entry in row x and column y is xy , meaning first y and then x .

	e	r	r^2	r^3	f	rf	r^2f	r^3f
e	e	r	r^2	r^3	f	rf	r^2f	r^3f
r	r	r^2	r^3	e	rf	r^2f	r^3f	f
r^2	r^2	r^3	e	r	r^2f	r^3f	f	rf
r^3	r^3	e	r	r^2	r^3f	f	rf	r^2f
f	f	r^3f	r^2f	rf	e	r^3	r^2	r
rf	rf	f	r^3f	r^2f	r	e	r^3	r^2
r^2f	r^2f	rf	f	r^3f	r^2	r	e	r^3
r^3f	r^3f	r^2f	rf	f	r^3	r^2	r	e

This table deserves ten minutes of close study. Look at the upper-left 4×4 block: composing rotations gives rotations, following the rule of adding angles and resetting to zero at 360° . It is an addition table in disguise. In the lower-right 4×4 block, composing two reflections produces a rotation. Looking in two mirrors with different axes has the effect of rotating through twice the angle between those axes. You can see this yourself with two small mirrors placed at an angle.

Two more observations matter. First, every entry is one of the same eight members. The actions are *closed* under composition: however many steps we take, we cannot escape them, so no ninth placement exists. Second, the table is *not symmetric* about its main diagonal: row r , column f contains rf , but row f , column r contains r^3f . Thus

$$fr = r^3f \neq rf.$$

This is the card experiment's algebraic form: composition is not commutative.

One-Minute Experiment

Verify one table entry physically. Label your card's corners 1, 2, 3, 4. Reflect first in the vertical midline, performing f , then rotate 90° anticlockwise, performing r . Record the corners' positions: you have performed rf . Reset the card and reverse the order: rotate 90° , then reflect, giving fr . Compare with the table: do r^3f , listed for fr , and rf indeed produce different corner positions? Check three more entries until the table has earned your trust.

Step back and consider the eight actions as a *whole*. They satisfy four properties: closure under composition (no outsiders in the table), an identity element e (composing with it leaves any action unchanged), inverses for every action (each row and column contains e exactly once; a 90° rotation pairs with 270° , a reflection with itself), and associativity of composition: $(ab)c = a(bc)$, because both mean first c , then b , then a . The brackets merely alter how we group the description.

Associativity deserves another word. It may sound obvious, but it is a foundation of the whole system. Because of it, the sequence first c , then b , then a can be written simply abc , without semicolons or brackets and without arguing over which portion to calculate first. Subtraction lacks this property: $(8 - 5) - 2 = 1$, while $8 - (5 - 2) = 5$, so repeated subtraction must keep its brackets. Composition is naturally associative. That is fortunate for us, and one reason the language of groups can be so concise.

A system with these four properties has a special name.

Terminology Card: Group

Intuition: a bag of actions that can be composed and undone, with every composition staying inside the bag. *Formal definition:* a set G equipped with an operation that is closed, has an identity element, gives each element an inverse, and is associative. Such a pair $(G, \text{operation})$ is called a *group*. *Example:* the eight symmetries of a square, with composition, form its symmetry group. It has 8 elements and is not commutative.

This is the group in our chapter title. It was forced into existence by our questions, rather than arriving as arbitrary terminology: investigating successive actions brought these four properties to the surface. Mathematicians later found that the symmetry systems worth studying—of figures, equations, crystals, and particles—all satisfy them. Groups became a unified language for symmetry.

6.5 What Is Symmetry Good For?

You know a new concept has become useful when it lets you do something previously out of reach. Groups and symmetry have contributed to at least three major achievements.

First: classifying patterns. Folding paper once before cutting gives one symmetry axis; folding twice usually gives two, sometimes more. Wallpaper, floor tiles, and fabrics have patterns repeating throughout the plane. Mathematicians asked how many types of repeating plane patterns exist when classified by their symmetries. The answer is astonishing: exactly 17. From ancient Egyptian murals to modern tiles, however elaborate the design, every repeating plane pattern has one of these 17 symmetry structures. In three dimensions, for arrangements of atoms in crystals, the answer becomes 230, and chemists still use this catalogue. The proofs go beyond our scope, but you have already met their counting tools: closure, inverses, and identity.

Connections

Looking back, Chapter 3's congruent triangles mean precisely that an isometry carries one triangle onto the other. Chapter 5's Vieta relations will reappear below as an expression of symmetry among equation roots. Looking ahead, Chapter 7 will show how rotations, reflections, and scalings can be recorded in numerical arrays called matrices and calculated directly. Symmetry connects algebra, geometry, and physics.

Second: solving equations. This achievement is especially beautiful and unexpected. Consider $x^2 - 5x + 6 = 0$, with roots 2 and 3. Perform an action: exchange the roots. Which expressions remain unchanged? $x_1 + x_2 = 5$ and $x_1x_2 = 6$ do, because swapping two numbers preserves their sum and product. But $x_1 - x_2$ changes sign. Thus $x_1 + x_2$ and x_1x_2 are *symmetric expressions* under exchanging roots. Vieta's relations say precisely that these expressions can be read from the equation's coefficients without first finding its roots.

Extending this idea to cubics and quartics, mathematicians discovered that solvability by radicals depends on the internal structure of a symmetry group of the roots. The group structure of the general quintic is too complicated, so it has no solution by radicals. The proof of this result, which shook mathematics, uses the very concept illustrated by our eight actions on the table. The young mathematicians who created the theory faced the doubts of the mathematical establishment. Their insight that symmetry determines solvability later became a main branch of algebra.

Third: taming the Rubik's cube. For a scrambled three-by-three cube, the number of possible colour configurations is 43,252,003,274,489,856,000, approximately 4.3×10^{19} , or roughly forty-three quintillion. Random turning could take longer than the age of the universe to solve it. But the cube's moves form a group: each turn is a transformation, and moves can be composed and undone. Group theory lets players describe an operation that exchanges just two edge pieces while leaving everything else fixed. Such operations affecting very few pieces are called *commutators*, the ingredients of cube algorithms. With computers, mathematicians also proved that any scrambled state can be solved in at most 20 moves, counting a 180° turn as one move. This number was established by using group structure to organize the astronomical number of

positions into equivalence classes and checking them systematically.

All three achievements use the same key: track *transformations between objects*, rather than every individual object, and ask about shared structure as well as shared shape.

What is shared structure? Take a small example. The square's four rotations e, r, r^2, r^3 form a closed group of their own. Elsewhere, consider addition modulo 4 on a clock with marks 0, 1, 2, 3: $1 + 2 = 3$, $2 + 3 = 1$, and $3 + 3 = 2$. One system consists of geometrical actions, the other arithmetic; their members are entirely different. But match r to 1, r^2 to 2, r^3 to 3, and e to 0, and their operation tables are identical. Rotating 90° then 180° gives 270° , just as $1 + 2 \equiv 3 \pmod{4}$. The tables have different labels but the same framework. We call the groups *structurally identical*, or isomorphic.

This is a substantial leap. Geometrical actions and integer addition look unrelated, yet structurally they are the same. Mathematics draws strength from this ability to change labels: whenever the structure matches, a theorem proved on one side transfers immediately to the other. For example, repeatedly adding 1 modulo 4 produces every element, so repeatedly composing r produces all four rotations. A group in which one member generates the whole family is called cyclic. The full eight-element symmetry group has no such member: r cannot produce reflections, and repeatedly applying f gives only itself and e . Looking at structure makes the difference between the eight-action family and its four-rotation subfamily clear.

Looking back over the chapter, we started from sameness of shape (congruence), passed through sameness under actions (symmetry transformations), and arrived at sameness of structure (groups and isomorphism). Each level is more abstract, but each lets us see more at once.

Keep Climbing

For further climbing, remember these signposts: *group theory*, the study of symmetry structures; *isomorphism*, the precise meaning of shared structure; *Galois theory*, using groups to decide solvability by radicals; *permutation groups*, the mathematical model of the Rubik's cube; *wallpaper groups and space groups*, the stories of 17 and 230 types; and *Lie groups*, continuous symmetries and a language of physics. For now these are only signposts, but they all lead out from a square card.

Questions to Think About

Observation

1. Replace the square with an equilateral triangle. Count its symmetries, separating rotations from reflections. Hint: imitate the theorem's proof. How many images can vertex 1 have? How many choices remain for its neighbour?
2. In the square's composition table, find and count all elements satisfying $x^2 = e$. Hint: besides the four reflections, a rotation also returns to the start when performed twice.

Proof

1. Prove that if a and b are symmetries of a figure F , then their composition ab is also a symmetry of F . Hint: return to the definition and check whether the figure still coincides with itself after two steps.
2. Prove the shoes-and-socks rule: $(ab)^{-1} = b^{-1}a^{-1}$. Hint: compose ab with $b^{-1}a^{-1}$ and use associativity to cancel step by step to e . Then consider why the order reverses: socks before shoes in the morning means shoes before socks in the evening.

Challenge

1. A Rubik's cube is played using turns of its six faces. How many *rotational* symmetries preserve the cube as a whole, with reflections forbidden? Hint: watch one face. How many destinations can it have? Once it is fixed, about which axis can the cube still rotate? The answer is a satisfying two-digit number.
2. Design a two-element system of actions that forms a group, and write its composition table. Hint: start with a reflection and the identity, or with adding odd and even numbers, and verify the four group properties.

The Door to the Next Chapter

This chapter turned rotations and reflections into objects we can tabulate and calculate. Yet you may feel a quiet dissatisfaction: checking a composition still means manipulating a card, and locating a point after rotation still requires drawing and measuring. Actions remain *handwork*.

Can transformations become arithmetic? If each plane point is located by (x, y) , can a 90° rotation about the origin become a fixed algorithm on those two numbers, immediately producing the new coordinates for any input? What about translating three centimetres rightward, or doubling the scale? If this works, composing transformations means running two algorithms in succession. Formulas could generate the composition table automatically, and the card could retire.

It can be done. We need a new protagonist: a quantity with both magnitude and direction, the *vector*, and a numerical array that manipulates it, the *matrix*. Next chapter, two-dimensional arrows will lead us into linear algebra, where this chapter's rotations, reflections, and scalings become neat rows of numbers.

Chapter 7 Vectors and Linear Algebra: From Two-Dimensional Arrows to Higher Dimensions

In earlier chapters we used symbols to represent numbers, equations to study relationships, and coordinates to sew algebra and geometry together. We also saw how translations, rotations, and reflections form families of symmetries. This chapter does something apparently small that changes the face of mathematics: draw an arrow for every movement in the plane, and ask whether arrows themselves can be added, multiplied, and studied like numbers.

They can indeed. Once they are, two-dimensional game screens, three-dimensional animation, web rankings with tens of thousands of dimensions, and even vibrating strings all become exercises in one language. That language is linear algebra, one of the most widely applied branches of modern mathematics. Its starting point is simply an arrow in the plane.

7.1 An Afternoon in a Game

Imagine playing an adventure game viewed from above. Your character stands on a grassland, and the screen gives no compass directions, only the character's own orientation. Two keys are available: one steps forward, the other steps left.

Your afternoon task is to reach a well on the map. Facing northeast, you take 3 steps forward and 2 to the left, realize you have overshot, then turn around and take 1 step forward. After all this, how far are you from the starting point, and in which direction?

If the question were simply how many steps you took, any primary-school student could calculate $3 + 2 + 1 = 6$. But that is clearly not what you want: 6 steps might bring you back to the start or carry you far away. You want the arrow that would take you straight from the starting point to the endpoint.

This puzzle plays out daily in the real world. A delivery rider travels 3 kilometres east and 4 north; how far is he from the depot? Wind pushes a sailboat southeast and a current pushes it northeast; where does it end up? A drone receives commands to rise and move forward simultaneously; what path does it actually follow? All these questions share a structure: several movements must be combined into one total movement.

Guess intuitively: after 3 kilometres east and 4 north, how far are you from the start? Many people first say 7 kilometres; others vaguely remember Pythagoras and guess 5. Which is right?

More importantly, can we invent notation that makes guessing unnecessary for this whole class of problems?

7.2 First Attempt: Are Numbers Alone Enough?

The most natural attempt is to give each movement one number, its distance travelled, and add the numbers.

Disaster arrives immediately. Suppose the rider goes 3 kilometres east, turns around, and goes 3 west. His travel-distance account records 6 kilometres and very real effort. But his distance-from-the-depot account places him right at its door, as if he had gone nowhere. Two identical entries of 3 kilometres combine to give zero.

Counterexample Alert

Someone claims that combining two movements gives a total displacement distance equal to the sum of their distances. Is this true? Consider 3 kilometres east followed by 3 west. $3 + 3 = 6$, but the traveller returns to the start, with total displacement 0. The mistake is that a distance contains no directional information. Opposite movements look identical in a distance ledger but cancel in reality. Any attempt to describe movement with a single number will fail on examples like this.

The failure gives an important clue: movement naturally carries two pieces of information, how far and in which direction. Discard direction, and no amount of adding distances produces the right answer. Our new concept must hold both.

A subtler difficulty remains. Even if we record phrases such as “3 kilometres east” and “4 kilometres north,” combining three or five movements becomes a verbal tangle: first east, then north, southeast, northwest, and so on. We need calculable notation, allowing addition, scaling, and equations like numbers, rather than a travel diary.

7.3 Inventing Arrows: The Birth of Vectors

The solution is surprisingly simple: since movement has distance and direction, represent it by an arrow. Its direction is the movement’s direction, and its length the movement’s distance.

Terminology Card: Vector

Intuition: an arrow with direction and length, representing a displacement, force, velocity, or any quantity with magnitude and direction.

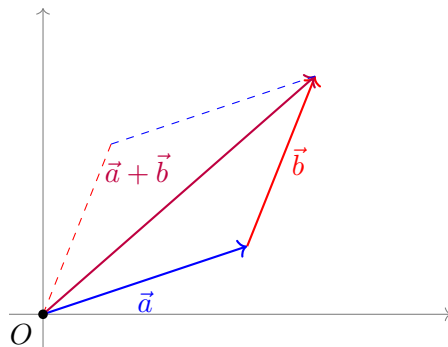
Formal definition: an object in the plane or space determined by an ordered pair or list of real numbers, written $\vec{v} = (x, y)$. Two vectors are equal if and only if their directions and lengths agree, regardless of where they are drawn.

Example: the arrow from the origin to $(3, 4)$ is the vector $(3, 4)$, of length $\sqrt{3^2 + 4^2} = 5$.

Notice “regardless of where they are drawn.” An arrow pointing 3 kilometres east is the same vector whether drawn from your home or your school. Vectors record how to move, not where

you stand. This initially odd convention makes them easy to calculate with: we can translate all arrows to a common starting point for comparison.

The remarkable thing about vector addition is that geometry forces it upon us. What is the total effect of moving by $\vec{a}$ and then $\vec{b}$? Attach the tail of $\vec{b}$ to the tip of $\vec{a}$. The arrow from the initial point to the final point is the answer, written $\vec{a} + \vec{b}$. This is the triangle rule. Alternatively, align the tails and complete a parallelogram; its diagonal is the sum.



With coordinates, addition is almost absurdly simple: add corresponding components.

$$(3, 1) + (1, 2.5) = (3 + 1, 1 + 2.5) = (4, 3.5).$$

Joining head to tail geometrically and adding components algebraically describe the same thing. Chapter 4's coordinate bridge is carrying traffic again.

Addition naturally leads to scaling: stretch an arrow to 2 times its length, shrink it to half, or reverse its direction. This is scalar multiplication, written $2\vec{v}$, $\frac{1}{2}\vec{v}$, and $-\vec{v}$. In coordinates, multiply every component by the same number: $2 \cdot (3, 4) = (6, 8)$.

Return to the game. Write $\vec{f}$ for one forward step and $\vec{l}$ for one leftward step. Your whole afternoon—3 steps forward, 2 left, then turning around and taking 1 forward—becomes

$$3\vec{f} + 2\vec{l} + (-1)\vec{f} = 2\vec{f} + 2\vec{l}.$$

A travel diary has become one line of algebra. An expression like $3\vec{f} + 2\vec{l}$, formed by multiplying vectors by numbers and adding, is a **linear combination**. Linear here means that we allow only addition and scalar multiplication, those two simplest operations, without multiplying vectors together, squaring them, or doing anything more elaborate. Do not underestimate the restriction: mathematics built from just these two operations is among the world's most powerful.

The next question determines the chapter's direction: which arrows should be our basic steps so that every possible movement in the plane is a linear combination of them?

One arrow is plainly insufficient: if you can only move forward, you can never reach a well to the side. What about two noncollinear arrows, such as $\vec{f}$ pointing east and $\vec{l}$ north? Intuitively, any movement can be divided into some amount east and some amount north. Two independent directions seem sufficient in the plane, while a 3rd arrow, say northeast, is redundant because it is already a linear combination of the first two.

Terminology Card: Basis and Dimension

Intuition: a basis is a collection of basic steps, with too few leaving destinations unreachable and too many creating redundancy. The dimension counts those steps.

Formal definition: a collection of vectors is a basis of a space if it is linearly independent (none is a linear combination of the others) and its linear combinations fill the entire space.

The number of vectors in a basis is the dimension of the space.

Example: $\vec{e}_1 = (1, 0)$ and $\vec{e}_2 = (0, 1)$ form a basis of the plane, which is two-dimensional. So do $(1, 0)$ and $(1, 1)$: a basis is not unique.

Dimension, a word we have used casually for years, finally has a precise meaning: how many independent basic directions are minimally needed to assemble every vector? A line has one dimension, a plane two, and our physical space three. Algebra is untroubled by larger numbers: a list of a hundred components, $(x_1, x_2, \dots, x_{100})$, can still be added and scaled, and is a vector in a hundred-dimensional space. We can no longer draw the arrow, but the arithmetic works as usual. This is the title's passage to higher dimensions: algebraic rules carry two-dimensional intuition unchanged into places our eyes cannot see.

7.4 The Central Theorem: Each Arrow Has Just One Expression

Why believe that two independent arrows can assemble everything in the plane? Intuition is not enough. Let us make it a theorem and prove it.

Theorem 7.1 (Uniqueness of Coordinates). *Let $\vec{e}_1$ and $\vec{e}_2$ be two noncollinear vectors in the plane. Every plane vector $\vec{v}$ can be written uniquely as*

$$\vec{v} = a\vec{e}_1 + b\vec{e}_2,$$

where a and b are real numbers. The representation always exists, and there is exactly one pair of coefficients.

Proof. First prove existence. Translate the tails of $\vec{e}_1$, $\vec{e}_2$, and $\vec{v}$ to a common point O . Through the tip of $\vec{v}$, draw a line parallel to $\vec{e}_2$, meeting the line containing $\vec{e}_1$ at P . Since $\vec{e}_1$ and $\vec{e}_2$ are noncollinear, these lines are not parallel and must meet. The arrow from O to P is collinear with $\vec{e}_1$, so it is some multiple $a\vec{e}_1$ of $\vec{e}_1$. The segment from P to the tip of $\vec{v}$ is collinear with $\vec{e}_2$, so it is some multiple $b\vec{e}_2$. The triangle rule gives $\vec{v} = a\vec{e}_1 + b\vec{e}_2$.

Now prove uniqueness. Suppose one arrow has two expressions:

$$\vec{v} = a\vec{e}_1 + b\vec{e}_2 = a'\vec{e}_1 + b'\vec{e}_2.$$

Subtracting gives $(a - a')\vec{e}_1 + (b - b')\vec{e}_2 = \vec{0}$. If $a - a' \neq 0$, then

$$\vec{e}_1 = -\frac{b - b'}{a - a'}\vec{e}_2,$$

which makes $\vec{e}_1$ a multiple of $\vec{e}_2$: the arrows would be collinear, contradicting the assumption. Thus $a = a'$, and similarly $b = b'$. There is just one representation. $\square$

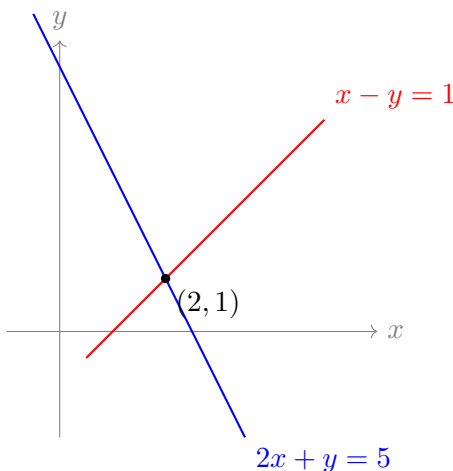
Pause to examine the theorem. Once a basis is chosen, every plane vector corresponds one to one with an ordered pair (a, b) of real numbers. Vectors are geometrical objects and coordinates algebraic objects; the theorem guarantees a perfect translation, with nothing added or lost. In Chapter 4 we wrote (x, y) every day while implicitly relying on this result. It has finally stepped out from backstage to receive a proof.

The same linear-combination viewpoint resolves a geometrical puzzle left from Chapter 2. What picture does the system

$$\begin{cases} 2x + y = 5, \\ x - y = 1 \end{cases}$$

draw? There are two views, both correct and important.

The first reads row by row: each equation is a line, and a solution is a shared point. Intersecting lines give one solution; distinct parallel lines give none; coincident lines give infinitely many. There are no other cases, because two plane lines have only these three relationships. Numerically, adding gives $3x = 6$, so $x = 2$; substitution gives $y = 1$, and the intersection is $(2, 1)$.



The second reads column by column, rewriting the system as

$$x \begin{pmatrix} 2 \\ 1 \end{pmatrix} + y \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}.$$

It asks which multiples of $(2, 1)$ and $(1, -1)$ assemble the target arrow $(5, 1)$. The solution is unique precisely because the two arrows are noncollinear: they form a basis, and the theorem guarantees one assembly. No solution corresponds to collinear arrows with the target off their line, unreachable by any combination. Infinitely many solutions correspond to all three arrows being collinear, allowing many combinations.

In three dimensions, each linear equation in three unknowns is a plane. Three planes may meet at one point (one solution), along a line (infinitely many), or pairwise like open book pages

without a common point (no solution). Algebra's questions of existence and number of solutions become geometry's questions of how planes are arranged. Linear algebra is essentially the art of translating between these pictures.

7.5 Matrices: Putting Transformations into Arrays

Chapter 6 studied actions such as rotation, scaling, and reflection in geometrical language: rotate the entire plane 90° about the origin, or multiply every distance from the origin by 2. With vectors and coordinates, we can ask something practical: can every action have an instruction sheet into which we insert any point's coordinates and calculate its new coordinates?

Start with a 90° anticlockwise rotation. Experiment on the basis arrows: $\vec{e}_1 = (1, 0)$ goes to $(0, 1)$, and $\vec{e}_2 = (0, 1)$ goes to $(-1, 0)$. By uniqueness of coordinates, an arbitrary vector $(x, y) = x\vec{e}_1 + y\vec{e}_2$ becomes

$$x \cdot (0, 1) + y \cdot (-1, 0) = (-y, x).$$

Check a specific point: $(3, 1)$ should become $(-1, 3)$. Draw it on squared paper: the length is unchanged and the direction has turned one right angle anticlockwise.

Place the two images $(0, 1)$ and $(-1, 0)$ side by side as columns in an array:

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

This array is a matrix. Define multiplication of A by a vector (x, y) using the row-times-column rule:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \cdot x + (-1) \cdot y \\ 1 \cdot x + 0 \cdot y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}.$$

The first row computes the first output component, and the second row the second. A rotation of the entire plane has been compressed into four numbers.

Now scale every distance from the origin by 2. The direction of $\vec{e}_1$ stays fixed as it lengthens to $(2, 0)$, and $\vec{e}_2$ becomes $(0, 2)$. The matrix is

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix},$$

which takes (x, y) to $(2x, 2y)$. Similarly, reflection in the x -axis is $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, and a horizontal

shear is $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Why can matrices hold these transformations? Rotations, scalings, reflections, and shears share a helpful temperament: they take lines to lines, fix the origin, and satisfy

$$T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v}), \quad T(c\vec{v}) = cT(\vec{v}).$$

A transformation with these two properties is **linear**. Their power is that once we know the destinations of two basis arrows, the destination of any vector $x\vec{e}_1 + y\vec{e}_2$ is forced: its image

must be the same linear combination of the basis images. Thus all information about a linear transformation is concentrated in the two basis images, the matrix's two columns. A matrix is a complete record of a transformation, not an arbitrary arrangement of numbers.

What about successive actions? Rotate 90° and then scale by 2. The combined matrix is exactly the product of the two matrices under the row-times-column rule. Calculate it explicitly:

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 \cdot 0 + 0 \cdot 1 & 2 \cdot (-1) + 0 \cdot 0 \\ 0 \cdot 0 + 2 \cdot 1 & 0 \cdot (-1) + 2 \cdot 0 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}.$$

Check $(3, 1)$: rotate 90° to get $(-1, 3)$, then scale by 2 to get $(-2, 6)$. Applying the product directly gives $(0 \cdot 3 + (-2) \cdot 1, 2 \cdot 3 + 0 \cdot 1) = (-2, 6)$, exactly the same. Watch the order, though. Scaling then rotating gives the same product here because these two transformations happen to commute. But rotating then reflecting and reflecting then rotating generally differ. Matrix multiplication is not always commutative, matching the importance of action order already seen in Chapter 6's discussion of the cube.

Chapter 6's composition was a geometrical operation; matrix multiplication is its shadow in coordinates. Your game character turning, the camera zooming, and a three-dimensional model appearing on screen all involve chains of matrix products. Much of a graphics chip's work every second is simply very fast matrix multiplication.

Connections

Looking back, Chapter 4 used numbers to describe points, and this chapter uses arrays to describe transformations: the same bridge has grown wider. Chapter 6's rotations and reflections have become explicit arrays, and the square's eight symmetries correspond to eight 2×2 matrices. Looking ahead, in Chapter 8 vectors and matrices will describe the line that a curve resembles near a point. In secondary school and university they will appear in physical forces and velocities, economic input-output models, and every frame of computer graphics. Linear algebra is a common native language of science and engineering.

7.6 Arrows That Keep Their Direction: Eigenvectors

A typical linear transformation turns most arrows in new directions. A 90° rotation is extreme: no arrow retains its original direction. But some transformations hide loyal arrows whose direction stays the same, or reverses exactly, while only their length changes.

Consider a specific example:

$$M = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Try a few arrows. $(1, 0)$ goes to $(2, 1)$, changing direction; $(0, 1)$ goes to $(1, 2)$, also changing direction. But $(1, 1)$ goes to

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

keeping its direction exactly and growing by a factor of 3! Now examine $(1, -1)$, along the northwest–southeast direction. Perhaps the transformation treats it more gently. Calculation gives

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

with both direction and length unchanged.

Terminology Card: Eigenvectors and Eigenvalues

Intuition: an arrow that a transformation does not turn aside. It keeps its direction or reverses, and is merely stretched, shortened, or left alone.

Formal definition: for a matrix A , if a nonzero vector $\vec{v}$ satisfies $A\vec{v} = \lambda\vec{v}$ for some number λ , then $\vec{v}$ is an eigenvector of A , with corresponding eigenvalue λ .

Example: the matrix M above has eigenvector $(1, 1)$ with eigenvalue 3, and eigenvector $(1, -1)$ with eigenvalue 1.

Why find these arrows? Eigenvectors are the transformation’s principal axes. Along them, a complicated transformation reduces to the simplest action: stretching or shrinking. The matrix M seems to mix up the plane, but it merely stretches the northeast–southwest direction by 3 and leaves the northwest–southeast direction unchanged. Understand those two axes and you understand the whole transformation. Mathematicians often say that the best way to understand a linear transformation is to find all its eigenvectors, like taking a photograph of its skeleton.

7.7 Another Perspective: Compression, Ranking, and Vibration

Linear algebra sounds abstract, yet it hides in three things you use every day.

Image compression. A grayscale photograph with 512×512 pixels is essentially a 512×512 matrix, with each entry recording brightness. Direct storage requires more than 26 times ten thousand numbers. But a photograph is not random: the sky’s values are almost identical, and a wall corner changes sharply in only a few directions. Linear algebra offers singular value decomposition, splitting the matrix into a sum of characteristic directions, with the most important few carrying over ninety percent of the visual information. Keep only the first few dozen and discard the rest, and the data may shrink to a tenth while the eye notices little difference. Every JPEG on your phone and every streamed animated episode relies on this kind of idea: approximating a whole matrix through a few important directions.

Web ranking. Imagine the Internet as a giant network: every page is a point and every link an arrow. How does a search engine judge importance? An ingenious idea gives each page an importance score equal to the sum of contributions from pages linking to it. Being cited by more important pages makes a page more important. Try a toy Internet of three pages: page A is cited by B and C, B by C, and C by A. Give them scores x , y , and z . The rules say that x receives contributions from B and C, y from C, and z from A: three unknowns and three equations. Arrange the scores as a three-dimensional vector $\vec{v} = (x, y, z)$ and the link rules in a

3×3 matrix A , and the equation is exactly $A\vec{v} = \vec{v}$.

Replace the toy with the real Internet, and tens of billions of pages mean tens of billions of unknowns. Frightening? It is still a linear system, specifically the problem of finding an eigenvector of a huge matrix. The score vector satisfies $A\vec{v} = \vec{v}$, so it is an eigenvector with eigenvalue 1. In practice, we avoid solving directly and use the trick in Experiment 2: begin with guessed scores and multiply repeatedly by A . The vector with billions of dimensions quickly leans toward the most important eigenvector, establishing the ranking. At the heart of one of the world's most profitable algorithms lies this chapter's equation $A\vec{v} = \lambda\vec{v}$.

Vibration. Pluck a guitar string fixed at both ends and it vibrates to make sound. Its constantly changing shape seems complicated, but physicists discovered that every vibration can be decomposed into standard modes: the low fundamental tone, the overtone an octave above, and so on. Each standard mode is an eigenvector of the string's equation of motion, and its eigenvalue determines the pitch. A string's rich tone is many eigenmodes singing together. Earthquake-resistant buildings, bridge design, and microphone noise reduction all use the same analysis: break complicated motion into simple motions along characteristic directions.

Mathematical Toolbox

This chapter's new tools:

- Vectors: an arrow, or coordinate pair, records magnitude and direction together. Addition joins head to tail geometrically and adds components algebraically.
- Linear combinations and bases: every vector is a linear combination of basis vectors with unique coefficients, making coordinates reliable.
- Dimension: the minimum number of independent directions needed to fill a space; the same rules work in a hundred dimensions.
- Matrices: complete records of linear transformations, with two columns giving the images of two basis arrows. Matrix multiplication corresponds to composition.
- The geometry of systems: intersections of lines and planes express the problem of assembling a target arrow from basis arrows.
- Eigenvectors: arrows retaining their direction reveal a transformation's principal axes, the common heart of compression, ranking, and vibration analysis.

One-Minute Experiment

Experiment 1: Building with Basis Arrows. On squared paper, use $\vec{e}_1 = (2, 1)$ and $\vec{e}_2 = (-1, 2)$ as basis arrows. Assemble $\vec{v} = (1, 8)$ by finding $a\vec{e}_1 + b\vec{e}_2 = \vec{v}$, or solving $2a - b = 1$ and $a + 2b = 8$. Calculate, then draw the triangle rule to check. Replace the basis with the collinear arrows $(2, 1)$ and $(4, 2)$ and try to assemble $(1, 8)$ again. Experience how collinearity leaves a direction missing.

Experiment 2: Following an Arrow's Fate. Take $M = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Draw $(1, 0)$, $(1, 1)$, $(1, -1)$, and $(2, 1)$, calculate their images under M , and draw them too. Which arrows keep their direction? Which stretches most? Apply M three times successively to $(1, 0)$ and watch the direction it increasingly approaches: the eigenvector with the largest eigenvalue. This is how web-ranking algorithms actually compute an eigenvector: repeated multiplication lets the most important direction emerge.

Keep Climbing

For further climbing, look for these terms: determinants, measuring area scaling and detecting whether a basis degenerates; inverse matrices, instructions for undoing transformations; computing eigenvalues through the characteristic equation; singular value decomposition, the main character in image compression; Markov chains, the probabilistic clothing of web ranking; and vector spaces and Hilbert spaces, extending vectors to functions and providing a home for Fourier series. Competition topics such as the fundamental theorem of plane vectors and points dividing segments in a given ratio are direct descendants of this chapter's story.

Questions to Think About

Observation

1. Draw any three arrows $\vec{a}$, $\vec{b}$, and $\vec{c}$ on squared paper. Use the triangle rule to draw $(\vec{a} + \vec{b}) + \vec{c}$ and $\vec{a} + (\vec{b} + \vec{c})$. What do you observe? Now compare $\vec{a} + \vec{b}$ and $\vec{b} + \vec{a}$. (Hint: carry out every addition head to tail, rather than trusting a visual impression.)

2. For the horizontal shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, calculate the images of the x -axis arrow $(1, 0)$ and the y -axis arrow $(0, 1)$. Are there eigenvectors? (Hint: an arrow keeps its direction only if its two components scale by the same factor.)

Proof

1. Prove that if $\vec{v}_1$ and $\vec{v}_2$ are eigenvectors of A with eigenvalue λ , every nonzero linear combination $a\vec{v}_1 + b\vec{v}_2$ is also an eigenvector with eigenvalue λ . (Hint: start from $A\vec{v}_1 = \lambda\vec{v}_1$ and $A\vec{v}_2 = \lambda\vec{v}_2$, and apply the two linearity properties to $A(a\vec{v}_1 + b\vec{v}_2)$.)

2. Prove that anticlockwise rotation through θ about the origin has matrix $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$.

(Hint: calculate only the images of $(1, 0)$ and $(0, 1)$; these are the columns. Recall the meanings of $\cos$ and $\sin$ from coordinates on the unit circle.)

Challenge

1. Can you find arrangements of three planes that intersect pairwise without a shared point, besides the open-book picture mentioned in the text? What algebraic reason for inconsistency corresponds to each arrangement? (Hint: intersect the first two planes in a line, then consider the third plane's possible relationships with that line.)

2. Let A act repeatedly on the plane: $\vec{v}_0 = (1, 0)$ and $\vec{v}_{n+1} = A\vec{v}_n$. For $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, calculate $\vec{v}_1$ through $\vec{v}_4$. Which arrow does the direction of $\vec{v}_n$ approach, and approximately what factor multiplies its length at each step? Can you explain why repeated multiplication by the same matrix always draws an arrow toward the eigenvector with largest eigenvalue? (Hint: express $\vec{v}_0$ as a linear combination of the two eigenvectors and count how often each term is multiplied by 3 or by 1.)

The Door to the Next Chapter

This chapter lived in a straight world: vectors were straight arrows, matrices sent lines to lines, and systems were solved at intersections of lines and planes. Linear algebra is powerful precisely because straight lines are the easiest geometrical objects to understand completely.

But the world is not always straight. Roller-coaster tracks bend, the water surface in a coffee cup curves, and Earth's surface is curved. Does a curved track still have a direction? Can an arrow sit on a sphere? Can a number measure where the bending is sharpest?

The mathematicians' answer fits this chapter's spirit surprisingly well. Curved objects may be complicated as wholes, but magnify enough and look locally enough, and each small piece is approximately straight. Near each point of a bend lies a line that best follows it; near each point of a surface lies a plane that best fits it. Lines and planes are exactly the objects we have just mastered. Next chapter, we will ask why a tiny part of a curve resembles a line and extend our linear world across the curved one.

Chapter 8 From Lines to Surfaces: A Preview of Local Approximation

8.1 A Bicycle Race in the Rain

The Tour of Qinghai Lake reaches a downhill section as light rain begins. A helicopter camera looks down on a mountain road winding like a grey ribbon across green slopes. The commentator keeps repeating, “The next bend is very sharp; the riders must slow down early.”

The riders do behave differently at different bends. At some, they barely touch the brakes and pass with a gentle lean. At others, they reduce speed well in advance and lean their bikes steeply, their tyres tracing arcs across the wet road. Clearly, sharp and gentle bends differ objectively; these are more than a commentator’s flourishes.

Now imagine being the race engineer. The organizers hand you a plan of the route: a winding line on a map, with a scale but no contours or other information. They ask two apparently simple questions.

First: which bend is sharpest? Can you devise a ruler for measuring how much a road bends?

Second: why, when the map is enlarged many times, does every little piece of even a winding road look almost straight?

Before reading on, guess intuitively. For the first question, most people say that a bend is sharp when direction changes quickly. True, but that only rephrases the question: what does quickly mean, and in what units? The second question often gives people pause. It seems self-evident until someone asks why, and then it is not so obvious. These two questions begin our chapter.

8.2 First Attempt: Some Unsuitable Rulers

The simplest way to measure bending is to borrow a straight line. Curves are unfamiliar, but lines are old friends, so perhaps we can measure how far a bend departs from a line.

First ruler: join the bend’s endpoints and measure the segment. A sharper bend should bring the endpoints closer and shorten the segment, right? Wrong. A long, gently curving arc and a short, abrupt hairpin can have exactly the same endpoint distance. That distance reflects how much of the road you selected, not how sharply it bends.

Second ruler: again join the endpoints, but measure the greatest distance from the curve

to this chord, the so-called sagitta. Does greater height mean greater bending? Still no. Select a large piece of the same bend and the height may be large; select a small central piece and it shrinks. A measure that changes arbitrarily with the chosen interval cannot describe a degree of bending. That should be the bend's own property, not something an observer can alter at will.

Both failures suggest an important clue: bending must be determined near a point. You cannot judge it by standing far away and enclosing a large stretch; you must look close enough that even an arbitrarily small piece has a definite degree of bending. Mathematicians call this near-a-point viewpoint **local**. This chapter's central idea is that curved things are locally straight. To study bending, first find the line a curve locally resembles, then examine how it gradually departs from that line.

But what exactly does looking closely mean? How close is close? We need a clearer picture.

8.3 Putting a Curve Under a Microscope

Begin with something familiar that you may never have examined carefully: Earth is round, yet the playground under your feet looks flat. The playground is not special; you are small and Earth is large. Let us calculate. Walking an arc length s on a circle of radius R changes your direction by s/R radians. Earth's radius is about 6 400 000 metres, so walking 1 metre along its surface changes direction by only

$$\frac{1}{6\,400\,000} \text{ radians} \approx 0.000\,008\,9^\circ.$$

This angle is too small for any protractor to measure, so everyday experience insists that the ground is flat. Ancient people believed in a flat Earth because its local flatness is so convincing. Only sufficiently precise observations—comparing noon shadows at two places, or watching a distant ship's hull disappear before its mast—revealed the curvature behind that local appearance. Local flatness and global curvature are compatible: they are two faces of the same object at different scales.

Reverse the idea. Focus on a small piece of a winding curve and magnify it, then magnify again and again. Provided the curve is sufficiently smooth, without breaks or sharp corners, enough magnification makes it look straight. This happens with circles, parabolas, and maps of mountain roads. It is no coincidence; it expresses the meaning of smoothness. Under a microscope, a smooth curve is straight.

Counterexample Alert

“Every curve looks straight if magnified enough.” False. Draw a V, the graph of $y = |x|$, and magnify its corner. Double it, enlarge it tenfold or a hundredfold: the same corner remains, with exactly the same angle. Magnification changes size, not angles. Looking locally straight is a privilege of smooth curves, unavailable to curves with corners or breaks. Mathematicians give this property a formal name, differentiability, which will be a main character in Part III. For now, remember that this chapter concerns smooth curves without sharp corners.

One-Minute Experiment

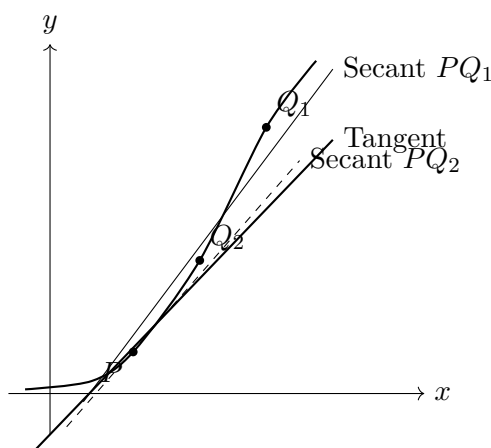
You can see magnification making a curve straight without special instruments. Draw a winding line and mark a fingernail-sized patch around one part with a pencil. Photocopy an enlargement centred on the mark, then enlarge the copy again. After two or three rounds, the marked portion will look almost like a straight segment. A magnifying glass focused on a small portion gives a similar effect. Repeat with a sheet bearing a V and compare the corner: the counterexample is in your hands.

8.4 The Tangent: A Curve's Stand-in at a Point

If a small piece of a smooth curve resembles a line, the next step is natural: find that line and make it the curve's stand-in near the point. Lines are familiar—a direction or slope can be described with one number. Translating a curved problem into a straight one immediately lowers the difficulty.

But which line is the best fit? A trap awaits. You might say that a tangent is a line touching the curve at only one point. That impression comes from circles and fails immediately for general curves. At the vertex of $y = x^2$, the vertical line also meets the parabola only once, but it is plainly the wrong stand-in: its direction does not follow the parabola at all. Conversely, a line fitting best locally may meet the curve again farther away. Meeting only once is a property of a circle's tangent, not the definition of a tangent.

The right idea grows from our failed rulers and repeated magnification. Fix a point P on the curve, choose a nearby point Q , and join them by the line PQ . This is a **secant**; it cuts into the curve and describes the average direction from P to Q . Now slide Q along the curve toward P . The secant turns, its turning becomes smaller, and its position becomes more stable. The position it settles toward is the **tangent** at P .



Definition 8.1 (An Intuitive Definition of the Tangent). Let P lie on a smooth curve, and choose another point Q on it. The definite line approached by the secant PQ as Q moves arbitrarily close to P along the curve is called the tangent at P .

A subtle point hides in this definition: Q can never actually equal P . Two points determine a line; once Q reaches P , the secant ceases to exist. A tangent is determined by a process that never reaches its endpoint. For now we accept it intuitively. The difficulties it creates, and how mathematicians tame it into a rigorous limit, belong to the next chapter.

Terminology Card: Tangent

Intuition: a curve's best straight-line stand-in at a point, the shape seen under a microscope.

Formal definition: the limiting position of the secant PQ as Q approaches P . Example: at every point of a circle, the tangent is perpendicular to the radius through that point.

That example deserves scrutiny because it checks the new idea against old knowledge. Elementary geometry says that a circle's tangent is perpendicular to its radius. Can approaching secants recover this? Yes. Let the centre be O , choose P on the circle, then choose Q and Q' symmetric about OP . Symmetry makes the chord QQ' perpendicular to OP . Slide Q and Q' simultaneously toward P around the circle. They remain symmetric at every step, so the secant remains perpendicular to OP . Its limiting position, the tangent, is therefore perpendicular to OP too. The new method makes the old conclusion a special case of a wider picture. That is an important test of a mathematical idea's reliability.

What use is this stand-in? Plenty. If a wheel suddenly loses its grip while turning, the vehicle shoots off in the tangent direction. A hammer thrower's hammer flies tangentially when released, and sparks from a grinding wheel do the same. A curve's intention at each instant is written in its tangent.

8.5 How Sharply Does It Bend? Curvature

With tangents, direction becomes precise: a curve's direction at a point is its tangent's direction. We can now build our ruler for bending.

Imagine being an ant crawling along a curve with a compass continuously recording the angle of your direction. Over equal distances, the compass barely moves on a gentle stretch but turns rapidly at a sharp bend. We can therefore define the degree of bending as how much direction changes per unit arc length.

Definition 8.2 (A Preliminary Definition of Curvature). The curvature at a point of a smooth curve is the rate at which the tangent's direction angle changes with arc length: near that point, the number of radians the tangent turns per unit arc length travelled. Greater curvature means sharper bending.

The definition is somewhat abstract, but one family of curves lets us calculate it completely: circles.

Theorem 8.3 (The Curvature of a Circle). *A circle of radius R has the same curvature everywhere, equal to $\frac{1}{R}$.*

Proof. Start at a point on the circumference and travel an arc length s to another point. Let the corresponding central angle be θ , measured in radians. The arc-length formula gives

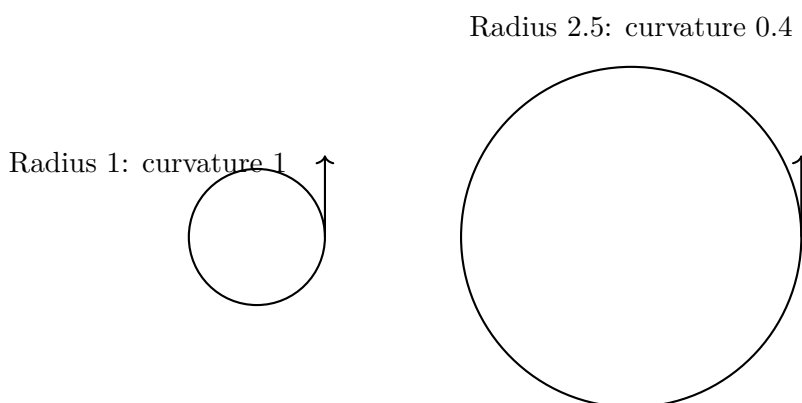
$$s = R\theta.$$

Now consider the change in direction. The radius is always perpendicular to the tangent. From start to finish, the radius turns through the central angle θ . Perpendicularity is preserved during this turning, so the tangent also turns through θ . Thus the ratio of direction change to arc length is

$$\frac{\theta}{s} = \frac{\theta}{R\theta} = \frac{1}{R}.$$

This ratio contains no information about the starting point, so it is the same everywhere on the circle, as required. $\square$

Example 8.4. On a roundabout of radius 10 metres, every 1 metre travelled changes direction by $\frac{1}{10}$ radian, about 5.7° . On a sharp bend of radius 2 metres, every 1 metre changes direction by $\frac{1}{2}$ radian, about 28.6° . The latter has five times the curvature. Sharper bending means smaller radius and greater curvature, matching our intuition.



Note. A straight line never turns, so its direction angle is constant and its curvature is 0. Through the formula $\frac{1}{R}$, we can regard it as a circle of infinite radius. This is a useful viewpoint: a line is a curve with zero bending, as naturally as 0 represents none.

Terminology Card: Curvature

Intuition: how sharply something bends, measured by the turn in direction per unit distance.
 Formal definition: the rate of change of the tangent's direction angle with arc length; a circle's curvature is the reciprocal of its radius. Example: a circle of radius 4 centimetres has curvature $\frac{1}{4}$ radian per centimetre.

Curvature is part of engineers' everyday work. A railway joining a straight track to a bend must not let curvature jump abruptly from 0 to $\frac{1}{R}$, or the train suffers a violent sideways jolt. A clever design inserts a transition in which curvature rises smoothly and gradually from zero to $\frac{1}{R}$, almost unnoticed by passengers. Motorway ramps, roller-coaster tracks, and the tightening

arcs at loop entrances all use the same principle: let the amount of bending change smoothly. An abstract definition of direction change per unit arc length ends up helping every passenger's seat belt.

A general curve has different curvatures at different places: mountain-road straights have curvature 0, gentle bends little curvature, and sharp bends much. Mathematicians use a natural and elegant approach: at each point find the circle that fits the curve best, neither too loose nor too tight. This is the **osculating circle**, and the point's curvature is the reciprocal of its radius. At a straight portion that radius is infinite; at a sharp bend it is small and tight. For now this is only an intuitive description. A rigorous definition needs derivatives and must wait for the next part. But notice our method: approximate a curve by a line, then approximate its departure from the line by a circle. Step by step, bending becomes increasingly precise local approximation, a path leading deep into differential geometry.

8.6 Another Perspective: Straight Lines on a Sphere

So far we have stayed in the plane. Change the stage and see how far these ideas travel.

Open a flight-tracking app and examine a flight from Beijing to New York. On an ordinary flat map, its route is not straight but a broad northward arc passing near the Arctic Circle. Why the detour? There is no detour. Flattening a sphere into a map distorts shapes; on the real sphere, this curved-looking route is almost the straightest path.

In the plane, straight and shortest describe two sides of the same property: a segment is the shortest connection between two points. Carry that property to a sphere to identify its straight lines, the shortest routes between two surface points. They follow circles cut by planes through the sphere's centre: **great circles**. The equator and every complete meridian circle are great circles. Other latitude circles are not, since their planes miss the centre, so following a latitude is not the shortest route.

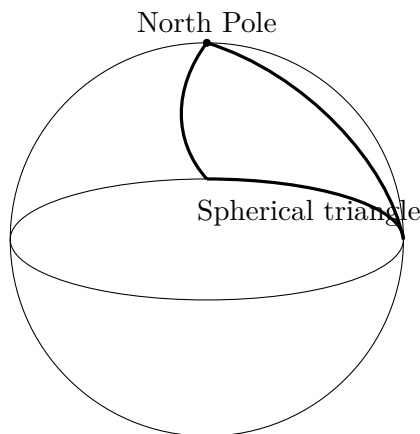
Terminology Card: Great Circle

Intuition: a straight line on a sphere, giving its straightest and shortest routes. Formal definition: a circle formed by intersecting a sphere with a plane through its centre, also called a geodesic. Example: the equator and every complete meridian circle are great circles; a short route from Beijing to New York follows one.

Something astonishing happens once great circles appear. Use their arcs as sides on a sphere and you can form spherical triangles. Consider one explicitly: start at the North Pole, follow a meridian south to the equator, travel east along one quarter of the equator, then follow another meridian north to the pole. All three sides are great-circle arcs, forming a standard spherical triangle. What is its interior angle sum?

Theorem 8.5 (A Triangle with Three Right Angles). *The spherical triangle just described has three angles of 90° , summing to 270° .*

Proof. There are two steps. First, examine the equatorial angles. At each intersection, the equator points east and the meridian north. East–west and north–south are perpendicular, so each meridian meets the equator at 90° : both base angles are right. Second, examine the polar angle. The angle between the meridians at the pole equals their longitude difference. Travelling one quarter of the equator gives exactly $360^\circ \div 4 = 90^\circ$ of longitude difference. All three angles are 90° , summing to 270° . $\square$



270° ! One of plane geometry's firmest laws says triangle angles always sum to 180° . On a sphere it fails, with a whole extra 90° . That extra amount is called the **spherical excess**. First see how large the triangle is. The two meridian circles divide the sphere from pole to pole into four wedges, and the equator halves each into north and south. Our triangle is one piece, or one-eighth of the sphere. Larger triangles have greater excess: this one-eighth-sphere triangle has 90° excess. Walk only one-eighth of the equator before returning, and the triangle shrinks, its polar angle becomes 45° , its angle sum 225° , and its excess halves. In the early nineteenth century, mathematicians distilled this observation into an exact theorem: a spherical triangle's excess is proportional to its area. On a sphere of radius R , the excess in radians equals the area divided by R^2 . We will not give the full proof, but the intuition is clear: a larger triangle encloses more curvature, while a tiny one nearly lies in a plane and its angle sum nearly returns to 180° .

Pause to appreciate that last point. An ant living on a sphere need not fly into space to see its world's curvature. It can draw a triangle on the ground and measure its angle sum. Curvature is written into the surface's own geometrical laws and can be detected with rulers, protractors, and light. Mathematicians call this an **intrinsic** property. Both the ground beneath us and the heavens above can be investigated from within in this way.

Connections

This chapter's ideas have relatives throughout the book and beyond it. Looking ahead, the limiting process in secants approaching a tangent is Chapter 9's main character; tangent slope is the geometrical face of the derivative in Chapter 10; and curvature and spherical geometry grow into differential geometry in Chapter 17. Beyond the book, satellite navigation uses great circles rather than straight map lines for surface distances. Einstein's general relativity goes further: gravity is the curvature of spacetime, and planets orbit the Sun along straight lines in that curved spacetime. Even our ant has real counterparts: surveyors measure triangle angle sums on the ground to investigate Earth's surface shape.

One-Minute Experiment

Find an orange, apple, or tennis ball, a marker, and a flexible tape measure. Draw an equator around its middle and two meridians through its poles, meeting at roughly a right angle. Their arcs enclose a spherical triangle. Tear a small paper wedge as an angle template, or hold a protractor against the surface to estimate the three angles. Each is near 90° , and the sum clearly exceeds 180° . Finally, try something more destructive: peel the orange and flatten a piece of peel on a table. It must either curl upward or split. A spherical surface cannot flatten without stretching or tearing. This is why every world map must distort something, and it is tangible evidence of curvature.

Mathematical Toolbox

This chapter adds five tools to your backpack.

- **Local approximation through magnification:** study bending by enlarging a small piece. Smooth curves locally resemble lines; smooth surfaces locally resemble planes.
- **From secants to tangents:** average direction approaches instantaneous direction. A tangent is the limiting secant, a curve's straight-line stand-in at a point.
- **Curvature:** direction change per unit arc length measures bending. A circle has curvature $= 1/R$, and a line has curvature 0.
- **Great circles and geodesics:** the straightest surface routes are defined through shortest paths; on a sphere they follow circles cut by planes through its centre.
- **Spherical excess:** the amount by which a spherical triangle's angle sum exceeds 180° , proportional to area. It makes curvature measurable from within the surface.

Keep Climbing

For further climbing, remember these terms for later encounters in school, university, or competition reading: limits and derivatives, making arbitrarily close rigorous; osculating circles and radii of curvature, second-order local approximations of curves; Gaussian curvature and Gauss's remarkable theorem, showing that surface curvature is determined entirely by measurements within the surface; geodesics and Riemannian geometry; and popular accounts of curved spacetime in general relativity. Behind each term stands an intuitive picture already met here.

Questions to Think About**Observation**

1. Find a familiar bend, such as your route to school, a track at the playground, or a roller-coaster diagram. Mark where you think curvature is greatest and least. Hint: imagine driving along it. Where must the steering wheel turn most sharply, and where hardly at all?
2. On a globe, examine the equator, meridian circles, and the latitude circle at 60° north. Which are great circles? Is flying due east from Beijing along 60° north to a European city at the same latitude the shortest route? Hint: imagine slicing Earth with a plane through its centre. Does the plane of that latitude circle pass through the centre?

Proof

1. Verify a circle's curvature using an entire semicircle instead of the arc-length formula $s = R\theta$. Travelling halfway around, how much does the tangent direction turn, and how far do you travel? Does their ratio still give $\frac{1}{R}$? Hint: the tangent directions at the semicircle's endpoints are opposite, and the arc length is πR .
2. Prove that the three-right-angle spherical triangle occupies exactly one-eighth of the sphere, and verify that its excess in radians equals its area divided by R^2 . Hint: the whole sphere has area $4\pi R^2$. Is one-eighth of the sphere made of four such triangles or eight? The excess is $270^\circ - 180^\circ$; remember to convert degrees to radians.

Challenge

1. Roll a rectangular sheet into a cylinder without stretching or tearing it. What happens to a straight segment drawn on the paper? Do triangle angles on the cylinder sum to more than 180° ? Hint: without stretching or tearing, every length and angle on the sheet is preserved. Could an ant living there distinguish a plane from a cylinder using our angle-sum method? How does this differ fundamentally from a sphere?

2. An ant carries a small arrow around a spherical triangle, keeping it as far as possible pointing forward without spinning along each edge, then changing direction with the triangle's corners. After one circuit, does the arrow point as it did initially? If not, by how much does it differ? Hint: try the plane first, where the answer should be unchanged, then the spherical right triangle. Add the corner turns and compare with spherical excess.

The Door to the Next Chapter

Looking back, we have essentially invited the infinitely small into mathematics. Magnified indefinitely, a smooth curve resembles a line; a tangent is where secants go as Q approaches P without bound; curvature is the rate of direction change on an infinitesimal piece; and sufficiently small spherical triangles have angle sums arbitrarily close to 180° . Every central concept rests on a process of endless approach without arrival.

Today we accepted all of this through intuition. But when Q slides toward P , how close is close enough? What distinguishes approaching indefinitely from arriving? If a theory's foundations use such vague verbs, why trust its conclusions?

More than two hundred years ago, mathematicians wrestled with these same questions and argued for over a century before finding an answer: the **limit**. It translates endless approach into precise, testable inequalities. Limits form the foundation of Part III and the starting point for derivatives, integrals, and the mathematics of change. Next chapter we confront the spectre behind every smooth curve and bend: is the infinitesimal a legitimate mathematical object?

Part III

From Change to Infinity—Calculus and Analysis

Chapter 9 Limits: Infinite Approach Need Not Be Vague

This chapter confronts one of mathematics' most slippery words: infinity. You have met it already: the dots in $0.333\cdots$, parallel lines that never meet, and the endless decimal tail of $\sqrt{2}$. Strangely, middle-school mathematics tends to dodge it. A teacher may say that a process continues forever, but rarely says exactly what continuing forever produces. Here we stop dodging. We will see how a bewildering paradox from more than two thousand years ago forced mathematicians to invent limits, and how the vague-sounding phrase infinitely close became a rule that can be checked rigorously. Infinity is not mysterious; infinity left unexplained is.

9.1 A Race That Can Never Finish?

Imagine an unusual race. Achilles, the legendary swift runner, runs at 10 metres per second. His opponent, a tortoise, manages only 1 metre per second. For fairness, though really to create trouble, the tortoise gets a 100-metre head start before Achilles leaves the starting line.

Can Achilles catch the tortoise?

Your intuition shouts yes: he is ten times faster, so catching up takes only seconds. Yet more than two thousand years ago, the philosopher Zeno offered an unsettling argument. Let us examine it step by step, he said.

First, Achilles reaches the tortoise's initial position, covering 100 metres. This takes 10 seconds. During those 10 seconds, however, the tortoise advances another 10 metres and reaches the 110-metre mark.

Second, Achilles runs another 10 metres to 110 metres, taking 1 second. The tortoise uses that time to move 1 metre farther, to 111 metres.

Third, Achilles runs 1 metre in 0.1 seconds, and the tortoise advances another 0.1 metre.

See the pattern? Whenever Achilles reaches the tortoise's previous position, the tortoise has moved a little farther. We can describe a fourth, fifth, or ten-thousandth step, endlessly. Zeno triumphantly concluded that because catching up requires infinitely many steps, each taking some time, Achilles can never catch the tortoise.

This is Zeno's famous paradox: its individual steps sound plausible, yet its conclusion is absurd. Everyone knows a fast runner can catch a slow tortoise. Ancient China had a similar saying in the Zhuangzi: take half of a one-chi stick each day, and it will not be exhausted in

ten thousand ages. Ancient thinkers in East and West had independently encountered the same monster: infinite subdivision.

9.2 Put Your Intuition on the Table

Before learning any tools, judge these three statements intuitively. Which are true and which false?

First, Zeno's infinitely many time intervals—10 seconds, 1 second, 0.1 seconds, 0.01 seconds, and so on—sum to infinity, so catching up takes infinitely long.

Second, $0.999\ldots$, with 9 repeating forever, is slightly less than 1. It approaches 1 endlessly but does not equal 1.

Third, after removing half a stick each day for infinitely many days, an infinitesimal piece remains instead of the stick being entirely removed.

Do not rush ahead. Record your guesses, and return to mark them at the end of the chapter.

All three questions ask the same thing: what results when infinitely many increasingly small quantities accumulate? A finite amount or an infinite one? Almost equality or exact equality? Guesswork cannot settle this, because intuition is notoriously unreliable here. We need a language that explains infinite approach precisely.

9.3 First Attempt: Adding One Term at a Time

The simplest method ignores infinity and just adds. Zeno's time intervals are

$$10 + 1 + 0.1 + 0.01 + 0.001 + \cdots$$

Cut off the dots and add only the first few terms:

$$10 = 10,$$

$$10 + 1 = 11,$$

$$10 + 1 + 0.1 = 11.1,$$

$$10 + 1 + 0.1 + 0.01 = 11.11,$$

$$10 + 1 + 0.1 + 0.01 + 0.001 = 11.111.$$

Interesting: each additional term appends another 1, climbing toward $11.111\ldots$. A table makes the trend clearer:

Terms added n	Partial sum S_n	Distance from $\frac{100}{9}$
1	10	$1.111\ldots$
2	11	$0.111\ldots$
4	11.11	$0.00111\ldots$
8	11.111111	About 1.1×10^{-7}

Every two added terms give the gap another 0 after the decimal point. The repeating decimal $11.111\ldots$ equals $\frac{100}{9}$, approximately 11.11 seconds. Surely that is the expected catching-up time? Relative speed confirms it: Achilles gains 9 metres each second, so closing a 100-metre gap takes $\frac{100}{9}$ seconds. Two approaches pointing to the same number are reassuring.

But do not celebrate yet. This method has two large holes.

First, we can never finish adding. Every finite result, 11.11 or 11.111, is only an approximation. Why must infinitely many terms give exactly $\frac{100}{9}$ rather than something slightly greater or smaller? We guessed the answer and checked it using relative speed. What if another problem offers no independent check? Guessing is not enough for mathematics.

The second hole is subtler: we smuggled in the belief that steadily moving toward $\frac{100}{9}$ means ultimately arriving at $\frac{100}{9}$. Between moving toward and arriving lies an entire infinity. Intuition wants to jump that gap; mathematics requires a bridge.

Counterexample Alert

Beware the claim that later terms become negligibly small, so we can simply omit them and keep the earlier sum. That may suffice for a decimal calculation, but it hides the real question. How small is negligible, and for whom? If only whole seconds matter, 0.001 seconds can be ignored; when scheduling a rocket, 0.001 seconds may correspond to tens of metres. Neglecting a quantity is an engineering choice, not a mathematical conclusion. Mathematics asks whether including every one of the infinitely many terms gives a result, and what that result is.

9.4 Defining Infinite Approach

A bridge needs foundations. The first is to study an endless ordered list of numbers as a single object.

Definition 9.1 (Sequence). Numbers arranged in order according to a rule and continuing indefinitely form a **sequence**. The number in position n is its n th term, written a_n . The sequence can be written

$$a_1, a_2, a_3, \dots, a_n, \dots$$

Our partial sums form a sequence:

$$10, 11, 11.1, 11.11, 11.111, \dots$$

Its n th term is the sum of the first n time intervals. Unlike a set, a sequence has order, permits repetitions, and always has a next term. To study an infinite process, divide it into stages, list the result at each stage, and ask where the sequence ends.

But ends itself needs clarification. A sequence has no last term, so how can it have an endpoint? We must change the question. This is the crucial leap of limits: **ask which number a sequence approaches arbitrarily closely, rather than where it finally arrives**. What does

arbitrarily closely mean? Close means a small distance; does infinitely close mean an infinitesimal distance? And what is infinitesimal? We cannot define vague words with other vague words.

Mathematicians found a clever replacement that can be checked word by word. Here is its everyday version:

However close you demand, I can get that close—and from some step onward, stay that close forever.

In mathematical language, this becomes the definition of a sequence's limit.

Definition 9.2 (The Limit of a Sequence: An Intuitive Version). Let $a_1, a_2, a_3, \dots$ be a sequence and L a number. If, for every positive number d , however small, there is a position N such that **all** terms after position N lie less than d from L , so that

$$|a_n - L| < d \quad (n > N),$$

then the sequence **converges** to L , and L is its **limit**. We write

$$\lim_{n \rightarrow \infty} a_n = L.$$

This is one of the book's central definitions and deserves to be unpacked sentence by sentence.

First, it is a challenge-and-response game. You choose a closeness requirement d : 0.1, 0.0001, a hundred-millionth, or anything smaller. I must prove that from some term onward, every term lies in the corridor $(L - d, L + d)$, centred at L with width $2d$, and never leaves it. If you choose a smaller d , I may need to move N farther out, perhaps from term 10 to term 10000. But whatever challenge you set, I must answer. If I can always win, the limit is L .

Second, the definition never uses infinitesimal or infinitely close. It uses only the ordinary inequality $|a_n - L| < d$, where d is an ordinary positive number. The magic lies in however small: a succession of finite, checkable challenges replaces direct talk about infinity. A procedure that never fails makes the vague idea precise.

This language was hard won. When calculus began, mathematicians calculated directly with infinitesimals. Their answers were often correct, but nobody could clearly explain what an infinitesimal was. Critics mocked them as ghosts of departed quantities: a quantity was nonzero when used as a divisor, then zero when discarded, apparently contradicting itself. Debate continued for over a century until nineteenth-century mathematicians replaced this elusive character with our corridor game of inequalities. Infinite approach became a rule that could be checked word for word rather than a feeling. The definition you have met is the outcome of that long argument.

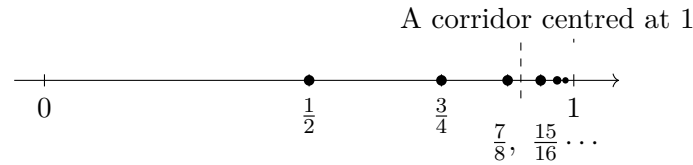
Terminology Card: Limit

Intuition: the destination a sequence heads toward. It need not arrive, but eventually stays within any specified small neighbourhood of that destination.

Formal definition: for every positive d , there exists N such that $n > N$ implies $|a_n - L| < d$.

Example: $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ converges to 0. If you choose $d = 0.01$, I answer $N = 100$, since every term from the 101st onward is below 0.01. For smaller d , choose N larger than $\frac{1}{d}$.

A number line makes it clearer. The black dots below are early terms of $\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \frac{15}{16}, \dots$. They crowd toward 1. A corridor centred at 1 between dashed boundaries can shrink as much as we like, but as long as it has positive width, all sufficiently late terms move in.



One common misunderstanding needs emphasis: convergence is not arrival. Every term of $\frac{1}{n}$ is greater than 0, and none equals 0, yet its limit is 0. A limit describes a trend and destination without requiring a finish-line crossing. Conversely, some sequences have no destination: $1, -1, 1, -1, \dots$ keeps jumping between two values. Any corridor narrower than 2, centred at either 1 or -1 , is left repeatedly. Such a sequence **diverges**; it has no limit.

Counterexample Alert

Consider a subtler false argument: if the distance from a_n to L decreases with every term, then L must be the limit. Take $\frac{1}{n}$, or $1, \frac{1}{2}, \frac{1}{3}, \dots$. Its distance from -1 is $1 + \frac{1}{n}$, which really does decrease! The argument would make -1 its limit, yet the limit is 0. Decreasing distance is too weak a requirement: $1 + \frac{1}{n}$ shrinks but always exceeds 1, so it never becomes arbitrarily small. It cannot meet every positive tolerance d . Thus **moving steadily closer to a target and getting arbitrarily close to it are different things**. Every word in the definition matters.

9.5 The Central Theorem: Infinitely Many Shrinking Terms Can Have a Definite Sum

With the language of limits, we can finally answer Zeno directly. First state a precise proposition.

Theorem 9.3 (The Sum of an Infinite Geometric Sequence). *Let $a, ar, ar^2, ar^3, \dots$ be a geometric sequence with first term a and common ratio r , where $|r| < 1$. Its first n terms give the partial sum*

$$S_n = a + ar + ar^2 + \dots + ar^{n-1}.$$

As n increases without bound, S_n converges and

$$\lim_{n \rightarrow \infty} S_n = \frac{a}{1-r}.$$

This limit is called the sum of the infinite geometric sequence, written

$$a + ar + ar^2 + \dots = \frac{a}{1-r}.$$

Notice the wording. The theorem does not ask us to perform infinitely many additions, an operation we cannot carry out. It says that the finite sums S_n form a convergent sequence, and we **define** the infinite sum as its limit. Infinite summation is another name for a limit. This distinction unlocks Zeno's paradox.

Proof. There are two steps. First obtain a compact expression for S_n using the familiar algebraic trick of shifting and subtracting. Starting from

$$S_n = a + ar + ar^2 + \cdots + ar^{n-1},$$

multiply both sides by r :

$$rS_n = ar + ar^2 + \cdots + ar^{n-1} + ar^n.$$

Subtract, cancelling every middle term and leaving only the ends:

$$S_n - rS_n = a - ar^n, \quad \text{so} \quad S_n = \frac{a(1 - r^n)}{1 - r}.$$

Second, apply the limit definition as n grows. Rewrite this as

$$S_n = \frac{a}{1 - r} - \frac{a}{1 - r} r^n.$$

Since $|r| < 1$, the absolute value of r^n shrinks as n increases. For example, with $r = 0.1$, r^n runs through $0.1, 0.01, 0.001, \dots$. Given any positive d , sufficiently large n makes $|r^n|$ small enough that $\left| \frac{a}{1 - r} \right| \cdot |r^n|$ is less than d . Thus the distance from S_n to $\frac{a}{1 - r}$ becomes arbitrarily small, so by definition

$$\lim_{n \rightarrow \infty} S_n = \frac{a}{1 - r}.$$

Both steps use ordinary algebra and inequalities, yet together they cross infinity. □

The condition $|r| < 1$ is essential. With $r = 2$, the sum $1 + 2 + 4 + 8 + \cdots$ grows without bound and its partial sums do not converge. With $r = -1$, the sum $1 - 1 + 1 - 1 + \cdots$ has partial sums alternating between 1 and 0, again without a limit. Both diverge, so $\frac{a}{1 - r}$ cannot be used. A theorem's conditions resemble an appliance's rated voltage: connect it incorrectly and even a beautiful formula burns out.

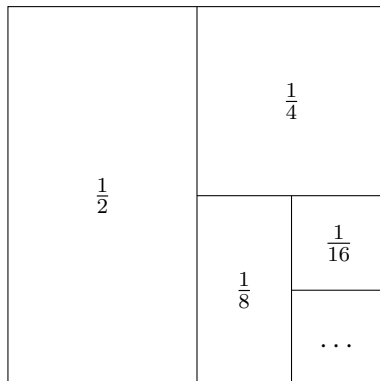
Pause at the proof's second step. Why trust it? Because every claim can be checked. Doubt that r^n can become arbitrarily small? Take $r = \frac{1}{2}$. If you demand $d = 10^{-9}$, I choose $n = 31$, giving $2^{-31} < 10^{-9}$. Demand a smaller d and I supply a larger n . No mysticism, just an answer to every challenge.

Corollary 9.4 (Resolving Zeno's Paradox). *The time Achilles needs to catch the tortoise is the limit of the sums of the stage times:*

$$10 + 1 + 0.1 + 0.01 + \cdots = \frac{10}{1 - 0.1} = \frac{100}{9} \text{ seconds.}$$

Infinitely many stages need not take infinitely long. Zeno's error was assuming that infinitely many positive numbers must sum to infinity. Our theorem supplies a counterexample.

A picture offers another beautiful view. Begin with a square of area 1, remove half, then half the remainder, and keep going. The successive areas removed are $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots$. You can see how these pieces fill the square:



The picture says $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots = 1$, matching $\frac{a}{1-r}$ with $a = \frac{1}{2}$ and $r = \frac{1}{2}$. This answers our third opening question too: the daily lengths removed from the stick sum to exactly 1 chi. In the infinite limit, the total removed is one whole stick, with no tiny remnant. Saying that half still remains after every day describes a process that never finishes; saying that the total is exactly 1 describes its limit. These statements concern different objects and do not conflict. Confusing process with limit is the heart of Zeno's mistake.

9.6 Another Perspective: Does the Number Line Have Holes?

So far we have asked what a limit equals. Now ask something deeper: must it exist, and what kind of number is it?

Consider

$$1, 1.4, 1.41, 1.414, 1.4142, \dots$$

Every term is a terminating decimal and therefore rational. The terms crowd together: beyond some point, any two can be made arbitrarily close, since their shared initial decimal digits grow longer. Intuitively they should crowd toward a point on the line. We know its length, the diagonal of a square of side 1, is $\sqrt{2}$, and $\sqrt{2}$ is not rational. If only rational numbers were allowed, the sequence would try desperately to converge with no destination available: its limit is outside the city.

It is like a lift rising steadily yet never reaching a floor, because the building lacks the required floor. The rational number line is full of such holes: mark off a diagonal with a compass, and its endpoint may fall into one.

Terminology Card: Completeness of the Real Numbers

Intuition: the real number line has no holes. Every sequence whose terms crowd arbitrarily closely together, later called a Cauchy sequence, has a real limit.

Formal definition: the real numbers satisfy the completeness axiom: every Cauchy sequence converges to a real number; equivalently, every set of real numbers bounded above has a least upper bound.

Example: $\sqrt{2}$ is the limit of $1, 1.4, 1.41, \dots$. We need not know $\sqrt{2}$ first and then find a sequence; such a sequence can instead define $\sqrt{2}$.

Completeness is an axiom, a property we require of the reals rather than prove here. Think of it as a building promise: whenever a sequence ought to have a limit, the corresponding point is guaranteed to be on the line. This promise supports limit theory and the entire structure of calculus.

Completeness has a simpler-looking form worth examining. Imagine a balloon growing toward a ceiling: it grows at every stage but never passes the ceiling. Intuition says it must approach some definite height. In sequence language, **a sequence whose terms never decrease and are all bounded above by one fixed number must converge**. This is the monotone convergence principle, another expression of the number line having no holes, like the Cauchy criterion. Experiment 3's sequence $\left(1 + \frac{1}{n}\right)^n$ relies on it: it can be proved increasing and always less than 3, so its limit exists. Mathematicians proved existence before naming the limit e . Remember that order: existence first, name afterward.

Now return to the second opening question: does $0.999\dots$ equal 1? Apply Theorem 9.3. The decimal $0.999\dots$ means the infinite sum

$$\frac{9}{10} + \frac{9}{100} + \frac{9}{1000} + \dots,$$

with first term $a = \frac{9}{10}$ and common ratio $r = \frac{1}{10}$, so

$$0.999\dots = \frac{\frac{9}{10}}{1 - \frac{1}{10}} = \frac{\frac{9}{10}}{\frac{9}{10}} = 1.$$

It is exactly 1, with no shortfall. If this feels uncomfortable, you may be picturing $0.999\dots$ as a number still climbing toward 1 without reaching it. But it denotes the limit of an infinite sum, whose value is 1, rather than the process. Thus 1 and $0.999\dots$ are two representations of the same real number, like $\frac{1}{2}$ and $\frac{2}{4}$ for the same fraction. Every terminating decimal has such a twin representation; for example, $0.5 = 0.4999\dots$.

Connections

Limits belong beyond pure mathematics. Ideal gases, smooth inclines, and instantaneous velocities in science are products of limits: real gas molecules have size, and an ideal gas lets that size tend to 0; real inclines have friction, and a smooth one lets friction tend to 0. Engineers simulating weather or designing chips use the reverse operation, dividing continuous change into finitely many computational steps, with more steps approaching reality more closely. Every weather forecast on your phone comes from finite steps approximating the infinite. Our limits underpin next chapter's derivative: the limit of average rates of change as a time interval tends to 0.

Mathematical Toolbox

This chapter's new tools:

- **Sequences:** divide an infinite process into stages and study the ordered list of results.
- **The limit definition, or corridor game:** for every positive d , there exists N such that $n > N$ implies $|a_n - L| < d$. Infinite approach becomes a checkable inequality.
- **Infinite geometric sums:** if $|r| < 1$, then $a + ar + ar^2 + \cdots = \frac{a}{1-r}$. An infinite sum is a limit of partial sums, not an addition performed infinitely many times.
- **Completeness:** the real line has no holes, so sequences that should converge have destinations.

9.7 Mathematical Experiments

One-Minute Experiment

Experiment 1: Adding Toward an Infinite Sum. Use a calculator to compute

$$\frac{1}{2}, \quad \frac{1}{2} + \frac{1}{4}, \quad \frac{1}{2} + \frac{1}{4} + \frac{1}{8}, \quad \dots$$

Record each result, continuing at least through $\frac{1}{2^{10}} = \frac{1}{1024}$. You obtain 0.5, 0.75, 0.875, 0.9375, $\dots$. Each term added halves the distance to 1. After how many terms is the distance to 1 less than 0.001? In the corridor game, which N corresponds to your demand $d = 0.001$?

Experiment 2: Building an Approximation Road to $\sqrt{2}$. Use only arithmetic, without the square-root key. Start with $[1, 2]$ and test the midpoint 1.5. Since $1.5^2 = 2.25 > 2$, $\sqrt{2}$ lies in $[1, 1.5]$. Test the next midpoint, 1.25, comparing its square with 2 to choose the left or right half. Each step halves the interval. Perform 8 steps and record the endpoints; you will obtain $\sqrt{2} \approx 1.414$. What does this nested-interval process share with the decimal approximations to $\sqrt{2}$, namely 1, 1.4, 1.41, $\dots$? If the number line had holes, would the experiment still guarantee a number trapped inside?

Experiment 3 (Optional): A Magical Number. Calculate $\left(1 + \frac{1}{n}\right)^n$. For $n = 1$ it is 2; for $n = 10$, about 2.594; for $n = 100$, about 2.705; and for $n = 1000$, about 2.717. It seems to converge to a mysterious number beginning 2.718. Record the observation; the exercises return to it.

Questions to Think About

Observation

- Examine $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \dots$, whose n th term is $\frac{n}{n+1}$. Calculate early terms to guess the limit, then find the distance between the n th term and that limit. **Hint:** rewrite $\frac{n}{n+1}$ as $1 - \frac{1}{n+1}$.
- Use this chapter's theorem to express $0.333\dots$ and $0.121212\dots$ as fractions. What is the common ratio for the second decimal? **Hint:** $0.121212\dots$ repeats every two digits, so it is a geometric sum with first term $\frac{12}{100}$ and ratio $\frac{1}{100}$.

Proof

- Use the corridor game to prove rigorously that $\frac{1}{n}$ converges to 0. For any positive d , give an explicit N , allowed to depend on d , and explain why $n > N$ implies $\left|\frac{1}{n} - 0\right| < d$. **Hint:** all you need is $\frac{1}{n} < d$. Take reciprocals to see how large n must be.
- Prove that a sequence cannot converge to two different numbers L and M . **Hint:** let the distance between L and M be $h > 0$. Choose $d = \frac{h}{2}$ and draw corridors centred at L and

M. They do not overlap. What contradiction arises if the sequence must eventually stay in both?

Challenge

1. Experiment 3's sequence $\left(1 + \frac{1}{n}\right)^n$ appears to converge. Investigate or think about its limit, the natural constant e , approximately 2.71828. Every term is rational, but e is irrational. Why is this unsurprising in light of completeness? **Hint:** recall $\sqrt{2}$. The types of the individual terms and the type of their limit are separate questions.
2. Zeno's arrow paradox says that a flying arrow occupies a definite position at each instant, so it is stationary at each instant and therefore never moves. Identify the gap using this chapter's language. **Hint:** does the leap from definite position to no motion again confuse a process with a limit? How should velocity at an instant be defined? That is next chapter's problem.

Keep Climbing

Further climbing brings these sights: replace d by the Greek letter ε and justify N rigorously to obtain the ε - N language of university analysis. The precise statement that terms crowding together have a limit is the Cauchy convergence criterion, equivalent to monotone bounded convergence as a form of completeness. General infinite series extend geometric sums, and deciding convergence or divergence begins series theory. The limit e of $\left(1 + \frac{1}{n}\right)^n$ returns in logarithms, compound interest, and differential equations. Competition methods using upper and lower bounds, including the squeeze theorem—both sides approach L , leaving the middle no escape—are variations on the corridor game.

The Door to the Next Chapter

We have tamed the accumulation of infinitely many quantities, but Zeno's arrow still hangs in the air with a trickier question: what is velocity at a single instant?

A car's dashboard says 60 kilometres per hour. Where does that number come from? Speed is distance divided by time, but in one instant the distance is 0 and the time is 0. What is $\frac{0}{0}$? It gives no answer. Intuition has failed again.

You have already glimpsed the way out: replace the instant with a short interval beginning there. Calculate average speeds over 1 second, 0.1 seconds, 0.01 seconds, and so on. If these averages form a convergent sequence, its limit deserves the name instantaneous velocity. Likewise, a curve's tangent slope is the limit of secant slopes as the two points approach each other.

The limit of average rates of change is the derivative. Next chapter, we bring the corridor game into the world of functions to measure change itself precisely.

Chapter 10 Derivatives: A Curve's Rate of Change at a Point

10.1 A Question on a Roller Coaster

A roller coaster is an amusement park's most thrilling attraction. It climbs a high, gentle slope, plunges downward, then sweeps into an almost vertical loop. Riders sense the differences clearly: some places are level, some steep, some turning, and the speed seems different at every instant.

Draw the track on coordinate paper as a curve $y = f(x)$, with horizontal distance x and height above ground y . Riders say things like “steepest here,” “the descent starts here,” and “almost level here.” Can mathematics express these descriptions precisely?

Consider a simpler scene. A car's odometer reads 120 kilometres at 10 a.m. and 180 kilometres at 11 a.m. Its average speed over that hour is 60 kilometres per hour; everyone agrees. But a traffic camera records it at 92 kilometres per hour at 10:23:15, in an 80-kilometre-per-hour zone. Notice the distinction: this is a speed at an instant, not an average over an interval. At an instant the time interval and distance increment are both zero. Is the speed not $\frac{0}{0}$?

The steepness of a track at a point and a car's speed at an instant are really the same mathematical problem. It troubled mathematicians for years. The tool that resolved it is this chapter's protagonist: the derivative.

10.2 First Guess: Does Instantaneous Velocity Exist?

Before continuing, answer a few questions intuitively:

- Does saying that a car's speed at one instant is 92 kilometres per hour have real meaning, or is it just convenient speech?
- Does it make sense to speak of a winding curve's steepness at one point? Is steepness not a property of a whole stretch?
- If instantaneous velocity exists, why does $\frac{0}{0}$ not destroy it?

Many people say that an instant must have a definite velocity because they can feel it while cycling. Others object that velocity is distance divided by time, so a zero interval makes the

division meaningless. Both sides are partly right. Mathematicians translate both intuitions into one precise, consistent definition. Last chapter's limits prepared the tools for that translation.

This translation actually happened in the seventeenth century, twice. Studying planetary motion, Newton needed velocity at each instant and called rates of change fluxions. Leibniz approached from geometry and tangents, inventing the still-used notation $\frac{dy}{dx}$. They independently grasped the same idea, then became embroiled in years of priority disputes that drew learned societies across Europe into taking sides. From today's perspective the dispute is unfortunate: motion and tangents looked like different questions but led to one concept. That shows how naturally the concept arises whenever change itself is questioned seriously.

10.3 First Attempt: A Very Short Time Interval

Start with the car. Suppose its distance travelled follows $s(t) = t^2$, in kilometres and hours. The numbers are unrealistic, but the reasoning is unchanged. What is its instantaneous velocity at $t = 2$?

Try a small interval, say $\Delta t = 0.1$ hours, and calculate the average:

$$\frac{s(2.1) - s(2)}{0.1} = \frac{4.41 - 4}{0.1} = 4.1 \text{ kilometres per hour.}$$

Now take $\Delta t = 0.01$:

$$\frac{s(2.01) - s(2)}{0.01} = \frac{4.0401 - 4}{0.01} = 4.01 \text{ kilometres per hour.}$$

With $\Delta t = 0.001$, the answer is 4.001; with $\Delta t = 0.0001$, it is 4.0001. Shorter intervals give answers closer to 4. We therefore guess that the instantaneous velocity at $t = 2$ is 4 kilometres per hour.

A table clarifies the approach. We can approach from the left too, using a small interval before $t = 2$, and check whether both sides head toward the same number:

Δt	1	0.1	0.01	0.001
From the right: $\frac{s(2+\Delta t)-s(2)}{\Delta t}$	5	4.1	4.01	4.001
From the left: $\frac{s(2)-s(2-\Delta t)}{\Delta t}$	3	3.9	3.99	3.999

The right-hand values approach 4 from above, and the left-hand values from below. Meeting at the same place is essential: if the other side approached a different number, the instantaneous velocity would not exist. The graph $y = |x|$ at the origin, considered in the exercises, behaves that way.

Notice that we never dare set Δt equal to 0, since both numerator and denominator would become 0. Instead, Δt can be arbitrarily small and the average velocity arbitrarily close to 4. This is exactly last chapter's language of limits. Instantaneous velocity is the limit of averages as the interval tends to zero, rather than an average from one very short interval.

In symbols, the limit of $\frac{s(t+\Delta t)-s(t)}{\Delta t}$ as Δt tends to 0 is the instantaneous velocity at time t .

Counterexample Alert

A widespread false proof says that instantaneous velocity uses a zero time interval, so substituting $\Delta t = 0$ gives $\frac{0}{0}$ and proves that instantaneous velocity does not exist.

The error is treating a limit as substitution. As Δt approaches 0, it becomes arbitrarily small while remaining nonzero. For $\Delta t \neq 0$, an expression such as $\frac{s(t+\Delta t)-s(t)}{\Delta t}$ can often be simplified first. For example, $\frac{(2+\Delta t)^2-4}{\Delta t} = \frac{4\Delta t+(\Delta t)^2}{\Delta t} = 4 + \Delta t$. Now let Δt tend to 0 and obtain exactly 4. At no stage did we divide by zero.

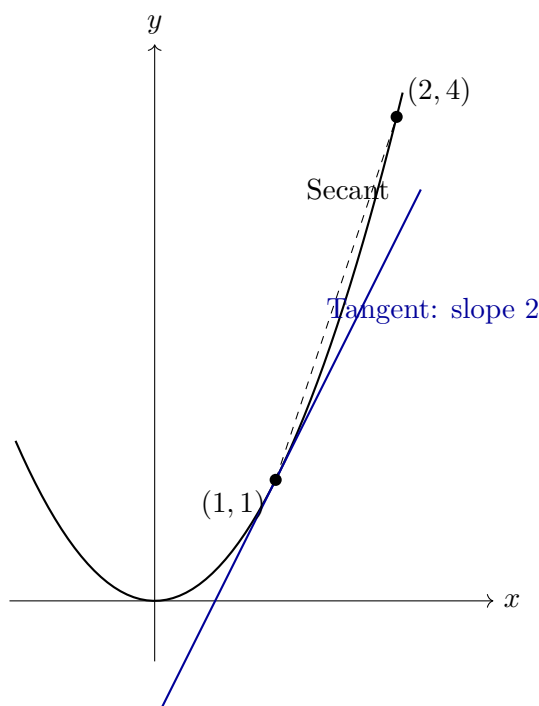
10.4 The Same Idea on a Curve: Secants and Tangents

Turn to geometry. How do we describe the steepness of $y = x^2$ at $(1, 1)$? A line has a slope, but where does a curved line get one?

Imitate the car problem: use two points. Choose a second point $(1+h, (1+h)^2)$ on the curve. The line joining them is a secant with slope

$$\frac{(1+h)^2 - 1}{h} = \frac{2h + h^2}{h} = 2 + h.$$

For $h = 1$, its slope is 3; for $h = 0.1$, 2.1; for $h = 0.01$, 2.01. As h tends to 0, the secant slope tends to 2. Geometrically, as the second point slides toward $(1, 1)$, the secant turns toward a definite line. That line is the tangent at the point, and its slope is 2.



The two problems have become one: instantaneous velocity equals the tangent slope on a distance graph. Pause at this union. Velocity in algebra and slope in geometry, formerly inhabitants of different worlds, are now the same number. That number is the derivative.

Terminology Card: Tangent

Intuition: the line that fits the curve best near the contact point, embodying its local direction. Meeting the curve only once is insufficient: a parabola's symmetry axis meets only its vertex but is not tangent there. Formal definition: the limiting position of secants through a fixed curve point and a nearby point as the nearby point approaches the fixed one. Example: the tangent to $y = x^2$ at $(1, 1)$ is $y = 2x - 1$, supporting the entire parabola from below.

Definition 10.1 (Derivative). Suppose f is defined near x . If the limit

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

exists, then f is differentiable at x . This limit is the derivative of f at x , written $f'(x)$, or often in physics $\frac{dy}{dx}$.

Terminology Card: Derivative

Intuition: approximately how much the vertical coordinate changes per small horizontal step at a point; geometrically, the tangent slope there. Formal definition: the limit of average rates of change as the interval length tends to zero. Example: $f(x) = x^2$ has derivative 2 at $x = 1$. Near $(1, 1)$, a tiny horizontal increase produces approximately twice that vertical increase.

10.5 Calculating Derivatives by Hand: x^2 , x^3 , and Linear Functions

We learn whether a definition is useful by calculating with it. Let us invent three derivatives directly from the definition.

First take $f(x) = x^2$. For any x ,

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h.$$

Let h tend to 0 to get $f'(x) = 2x$. One formula now holds every tangent slope on the parabola: at $x = 1$ it is 2, matching our secant experiment; at $x = 0$ it is 0, showing a horizontal bottom; at $x = -3$ it is -6 , whose negative sign indicates descent.

Next take $f(x) = x^3$, using a familiar algebraic identity:

$$(x+h)^3 = x^3 + 3x^2h + 3xh^2 + h^3.$$

Thus

$$\frac{(x+h)^3 - x^3}{h} = \frac{3x^2h + 3xh^2 + h^3}{h} = 3x^2 + 3xh + h^2.$$

As h tends to 0, the last two terms disappear, giving $f'(x) = 3x^2$. Notice that $3x^2$ is always nonnegative, so x^3 rises or levels off everywhere. It is flattest at the origin, with slope 0, but that

point is neither peak nor valley, merely a resting place. Compare how the curves grow steeper: the slope $2x$ of x^2 increases uniformly with x , while the slope $3x^2$ of x^3 increases ever faster. At $x = 10$, the parabola's slope is 20 and the cubic's already 300. Derivatives tell us both steepness at a point and how steepness itself changes, an idea that will become another object of study.

Finally, for $f(x) = kx + b$,

$$\frac{k(x+h) + b - (kx + b)}{h} = \frac{kh}{h} = k,$$

so its derivative is k . As expected, a straight line is equally steep everywhere, and its tangent is itself.

Example 10.2. A rocket's height after ignition is $h(t) = 5t^2$ metres. Find its instantaneous velocity at $t = 10$ seconds.

Since $h(t) = 5t^2$, use the derivative of x^2 , carrying along the constant factor as explained below, to get $h'(t) = 10t$. At $t = 10$ seconds the instantaneous velocity is 100 metres per second. We calculated no interval average, yet obtained an answer more precise than any such average: it describes the exact instant ten seconds after ignition.

10.6 Why Derivatives Matter: Local Linear Approximation

The derivative's deepest meaning is the best straight-line stand-in for a curve near a point.

Return to $f(x) = x^2$ at $x = 1$, with tangent $y = 2x - 1$. At $x = 1.02$, the curve gives $1.02^2 = 1.0404$, while the tangent gives $2 \times 1.02 - 1 = 1.04$. The difference is only 0.0004! Near $x = 1$, we can approximate x^2 by $2x - 1$ with a very small error. Zoom close enough and every smooth curve becomes a line. Chapter 8's promise of local straightness now has a precise measure: the line is the tangent and its slope the derivative.

An immediate application is estimation. Estimate $\sqrt{4.1}$. The derivative of $g(x) = \sqrt{x}$ is $\frac{1}{2\sqrt{x}}$, obtained from the limit definition by rationalizing the numerator. At $x = 4$ it is $\frac{1}{4}$. The tangent approximation gives

$$\sqrt{4.1} \approx \sqrt{4} + \frac{1}{4} \times 0.1 = 2 + 0.025 = 2.025.$$

The actual value is about 2.02485, an error below 0.0002. Without a calculator, the derivative has supplied three significant figures.

Examine the general approximation on its own:

$$f(x_0 + \Delta x) \approx f(x_0) + f'(x_0) \Delta x.$$

The left side is a value on the curve; the right is a value on the tangent. The structure is current value plus rate times step. A larger rate gives a larger change for the same step, matching our experience that each step on a steep slope takes more effort. Remember the scope: this is reliable only for small Δx , and stronger bending, or a larger second derivative, makes it fail sooner. Approximation is a tradeoff whose error must be acknowledged.

Mathematical Toolbox

This chapter's new tools:

- **Derivative:** a limit of average rates, geometrically a tangent slope and physically an instantaneous velocity.
- **The power rule:** $(x^n)' = nx^{n-1}$, established here for $n = 2, 3$.
- **Local linear approximation:** $f(x_0 + \Delta x) \approx f(x_0) + f'(x_0) \Delta x$, improving as we approach x_0 .
- **Reading derivative signs:** $f' > 0$ means rising, $f' < 0$ falling, and $f' = 0$ a possible peak, valley, or resting place.

10.7 Differentiating Assembled Functions: Product and Chain Rules

Knowing x^2 and x^3 is not enough. Real functions are often assembled by multiplying two functions or putting one inside another. Can derivatives be assembled too?

Addition is easiest: $(f + g)' = f' + g'$, because the change in a sum is the sum of the changes. The definition proves this directly. Constants behave similarly: $(cf)' = cf'$.

Multiplication is trickier. Many people's first guess is that a product's derivative is the product of its derivatives. A familiar example immediately refutes it: $x^3 = x \cdot x^2$. The derivative of x^3 is $3x^2$, while those of x and x^2 are 1 and $2x$, whose product is only $2x$. The correct rule is a little more complicated.

Theorem 10.3 (The Product Rule). *If f and g are differentiable at x , then*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

Proof. Examine the difference quotient of the product, adding and subtracting the same term in the numerator:

$$\frac{f(x+h)g(x+h) - f(x)g(x)}{h} = \frac{f(x+h)g(x+h) - f(x)g(x+h)}{h} + \frac{f(x)g(x+h) - f(x)g(x)}{h}.$$

Factor out $g(x+h)$ from the first part to get $\frac{f(x+h)-f(x)}{h} g(x+h)$, and $f(x)$ from the second to get $f(x) \frac{g(x+h)-g(x)}{h}$. As h tends to 0, the difference quotients tend to $f'(x)$ and $g'(x)$, while $g(x+h)$ tends to $g(x)$, since differentiability implies continuity, as the definition directly verifies. The limit is therefore $f'(x)g(x) + f(x)g'(x)$. $\square$

The geometry is vivid: regard f and g as rectangle side lengths, with area fg . When both sides grow a little, area changes in two main pieces. Growth in length contributes current width times the rate of length change; growth in width contributes current length times the rate of

width change. Strictly, a tiny corner also contributes the product of the two increments, which becomes negligible in the limit.

Check our counterexample: for $x^3 = x \cdot x^2$, the product rule gives $1 \cdot x^2 + x \cdot 2x = x^2 + 2x^2 = 3x^2$, exactly as required.

Now put functions inside functions, as in $y = (x^2 + 1)^3$. The inner layer is $u = x^2 + 1$, the outer layer u^3 . Intuitively, x changes and u follows; the change in u then changes y . The rates should multiply along the chain.

Theorem 10.4 (The Chain Rule). *If $u = g(x)$ is differentiable at x and $y = f(u)$ at the corresponding u , then $y = f(g(x))$ is differentiable at x , with*

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

We omit the full proof, which involves technical details about interchanging limiting processes, but explain the intuition. For small Δx , we have $\Delta u \approx g'(x)\Delta x$ and $\Delta y \approx f'(u)\Delta u$. Substitution gives $\Delta y \approx f'(u)g'(x)\Delta x$. The approximation errors disappear in the limit.

Apply it to $y = (x^2 + 1)^3$. The derivative of the outer u^3 is $3u^2$, and that of the inner $x^2 + 1$ is $2x$. Thus

$$y' = 3(x^2 + 1)^2 \cdot 2x = 6x(x^2 + 1)^2.$$

To appreciate the saving, expand $(x^2 + 1)^3$ as $x^6 + 3x^4 + 3x^2 + 1$ and differentiate term by term. You get $6x^5 + 12x^3 + 6x$, identical after factoring. But the chain rule takes one line and reveals which stage contributes each rate.

10.8 Another Perspective: Error, Optimization, and Marginal Quantities

Once derivatives became available, many trial-and-error problems gained systematic solutions. Here are four perspectives.

Error estimates. You measure a square's side as 10 centimetres, but ruler error means the true length could be 10 ± 0.05 centimetres. How uncertain is its area? With $S = x^2$ and $S' = 2x$, the derivative at $x = 10$ is 20. Linear approximation says a side error of 0.05 gives an area error of approximately $20 \times 0.05 = 1$ square centimetre. The derivative measures how much measurement error is amplified. Engineers use this as an early check of robustness: a large derivative can turn tiny manufacturing errors into serious consequences.

Optimization. Use 40 metres of fencing to enclose a rectangular vegetable garden against a wall, which forms one long side. What dimensions maximize area? If the width is x metres, the length is $40 - 2x$ metres and

$$S(x) = x(40 - 2x) = 40x - 2x^2.$$

Previously we completed squares or guessed from a graph. Now $S'(x) = 40 - 4x$. A positive derivative means area grows with x ; a negative one means it shrinks. The switch from growth to

decline occurs where the derivative vanishes: $40 - 4x = 0$, so $x = 10$. The length is 20 metres and maximum area 200 square metres. Peaks and valleys share a horizontal tangent, or zero derivative, giving our first map to extrema. Boundaries and resting places must still be checked, as the counterexample below explains.

Marginal quantities. Economists often ask how much one more item increases cost. The derivative of cost $C(x)$ at production level x is marginal cost. If a factory has total cost $C(x) = 1000 + 20x + 0.01x^2$ yuan, its marginal cost at 100 items is $C'(100) = 20 + 0.02 \times 100 = 22$ yuan. The 101st item therefore adds approximately 22 yuan. Approximately matters: a derivative is a continuous local approximation, whereas products come in discrete units. Stating a model's scope is good mathematical practice.

Motion. Return to the car. Distance's derivative is velocity; what is velocity's derivative? Acceleration, the rate at which velocity changes. The slight sinking sensation when a lift starts, or the sideways pull in a turn, comes from acceleration rather than velocity itself. If free-fall distance is $s(t) = 4.9t^2$ metres, then velocity is $v(t) = s'(t) = 9.8t$ metres per second. Differentiating again gives the constant $v'(t) = 9.8$. This is the fact Galileo found with inclined-plane experiments: falling acceleration is almost unchanged by weight or time spent falling. Two derivatives compress crates of experimental data into one formula. The derivative of a derivative is the second derivative, describing how quickly a curve bends; it will reappear with Chapter 8's curvature.

Connections

Derivatives connect our earlier chapters. Chapter 2's slope is the constant-function special case; Chapter 8's promise of local straightness has become tangents and derivatives; and Chapter 9's limits form the definition's framework. Looking ahead, Chapter 12 will rigorously establish the role of zero derivatives as candidates for extrema, while Chapter 17 upgrades tangent lines to tangent planes. In physics, economics, and biology, wherever a rate appears—population growth, radioactive decay, or disease spread—derivatives are the first language.

One-Minute Experiment

Experiment 1: A Tangent Through the Telescope. Draw $y = x^2$ on squared paper using integer x from -3 to 3 . At $(2, 4)$, gently draw a ruler line following the curve that touches without crossing. Measure its slope by counting rise over run. Theory gives $2x = 4$. How far is your hand-drawn value from 4?

Experiment 2: Estimating with a Tangent. Without a calculator, use the tangent to $f(x) = x^2$ at $x = 5$ to estimate 5.1^2 . The tangent is $y = 10x - 25$; substituting $x = 5.1$ gives 26. The actual value is 26.01, an error of 0.01. Estimate 5.5^2 similarly and compare the error, seeing how approximation worsens farther from the contact point.

Experiment 3: Detecting Slope Signs. Draw a smooth curve, perhaps rising, falling, then rising again. Without calculating, label several points with positive, negative, or zero derivative. Swap with a classmate to check. This trains your eye to see derivatives.

10.9 Another Look at Peaks: Zero Derivatives and Extrema

The fencing problem used the claim that a maximum has zero derivative. State it formally.

Theorem 10.5 (A Necessary Condition for an Extremum). *If f attains a maximum or minimum at a differentiable point x_0 in the interior of an interval, then $f'(x_0) = 0$.*

Proof. Consider a maximum. Since $f(x_0)$ is maximal, every nonzero h with $x_0 + h$ still in the interval satisfies $f(x_0 + h) \leq f(x_0)$. For $h > 0$, the quotient $\frac{f(x_0+h)-f(x_0)}{h} \leq 0$, giving $f'(x_0) \leq 0$ in the limit. For $h < 0$, the negative denominator reverses the inequality, so the quotient is ≥ 0 and the limit gives $f'(x_0) \geq 0$. A number both at most and at least 0 must be 0. $\square$

The theorem is carefully restrained: every peak has a horizontal tangent, but a horizontal tangent need not be a peak. The distinction is crucial.

Counterexample Alert

Someone claims that $f'(x_0) = 0$ proves a maximum or minimum at x_0 . The example $f(x) = x^3$ refutes this: $f'(x) = 3x^2$ vanishes at the origin, but function values are negative to its left and positive to its right. The curve increases throughout; the origin is neither peak nor valley, just the resting place mentioned earlier, a stationary point and in this case an inflection point.

A second trap concerns the interval's interior. On $[0, 1]$, the function $f(x) = x$ reaches its maximum at $x = 1$, where the derivative is 1, not 0. Extrema on closed intervals can lie at endpoints. The complete search therefore includes points of zero derivative, points where the derivative does not exist, and endpoints, then compares their function values. No candidate may be omitted.

Corollary 10.6. *If a differentiable function f satisfies $f'(x) = 0$ throughout an interval, then f is constant there.*

This apparently ordinary result is one of calculus's best detective tools. To determine whether two functions differ by a constant, differentiate their difference and check whether it is identically zero. A rigorous proof needs Chapter 12's mean value theorem; for now we record it as a promise.

10.10 Challenge: Calculating a Bending Track by Hand

Combine our tools in a complete problem. A park designs a slide whose height in metres depends on horizontal distance x in metres as $y = x^3 - 6x^2 + 9x + 1$, for $0 \leq x \leq 4$. Where does it descend most steeply, and where are its highest and lowest points?

Differentiate term by term: $(x^3)' = 3x^2$, $(-6x^2)' = -12x$ by the constant-multiple rule, $(9x)' = 9$, and the constant term has derivative 0. Thus

$$y' = 3x^2 - 12x + 9 = 3(x^2 - 4x + 3) = 3(x - 1)(x - 3).$$

The derivative vanishes at $x = 1$ and $x = 3$. For $x < 1$, both factors are negative, so the derivative is positive and the slide rises. For $1 < x < 3$ it descends, and for $x > 3$ it rises again. Thus $x = 1$

is a peak, with $y(1) = 1 - 6 + 9 + 1 = 5$ metres, and $x = 3$ a valley, with $y(3) = 27 - 54 + 27 + 1 = 1$ metre.

But the problem asks for extrema on $[0, 4]$, so check the endpoints too: $y(0) = 1$ and $y(4) = 64 - 96 + 36 + 1 = 5$. Comparing $\{5, 1, 1, 5\}$, the maximum is 5 metres at $x = 1$ and $x = 4$, and the minimum 1 metre at $x = 0$ and $x = 3$. Looking only at zero derivatives would miss the endpoint answers, exactly as our warning predicted.

The steepest descent occurs where the derivative is most negative. Since $y' = 3(x-1)(x-3) = 3(x-2)^2 - 3$ is an upward-opening parabola centred at $x = 2$, its minimum occurs at $x = 2$, with $y'(2) = -3$. The steepest descent is therefore at $x = 2$ metres, with slope -3 , exactly at the midpoint of the line joining the peak and valley. This coincidence holds generally for curves of the x^3 family; its reason lies in the derivative of the derivative, a direction for further climbing.

Keep Climbing

This is only calculus's first step. Further signposts include **second derivatives**, rates of change of derivatives describing concavity and acceleration; the **mean value theorem**, linking function values to derivatives and justifying that zero derivative implies constancy; **L'Hopital's rule**, using derivatives for limits of the form $\frac{0}{0}$; **Taylor expansions**, upgrading local linear approximation to quadratic, cubic, and infinitely many terms; and **multivariable functions and gradients**, describing a whole collection of directional slopes at a surface point. Competition proofs of inequalities by constructing and differentiating functions use these same tools.

Questions to Think About

Observation

1. Draw a curve of your own with three zero-derivative points. Must they all be maxima or minima? *Hint: look at x^3 near the origin.*
2. The graph of $f(x) = |x|$ has a corner at $x = 0$. Calculate the difference-quotient limits separately for $h > 0$ and $h < 0$. What do you obtain? *Hint: what does disagreement between the left and right limits mean?*

Proof

1. Prove $(x^4)' = 4x^3$ from the definition. *Hint: expand $(x+h)^4$. After forming the quotient, terms involving h^2 and higher powers in the expansion disappear in the limit. Conjecture and check the general rule $(x^n)' = nx^{n-1}$.*
2. Show that the constant-multiple rule $(cf)' = cf'$ is a special case of the product rule. *Hint: what is the derivative of a constant function?*

Challenge

1. A factory's profit in yuan from x items is $L(x) = -2x^2 + 80x - 300$. Find the output maximizing profit, and explain why producing more then becomes financially worse. *Hint: solve $L'(x) = 0$ and interpret the derivative's signs on either side.*
2. Prove that the derivative of a differentiable even function, satisfying $f(-x) = f(x)$, is odd. *Hint: differentiate both sides of $f(-x) = f(x)$ using the chain rule, or compare $f'(-a)$ and $f'(a)$ directly with difference quotients.*

The Door to the Next Chapter

We have learned to move from accumulated quantities to rates: a distance function gives velocity at every instant, and a cost function gives marginal cost. The process uses a division and a limit.

Reverse the question, however, and division no longer helps: *if I know only a car's speedometer reading at every instant, can I calculate the total distance travelled?* Or geometrically: *given $y = x^2$, can I calculate the area between it and the x -axis?* We know areas of straight-edged figures, but this boundary is curved.

Intuition suggests dividing the journey into countless tiny intervals, treating velocity as nearly constant on each, approximating distance by velocity times time, and adding the pieces. Finer divisions improve the approximation. This is another limit, but in the opposite direction: this chapter used limits of difference quotients; the next uses limits of sums. Astonishingly, these directions are inverse operations, undoing one another like multiplication and division. Differentiating an accumulated quantity gives its rate; summing the rate recovers the accumulation. This profound duality is the fundamental theorem of calculus, next chapter's story.

Chapter 11 Integration: Adding Countless Small Quantities

11.1 An Odometer Accumulates Distance

Sit in a car and look at two instruments: the speedometer shows how fast you are moving now, and the odometer shows how far you have travelled altogether. Last chapter asked how a changing odometer reading determines velocity at each instant. The answer was differentiation: divide the distance increase over a short interval by that interval, then let the interval shrink toward zero.

Now reverse the question. Suppose the odometer is broken but the speedometer works, recording the current speed every second. Can those records alone tell you the total distance travelled?

Intuition suggests this: during the first second, speed is about 20 metres per second, so distance is about 20 metres; during the second, speed is about 22 metres per second, so distance is about 22 metres. Add these distances for an approximate total. Record every tenth of a second instead, and speed changes less within each interval, so the sum should be more accurate.

Such instrument pairs are everywhere. A tap's flow is a rate; the water meter records an amount. Electrical power is a rate; the electricity meter records energy. Network speed is a rate; a phone's data-usage counter records an amount. Every total-amount meter quietly accumulates an instantaneous rate continuously. This chapter asks exactly what operation that accumulation is.

The simplest idea—*divide a process into many small intervals, treat each as approximately uniform, and add all their contributions*—is the starting point of integration. The same idea finds areas bounded by curves, volumes of spheres and cones, work done by varying forces, and probabilities of random events. It also answers a remarkable question: when can adding infinitely many numbers produce a finite answer?

11.2 A Problem That Stalled Mathematics for Two Thousand Years

Begin with a more geometrical question. The parabola $y = 1 - x^2$ and the x -axis enclose a little hill, with x from -1 to 1 . What is its area?

You can calculate triangles and rectangles, and you have memorized a circle's area formula. But this hill has a curved boundary and is neither polygon nor circle. Guess intuitively: its enclosing rectangle has width 2, height 1, and area 2. Its largest inscribed triangle has base 2, height 1, and area 1. So the answer lies between 1 and 2. Can you narrow it? Some guess 1.5, others near 1.3. Intuition alone has run out.

Ancient mathematicians had impressive successes with curved boundaries. To find a circle's area, Liu Hui in China's Wei–Jin period used polygonal exhaustion: inscribe a regular hexagon, then a dodecagon, then a twenty-four-sided polygon, and keep doubling. Each polygon fits the circle more closely. He explained that finer cutting loses less, and repeated cutting ultimately coincides with the circumference without loss. This is a mature idea of limits, more than a thousand years before comparable European approaches.

Archimedes in ancient Greece tackled the parabolic hill. He filled it with successive layers of shrinking triangles: one large triangle, then two smaller ones, then four smaller still. He proved that each layer's total area is exactly one quarter of the preceding layer, giving

$$S = 1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \cdots$$

He also proved that the infinite sum is $\frac{4}{3}$. Thus the parabolic region has exactly $\frac{4}{3}$ times the area of its large inscribed triangle. Without algebraic notation or coordinates, this was a feat like climbing a cliff barehanded.

Yet Archimedes' method has an unsettling feature: every new curve demands a newly invented cutting technique, like forging a separate key for every lock. Circles need one method, parabolas another, spirals another, each relying on a construction visible only to genius. Could there be a *master key*, one routine for every curve? The question stalled mathematicians for nearly two thousand years.

11.3 First Attempt: Too Few Rectangles, Too Much Error

The natural attempt uses rectangles. Divide $[-1, 1]$ into intervals, erect a rectangle on each with a height taken from the curve, and add their areas to approximate the hill.

Try two intervals. On each half, height varies from 0 to 1. Using the maximum height 1 gives rectangles of area $1 \times 1 = 1$ each and total 2, the enclosing rectangle, visibly too large at the corners. Using the minimum gives height 0 at the left endpoint and 0 at the right endpoint, for total 0, absurdly small. The true area lies between 2 and 0, but the gap is too large to tell us much.

Four intervals improve matters, ten improve them further, and a hundred further still. A reassuring rule appears: the upper sum using maxima is never below the true area; the lower sum using minima is never above it. Doubling the divisions pushes the upper sum down and the lower sum up, squeezing their gap. The true area is like a prisoner between approaching walls. Yet finitely many flat-topped rectangles leave a gap against a curved boundary. We cannot call

100 rectangles the answer, only say that 100 rectangles give an approximation near it. What is the answer itself? Must we forever approach without arriving?

Counterexample Alert

“More rectangles give greater accuracy, so a million rectangles give the exact value.” This sounds plausible but confuses concepts. A million is still finite, the tops are still flat, and the result remains an approximation with a small error. Excellent approximation and exact equality differ. Integration’s crucial step goes beyond every finite subdivision: let the number of pieces grow without bound and take the *limit* of the approximations. Recall Chapter 9: the limit is the exact destination.

11.4 Turning Finer Subdivision into a Definition

The way out is to recognize that we need a limiting process, not a larger finite number. Divide $[a, b]$ into n intervals, each of width $\Delta x = \frac{b-a}{n}$. Choose any point in interval i and record the curve’s height $f(x_i)$. Its rectangle contributes approximately $f(x_i) \Delta x$. Add them:

$$S_n = f(x_1) \Delta x + f(x_2) \Delta x + \cdots + f(x_n) \Delta x.$$

This is a *Riemann sum*. As n grows and Δx shrinks, suppose S_n approaches a definite S , independent of the sample points chosen. Then f is integrable on $[a, b]$, and S is its definite integral.

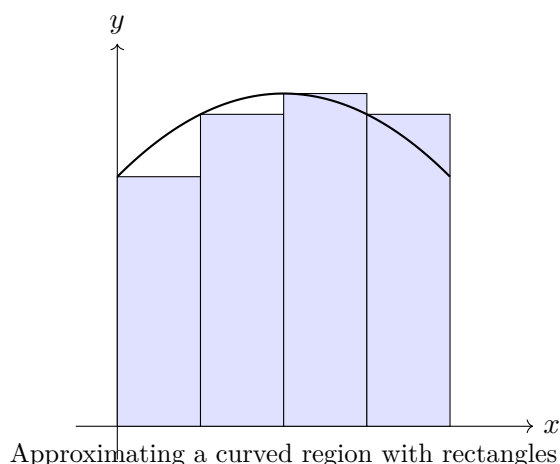
Definition 11.1 (Definite Integral). Let f be defined on $[a, b]$. Divide $[a, b]$ into n equal intervals of width $\Delta x = \frac{b-a}{n}$, and choose a point x_i in each. If, as $n \rightarrow \infty$, the Riemann sum

$$S_n = \sum_{i=1}^n f(x_i) \Delta x$$

has a limit independent of the chosen points, that limit is the *definite integral* of f on $[a, b]$, written

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x.$$

Note. Not every imaginable function is integrable. Fortunately, the functions encountered here, including continuous functions and those with only a few jump discontinuities, are integrable. Investigating the full boundary of integrability belongs to university real analysis. Independence from sample points matters: if changing the points changes the limit, the function is too wild to yield a definite area this way.



What does $\int$ accomplish? It compresses the entire limiting process of cutting into infinitely many tiny pieces and adding them into one symbol. This elongated S, for sum, reminds us that integration is addition carried to a limit. The expression $f(x) dx$ resembles an infinitesimal rectangle's area, with height $f(x)$ and width dx .

Terminology Card: Definite Integral

Intuition: the exact area of a curved region, approached by sums of increasingly narrow rectangles.

Formal definition: the limit of Riemann sums as the subdivision becomes arbitrarily fine, written $\int_a^b f(x) dx$.

Example: $\int_0^2 x dx = 2$, exactly the area of a triangle with base 2 and height 2.

Return to the odometer. Divide the journey into n short intervals. In interval i , approximate velocity by $v(t_i)$ and distance by $v(t_i) \Delta t$, making the total a Riemann sum. Let $n \rightarrow \infty$ to get

$$\text{total distance} = \int_a^b v(t) dt.$$

On a velocity-time graph, $v(t_i) \Delta t$ is a thin rectangle's area, so *distance is the area under the velocity curve*. Integration connects a rate with an amount. At constant speed, the graph is horizontal and its area a rectangle, recovering the primary-school rule of distance equals speed times time. Integration extends that rule to arbitrary variation in speed.

11.5 The Master Key: The Fundamental Theorem of Calculus

We have a definition, but calculating every integral by a Riemann-sum limit would be little better than forging a new key each time. Calculus offers a magnificent gift: to calculate a definite integral, find a function whose derivative is f , and no direct limit calculation is needed.

Theorem 11.2 (The Fundamental Theorem of Calculus). *Let f be continuous on $[a, b]$, and let F be an antiderivative of f , meaning $F'(x) = f(x)$. Then*

$$\int_a^b f(x) dx = F(b) - F(a).$$

Why is this true? Distance and velocity make it clearest. If $s(t)$ is the odometer reading, then $v(t) = s'(t)$ by the definition of derivative. Between times a and b , the total odometer increase is $s(b) - s(a)$. But our integral calculation says the same distance is $\int_a^b v(t) dt$. Therefore

$$\int_a^b v(t) dt = s(b) - s(a).$$

Replace v by f and s by F , and this is the theorem. Let us make the intuition more like a proof.

Proof. Divide $[a, b]$ into n equal intervals with endpoints $a = t_0 < t_1 < \cdots < t_n = b$ and width Δt . The total increment is a sum of smaller increments:

$$F(b) - F(a) = [F(t_1) - F(t_0)] + [F(t_2) - F(t_1)] + \cdots + [F(t_n) - F(t_{n-1})],$$

where the middle terms cancel in pairs, leaving the endpoints. By the meaning of derivative, for small Δt each interval satisfies

$$F(t_i) - F(t_{i-1}) \approx F'(t_{i-1}) \Delta t = f(t_{i-1}) \Delta t,$$

so $F(b) - F(a)$ approximates the Riemann sum $\sum f(t_{i-1}) \Delta t$. Let $n \rightarrow \infty$: approximation becomes an exact limit. The left side $F(b) - F(a)$ is independent of subdivision, while the right tends to $\int_a^b f(t) dt$, so they are equal. $\square$

The crucial step is the cancellation of the long chain of intermediate terms, leaving $F(b) - F(a)$. Through that cancellation, integration's total accumulation fits exactly with differentiation's local rate of change. This is why *differentiation and integration are inverse operations*.

Note. Antiderivatives are not unique: x^2 has derivative $2x$, and $x^2 + 7$ also has derivative $2x$, since a constant disappears under differentiation. This does not affect the theorem. Two antiderivatives differ by a constant, which cancels in $F(b) - F(a)$. Choose whichever antiderivative is most convenient.

Try a numerical example. For the parabolic hill, calculate $\int_{-1}^1 (1 - x^2) dx$. Which function differentiates to $1 - x^2$? Try $F(x) = x - \frac{x^3}{3}$; indeed, $F'(x) = 1 - x^2$. Thus

$$\int_{-1}^1 (1 - x^2) dx = F(1) - F(-1) = \left(1 - \frac{1}{3}\right) - \left(-1 + \frac{1}{3}\right) = \frac{4}{3}.$$

$\frac{4}{3}$! Exactly Archimedes' answer from infinitely many triangles. Two thousand years later, the master key agrees with the old locksmith, one of mathematics' most reassuring moments.

Mathematical Toolbox

This chapter's new tools:

- **Riemann sums:** divide intervals finely and approximate curved areas with rectangle sums.
- **Definite integral** $\int_a^b f(x) dx$: a limit of Riemann sums, geometrically the area under a curve.
- **The fundamental theorem of calculus:** find an antiderivative F , then the integral is $F(b) - F(a)$. Differentiation and integration undo each other.
- **Slicing:** volume is the integral of cross-sectional area; solids with equal cross-sections at every height have equal volumes.
- **Series tests, intuitively:** terms tending to 0 are necessary but insufficient for convergence. Geometric series converge when the common ratio's absolute value is below 1.

Connections

This theorem sews three chapters together: Chapter 9's limits give the integral's definition, Chapter 10's derivatives supply its calculation key, and integration reconstructs total amounts from rates. Chapter 12 will show how scrutinizing conditions such as continuity and existence of limits gave rise to rigorous analysis.

11.6 Slicing Bread: From Area to Volume

Integration does more than area. How do we find a cone's or sphere's volume? Middle-school formulas say that a cone has one third the volume of a cylinder with the same base and height, and a sphere has volume $\frac{4}{3}\pi r^3$. Then we could only memorize them; now we can ask why.

Use the same method: cut. Imagine a sphere as a loaf sliced perpendicular to a horizontal axis. At position x from the centre, with the centre at 0 and ends at $-r$ and r , the section is a circle. Pythagoras gives squared radius $r^2 - x^2$ and area $\pi(r^2 - x^2)$. A slice of thickness Δx has approximate volume $\pi(r^2 - x^2)\Delta x$. Add and take the limit:

$$V = \int_{-r}^r \pi(r^2 - x^2) dx.$$

The integrand is only quadratic, so an antiderivative is easy: $F(x) = \pi\left(r^2x - \frac{x^3}{3}\right)$. The fundamental theorem gives

$$V = F(r) - F(-r) = \pi\left(r^3 - \frac{r^3}{3}\right) - \pi\left(-r^3 + \frac{r^3}{3}\right) = \frac{4}{3}\pi r^3.$$

The formula once memorized is the result of one integral.

Legend says Archimedes was proudest of the relation between a sphere and its enclosing cylinder, of base radius r and height $2r$: the sphere has two thirds its volume. He requested the figure on his tomb. Without integral notation, he compared slices: at each height, the sphere's section plus a cone's section equals the cylinder's section. In ancient China, Zu Geng, son of Zu Chongzhi, summarized the same idea: equal cross-sectional areas at every equal height force equal volumes. The principle later became known in Europe under Cavalieri's name, more than a thousand years after Zu Geng. Its independent growth in two civilizations suggests how naturally slicing approaches volume.

The cone's one-third factor follows similarly. Stand a cone of height h and base radius r with its tip upward. At distance x below the vertex, similar triangles give section radius $\frac{rx}{h}$ and area $\pi \frac{r^2 x^2}{h^2}$. Thus

$$V = \int_0^h \pi \frac{r^2 x^2}{h^2} dx = \frac{\pi r^2}{h^2} \cdot \frac{h^3}{3} = \frac{1}{3} \pi r^2 h.$$

Pull the constant $\frac{\pi r^2}{h^2}$ outside, leaving the familiar $\int_0^h x^2 dx = \frac{h^3}{3}$. Volume formulas become consequences of one statement: volume is the integral of cross-sectional area.

11.7 Another Perspective: Density, Probability, and Soup

Wherever a rate accumulates into an amount, integration appears. Consider three examples beyond area and volume.

Mass and density. An uneven metal rod is less dense at the left and denser at the right. Its linear density at position x is $\rho(x)$, mass per unit length. Divide it into n pieces. Piece i is approximately uniform, with mass $\rho(x_i) \Delta x$. Summing and taking the limit gives

$$\text{total mass} = \int_a^b \rho(x) dx.$$

In a pot of soup with uneven salt concentration, total salt is likewise the integral of concentration over volume, a three-dimensional integral studied at university.

Electricity and work. Current is the rate of charge flow, so total charge over an interval is the integral of current over time. If current rises as $I(t) = 2t$ amperes, charge flowing in the first three seconds is

$$\int_0^3 2t dt = 3^2 - 0^2 = 9 \text{ coulombs.}$$

Similarly, work over a displacement is the integral of force with respect to displacement. A spring becomes harder to stretch as its extension grows; integration calculates the total work of that changing force.

Probability and density. You wait at a station, with the bus equally likely to arrive at any time during one hour. What is the probability of arrival between 10 o'clock and 10:15? Intuition gives $\frac{1}{4}$, since the interval occupies a quarter of the hour. But if arrivals are more frequent toward evening, their likelihood varies with time. A *density curve* $p(x)$ describes the

probability concentration per unit time near x . Then

$$P(\text{arrival between } c \text{ and } d) = \int_c^d p(x) dx,$$

so probability is area under the density curve. Total probability is 1, meaning that the integral over the full range is 1. In our uniform example the density is $\frac{1}{60}$ per minute, a horizontal line, and probability is rectangle area. Densities for random quantities such as height and measurement error are bell-shaped, high in the middle and low at both ends; Chapter 18 studies them.

Terminology Card: Probability Density

Intuition: a curve whose area over an interval gives the probability that a random quantity falls in that interval.

Formal definition: a nonnegative function $p(x)$ whose integral over the entire range equals 1, with $P(c \leq X \leq d) = \int_c^d p(x) dx$.

Example: $p(x) = \frac{1}{60}$ for $0 \leq x \leq 60$ minutes describes uniform arrival over an hour. The probability of arriving within a quarter-hour is $\frac{15}{60} = \frac{1}{4}$.

A counterintuitive feature matters: a density curve's height at one point is not a probability. Arriving exactly at 10:30:00 has probability 0, since a single point has zero interval width and zero area. Probability belongs to intervals; density describes concentration. This is the same distinction as velocity at an instant versus distance. Integration accumulates a rate into an amount.

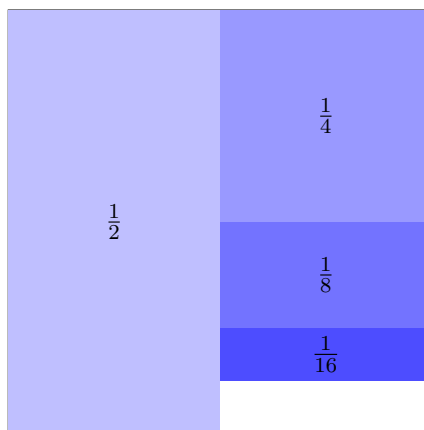
11.8 When Does Infinite Addition Give a Finite Result?

Archimedes' area calculation contains another treasure: $1 + \frac{1}{4} + \frac{1}{16} + \cdots = \frac{4}{3}$. Infinitely many positive numbers produce a finite total. Intuition protests: should infinitely many things not add to infinity?

Not necessarily. Chapter 9's Zeno discussion already planted the answer. To travel 1 kilometre, first travel $\frac{1}{2}$, then $\frac{1}{4}$, then $\frac{1}{8}$, and continue. Infinitely many stages still give a finite distance:

$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1.$$

Geometry makes this clearer. Colour half a unit square, then half the remainder, then half again. Each step approaches complete coverage while leaving a little uncoloured. The total coloured area tends to 1 without exceeding it.



Half, then half again: total area tends to 1

A sum with the same ratio between successive terms is a *geometric series*. If its ratio q satisfies $|q| < 1$, it has a finite sum:

$$a + aq + aq^2 + \cdots = \frac{a}{1 - q}.$$

For Zeno's example, $a = \frac{1}{2}$ and $q = \frac{1}{2}$, giving $\frac{1/2}{1 - 1/2} = 1$. The proof is simple: set $S = a + aq + \cdots + aq^n$ and multiply by q to get $qS = aq + \cdots + aq^{n+1}$. Subtract and cancel the middle terms: $(1 - q)S = a - aq^{n+1}$. When $|q| < 1$ and $n \rightarrow \infty$, $q^{n+1} \rightarrow 0$, so $S \rightarrow \frac{a}{1 - q}$. Again *cancellation* leads the way, echoing the fundamental theorem's proof.

Repeating decimals hide geometric series too. Does $0.999\dots$ equal 1? Expand it:

$$0.999\dots = 0.9 + 0.09 + 0.009 + \cdots = \frac{0.9}{1 - 0.1} = 1.$$

It *is* 1, rather than merely approaching 1, wearing an infinitely long disguise. The limit itself is the destination, not the unfinished process.

Some series converge extremely quickly. A famous example appears in the natural logarithm's base e :

$$e = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} + \cdots$$

The denominators are successive products: $2 = 1 \times 2$, $6 = 1 \times 2 \times 3$, $24 = 1 \times 2 \times 3 \times 4$, and so on. Each denominator gains another growing factor, making terms shrink faster than any geometric series. The first seven terms give $e \approx 2.718$, accurate to two decimal places. Compound interest, population growth, and radioactive decay all involve this number, whose face turns out to be a rapidly converging infinite sum. Next chapter's power-series expansions return to it.

But do not infer that shrinking terms guarantee a finite sum. The best-known counterexample is the *harmonic series*:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots$$

Its terms tend to 0, yet its sum is infinite.

Theorem 11.3 (The Harmonic Series Diverges). *The sum of $1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$ exceeds every prescribed number.*

Proof. Group the terms: first 1, then $\frac{1}{2}$, then $\frac{1}{3} + \frac{1}{4}$. Each term in that third group is at least $\frac{1}{4}$, so its sum exceeds $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$. The fourth group, $\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}$, has four terms each at least $\frac{1}{8}$, so it exceeds $4 \times \frac{1}{8} = \frac{1}{2}$. The next 8 terms are each at least $\frac{1}{16}$ and again total more than $\frac{1}{2}$. Each doubling of the term count supplies at least another $\frac{1}{2}$. Adding $\frac{1}{2}$ infinitely often exceeds every bound. $\square$

This proof, first given by a medieval scholar, is surprisingly simple but profound. Terms tending to 0 are a *necessary condition* for convergence, far from sufficient. Convergence depends on shrinking quickly enough. The sum of all $\frac{1}{n^2}$ converges, in fact to $\frac{\pi^2}{6}$: pi emerges from a collection of squares! Euler discovered this as a young man, astonishing the mathematical world. The sum of all $\frac{1}{n}$ diverges. What about $\frac{1}{n^{1.5}}$ or $\frac{1}{n^{1.001}}$? Locating the boundary between fast and slow is one of analysis's most delicate questions.

Counterexample Alert

“The terms tend to 0, so the sum is finite.” The harmonic series refutes this directly. The reverse implication does hold: a finite sum forces terms to tend to 0, but terms tending to 0 do not imply a finite sum. Necessary and sufficient conditions differ by more than a word.

11.9 Approximating an Integral by Hand

One-Minute Experiment

Take squared paper and check the fundamental theorem yourself.

1. Plot $y = x^2$ on $0 \leq x \leq 2$. Divide $[0, 2]$ into 4 intervals of width 0.5, erect rectangles using *right-endpoint* heights, and add their areas:

$$0.5 \times (0.5^2 + 1^2 + 1.5^2 + 2^2) = 0.5 \times 7.5 = 3.75.$$

Repeat with *left-endpoint* heights: $0.5 \times (0 + 0.25 + 1 + 2.25) = 1.75$. The true area lies between these values.

2. Find the exact value using the fundamental theorem. An antiderivative of x^2 is $\frac{x^3}{3}$, so

$$\int_0^2 x^2 dx = \frac{2^3}{3} - 0 = \frac{8}{3} \approx 2.67.$$

It does lie between 1.75 and 3.75.

3. Increase to 8 intervals of width 0.25. Recalculate both rectangle sums, record them below, and observe how they squeeze toward $\frac{8}{3}$:

Intervals n	Left-endpoint sum	Right-endpoint sum	Exact value $\frac{8}{3}$
4	1.75	3.75	2.666...
8	2.19	3.19	2.666...

If you wish, calculate $n = 16$ with a calculator. How much does each approximation improve? Guess the rate at which the error shrinks.

Keep Climbing

Further signposts include *integration by substitution* and *integration by parts*, two tools for calculating integrals; *improper integrals*, handling infinite intervals or unbounded integrands, such as $\int_1^\infty \frac{1}{x^2} dx = 1$; *numerical integration*, using trapezoids or parabolas for efficient computation; the *Riemann ζ function*, where the convergence boundary of $\sum \frac{1}{n^s}$ leads to Chapter 15's primes; and the more general *Lebesgue integral*, which replaces vertical strips with horizontal layers to handle stranger functions.

Questions to Think About

Observation

1. A speedometer always shows a positive reading, but the driver goes forward and then reverses to the start. Are the area under the velocity curve and total distance still the same? Hint:

should reverse motion be recorded as positive or negative velocity? What do the integral and the actual distance wearing down the odometer each represent?

2. Examine a water meter, electricity meter, and phone data-usage counter. Does each record a rate or an amount? What is its partner quantity? Hint: each rate-amount pair hides an integral and a derivative.

Proof

1. Without consulting a formula, prove $\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots = \frac{1}{2}$. Hint: multiply by q and subtract as in the text, or draw a square and repeatedly take thirds.
2. Use the fundamental theorem to explain why $\int_a^b x \, dx = \frac{b^2 - a^2}{2}$ equals the area of a trapezoid. Hint: what figure does $y = x$ enclose over $[a, b]$, and how long are the two parallel sides?

Challenge

1. The harmonic series diverges, but deleting all terms whose denominators contain the digit 9, such as $\frac{1}{9}$, $\frac{1}{19}$, and $\frac{1}{92}$, leaves a convergent sum. Can you estimate why? Hint: how many one-digit, two-digit, and three-digit numbers omit 9? What bounds the sum of their reciprocals in each group? Try summing those bounds geometrically.
2. The proof of $\int_a^b f(x) \, dx = F(b) - F(a)$ assumes that F has derivative f . If we know only that f is continuous, can we *construct* an antiderivative? Hint: consider $F(x) = \int_a^x f(t) \, dt$ and estimate $\frac{F(x+h) - F(x)}{h}$ for small h . Next chapter makes this idea rigorous.

The Door to the Next Chapter

We have moved quickly enough that our footing may feel uncertain. What does existence of a Riemann-sum limit mean? Why must approaching upper and lower sums squeeze out exactly one number rather than leave a permanent gap? The fundamental theorem requires f to be continuous, but can drawing without lifting a pencil be a definition? Behind the divergence of the harmonic series and convergence of $\sum \frac{1}{n^2}$, is there one common measuring tool?

These questions are easy to understand and beautifully difficult to answer. Nineteenth-century mathematicians forged everyday words such as approach, continuity, and convergence into precise definitions, creating *analysis*. Next chapter we enter that workshop to see intuition forged into theorems, and meet an old friend there: constructing any complicated sound from infinitely many simple waves.

Chapter 12 Analysis: Turning Intuition into Reliable Theorems

12.1 A Strange Question on a Mountain Path

During the summer holidays, your family climbs a mountain. The foot is 100 metres above sea level and the summit 1500 metres. You leave at eight in the morning and arrive at half past twelve, stopping for rests and even backtracking to find a dropped hat. Your speed changes constantly, with no regular pattern.

Back home, someone asks a strange question: can you guarantee that at some instant during the climb, your altitude was exactly 618 metres?

Intuitively you say yes. Altitude changes continuously from 100 to 1500 metres; surely you cannot jump from 617 to 619 metres.

The intuition sounds airtight. Before agreeing too quickly, consider two more questions:

- Your record shows an altitude of 500 metres at nine and 800 metres at eleven. Must some moment have had altitude exactly 600.001 metres?
- A thermometer rises from 10°C at dawn to 25°C at noon. Must the temperature at some instant have been exactly $\sqrt{500}$ degrees Celsius? Remember that $\sqrt{500}$ has a nonterminating, nonrepeating decimal expansion, unreadable on any thermometer's scale.

You may still answer yes to the first, but with less confidence: can altitude really be discussed with infinite precision? The second is harder. How can a value no thermometer can display have been passed through exactly?

This chapter forges the feeling that continuous change skips no intermediate value into a theorem that can be proved and trusted. The craft is **analysis**: its aim is to make intuition reliable, rather than merely speed up calculation. Mathematicians spent two centuries on it, having been fooled repeatedly by their intuitions, sometimes spectacularly.

12.2 Intuition Is Fooled for the First Time

The simplest picture of continuity is a graph drawable without lifting a pencil. Useful, but unreliable: historically it failed in at least three ways.

First, if skipping no intermediate value seems obvious, consider

$$f(x) = \begin{cases} 1, & x \geq 0, \\ -1, & x < 0. \end{cases}$$

It takes value -1 at $x = -1$ and 1 at $x = 1$, but never the intermediate value 0 . The graph jumps at $x = 0$. Passing through intermediate values really does depend on the no-jumps condition.

Second, something subtler: since $1^2 = 1 < 2$ and $2^2 = 4 > 2$, the solution $\sqrt{2}$ of $x^2 = 2$ ought to be passed through between 1 and 2 . Yet $\sqrt{2}$ is irrational. In a world containing only rational numbers, $f(x) = x^2 - 2$ is negative at $x = 1$ and positive at $x = 2$, but **never equals zero over the rationals**. The supposedly encountered middle value has no place to occur. Here intuition fails because **the number system itself has holes**. The rationals resemble a shirt full of pinholes: seemingly complete until inspected closely.

Counterexample Alert

Warning 1: over the rationals, $f(x) = x^2 - 2$ changes from negative to positive without reaching zero. Half the intermediate value theorem's secret belongs to the function, the other half to the real numbers. Their lack of holes is completeness.

Warning 2: continuous graphs need not be smooth. Weierstrass constructed a function continuous everywhere with no tangent anywhere, an eternally bristly curve that never magnifies into a line. It needs no pencil lift yet has no smooth point. Continuity is much weaker and subtler than smoothness.

Third, even without jumps or holes in the numbers, infinity can defeat intuition. The function $g(x) = 1/x$ behaves well at every point of $(0, 1]$, yet as x approaches 0 , $g(x)$ grows without bound. A finite horizontal range contains unbounded height. The intuition that a finite stretch of curve must have a highest point fails again.

All three failures teach the same lesson: every statement in analysis must withstand the most searching objection. Let us repair the gaps one by one.

12.3 Making Continuity Precise

Repair the first gap: what exactly is continuity? We want **small input changes to produce small output changes**. More precisely:

Definition 12.1 (Continuity: An Intuitive Version). A function f is **continuous** at a if $f(x)$ approaches $f(a)$ as x approaches a . Whatever output-error requirement you choose, such as deviation below 0.001 , there is a small input interval around a in which every x gives an output meeting your requirement. Continuity on an interval means continuity at every point of it.

Notice the challenge structure: **you** choose the tolerance, however severe, and **I must** provide a response. A feeling of closeness becomes a testable agreement. The jump function $f(x)$ fails at $x = 0$. Demand output deviation below $1/2$; however tightly I restrict inputs near 0 , positive and negative inputs remain, giving outputs 1 and -1 , a difference of 2 that exceeds the demand.

Terminology Card: Continuous Function

Intuition: a graph drawn without lifting the pencil; small input tremors cause only small output tremors.

Formal statement: as x tends to a , $f(x)$ tends to $f(a)$, so the limit equals the function value.

Example: $f(x) = x^2$ is continuous everywhere, since x near a makes x^2 near a^2 . The jump function is discontinuous at $x = 0$.

The reals repair the second gap. Their precious property is that **every process approaching a position arbitrarily closely finds that position among the real numbers**. This is **completeness**. The rationals lack it: 1, 1.4, 1.41, 1.414, ... approaches $\sqrt{2}$, but $\sqrt{2}$ is not rational. The reals have no such defect. Once we accept this, intuition's central claim can be proved.

12.4 The Intermediate Value Theorem: No Jumps, No Skipped Values

Theorem 12.2 (The Intermediate Value Theorem). *Suppose f is continuous on $[a, b]$, with $f(a) < 0$ and $f(b) > 0$, or conversely. Then there is at least one point c in the interval with $f(c) = 0$.*

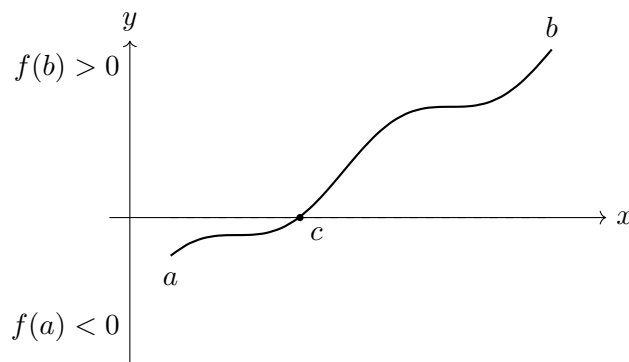
More generally, f takes every value between $f(a)$ and $f(b)$. Altitude changing continuously from 100 to 1500 metres must pass exactly through 618 metres, and even through the altitude corresponding to $\sqrt{500}$ metres.

The proof resembles binary search in computing: repeatedly halve the suspect interval, leaving the zero nowhere to escape.

Proof. Suppose $f(a) < 0 < f(b)$. Examine the midpoint $m_1 = (a + b)/2$. If $f(m_1) = 0$, we have found the zero. If $f(m_1) > 0$, the zero must lie in the left half $[a, m_1]$; if $f(m_1) < 0$, in the right half $[m_1, b]$. Either way, we obtain a half-length interval with negative value at the left and positive value at the right.

Repeat. After step n , the suspect interval has length $(b - a)/2^n$, with left endpoint a_n satisfying $f(a_n) \leq 0$ and right endpoint b_n satisfying $f(b_n) \geq 0$. The sequence $a_1, a_2, a_3, \dots$ is nondecreasing and bounded above by b ; the right endpoints are nonincreasing and bounded below by a . The gap tends to 0. Completeness gives both endpoint sequences the same limiting position c .

Finally use continuity: as x tends to c , the values tend to $f(c)$. All left-endpoint values are at most 0, so their limiting value $f(c)$ is at most 0. All right-endpoint values are at least 0, so $f(c)$ is at least 0. Together these force $f(c) = 0$. $\square$



Look again at the proof: it establishes **existence** and also **calculates** the zero. Each cut halves the error; twenty reduce it to about a millionth. For $\sqrt{2}$, start from $1^2 < 2 < 2^2$. The midpoint 1.5 has square $2.25 > 2$, so take the left half; 1.25 has square $1.5625 < 2$, so take the right half. After ten steps you have 1.414, matching manual square-root calculation. Analysis's first theorem immediately becomes a calculator.

12.5 Highest Points and Uniform Temperaments

The intermediate value theorem governs values in between; its sibling governs extreme values.

Theorem 12.3 (The Extreme Value Theorem). *A continuous function on a closed interval $[a, b]$ attains both a maximum and a minimum. There are points p and q in the interval such that $f(p)$ is at least every function value and $f(q)$ at most every function value.*

Both hypotheses matter. The function $g(x) = 1/x$ is continuous on $(0, 1]$, but that interval is not closed and there is no maximum. A discontinuous function on a closed interval can jump away from its highest prospective value. This theorem underpins optimization: asking for least material or greatest profit makes sense because suitable functions on suitable domains have answers. Finding them is Chapter 10's work for derivatives.

A finer concept, less common in exams but central to analysis, is **uniform continuity**. Ordinary continuity gives each point its own response to a tolerance. Uniform continuity requires **one response for the whole interval**: after you set an output tolerance, I produce a single input-length ruler that works wherever it is placed in the interval.

Why might this fail? For $g(x) = 1/x$ on $(0, 1]$, the curve becomes steeper near 0. An input gap of 0.001 changes output by about 0.001 near $x = 1$, but by thousands near $x = 0.001$. Any fixed ruler eventually fails farther left. By contrast, $f(x) = x^2$ behaves well on every **closed** interval $[a, b]$: its slope is bounded by 2 times the larger endpoint's absolute value, so a uniform rule is available. Every continuous function on a closed interval can be proved uniformly continuous. Intuition says that a finite stage cannot host infinite unruliness; this time it wins, but only with proof.

12.6 Cauchy Sequences: Proving an Answer Exists Before Knowing It

A deeper question hides in the bisection proof. We construct $a_1, a_2, \dots$ and claim that it approaches some position c . But c is what we seek. How can we know it exists **before** finding it? Could the sequence approach an empty position, like rationals approaching $\sqrt{2}$?

The nineteenth-century mathematician Cauchy gave an elegant answer: **we need not know the destination in advance** to decide convergence. We need only inspect how closely the sequence's terms stick together.

Definition 12.4 (Cauchy Sequence: An Intuitive Version). A sequence $a_1, a_2, a_3, \dots$ is **Cauchy** if, for any specified tolerance, the distance between **any two terms** sufficiently far along is no greater than that tolerance.

Every convergent sequence is Cauchy: terms gathering around one destination necessarily gather near each other. The deeper converse asks whether gathering terms always have a destination. Not among rationals: approximations to $\sqrt{2}$ refute it. Among reals, **they always do**. Every real Cauchy sequence converges: this standard formulation of completeness is analysis's foundation stone.

Terminology Card: Cauchy Sequence

Intuition: a sequence whose later members cluster as closely together as demanded.

Formal statement: for every tolerance $\varepsilon > 0$, there is a term after which the difference between any two terms is less than ε .

Example: 1, 1.4, 1.41, 1.414, ..., the successive decimal truncations of $\sqrt{2}$, is Cauchy. Beyond decimal place n , any two terms differ by at most 10^{-n} . Its destination is $\sqrt{2}$, a limit available among reals but not rationals.

We now have a new answer to whether an equation has a solution: construct increasingly close approximate solutions, use completeness to guarantee a limit, then continuity to verify that it solves the equation. This procedure proves existence for many equations even when no exact expression for the solution can ever be written.

12.7 Power Series: Functions as Infinitely Long Polynomials

The preceding sections defended intuition by repairing its gaps; the next three use infinite processes to create remarkable new objects. You have already glimpsed the first tool in infinite decimals. The expansion $\pi = 3.14159\dots$ splits π into 3 plus 0.1 plus 0.04 and further corrections. Power series do the same for **functions**.

The simplest example is geometric summation. We know $1 + x + x^2 + \dots + x^n = \frac{1 - x^{n+1}}{1 - x}$. For $|x| < 1$, the term x^{n+1} tends to 0, giving

$$\frac{1}{1 - x} = 1 + x + x^2 + x^3 + \dots \quad (|x| < 1).$$

Check $x = 1/2$: the right side is $1 + 1/2 + 1/4 + \dots = 2$, and the left is $1/(1 - 1/2) = 2$. They agree.

Read this in a new way: $\frac{1}{1-x}$ equals an infinitely long polynomial. Why split it this way? Polynomials need only addition and multiplication. Decomposing a difficult function into powers gives a recipe whose number of terms can be chosen to match the required accuracy.

The best-known recipe belongs to the exponential:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

It holds for **all** x , unlike $\frac{1}{1-x}$, whose recipe works only for $|x| < 1$. Each recipe has its own radius of convergence. At $x = 1$, the first six terms give

$$1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} \approx 2.7167,$$

whereas $e \approx 2.71828$: five terms already approximate the first three decimal places. Your calculator's e key probably uses a series of this kind internally.

Why do convergence radii differ? Look at $\frac{1}{1-x}$: it blows up at $x = 1$ because the denominator vanishes, so a recipe centred one unit from that point can extend only a distance 1. The function e^x remains well behaved along the whole real line. Roughly, **a power series reaches as far as the function's nearest bad point**. Complex analysis makes this precise; sometimes bad points hide away from the real axis, one reason mathematicians had to study complex functions.

Power series also reveal a beautiful derivative property. In the recipe for e^x , differentiating each term gives its predecessor: $\frac{x^3}{3!}$ gives $\frac{x^2}{2!}$, $\frac{x^2}{2!}$ gives x , and the constant 1 gives 0. Differentiating the entire series **leaves it unchanged!** This is the recipe version of Chapter 10's claim that the derivative of e^x is itself.

12.8 Fourier Series: Every Sound Is a Collection of Pure Tones

The second tool is more romantic. Power series build with $1, x, x^2, \dots$; Fourier proposed **sine waves** as different building blocks. He claimed that almost every periodic function can be assembled from sine waves of different frequencies. Many leading mathematicians objected: making sharp corners from smooth waves sounded impossible.

Consider the famous recipe for a square wave, alternating between 1 and -1 over 0 to 2π :

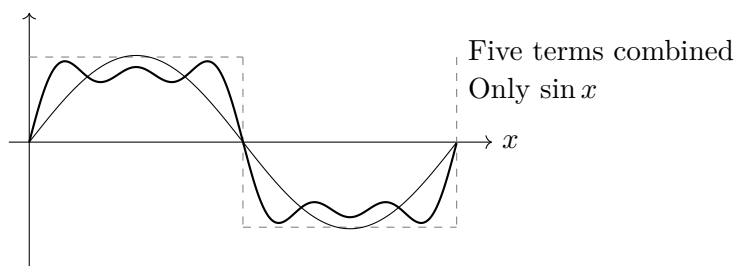
$$\text{square wave}(x) = \frac{4}{\pi} \left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \frac{\sin 7x}{7} + \dots \right).$$

The first term alone is a gentle sinusoid, far from square. Add the second and its peaks flatten; with five terms, it looks convincingly rectangular apart from little ripples at the edges. More terms make the ripples smaller and the corners sharper.

A simple pattern lies in the recipe: frequencies follow successive odd multiples, while amplitudes decay as their reciprocals. Triple frequency gets one-third height, quintuple frequency one-fifth. Why are high-frequency contributions smaller? Most of the square wave's body is flat; only its corners need fine high-frequency carving. Details occupy less than the main body.

Across Fourier decompositions, **low frequencies determine broad shape, high frequencies determine detail**. Remember this for the experiments and connections below.

When Fourier submitted these ideas to the Paris Academy, his mathematical reviewers, including the exacting Lagrange, refused to sign off. They first wanted to know whether the infinite sum converged and to what. Those are precisely the questions Cauchy sequences and completeness address. History came full circle: judging Fourier's waves forced mathematicians to clarify continuity and convergence, driving much of the demand for rigorous analysis.



Substitute a particular value for an enduring surprise. At $x = \pi/2$, the square wave equals 1, while $\sin(\pi/2) = 1$, $\sin(3\pi/2) = -1$, and $\sin(5\pi/2) = 1$ alternate in sign. Thus

$$1 = \frac{4}{\pi} \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots \right), \quad \text{so} \quad \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots.$$

π , hiding in a circle's circumference, can be calculated by alternating reciprocals of all odd numbers. After 100 terms you get about two decimal places of π . Convergence is slow, but the identity is beautiful.

One-Minute Experiment

Build a Square Wave by Hand. On squared paper, draw $y = \sin x$ for x from 0 to 2π , calculating points at $x = 0, 0.5, 1, \dots$. Draw $\frac{1}{3} \sin 3x$, with triple frequency and one-third height. **Add** their heights at each x and plot the new curve. Peaks flatten and troughs rise, revealing the beginnings of a rectangle. If you wish, add $\frac{1}{5} \sin 5x$. Three colours on one graph recreate Fourier's view two centuries ago.

Another Experiment: hum a sustained note into a phone spectrum app or music equalizer. Several bars appear: the lowest, brightest one is the fundamental, with overtones above. Your voice and a piano playing the same note sound different because their overtone bars have different relative heights. Timbre is written in Fourier coefficients.

Connections

Fourier series live in your pocket. Music compression such as MP3 divides sound into short segments, decomposes each into frequency components, and discards inaudible high-frequency contributions, reducing data with little audible change. JPEG uses the related cosine transform: low-frequency coefficients describe smoothly varying regions, while details need high frequencies. Telephone noise reduction, seismic analysis, and medical imaging are descendants of Fourier's idea. A once-ridiculed claim now handles humanity's images and sounds daily.

12.9 The Complex Plane and Euler's Formula

The third tool comes from our old friend i , with $i^2 = -1$. You may have regarded i as a temporary helper brought in to solve quadratics. Joined with analysis, it produces one of mathematics' most celebrated formulas.

Place complex numbers in the plane: $a+bi$ has horizontal coordinate a and vertical coordinate b . Addition is vector translation. Multiplication is more surprising: multiplying by i means **rotating ninety degrees anticlockwise**. Multiplying 1 by i gives i , moving $(1, 0)$ to $(0, 1)$; another multiplication by i gives -1 , another ninety-degree turn. Two turns give a half-turn, or multiplication by -1 . Geometrically, $i^2 = -1$ says two quarter-turns make a half-turn. Complex multiplication is **rotation combined with scaling**.

Now invite power series in. Euler tried something apparently wild: insert the purely imaginary $i\theta$ into the recipe for e^x . The cycle $i^2 = -1$, $i^3 = -i$, $i^4 = 1$ separates odd and even terms into two teams:

$$e^{i\theta} = \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \cdots\right) + i\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right).$$

The bracketed series are precisely the recipes for $\cos \theta$ and $\sin \theta$, checkable using Chapter 10's derivatives: all derivative values at 0 match. Thus:

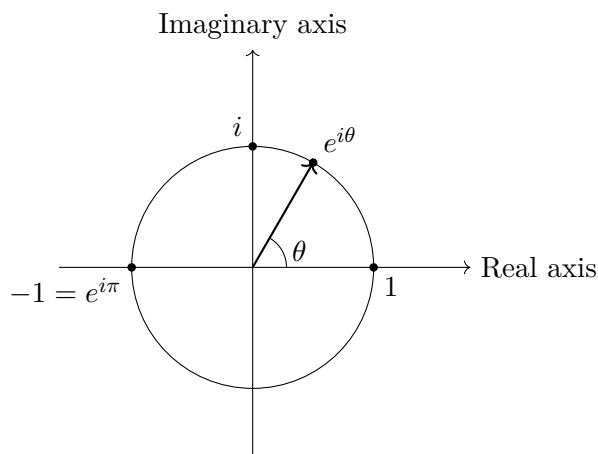
Theorem 12.5 (Euler's Formula). *For every real θ ,*

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

Its geometry is breathtaking. The right side is the unit-circle point at angle θ ; the left is e raised to a purely imaginary power. As θ increases, $e^{i\theta}$ moves uniformly around the unit circle. Set $\theta = \pi$: since $\cos \pi = -1$ and $\sin \pi = 0$,

$$e^{i\pi} + 1 = 0.$$

Five fundamental constants—0, 1, π , e , and i —stand together, joined by addition, multiplication, and exponentiation. This identity rarely misses a mathematician's list of the most beautiful formulas.



Euler's formula is also Fourier series' official language. With $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$ and $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$, sums of sines and cosines become sums of $e^{ik\theta}$, turning trigonometric identities into simple exponent arithmetic. Engineers doing Fourier analysis often write scarcely any sin or cos, instead using rotating arrows $e^{i\theta}$.

One step farther lies **complex analysis**: replace x with complex z and study $f(z)$. Something extraordinary happens. Complex differentiability is so strong that a function differentiable throughout a region is automatically smooth, automatically has power-series expansions, and has its values fixed by boundary values. It resembles a red-hot metal plate where disturbing one part affects the whole. This rigidity makes complex analysis elegant and powerful, and leads to some of mathematics' deepest questions.

12.10 What We Have Gained

Mathematical Toolbox

This chapter's new tools:

- **Continuity, intuitively**: you choose an error requirement, I supply a response. Close enough inputs give close enough outputs.
- **Completeness of the reals**: Cauchy sequences, whose later terms cluster, have real limits. The rationals lack this property.
- **The intermediate value theorem**: continuous functions take every value between their endpoint values. Bisection both proves this and approximates zeros.
- **The extreme value theorem**: a continuous function on a closed interval attains maximum and minimum; neither hypothesis can be dropped.
- **Uniform continuity**: one response works throughout the interval. Continuous functions on closed intervals have it.
- **Power series**: infinitely long polynomial recipes, such as $e^x = \sum x^n/n!$, each with its own convergence radius.
- **Fourier series**: periodic functions as sums of sine waves; the square-wave expansion also yields $\pi/4 = 1 - 1/3 + 1/5 - \dots$.
- **Euler's formula**: $e^{i\theta} = \cos \theta + i \sin \theta$, expressing complex multiplication as rotation and scaling, with special case $e^{i\pi} + 1 = 0$.

Keep Climbing

Signposts for further climbing:

- **ε - δ language:** our challenge-and-response agreement in precise symbols, the first lesson of university analysis.
- **The Weierstrass function:** an everywhere-continuous, nowhere-differentiable bristly curve, rigorously constructed using a uniformly convergent series of functions.
- **Taylor series:** the general theory of power-series recipes, with coefficients determined by successive derivatives.
- **The fast Fourier transform (FFT):** an algorithm dramatically reducing the work of Fourier decomposition, often counted among the twentieth century's most important algorithms.
- **The Riemann ζ function and Riemann hypothesis:** a jewel of complex analysis connecting our series to next chapter's primes, and an unsolved million-dollar problem.

Questions to Think About

Observation

1. One day you climb from foot to summit between 12 noon and 4 p.m.; the next day you descend the same route between 12 noon and 4 p.m. Use the intermediate value theorem to prove that at some clock time on both days you are at the same altitude. *Hint:* plot both altitude functions together and consider their difference.
2. Does $x^3 + x - 1 = 0$ have a solution between 0 and 1? Perform three bisections by hand to narrow its location to an interval of length $1/8$. *Hint:* evaluate at $x = 0, \frac{1}{2}, 1$, then halve.

Proof

3. Prove that a continuous f satisfying $f(a) \cdot f(b) < 0$ has a zero in (a, b) , the intermediate value theorem. Complete the bisection proof's explanation that nondecreasing, bounded left endpoints approach some c . *Hint:* what happens to a sequence moving steadily rightward without ever passing a wall?
4. For $a_n = \frac{n}{n+1}$, or $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \dots$, verify the Cauchy property. How small is $|a_m - a_n|$ when m, n both exceed some N ? *Hint:* $|a_m - a_n| = \left| \frac{1}{n+1} - \frac{1}{m+1} \right|$; bound each fraction separately.

Challenge

5. Use $e^{i\theta} = \cos \theta + i \sin \theta$ to calculate $e^{i\alpha} \cdot e^{i\beta}$. Expand once using exponent laws and once using complex multiplication, then compare real and imaginary parts. Which two trigonometric

identities emerge for free? *Hint*: what is the real part after expanding $(\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta)$?

6. The series $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$ converges slowly to $\pi/4$. Approximately how large is the error after 100 terms? Estimate by comparing consecutive partial sums. *Hint*: alternating partial sums fall on opposite sides of the limit, with error bounded by the first omitted term.

The Door to the Next Chapter

End with one suggestive look at the harmonic series:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots .$$

Its terms tend to 0, but its sum **grows without bound**. Grouping gives $\frac{1}{3} + \frac{1}{4} > \frac{1}{2}$, $\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} > \frac{1}{2}$, and the eight terms from $\frac{1}{9}$ to $\frac{1}{16}$ again exceed $\frac{1}{2}$. Infinitely many halves make the sum unstoppable.

Euler asked a history-changing question about this small fact: what does harmonic divergence have to do with **prime numbers**? He found an astonishing identity: the sum of reciprocals of all positive integers equals an infinite product made only from primes. If there were only finitely many primes, the product would be finite, contradicting the divergent harmonic sum. Analysis thus reproved Euclid's two-thousand-year-old theorem of infinitely many primes, with a sharper tool.

Beyond this door lies a new world: integer multiplication and division hide series, functions, and even the complex plane. To read it, return first to simple questions. What are primes? What patterns govern them? Why is a factorization such as $7 \times 11 \times 13 = 1001$ unique? Next chapter we enter the world of integers, congruences, and the modern cryptography they protect.

Part IV

Branching from Integers—Number Theory and Analytic Number Theory

Chapter 13 Elementary Number Theory: Primes, Congruences, and the Mechanisms of Integers

13.1 A Class That Will Not Split: Starting with a Line-Up

At the school sports day, the PE teacher wants to arrange the second-year middle-school pupils in rows, with equally many in each row and nobody left over. Three sixth-grade classes report their numbers: Class A has 48 pupils, Class B has 45, and Class C has 47.

Class A sorts itself out quickly: 48 pupils can form 6 rows and 8 columns, or 4 rows and 12 columns, with plenty of other options. Class B has no trouble either: 45 pupils can form 5 rows and 9 columns. Class C tries for ages: 2 per row? One left over. 3 per row? One left over. 4, 5, 6 per row... always one left over. The number 47 is like an unbreakable stone: it cannot be split into a product of two smaller integers.

Make an intuitive guess first: are these “stone numbers” rare oddities, or are they scattered everywhere? Do they eventually run out—could every integer beyond some point be breakable? It sounds like a child’s casual question while counting, yet it has challenged some of history’s sharpest minds. In this chapter, we start with this “stone” and explore the mechanisms hidden inside integers.

13.2 A First Attempt: Sieves and Misconceptions

When 47 refuses to split, the simplest approach is to try possible factors one by one. 2 fails (47 is odd), 3 fails ($4 + 7 = 11$ is not a multiple of 3), and 5 fails (the last digit is neither 0 nor 5). What about 7? $7 \times 6 = 42$ and $7 \times 7 = 49$, so that fails too. We can actually stop here—why? Because $7 \times 7 = 49$ already exceeds 47. If 47 were a product of two integers both greater than 1, at least one would be no larger than its square root, $\sqrt{47} \approx 6.9$. Testing up to 6 is enough; there is no need to reach 46. This is our first mechanism: to test whether a number can be split, we need only check up to its square root.

Following this idea, someone makes a bold conjecture: “The unbreakable numbers are the odd numbers.” That sounds reasonable: surely every even number has a factor of 2? But someone immediately objects: 9 is odd, yet equals 3×3 ; 15, 21, and 25 all split too. Conversely, 2 is even but cannot be split. Evidently, parity and factorability are different matters.

Counterexample Alert

“All odd numbers are unbreakable” is false. $9 = 3 \times 3$, $15 = 3 \times 5$, and $21 = 3 \times 7$ are odd, yet each is a product of smaller integers. The only reliable way to decide whether a number can be split is to check every possible factor up to its square root, not to inspect its final digit or parity.

Over two thousand years ago, ancient Greek mathematicians devised a systematic method now called a “sieve”: write out the numbers from 2 to 100, circle 2, and cross out all multiples of 2; find the next uncrossed number, 3, circle it, and cross out all multiples of 3; then find 5, 7, and so on. Every crossed-out number is a “composite product” divisible by something smaller; the survivors are the unbreakable stones. The method is laborious but honest: it guarantees that nothing is missed.

There is another easily overlooked shortcut: after crossing out multiples of 7, we can stop. By the square-root bound, every composite number up to 100 has a prime factor no larger than $\sqrt{100} = 10$, so it was already eliminated when we crossed out multiples of 2, 3, 5, 7. Every survivor after these four rounds must be unbreakable. Sieving the numbers up to 100 leaves this list:

2	3	5	7	11
13	17	19	23	29
31	37	41	43	47
53	59	61	67	71
73	79	83	89	97

Count them: there are 25. Look at the table for a while and questions emerge. Why is every entry except 2 odd? Why do the gaps seem to widen farther down the list? Why are some close together (41 and 43), while others are far apart? We will soon return to these questions one by one. Some have answers; others still do not.

13.3 Inventing a Concept: Primes Are the “Atoms” of Integers

The numbers that survive the sieve deserve an official name.

Definition 13.1 (Prime and composite numbers). An integer greater than 1 with no positive divisors other than 1 and itself is called a *prime number* (or simply a prime); otherwise it is called a *composite number*. The number 1 is neither prime nor composite.

The question most likely to catch us out is: “Why exclude 1 from the primes? Isn’t 1 divisible only by 1 and itself?” This is a genuinely good question. The answer lies not in the definition itself, but in a theorem to come: mathematicians want every integer to have a unique factorisation. If 1 were prime, then 6 could be written as 2×3 , $1 \times 2 \times 3$, $1 \times 1 \times 2 \times 3$, and so on, destroying uniqueness. Excluding 1 gives the multiplicative world of integers a clean fundamental law. Definitions are not handed down from heaven; they are carefully designed to make theorems elegant.

Terminology Card: Prime number

Intuition: An unbreakable “atom” in the integer world, made only from 1 and itself, not from a product of smaller integers.

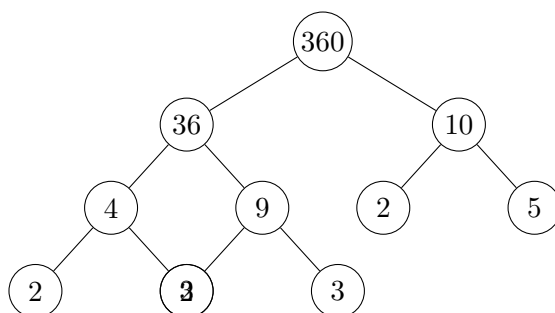
Formal definition: An integer greater than 1 whose only positive divisors are 1 and itself.

Example: 47 is prime because none of 2, 3, 5 divides it, and $7 > \sqrt{47}$, so testing can stop there.

Chemists discovered that matter consists of just over a hundred kinds of atoms; number theorists discovered that every integer greater than 1 is “built” from primes. $48 = 2 \times 2 \times 2 \times 2 \times 3$, $45 = 3 \times 3 \times 5$, while 47 is itself an atom. More remarkably, there is only one way to build each number.

Theorem 13.2 (Fundamental theorem of arithmetic). *Every integer greater than 1 can be uniquely factored into a product of primes: apart from the order in which the primes are written, the factorisation is unique.*

First consider existence, which is almost as natural as a tree growing. Take any composite number and split it into two smaller numbers; if either is still composite, keep splitting. Each split strictly decreases the numbers, and integers cannot decrease forever, so the process must stop. The numbers at its ends cannot be split any further: they are primes. The diagram shows a factor tree for 360.



Multiply all the “leaves”: $360 = 2^3 \times 3^2 \times 5$. Take another route (perhaps starting with 12×30), and the leaves are the same primes, merely encountered in a different order.

Uniqueness takes more thought. Why cannot two completely different collections of primes have exactly the same product? The key is the following lemma:

Corollary 13.3 (Prime divisor lemma). *If a prime p divides a product ab , then p divides a or b .*

This is false for composite numbers (6 divides $3 \times 4 = 12$, but 6 divides neither 3 nor 4). Its proof uses the greatest common divisor from the next section, so we leave the proof idea there for now. With the lemma, uniqueness is straightforward. Suppose a number has two prime factorisations and write them on opposite sides of an equality. The first prime on the left divides the product on the right; by the lemma, it divides one of the primes on the right. Since a prime is divisible only by itself and 1, these must be the same prime. Cancel it from both sides and repeat for what remains. Eventually, the primes on both sides must match exactly. Our intuition that “there is only one factorisation” is correct, and now rigorously proved.

13.4 Repeated Division: A Two-Thousand-Year-Old Machine

Now we supply the tool that the lemma was waiting for. Given two integers, say 252 and 105, what is the largest factor they share? This is called their greatest common divisor. The cumbersome method lists all divisors of each number and finds the intersection. But an astonishingly fast old method, recorded in ancient Greek mathematics books over two thousand years ago, still runs inside your phone's chips today.

Definition 13.4 (Greatest common divisor). The largest common divisor of integers a and b (not both 0) is denoted by $\gcd(a, b)$. If $\gcd(a, b) = 1$, then a and b are called *coprime*.

One-Minute Experiment

Take a sheet of paper and calculate $\gcd(252, 105)$. First, divide 252 by 105: quotient 2, remainder 42. Write $252 = 2 \times 105 + 42$. Next, divide the previous divisor 105 by the previous remainder 42: $105 = 2 \times 42 + 21$. Finally, divide 42 by 21: $42 = 2 \times 21 + 0$. The remainder is 0, so stop. The last nonzero remainder, 21, is the answer: $\gcd(252, 105) = 21$. Just three divisions, without needing to know any factors of 252 or 105.

This is the Euclidean algorithm, or repeated division. Why does it work? Its core is one statement: if $a = bq + r$, then the common divisors of a and b are exactly the common divisors of b and r . Any common divisor of b and r must divide $bq + r = a$; conversely, any common divisor of a and b must divide $a - bq = r$. The collection of common divisors is unchanged, so its largest member is unchanged too:

$$\gcd(a, b) = \gcd(b, r).$$

The numbers on the right get smaller. The remainders decrease but cannot fall below 0, so after finitely many steps the remainder must be 0, and the smaller number at that stage is the answer. No prime factorisation is needed. This is precisely its strength, because factoring large integers is difficult (a fact that will later come to cryptography's rescue).

The algorithm has a beautiful geometric version. Tile a 252×105 rectangle with the largest squares that fit. First place two 105×105 squares, leaving a 42×105 strip. Tile that strip with 42×42 squares: two fit, leaving 42×21 . Finally, 21×21 squares fill the remainder exactly. The final squares fit without any leftover edges, and their side length is the greatest common divisor. The uncovered portion shrinks just as the remainders shrink.

There is a by-product: reading the divisions backwards expresses the greatest common divisor as an “integer linear combination” of the original numbers. In our example:

$$21 = 105 - 2 \times 42 = 105 - 2 \times (252 - 2 \times 105) = 5 \times 105 - 2 \times 252.$$

In general, there are always integers x, y such that $\gcd(a, b) = ax + by$. Do not underestimate this line: the previous section's prime divisor lemma depends on it. Suppose a prime p divides ab but not a . Then $\gcd(p, a) = 1$ (the only positive divisors of p are 1 and p), so integers x, y satisfy $px + ay = 1$. Multiplying by b gives $pbx + aby = b$. The first term on the left is clearly divisible

by p , and by assumption ab in the second term is also divisible by p . Therefore b is divisible by p . The lemma is proved, tightening the final screw in the fundamental theorem of arithmetic.

The greatest common divisor also has an everyday use: reducing fractions. To put $\frac{252}{105}$ in lowest terms, divide numerator and denominator by $\gcd(252, 105) = 21$, obtaining $\frac{12}{5}$ in one step, without repeated trial cancellations. Its partner is the least common multiple: for two integers, the least common multiple times the greatest common divisor equals their product, $\text{lcm}(a, b) \cdot \gcd(a, b) = ab$. Thus the least common multiple of 252 and 105 is $252 \times 105 \div 21 = 1260$. Here is where it appears: two neon lights flash every 252 seconds and every 105 seconds respectively. After flashing together, they next coincide exactly 1260 seconds later. All such “meeting again” problems—gears realigning, planets lining up again, or two buses arriving together—have this same number behind them.

13.5 Parity, Invariants, and Congruence: Another View of Integers

Factorisation concerns multiplicative structure. Now change perspective: instead of asking which number something is, ask which class it belongs to. The simplest classification is odd or even: remainder 1 or 0 on division by 2. Do not dismiss this as too simple; it is a powerful weapon, formally called an *invariant*.

A classic scene: remove two diagonally opposite squares from an 8×8 chessboard. Can 31 dominoes of size 1×2 tile what remains? You could spend an afternoon trying, always nearly succeeding. A mathematician does not try placements, but looks at colours. Every domino, horizontal or vertical, covers one black and one white square, so a tileable region must have equal numbers of each. The two removed corners have the same colour, leaving two more black squares than white, or vice versa. 31 dominoes cover 31 black and 31 white squares, never 30 and 32. Without a single attempt, the answer is “impossible”. Parity remains unchanged during tiling; that invariance settles the matter.

Parity is the remainder on division by 2. Extending this idea to division by any number gives one of number theory’s most useful tools.

Definition 13.5 (Congruence). Let m be a positive integer. If integers a and b leave the same remainder when divided by m (equivalently, if m divides $a - b$), we say that a and b are *congruent modulo m* , written

$$a \equiv b \pmod{m}.$$

Terminology Card: Congruence

Intuition: Arithmetic on a clock face. It is 3 in the afternoon; what time will it be in 14 hours? $3 + 14 = 17$ and $17 \equiv 5 \pmod{12}$, so it will be 5 in the morning. On the clock face, 17 and 5 occupy the same position.

Formal definition: $a \equiv b \pmod{m}$ if and only if m divides $a - b$.

Example: $100 \equiv 2 \pmod{7}$, so the weekday 100 days from today depends only on 2: count two days ahead.

The beauty of congruence is that it behaves much like equality: we can add, subtract, or multiply both sides by the same integer. When dealing with enormous numbers, we can therefore replace them whenever convenient by smaller stand-ins modulo m . Want the last digit of 7^{100} ? The last digit is the remainder modulo 10. Since $7^2 = 49 \equiv 9 \pmod{10}$ and $7^4 \equiv 9^2 = 81 \equiv 1 \pmod{10}$,

$$7^{100} = (7^4)^{25} \equiv 1^{25} = 1 \pmod{10}.$$

The final digit is 1. We never calculated the eighty-odd-digit giant 7^{100} , yet we know its last digit. That is the power of looking only at classes.

Congruence also explains familiar rules whose reasons you may never have considered. For example: “A number is divisible by 3 if and only if its digit sum is divisible by 3.” Why do the digits cooperate? The secret is $10 \equiv 1 \pmod{3}$ (and likewise $10 \equiv 1 \pmod{9}$). Every power of 10 is therefore 1 modulo 3. Expanding a four-digit number gives:

$$\overline{abcd} = 1000a + 100b + 10c + d \equiv a + b + c + d \pmod{3}.$$

In other words, decimal notation itself puts a number and its digit sum in the same class. Try 123 456: its digit sum is $1 + 2 + 3 + 4 + 5 + 6 = 21$, a multiple of 3 but not 9, so 123 456 is divisible by 3 but not 9, without performing division. Old-fashioned accountants also used “casting out nines” to check calculations. To check the plausibility of $47 \times 62 = 2914$, repeatedly add the digits of 47, 62, and 2914 until each becomes a single digit ($47 \rightarrow 11 \rightarrow 2$, $62 \rightarrow 8$, $2914 \rightarrow 16 \rightarrow 7$), then see whether $2 \times 8 = 16 \rightarrow 7$ matches the product’s digital root. A match does not guarantee correctness, but a mismatch guarantees an error. This ancient “error-correcting code” rests on congruence modulo 9.

Developing congruence further brings us to a small gem discovered by the seventeenth-century French mathematician Fermat, famous for jotting remarkable assertions in book margins:

Theorem 13.6 (Fermat’s little theorem). *Let p be prime and let the integer a not be a multiple of p . Then*

$$a^{p-1} \equiv 1 \pmod{p}.$$

Proof. Consider the remainders of the $p-1$ numbers $a, 2a, 3a, \dots, (p-1)a$ on division by p . First, no two are congruent: if $ia \equiv ja \pmod{p}$ (say $i > j$), then p divides $(i-j)a$. By the prime divisor lemma, p divides $i-j$ or a , but a is not a multiple of p , and $0 < i-j < p$, so neither is possible. These $p-1$ remainders are distinct and none is 0 (otherwise p would divide some ia , giving the

same contradiction). They are therefore a rearrangement of $1, 2, \dots, p-1$. Multiply all $p-1$ numbers. On the one hand,

$$a \cdot 2a \cdots (p-1)a = a^{p-1} \cdot (p-1)!,$$

and on the other, modulo p , the product of their remainders is $1 \cdot 2 \cdots (p-1) = (p-1)!$. The two sides are congruent:

$$a^{p-1} \cdot (p-1)! \equiv (p-1)! \pmod{p}.$$

Since $(p-1)!$ is coprime to p , we can cancel it to obtain $a^{p-1} \equiv 1 \pmod{p}$. $\square$

For a numerical example, take $p = 11$ and $a = 2$. Since $2^5 = 32 \equiv -1 \pmod{11}$, we have $2^{10} \equiv (-1)^2 = 1 \pmod{11}$. The theorem's prediction is exact. What looks like a pure game became a foundation of banking three centuries later. The next section reveals how.

Note. The converse is false: some composite numbers n satisfy $a^{n-1} \equiv 1 \pmod{n}$ for certain a . For example, $341 = 11 \times 31$ “fools” the test with base 2. Such numbers are called pseudoprimes. Fermat's little theorem can therefore declare “not prime” with certainty when the congruence fails, but can only say “probably prime” probabilistically. Modern computer tests for large primes grew into a whole family of techniques through this opening.

13.6 Primes Never Run Out: A Proof Over Two Thousand Years Old

Return to our opening question: do Class C's “stone numbers” eventually disappear? When factoring enormous integers, might we ultimately draw only on a finite stock of primes? The ancient Greeks answered this with what is widely regarded as one of the most beautiful proofs in mathematics.

Theorem 13.7 (There are infinitely many primes). *There is no “largest prime”: given any prime, a larger prime can always be found.*

Proof. Argue by contradiction. Suppose there are only finitely many primes and list them all: $p_1, p_2, \dots, p_n$. Construct a new integer

$$N = p_1 p_2 \cdots p_n + 1,$$

by multiplying “all the primes” and adding 1. Since N is greater than 1, the fundamental theorem of arithmetic says that some prime divides N . Which prime? Not p_1 , because division of N by p_1 leaves remainder 1. Nor any of $p_2, p_3, \dots, p_n$, since division of N by each leaves remainder 1. A prime outside the list must therefore divide N , contradicting the claim that the list contains every prime. The assumption was false: there cannot be only finitely many primes. $\square$

Notice the proof's structure: it does not tell us where the next prime lies, only that the list can never end. Proof by contradiction—assuming the opposite and deriving an inconsistency—is

one of the sharpest tools in a mathematician's toolbox. Also notice that N itself need not be prime. For example, $2 \times 3 \times 5 \times 7 \times 11 \times 13 + 1 = 30031$, which equals 59×509 and is composite. But it is indeed divisible by new primes 59 and 509 outside the list. That is all the proof needs, not the primality of N itself.

Incidentally, although primes are infinite, their distribution becomes sparser and is highly irregular. Between 1 and 100 there are 25 primes, a quarter of all numbers; near 1000, only about one and a half in every ten numbers are prime. Larger numbers are less likely to be prime. This fits our intuition: larger numbers have more potential factors, so remaining unbreakable is harder. Mathematicians can even produce a “prime desert” of a thousand consecutive integers: $1001! + 2, 1001! + 3, \dots, 1001! + 1001$ are divisible by $2, 3, \dots, 1001$ respectively, giving a thousand consecutive composite numbers.

On the other hand, primes refuse to drift entirely apart. “Twin primes” such as 3 and 5, 17 and 19, or 101 and 103 differ by just 2; the largest known pair has hundreds of thousands of digits. Are there infinitely many such pairs? This famous twin prime *conjecture* has neither been proved nor disproved. What has been proved is an approximate version: infinitely many prime pairs differ by no more than some fixed bound. Sparser and more irregular, yet apparently never entirely isolated: the distribution of primes is one of mathematics' great unsolved mysteries, which we leave for “Further Climbing”.

13.7 Another Perspective: How These Mechanisms Secure the Modern World

Number theory was once called the “queen of mathematics”, proudly pure and useless. In the second half of the twentieth century, the situation reversed dramatically: today, every on-line purchase, encrypted message, and public-transport card transaction calls on this chapter's mechanisms.

First mechanism: error-correcting codes. The International Standard Book Number (ISBN) on a book's back cover is a string of digits whose last digit is a “check digit”, calculated from the others using modular arithmetic. In one classic scheme, the first nine digits $a_1, a_2, \dots, a_9$ receive weights $10, 9, \dots, 2$. Choose a check digit x so that the weighted sum is divisible by 11:

$$10a_1 + 9a_2 + \dots + 2a_9 + x \equiv 0 \pmod{11}.$$

Suppose the first nine digits are 9, 7, 8, 7, 0, 4, 0, 0, 3. Their weighted sum is

$$90 + 63 + 64 + 49 + 0 + 20 + 0 + 0 + 6 = 292,$$

and $292 = 11 \times 26 + 6$, so take $x \equiv -6 \equiv 5 \pmod{11}$: the check digit is 5. If a digit is mistyped, the weighted sum is no longer divisible by 11, and the system immediately flags an error. Even swapping two digits can be detected. Why modulo 11, not 10? The trick is that 11 is prime. Swapping adjacent digits changes the weighted sum by the difference of their weights times the difference of their digits, namely $\pm(a_i - a_{i+1})$. If the digits differ, this change contains a factor

between ± 1 and ± 9 ; the prime 11 cannot divide such a small integer, so the error must be exposed. Modulo 10, swaps of digits differing by 5 would escape detection. Bank-card numbers, parcel-tracking numbers, and QR codes use the same idea: deliberately store extra information predictable by a rule, leaving errors nowhere to hide. QR codes go further, storing blocks of redundancy rather than a single check digit. A phone can therefore read a code with a stained corner or even a small logo printed on it. That is a stronger error-correcting code, still built on the arithmetic of congruences.

Second mechanism: public-key cryptography. Traditional encryption resembles a padlock that uses the same key to lock and unlock, making the transfer of the key itself a vulnerability. In the 1970s, mathematicians proposed a seemingly outlandish scheme: give everyone the lock, but keep the key. Anyone can lock a box and send it to you, yet only you can open it. Primes, congruences, and Fermat's little theorem make this possible. Here is a simplified model (real systems use numbers hundreds of digits long, with the same principle):

1. Secretly choose two primes $p = 3$ and $q = 11$. Publish their product $N = 33$, but keep p and q secret.
2. The ideas behind Fermat's little theorem show that suitable "public" and "private" exponents, $e = 3$ and $d = 7$, satisfy $(m^e)^d \equiv m \pmod{N}$ for every message m . They obey $ed \equiv 1 \pmod{(p-1)(q-1)}$, namely $21 \equiv 1 \pmod{20}$.
3. To send the number $m = 4$, a friend encrypts with the public (N, e) : $4^3 = 64 \equiv 31 \pmod{33}$, and sends 31.
4. On receiving 31, decrypt with the secret d : compute $31^7 \pmod{33}$. Check by hand: $31 \equiv -2 \pmod{33}$, $(-2)^7 = -128$, and $-128 \equiv 4 \pmod{33}$. The original message 4 returns unchanged.

An eavesdropper sees $N = 33$, $e = 3$, and the ciphertext 31. Recovering d essentially requires knowing $(p-1)(q-1)$, hence factoring N back into 3×11 . Anyone can factor 33, of course, but replace N with the product of two three-hundred-digit primes, and even humanity's fastest computer could not finish within the age of the universe. Encryption is easy; factorisation is hard. Security rests not on keeping the algorithm secret, but on the difficulty of factoring large integers. Fermat's once "utterly useless" little theorem now protects astronomical flows of money every day.

Connections

This chapter traces a surprising chain of consequences: Euclid's repeated division, over two thousand years old, supports uniqueness in the fundamental theorem of arithmetic; Fermat's small seventeenth-century theorem, combined with the still-unbroken practical difficulty of factoring large integers, becomes a lock for the modern internet. In mathematics, the journey from "useless" to "useful" can take centuries. Judging a theory's value calls for that kind of patience.

Mathematical Toolbox

The number-theoretic tools added in this chapter:

- **Square-root bound:** To test whether n is prime, trial division need only reach $\sqrt{n}$.
- **Prime factorisation:** Every integer has a unique expression as a product of prime powers (the fundamental theorem of arithmetic).
- **Euclidean algorithm:** $\gcd(a, b) = \gcd(b, r)$, where $a = bq + r$; working backwards also gives $\gcd(a, b) = ax + by$.
- **Invariants:** Find a quantity unchanged during a process (such as parity or the difference between black and white square counts), and use it to prove impossibility.
- **Congruence:** $a \equiv b \pmod{m}$; replace large numbers by small stand-ins whenever convenient. Addition, subtraction, and multiplication preserve congruence.
- **Proof by contradiction:** Assume the conclusion false and derive a contradiction. The infinitude-of-primes proof is a model example.

Keep Climbing

To climb further, look up these keywords: **prime number theorem** (there are about $x/\ln x$ primes no greater than x : infinitely many, but increasingly sparse); **Goldbach's conjecture** (every even number greater than 2 is a sum of two primes, still unproved); **Riemann hypothesis** (the ultimate code behind prime distribution, with a million-dollar prize); **quadratic residues and reciprocity**; **Chinese remainder theorem** (the complete theory behind “Han Xin counting his soldiers”); **elliptic-curve cryptography** (a new generation of public-key systems after RSA).

Questions to Think About

Observation

1. List all twin prime pairs between 2 and 50 (prime pairs differing by 2). How many pairs are there? Hint: first sieve all primes up to 50, then find neighbouring primes whose difference is 2. Look for pairs such as (3, 5), (5, 7), and (11, 13).
2. Examine the values of $n^2 + n + 41$ for $n = 0, 1, 2, 3, 4$. They seem to be prime. Hint: do not rush to a conclusion; try some larger values of n , especially one “obviously special” value.

Proof

1. Prove that every prime other than 2, 3 has the form $6k + 1$ or $6k + 5$, where k is an integer. Hint: an integer has only six possible remainders on division by 6. Check why $6k$, $6k + 2$, $6k + 3$, and $6k + 4$ are composite.

2. Prove that if $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then $ac \equiv bd \pmod{m}$. Hint: start from the definition, substitute $a = b + sm$ and $c = d + tm$, expand the product, and identify terms with a factor of m .

Challenge

1. Find the remainder when 2^{2025} is divided by 7. Hint: calculate the remainders of $2^1, 2^2, 2^3, \dots$ modulo 7 and look for a cycle (or use Fermat's little theorem, since 7 is prime).
2. Our public-key example used $p = 3$, $q = 11$, $e = 3$, and $d = 7$. Build your own system with $p = 5$ and $q = 7$: calculate N and $(p - 1)(q - 1)$, choose e coprime to $(p - 1)(q - 1)$, and find d satisfying $ed \equiv 1 \pmod{(p - 1)(q - 1)}$ by reversing the Euclidean algorithm. Finally, encrypt and decrypt a message of your choice to verify recovery. Hint: take e to be 5; finding d means solving $5d \equiv 1 \pmod{24}$.

The Door to the Next Chapter

Throughout this chapter, we repeatedly did one simple thing: *count*. The sieve counts primes; the chessboard problem counts black and white squares; the proof of Fermat's little theorem "counts twice" the same $p - 1$ numbers, first multiplying the original numbers, then multiplying their rearranged remainders. Two counts give the same result, and the theorem emerges from the equality. This last technique deserves particular attention: counting the same objects differently can reveal a thoroughly non-obvious conclusion.

Yet once we take counting seriously, questions multiply. The prime number theorem says there are about $x/\ln x$ primes: how is that approximation counted? How many shuffles of a deck of cards are possible? In how many ways can 8 non-attacking queens be placed on an 8×8 chessboard? How many Morse-code signals consist of five dots and dashes? These objects are no longer isolated integers, but collections of arrangements, selections, and placements: discrete things, counted one by one. How can we count them systematically without duplication or omission? This is the subject of the next chapter: combinatorics and discrete structures. We will see that counting accurately is itself a discipline with theorems, techniques, and traps, and that this chapter's integer mechanisms—parity, congruence, and coprimality—will appear again and again.

Chapter 14 Combinatorics and Discrete Structures: Counting, Connections, and Algorithms

14.1 A Counting Puzzle Before a Class Party

Class 3 in the second year of middle school is organising a party, and class representative Xiaolin has two jobs. First, all 25 pupils will stand in a row on stage for a photograph. Their teacher wants to know how many different arrangements are possible before deciding whether the programme should say “line up randomly”. Second, the party ends with a traditional activity: every pupil shakes hands with every other pupil exactly once. To keep to schedule, the host needs to know the total number of handshakes.

Xiaolin thinks the first job is easy: surely 25 people can line up in a few dozen or a few hundred ways? For the second, he estimates that each person shakes 24 hands, so 25 people give $25 \times 24 = 600$ handshakes.

These apparently tiny problems open every door in this chapter. The first answer is startling: not dozens or hundreds, but a 26-digit number. The intuitive calculation 25×24 for the second hides a mistake: it counts every handshake twice. Counting, something we have done since nursery school, becomes a branch of mathematics requiring methods and theorems as soon as the scale or structure grows. This branch is *combinatorics*. Together with its close relatives *graph theory* and *algorithms*, it forms “discrete mathematics”: mathematics about objects that come one by one and can be clearly distinguished.

Why give the discrete world its own branch? Most of this book so far has occupied a continuous stage: points on the real line, smooth curves, continuously varying speeds. But much of life is not continuous: pupils come individually, handshakes happen one at a time, road networks link junction to junction, passwords consist of separate characters, and computer data consists of strings of 0s and 1s. These countable objects need another toolkit. This chapter puts that toolkit in your hands: counting without omissions or repetition, drawing relationships as points and lines, and getting a machine that only follows instructions to navigate those points and lines efficiently.

14.2 Make a Guess, Then Another Mistake

Return to the handshakes. Xiaolin’s calculation is: each person shakes 24 hands, so 25 people make $25 \times 24 = 600$ handshakes. What does your intuition say: is this too high or too low?

Test a smaller class. Suppose it has only 3 people, A, B, and C. Xiaolin’s method gives $3 \times 2 = 6$ handshakes. But count them: AB, AC, BC—only 3. The answer 6 is exactly twice 3. With 4 people, the pairs AB, AC, AD, BC, BD, CD give 6 handshakes, while Xiaolin’s calculation gives $4 \times 3 = 12$, again twice as many.

What went wrong? “A shakes B’s hand” and “B shakes A’s hand” are the same event, but counting 24 handshakes per person records it once for A and once for B. Every handshake has two hands, doubling the ledger. Divide 600 by 2 to obtain the correct answer, 300.

Counterexample Alert

A plausible claim: If n people all shake hands with one another, each shakes hands $n - 1$ times, so the total is $n(n - 1)$.

The mistake: Every handshake belongs to two people, so counting person by person necessarily counts it twice. Whenever we count ways for something to happen, first ask: could different entries in my count actually record the same event? Confusing ordered and unordered choices, and overlooking repeated counting, are among combinatorics’ commonest pitfalls. The same mistake occurs in a round-robin tournament with 20 teams, where each pair plays once: there are not $20 \times 19 = 380$ matches, but $380 \div 2 = 190$.

Now consider lining up. How many arrangements can 25 people make in a row? Guessing “a few hundred” falls into another trap: underestimating multiplication. Work step by step. There are 25 choices for the first position; once that choice is made, 24 remain for the second; then 23, 22, and so on, until only 1 remains for the last. The choices combine in sequence, giving

$$25 \times 24 \times 23 \times \cdots \times 2 \times 1.$$

This product is large enough to deserve its own notation: $25!$, read “25 factorial”. Its value is 15511210043330985984000000, a 26-digit number. Trying one arrangement per second would take hundreds of billions of times the age of the universe to exhaust them all. Factorial growth will play a central role when we ask later whether an algorithm can actually finish.

14.3 Two Counting Rules and Two Weapons

These two failures suggest that counting needs structure, not brute-force lists. Combinatorics has just two basic rules.

Addition principle: If an activity can be done in disjoint categories of ways, with a ways in the first category and b in the second, there are $a + b$ ways altogether. **Multiplication principle:** If an activity has successive steps, with a ways to do the first and b ways to do the second, there are $a \times b$ ways altogether. Lining up uses multiplication because the 25 positions occur in sequence.

Choosing either one of 3 buses to school or one of 2 cycling routes uses addition because bus and bicycle are alternative categories.

Beneath these rules, the two most frequently used weapons are *permutations* and *combinations*.

Select k people from n and arrange them in a row, keeping track of order. The number of arrangements is denoted by

$$A_n^k = n(n-1)(n-2) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

For example, assigning first, second, and third places among 8 pupils gives $A_8^3 = 8 \times 7 \times 6 = 336$ possible outcomes. If we ignore ranks and simply choose 3 people for an organising committee, each three-person group has been counted $3! = 6$ times among the permutations (its members can line up in 6 ways), so we divide by 6. The resulting number is written

$$\binom{n}{k} = \frac{n!}{k!(n-k)!},$$

read “ n choose k ”, and called a *combination number* or *binomial coefficient*. There are $\binom{8}{3} = 56$ committees of 3 from 8 people. It solves the handshake problem in one line: among 25 people, there are $\binom{25}{2} = \frac{25 \times 24}{2} = 300$ handshakes.

Terminology Card: Binomial coefficient

Intuition: How many ways can we take k objects from n without caring about their order?

Formal definition: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$, where $0 \leq k \leq n$ and, by convention, $0! = 1$.

Example: $\binom{5}{2} = \frac{5 \times 4}{2} = 10$: five basketball teams playing each other once produce 10 matches.

Binomial coefficients have a famous “map”. Arrange them in rows, with 1 at both ends of each row and each interior entry equal to the sum of the two above its shoulders:

$$\begin{array}{ccccccc} & & & & 1 & & & \\ & & & & & & 1 & \\ & & 1 & & 2 & & 1 & \\ & 1 & & 3 & & 3 & & 1 \\ & & 1 & & 4 & & 6 & \cdots \end{array}$$

Row 4, counting from row 0, is 1, 4, 6, 4, 1: exactly $\binom{4}{0}, \binom{4}{1}, \binom{4}{2}, \binom{4}{3}, \binom{4}{4}$. Known in China as Yang Hui’s triangle and in the West as Pascal’s triangle, this array was independently rediscovered many times. Its construction rule is itself a counting argument. Choosing k objects from n , either a particular object is included (choose the remaining $k-1$ from $n-1$) or it is excluded (choose all k from $n-1$). Thus

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Notice the flavour of this identity: a smaller problem of the same kind answers the current problem. This idea of calling on a smaller version of itself is a *recurrence*, our third weapon.

The classic recurrence example is climbing stairs. A staircase has n steps, and each move climbs either 1 or 2. How many routes are possible? Let their number be f_n . The final move has only two possibilities: one step from stair $n - 1$, or two steps from stair $n - 2$. Therefore

$$f_n = f_{n-1} + f_{n-2}, \quad f_1 = 1, f_2 = 2.$$

We obtain $f_3 = 3$, $f_4 = 5$, $f_5 = 8$, $f_6 = 13$, and so on: the famous Fibonacci sequence. Notice the resemblance to Pascal's triangle. A daunting problem apparently requiring every route to be listed splits into two smaller problems simply by asking where the last step came from. This decomposition will return in our discussion of recursive algorithms.

Why are combination numbers also called binomial coefficients? Expand $(a + b)^n$. For $(a + b)^2 = a^2 + 2ab + b^2$, the coefficients are 1, 2, 1; for $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$, they are 1, 3, 3, 1—rows of Pascal's triangle. This is no coincidence: in expanding $(a + b)^n$, a term $a^k b^{n-k}$ comes from choosing k of the n factors to contribute a , with the rest contributing b . There are exactly $\binom{n}{k}$ such choices. We obtain a central result.

Theorem 14.1 (Binomial theorem). *For any numbers a, b and nonnegative integer n ,*

$$(a + b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{n}b^n.$$

Proof. Write $(a + b)^n$ as a product of n factors $(a + b)$. In the expansion, each factor contributes either a or b . To obtain $a^k b^{n-k}$, exactly k of the n factors must contribute a . There are $\binom{n}{k}$ such choices, so the term's coefficient is $\binom{n}{k}$. Adding all terms for $k = 0, 1, \dots, n$ gives the formula. $\square$

Set $a = b = 1$ for an immediate numerical example: $\binom{4}{0} + \binom{4}{1} + \binom{4}{2} + \binom{4}{3} + \binom{4}{4} = 2^4 = 16$. A group of 4 people therefore has 16 possible “subset delegations”, from sending nobody to sending all four. Check directly: $\binom{4}{0} = 1$, $\binom{4}{1} = 4$, $\binom{4}{2} = 6$, $\binom{4}{3} = 4$, and $\binom{4}{4} = 1$ indeed sum to 16.

Our final counting weapon is so simple that it hardly sounds like a theorem, yet it is extraordinarily powerful.

Theorem 14.2 (Pigeonhole principle). *If $n + 1$ pigeons are placed in n pigeonholes, at least one hole contains two or more pigeons.*

Proof. Suppose instead that every hole contains at most one pigeon. Then n holes contain at most n pigeons, contradicting the total of $n + 1$. $\square$

This one-sentence theorem solves problems that seem to offer no foothold. Among 400 pupils, two must share a birthday: there are at most 366 possible birthdays, so pupils outnumber holes. Another example: among any 6 people, there are either 3 mutual acquaintances or 3 mutual strangers. Use the pigeonhole principle: fix person A. A's relationships with the other 5 people have only two categories, acquaintance or stranger. Five pigeons in two holes put at least 3 in one category. Then examine the relationships among those three (details are left as an exercise).

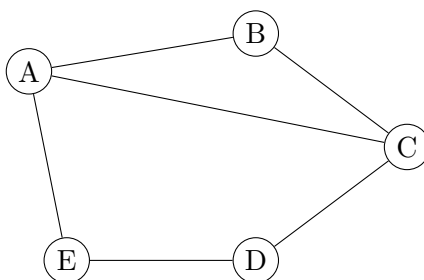
The deeper meaning is that *when objects outnumber categories, collisions are unavoidable*. This simple idea is the prototype of hash-collision arguments in modern cryptography and pigeonhole arguments in computer science.

14.4 Draw Each Team as a Point: Graphs, Trees, and Paths

Look at the round-robin tournament differently. With 20 teams playing in pairs, our concern is not who wins but who is connected to whom. Draw each team as a point and each match as a line between two points: the schedule becomes a “point-and-line diagram”. Such structures are everywhere: cities and flights, people and friendships, web pages and hyperlinks, junctions and streets, chip components and wires. Mathematicians abstract them into a concept.

Definition 14.3 (Graph). A *graph* consists of a collection of *vertices* and a collection of *edges*, each joining two vertices. If edges have directions, as with one-way streets, it is a directed graph; if edges carry numbers, such as distances or costs, it is a weighted graph. The number of edges incident to a vertex is its *degree*.

Notice how the definition deliberately discards geometry: the points have no positions, and the lines have no lengths or curvature. The only information is *which vertices are connected*. This is purposeful forgetting: discard what does not matter to see the structure itself.



In this small graph, vertex A has degree 3, being joined to B, C, and E, while vertex E has degree 2. Every graph obeys the following law.

Theorem 14.4 (Handshaking lemma). *In any graph, the sum of all vertex degrees is twice the number of edges.*

Proof. Count edge by edge. Every edge has two endpoints and contributes 1 to the degree of each. The degree sum is therefore exactly twice the edge count. $\square$

An immediate corollary is that *every graph has an even number of odd-degree vertices*. The degree sum is even, and odd summands must occur in pairs to give an even sum. In party language, the number of guests who shake an odd number of hands is always even. Quietly check at your next party: this is a theorem, not a coincidence.

One especially clean class of graphs is *trees*: connected graphs, with a path between every pair of vertices, containing no cycles. Like a real tree, a tree offers no route that loops back to its starting point. A basic fact is that a tree with n vertices has exactly $n - 1$ edges. Every edge is

indispensable: deleting any one disconnects the graph. Family trees, computer folder hierarchies, and knockout-tournament brackets are trees. One important use is connecting everyone with as few edges as possible. Connecting 6 villages needs at least 5 roads; to minimise total construction cost, select the cheapest tree from the candidate edges. This is the minimum spanning tree problem, which admits simple, efficient greedy algorithms.

A more immediately practical problem is the *shortest path*. Given a weighted graph with junctions as vertices, streets as edges, and lengths as weights, which route from home to school has minimum total length? You might suggest listing all routes and choosing the shortest, but as junctions multiply, the number of routes can explode like a factorial. In the 1950s, computer scientist Dijkstra devised a greedy algorithm: start at the origin, always process the vertex with the smallest currently known distance, and use it to update its neighbours' distance records. The search spreads out like ripples until reaching the destination. Each step makes only local comparisons, yet finds the globally shortest route, provided all edge lengths are nonnegative. Modern navigation software still uses enhanced versions of this central idea.

Replace lengths with capacities and we enter another world: *network flow*. Imagine the graph as water pipes, each with a maximum flow rate determined by its size. What is the most water that can reach your house from the waterworks per unit time? Intuitively, bottlenecks limit the answer. Draw a line on the map separating the waterworks from your house; the total capacity of pipes crossing it bounds how much can pass through. The tightest of all such cuts determines the true limit. This is the celebrated *max-flow min-cut theorem*: a network's maximum flow equals the capacity of its minimum cut. We will not prove it, but savour its form: a maximisation problem has exactly the same answer as a minimisation problem. Such "maximum equals minimum" duality recurs throughout mathematics, each time revealing a deep structural symmetry. Parcel sorting, power-grid scheduling, and even tournament scheduling all bear traces of network flow.

14.5 Algorithms: Giving Machines a Route to Follow

So far, we have solved problems with our own hands and minds. Real graphs, however, can have millions of vertices: a country's roads, or the links of the entire internet. We then need an *algorithm*: a precise, mechanical list of steps that anyone, or any machine, can follow to obtain the answer. Algorithms are the meeting point of discrete mathematics and computer science.

There are two basic approaches to algorithm design, both already glimpsed in this chapter. The first is *recursion*: break a problem into smaller problems of the same kind until they are small enough to answer directly. The stair-counting formula $f_n = f_{n-1} + f_{n-2}$ is recursive, as is traversing computer folders: "list this folder's contents" means "repeat the listing procedure for every subfolder". The second is *search*: organise all possibilities into a decision tree and walk through it systematically. Solving a maze is a search: junctions are vertices, corridors are edges, and finding the exit means finding a path in this graph.

Search has two personalities. *Depth-first search* is a single-minded explorer: follow one route to its end, then backtrack at a wall. *Breadth-first search* spreads like ripples: first inspect

everything reachable in one step, then in two. For shortest paths, this guarantees that the first exit found is nearest. Dijkstra’s algorithm is essentially an upgraded breadth-first search for weighted graphs.

But “able to compute an answer” and “able to afford the computation” are different things. This calls for our final tool: an intuitive account of *algorithmic complexity*.

Terminology Card: Big O notation

Intuition: Focus on how an algorithm’s workload grows as the problem size n increases, ignoring constant factors and minor details.

Formal description: If the workload is bounded by a fixed multiple of n , write $O(n)$; if bounded by a fixed multiple of n^2 , write $O(n^2)$; and so on.

Example: Finding the largest of n numbers takes $O(n)$ work, since each number need only be inspected once; listing all line-ups of n people is $O(n!)$, hopelessly slow.

Revisit the chapter through big O notation. Dijkstra’s algorithm on a graph of n vertices takes roughly $O(n^2)$ work: ten times as many junctions means about a hundred times as long, which is manageable. Checking every pairwise relationship among n people is also $O(n^2)$, again manageable. But $O(2^n)$ or $O(n!)$ tells a different story: 2^{40} already exceeds a trillion, and $25!$ is astronomical. A problem size of only a few dozen can keep a machine running indefinitely. Famous hard problems include the travelling salesman problem: given cities, find the shortest tour visiting every city and returning to the start. Nobody yet knows whether a general algorithm can be fundamentally much faster than near-enumeration. This is one of computer science’s greatest mysteries, usually written as the “P versus NP problem”, with a million-dollar prize awaiting an answer.

Discrete mathematics therefore leaves us with a two-sided lesson. Graphs, recurrences, and greedy methods bring seemingly impossible large problems within computational reach. Complexity theory, meanwhile, marks the boundaries unsentimentally: some problems may be not merely beyond today’s resources, but inherently intractable. Knowing which is which is itself deep mathematics.

14.6 Discrete Structures, Cryptography, and Number Theory: Why They Belong Together

This chapter appears to concern counting and connections, yet it stands alongside Chapter 13’s number theory and the cryptography in your phone. There are three layers to the connection.

First, *a computer is itself a discrete machine*. It stores strings of 0s and 1s and executes one instruction at a time, naturally inhabiting a discrete world. Even continuous quantities such as a circle’s area must be broken into finite decimal representations inside a computer. The mathematics of discrete structures—combinatorics, graphs, logic, and recurrences—is therefore computer science’s native language.

Second, *cryptographic security comes from hard discrete problems*. Chapter 13 explained

that multiplying two large primes is easy, while splitting their product back into primes is extremely difficult: this underpins RSA. Notice that the asymmetry is exactly a complexity issue: the forward operation is $O(\text{polynomial})$ multiplication, while the reverse currently has only slow, near-enumerative algorithms. Discrete logarithms and shortest vectors in lattices likewise underpin modern cryptography. At the other end, the pigeonhole principle watches over hash functions, which compress messages of arbitrary length into fixed-length fingerprints. Infinitely many messages and finitely many fingerprints guarantee collisions. Cryptographers must make *finding* a collision computationally infeasible.

Third, *number theory is itself discrete mathematics*. Integers come one at a time; primes are scattered points on the number line. Combinatorial tools continually serve number theory. Binomial coefficients conceal traces of primes: when p is prime, $\binom{p}{1}, \binom{p}{2}, \dots, \binom{p}{p-1}$ are all divisible by p ; try $p = 5$ yourself. Divisibility in recurrent sequences is a staple of competitions and research, and graph theory even helps investigate distances between primes.

Connections

Looking ahead: The multiplication principle and factorials return in Chapter 18's probability: an equally likely sample space is really a combinatorial count. Graph connectivity and cycles are generalised through the Euler characteristic in Chapter 16's topology. Chapter 19 explores algorithms and the limits of computability further.

Looking back: Chapter 7's matrices can record a graph's connections in an adjacency matrix. Multiplying the matrix by itself counts two-step paths: linear algebra and graph theory shake hands here. Chapter 13's congruences and primes jointly underpin divisibility properties of binomial coefficients and cryptographic applications.

14.7 Mathematical Experiments

One-Minute Experiment

Experiment 1: Counting trees. Draw 4 vertices and try to draw every non-isomorphic tree, meaning fundamentally different connection patterns. How many are there? Repeat with 5 vertices. (There are 2 with 4 vertices and 3 with 5.) Notice how quickly the sequence grows: with 10 vertices there are already 106.

Experiment 2: Shortest paths by hand. Draw a small graph with 5 junctions: home, school, park, bookshop, and station. Add seven or eight edges as you please, labelled with lengths. Following Dijkstra's idea, start at home and mark each junction with its currently known shortest distance. Each time you process a junction, update its neighbours. Compare your result with the route you guessed by eye.

Experiment 3: Binomial coefficients and primes. Calculate $\binom{5}{1}, \binom{5}{2}, \binom{5}{3}, \binom{5}{4}$ and check whether all are divisible by 5. Repeat for $n = 6$ and see whether the conclusion survives. What makes the difference?

Mathematical Toolbox

New tools for our collection:

- **Addition and multiplication principles:** Add across categories, multiply across steps; first ask whether anything is counted twice.
- **Permutations and combinations:** For ordered choices use $A_n^k = \frac{n!}{(n-k)!}$; for unordered choices use $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.
- **Recurrences:** Define the current problem using smaller problems of the same kind, as in $f_n = f_{n-1} + f_{n-2}$.
- **Binomial theorem:** The coefficients of $(a + b)^n$ are combination numbers; setting $a = b = 1$ gives 2^n subsets.
- **Pigeonhole principle:** More pigeons than holes force collisions, the prototype of unavoidable-collision arguments.
- **Graph language:** Vertices, edges, and degrees; the handshaking lemma $\sum \text{degrees} = 2 \times \text{number of edges}$.
- **Trees, shortest paths, network flow:** Connected, edge-efficient skeletons; greedy expansion; maximum flow equals minimum cut.
- **Big O notation:** Use growth rates ($O(n)$, $O(n^2)$, $O(2^n)$, $O(n!)$) to judge whether an algorithm can finish.

Keep Climbing

Directions for further climbing:

- **Generating functions:** Compress a sequence $a_0, a_1, a_2, \dots$ into a function and solve recurrences algebraically; the Fibonacci closed formula can be derived this way.
- **Ramsey theory:** Three mutual acquaintances or strangers among six people is only the beginning. More generally, sufficiently large structures inevitably contain sufficiently orderly substructures: the mathematics of the impossibility of total disorder.
- **P versus NP:** The gap between verifying and solving, one of the Clay Mathematics Institute's seven Millennium Prize Problems.
- **Spectral graph theory:** Study connectivity through matrix eigenvalues, supporting search-engine page ranking.
- **Extremal graph theory:** How many edges can a graph have under a restriction such as containing no triangles?

Questions to Think About

Observation

1. Calculate $\binom{10}{2}$ and $\binom{10}{8}$: they are equal. Explain why $\binom{n}{k} = \binom{n}{n-k}$ always holds by comparing choosing 2 people to attend a meeting with choosing 8 to stay behind. *Hint: every choice of k objects simultaneously leaves $n - k$ objects behind.*
2. In your group or family, count how many other members each person knows. Add the degrees, check that the sum is even, and explain using the handshaking lemma. *Hint: draw the acquaintance relationships as a graph.*

Proof

1. Prove that a tree with n vertices has exactly $n - 1$ edges. *Hint: induct on the number of vertices. A tree always has a leaf, a vertex of degree 1. Remove it and its incident edge to leave a smaller tree.*
2. Prove that among any 6 people, there are 3 pairwise acquaintances or 3 pairwise strangers. *Hint: the text took us halfway. Fix one person and use the pigeonhole principle to find three sharing the same relationship with that person. Then distinguish whether any pair among the three has that same relationship.*

Challenge

1. Place n equally spaced points on a straight line and join every pair by a segment. At least how many distinct midpoints do all these segments have? *Hint: draw and count for $n = 3, 4, 5$. Put the points at coordinates $1, 2, \dots, n$; a midpoint has coordinate $\frac{i+j}{2}$. Which values can this take?*
2. Among 8 identical-looking gold coins, 1 is a lighter counterfeit. What is the minimum number of weighings on a balance without weights that guarantees finding it? If the counterfeit might be either lighter or heavier, how many weighings are needed for 12 coins? *Hint: each weighing has three outcomes: left heavier, right heavier, or balance. Thus k weighings distinguish at most 3^k possibilities. View each weighing as a branch in a decision tree.*

The Door to the Next Chapter

We have learnt to count discrete objects: people in line, handshakes, edges in graphs. One phenomenon kept recurring: local rules for discrete objects, such as the divisibility of $\binom{p}{k}$ by a prime p , are often controlled by primes behind the scenes. But zoom out and ask a global question: how densely do these discrete prime points cover the entire number line?

Specifically, there are 25 primes up to 100, 168 up to 1000, and 78498 up to a million. Is there a formula for these numbers? Primes grow sparser, but at a uniform rate? Curiously, the strongest tools for answering this purely discrete question come from the most continuous world:

calculus, infinite series, and even functions on the complex plane. Mathematicians discovered that a continuous shadow seems to guide prime distribution with precision. Its secret is concentrated in an object called the Riemann ζ function, which also carries mathematics' most famous unsolved mystery.

Discrete and continuous: these two apparently diverging rivers meet in the next chapter.

Chapter 15 Analytic Number Theory:

Continuous Tools for Discrete Primes

15.1 A Counting Game That Gets Harder

During break, four pupils play a counting game: the first says “2”, and each following player names the next prime: 3, 5, 7, 11, 13, and so on. At first they compete to answer; near fifty they start counting on their fingers; near a hundred someone reaches for scrap paper to try divisions. The game has a curious feel: initially primes are densely packed, appearing every one or two numbers, but farther along, the empty stretches widen and the waiting grows longer.

Another scene unfolds out of sight. Whenever you pay by phone, your browser and the bank’s server exchange keys. One of the most popular public-key schemes begins by finding two large primes hundreds of digits long. How does a computer find them? Not by memorising a table: listing every prime of that size would require more than all the atoms in the universe. Its method is charmingly simple: choose a large number at random, test whether it is prime, and if not, try another until one works. This relies on a background fact: even hundreds of digits up the number line, primes are sparse but not vanishingly so. On average, a few hundred attempts find one.

Two questions emerge. We have already answered the first: do primes run out? No; Euclid proved their infinitude over two thousand years ago. The second is much subtler, and explains why our game grows harder:

Primes become sparser farther along. But how sparse? Is there a pattern to their thinning? Can a formula tell us roughly the density of primes near a number x ?

This sounds more like statistics or physics than a familiar integer problem. Integers jump one by one; they are discrete. Density, patterns, and formulas belong to the world of continuous variation. Analytic number theory does precisely this apparently unreasonable thing: it uses continuous tools—logarithms, integrals, infinite series, and even complex numbers—to study utterly discrete primes. Remarkably, this route not only works, but leads to mathematics’ most famous unsolved mystery.

15.2 A First Guess: Why a Fixed Share Does Not Work

Begin with intuition. There are 25 primes below a hundred, a quarter of the numbers. The natural guess is that primes occupy about a quarter of the integers farther along too, perhaps gradually a little less. In other words, prime density is approximately a fixed proportion.

Count a little farther and this guess collapses. Patiently extend the prime table, or consult one already made: up to 100 there are 25 primes, density 25%; up to 1000 there are 168, density 16.8%; up to 10000 there are 1229, only 12.3%. Density is not fluctuating around a fixed value, but steadily declining. The table rejects every fixed-proportion guess.

A second intuition asks whether density falls like $\frac{1}{x}$: near x , perhaps there is about 1 prime in every x numbers. This falls too quickly. It predicts only 1 prime per 1000 numbers near 1000, yet there are actually 168 primes up to 1000, averaging 1 in fewer than 6 numbers. The real decline is much slower than $\frac{1}{x}$, but much faster than a fixed proportion. It lies somewhere in between.

Counterexample Alert

“Primes grow sparser, so far enough along there will be extremely long prime-free stretches, longer than any prescribed length. That means primes are effectively finite.”

The first part is true: we will soon prove that arbitrarily long runs of composite numbers exist. The second is false. Sparsity is not disappearance. Euclid’s proof guarantees primes ahead however far you travel. Intuitively “too sparse to see” and logically “none remain” are different claims, a distinction we must watch carefully throughout this chapter.

A fixed proportion fails, and so does $\frac{1}{x}$. We need a new, slowly decaying function as a candidate density. You have already met it in Part III: the logarithm.

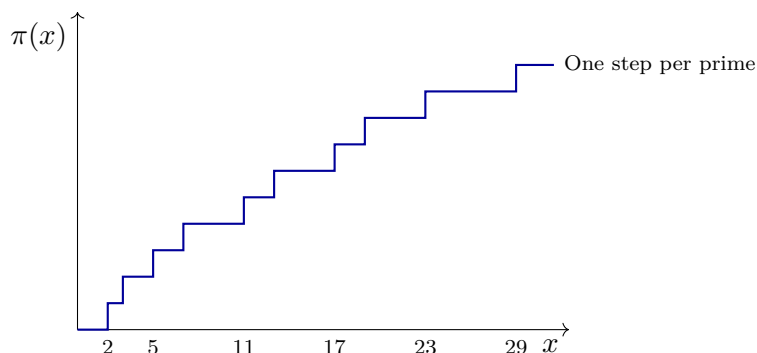
15.3 Building a Prime Counter

To discuss density, first choose a measuring tool. Our first new invention is a counting function:

Definition 15.1 (Prime-counting function). For any positive real number x , let $\pi(x)$ denote the number of primes no greater than x . For example, $\pi(10) = 4$ (the primes are 2, 3, 5, 7), and $\pi(100) = 25$.

This π is not the circle constant; it borrows the Greek letter corresponding to “p”, for prime. The notation solves a problem: instead of vaguely saying that primes seem scarcer, we can say $\pi(100) = 25$ and $\pi(1000) = 168$, turning impressions into numbers that can be compared and plotted.

What does $\pi(x)$ look like? A staircase: it stays flat as x passes composite numbers and jumps one step at every prime. Moving right, it only rises, never falls; because primes are infinite, the stairs continue forever without reaching a top.



This counter is purely discrete: its uneven steps reveal no smooth pattern. Yet something magical happens from a distance: the jagged staircase follows a smooth curve. To describe that curve we need our second protagonist, the natural logarithm $\ln x$.

Recall where logarithms come from. The exponential relations $10^2 = 100$ and $10^3 = 1000$ tell us that $\lg 100 = 2$ and $\lg 1000 = 3$. Logarithms translate multiplication into addition and grow extremely slowly: multiplying x by ten adds only one to $\lg x$. Mathematicians prefer the natural logarithm $\ln x$, with base $e \approx 2.718$, because it is especially convenient in calculus. The area under $y = \frac{1}{t}$ from 1 to x is exactly $\ln x$. This identity as an area will soon prove useful.

Our third tool is a famous series:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots$$

This is the harmonic series. Its terms shrink towards zero, yet their sum is infinite! A clever grouping reveals why. Put $\frac{1}{2}$ alone; group $\frac{1}{3} + \frac{1}{4}$, whose two terms are each at least $\frac{1}{4}$, giving at least $\frac{1}{2}$; group the four terms from $\frac{1}{5}$ to $\frac{1}{8}$, each at least $\frac{1}{8}$, again giving at least $\frac{1}{2}$; then take eight terms, sixteen, and so on. Every group contributes at least $\frac{1}{2}$. Infinitely many contributions of $\frac{1}{2}$ clearly diverge to infinity. The lesson: tending to zero does not mean having a finite sum.

A crucial quantitative fact is that the first n terms sum to approximately $\ln n$. This is unsurprising: $\ln n$ is the area under $y = \frac{1}{t}$, and $1 + \frac{1}{2} + \cdots + \frac{1}{n}$ adds rectangles of width 1 and height $\frac{1}{k}$ covering that area. The rectangular sum cannot differ from the area by much. Roughly:

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} \approx \ln n.$$

Our three tools are on the table: the staircase counter $\pi(x)$, the slow logarithm $\ln x$, and the area bridge connecting logarithms with sums of reciprocals. It is time for the chapter's central story.

15.4 The Prime Number Theorem: Order in Sparsity

Late in the eighteenth century, the young Gauss received an intriguing gift: a logarithm table with a prime table in its appendix. He reportedly compared the two repeatedly and noticed something strange: the rhythm of prime density matched the rhythm of logarithmic growth. He

guessed that near x , every $\ln x$ consecutive integers contain about 1 prime. In other words, prime density near x is approximately $\frac{1}{\ln x}$.

If density is $\frac{1}{\ln x}$, the total primes up to x should result from accumulating this density from 2 to x . In integral language, the smooth curve is $\int_2^x \frac{dt}{\ln t}$. A rougher but easier approximation is $\frac{x}{\ln x}$. Gauss's guess was eventually proved correct, becoming the prime number theorem:

Theorem 15.2 (Prime number theorem). *As x grows larger,*

$$\pi(x) \approx \frac{x}{\ln x},$$

meaning precisely that the ratio $\pi(x)/\frac{x}{\ln x}$ tends to 1 as x increases without bound.

Note. Gauss guessed this law in the late eighteenth century, but its proof took another hundred years. Only in 1896 did Hadamard and de la Vallée Poussin independently prove it. Their weapons were not integer tricks but the theory of functions on the complex plane, the subject of this chapter's second half. The proof lies beyond this book, but we can test the theorem ourselves in a table.

Try it: $\ln 100 \approx 4.6$, $\ln 1000 \approx 6.9$, $\ln 10^6 \approx 13.8$, and $\ln 10^9 \approx 20.7$. Substitute into $\frac{x}{\ln x}$ and compare with the true $\pi(x)$:

x	$\pi(x)$	$x/\ln x$ (rounded)	Ratio $\pi(x)/(x/\ln x)$
10^2	25	22	1.15
10^3	168	145	1.16
10^4	1229	1086	1.13
10^5	9592	8686	1.10
10^6	78498	72382	1.08
10^9	50847534	48254942	1.05

Look at the rightmost column: 1.15, 1.16, 1.13, 1.10, 1.08, 1.05, inching slowly towards 1. This is what a ratio tending to 1 looks like in practice: unhurried, but never stopping or turning back. A table is not a proof, as we will emphasise again, but it turns the claim that prime sparsity follows a precise law into visible numbers.

Back in our counting game, the prime number theorem says that the average gap near x is roughly $\ln x$. Near 100, a prime appears about every 4.6 numbers; near 10^6 , about every 13.8; at astronomical numbers hundreds of digits long, every few hundred. This explains why a computer finds a large prime after a few hundred random trials. Cryptographers are not merely lucky: the prime number theorem supports them.

Terminology Card: Prime number theorem

Intuition: Primes grow sparser, but $\ln x$ precisely controls the rhythm: density near x is about $1/\ln x$.

Formal statement: $\pi(x)/(x/\ln x) \rightarrow 1$ as x increases without bound.

Example: At $x = 10^6$, $x/\ln x \approx 72382$, while the true count is $\pi(x) = 78498$. The difference is only about 8%, and the percentage shrinks as x grows.

Be careful: an average gap of roughly $\ln x$ is an average, not a rule for every stretch. Actual gaps fluctuate wildly. Some primes differ by only 2, such as 29 and 31, or 41 and 43, called twin primes. Elsewhere hundreds or thousands of composites occur consecutively. In fact, the empty stretches can be arbitrarily long:

Theorem 15.3 (Prime gaps can be arbitrarily large). *For every positive integer n , there exist n consecutive positive integers none of which is prime.*

Proof. Choose the large number $N = (n+1)!$, the product $1 \times 2 \times 3 \times \cdots \times (n+1)$. Consider these n numbers just beyond N :

$$N+2, N+3, N+4, \dots, N+(n+1).$$

For $N+2$, the number N contains a factor 2, since the factorial includes 2, so $N+2$ is divisible by 2 and composite. Similarly, for $N+3$, the number N contains a factor 3, so $N+3$ is divisible by 3 and composite. In general, for every k from 2 to $n+1$, the number N contains k as a factor. Thus $N+k$ is divisible by k , and $N+k$ is clearly larger than k , hence composite. All n consecutive integers from $N+2$ to $N+(n+1)$ are composite. $\square$

This is a purely constructive proof: make the factorial large enough for the desired gap. For 1000 consecutive composites, take $1001!$ plus 2 through $1001!$ plus 1001, a “prime desert” of astronomical length. Yet twin primes seemingly keep appearing forever: 3 and 5, 11 and 13, $\dots$, 101 and 103, $\dots$. Mathematicians conjecture infinitely many pairs: the famous twin prime conjecture remains neither proved nor disproved. In 2013, Yitang Zhang proved infinitely many prime pairs differ by at most seventy million; worldwide collaboration subsequently reduced the bound to 246. From 246 to 2 seems a short step, but nobody has crossed it.

The picture is therefore this: gaps grow on average, controlled by $\ln x$; individual gaps can be arbitrarily wide; and twins separated by 2 seem never to disappear. Average regularity and individual unpredictability coexist. To know how orderly primes really are, $\pi(x) \approx x/\ln x$ is not enough. We must ask about the error: is the difference between $\pi(x)$ and $\frac{x}{\ln x}$ erratic, or governed by something deeper? This brings us to mathematics’ most mysterious doorway.

15.5 A Series Machine That Tunes In to Primes

Turn from the table to a machine. In the eighteenth century, Euler studied the infinite series

$$\zeta(s) = 1 + \frac{1}{2^s} + \frac{1}{3^s} + \frac{1}{4^s} + \cdots$$

For now, take s to be real and greater than 1. At $s = 1$, this is the divergent harmonic series; at $s = 2$, it converges to $\frac{\pi^2}{6}$, itself a legendary result: the circle constant emerges from reciprocal squares of integers. Euler's key observation was that this seemingly unrelated series is entirely determined by primes. The reason is the fundamental theorem of arithmetic: every integer factors uniquely into primes. Factor each denominator, regroup the terms, and use geometric-series summation to turn addition into multiplication:

$$1 + \frac{1}{2^s} + \frac{1}{3^s} + \frac{1}{4^s} + \cdots = \frac{1}{1 - 2^{-s}} \times \frac{1}{1 - 3^{-s}} \times \frac{1}{1 - 5^{-s}} \times \frac{1}{1 - 7^{-s}} \times \cdots$$

The right-hand side is an infinite product over every prime. This Euler product formula is the chapter's deepest line of poetry: all integers sing on the left, while each prime plays its own instrument on the right. The additive world of integers and the multiplicative world of primes are two expressions of the same machine.

A corollary follows immediately. Let s tend to 1: the left-hand series diverges to infinity. If primes were finite in number, the right-hand side would be a finite product of finite fractions, a contradiction. Thus primes must be infinite: Euler's product provides a proof with a completely different flavour. A finer calculation shows that the sum of all prime reciprocals,

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \frac{1}{11} + \cdots$$

also diverges, as slowly as roughly $\ln(\ln x)$. Savour that result: primes have enough weight among the integers that even their reciprocal sum cannot be contained. Compared with squares, whose $\sum \frac{1}{n^2}$ converges, primes are much denser. Their infinitude is not a marginal affair; it has real substance.

15.6 Riemann's Astonishing Leap: Sending s into the Complex Plane

Euler's machine requires $s > 1$; otherwise the series will not converge. In 1859, Riemann published just one short, eight-page paper, yet changed mathematical history by asking whether $\zeta(s)$ could be extended to the whole complex plane.

Terminology Card: Riemann ζ function

Intuition: A machine taking complex inputs, driven for $s > 1$ by $1 + 2^{-s} + 3^{-s} + \dots$. Analytic continuation extends it to almost the entire complex plane.

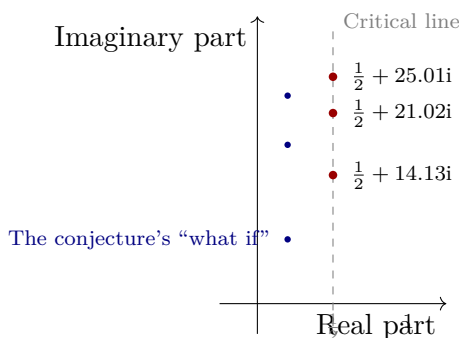
Part of the formal definition: $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$ when the real part of s exceeds 1, then uniquely extended elsewhere, “blowing up” only at $s = 1$.

Example: $\zeta(2) = \frac{\pi^2}{6}$; under continuation, $\zeta(-1)$ receives the value $-\frac{1}{12}$. This does not mean that $1 + 2 + 3 + \dots$ genuinely equals $-\frac{1}{12}$: it is the continued function’s value there. These must not be confused.

The difficulty is that the series converges only where the real part exceeds 1; outside that region, $1 + 2^{-s} + 3^{-s} + \dots$ ceases to behave. Mathematicians use ingenious methods—roughly, rewriting the series indirectly into a convergent form, then proving that the resulting function is the unique appropriate extension—to give ζ a definite meaning everywhere except $s = 1$. This technique, analytic continuation, belongs to complex analysis and lies beyond this book. Remember its valuable uniqueness: like extending a circular arc along its own curvature, there is just one natural continuation.

With the function in hand, Riemann asked which inputs produce zero: where are the solutions of $\zeta(s) = 0$? Zeros are a function’s “genes”: just as roots determine a polynomial, zeros largely determine an analytic function. Calculation shows that the negative even integers $-2, -4, -6, \dots$ are zeros, called trivial zeros, with little surprise attached. The real secret lies in the vertical critical strip, where real parts lie between 0 and 1. It contains infinitely many nontrivial zeros, and Riemann calculated that the first few have real part exactly $\frac{1}{2}$:

$$\frac{1}{2} + 14.13i, \quad \frac{1}{2} + 21.02i, \quad \frac{1}{2} + 25.01i, \quad \dots$$



Why do zeros concern primes? Through Euler’s product, ζ contains information about every prime, and its zeros are knobs for extracting that information. Riemann’s eight-page paper gave a remarkable formula, whose proof was completed later: $\pi(x)$ can be expressed as a smooth main term plus a correction wave from each nontrivial zero. The larger a zero’s real part, the larger its oscillation, and the harder it is to control the deviation of $\pi(x)$ from the smooth curve. The question of prime regularity becomes precisely the geometric question of where zeros lie.

Riemann turned his computed observations into a conjecture:

Riemann hypothesis (1859, still unproved): Every nontrivial zero of ζ has real part $\frac{1}{2}$: they all lie neatly on one vertical line.

Notice the wording: a **conjecture**, not a theorem. Over more than 160 years, computers have checked over ten trillion nontrivial zeros, all on this line. Countless mathematicians have sought a proof, accumulating mountains of equivalent statements and consequences, yet no proof has appeared. It is one of the Clay Mathematics Institute’s seven Millennium Prize Problems, with a million-dollar reward.

Why does the hypothesis matter? It governs the prime number theorem’s error. Mathematicians have proved that the deviation between $\pi(x)$ and the smooth curve is controlled entirely by the real parts of nontrivial zeros. If the hypothesis holds, this deviation is squeezed to its minimum, roughly bounded by the order of $\sqrt{x} \ln x$. Relative to x , this is like measuring the Earth–Moon distance with an error below a kilometre. Conversely, even one zero off the critical line would create much larger-than-expected irregular fluctuations in prime distribution. In everyday language: **primes are mischievous, but within limits; one straight line keeps their randomness in check.**

Counterexample Alert

“Computers have checked ten trillion zeros on the critical line, so the Riemann hypothesis is essentially settled; only a formality remains.”

This is the classic mistake of treating experiment as proof. There are infinitely many nontrivial zeros; however many we check, they are a cup of water from an infinite ocean. Mathematics offers harsh lessons: some statements hold for unimaginably large numbers, then first fail farther beyond human reach. Prime-distribution research has genuinely encountered reversals only at enormous scales. Experiments inspire conjectures, build confidence, and find counterexamples; only proof guarantees every case. This is the weight of “analytic” in the chapter title: the demand is for argument, not statistics.

Connections

Looking back, we have braided together several earlier threads: Chapter 9’s limits, with a ratio tending to 1; Chapter 11’s series and integrals, including the harmonic series and the area meaning of $\ln x$; Chapter 12’s complex numbers, used to continue ζ ; Chapter 13’s fundamental theorem of arithmetic, underpinning Euler’s product; and Chapter 14’s computational experiments, including prime tables and sieves. Looking ahead, the Riemann hypothesis’s claim that all zeros lie on one line already gestures towards Chapter 16’s topology through shape and position. Public-key cryptography is prime distribution’s most concrete practical application: it relies both on finding primes easily, with density guaranteed by the prime number theorem, and on factoring large numbers being difficult, with no fast algorithm yet known.

15.7 Count for Yourself: What Experiments Can and Cannot Do

One-Minute Experiment

Prepare a grid of the numbers 1 to 100 and perform the sieve of Eratosthenes by hand: circle 2 and cross out its multiples; circle the next survivor, 3, and cross out its multiples; repeat for 5 and 7. Up to 100, sieving only to $\sqrt{100} = 10$ suffices—why? The survivors are all primes up to 100. Count them: are there 25?

Record consecutive prime gaps: $3 - 2 = 1$, $5 - 3 = 2$, $7 - 5 = 2$, $11 - 7 = 4$, and so on. Put them in a row and answer:

1. What is the largest gap, and where does it occur?
2. How many twin prime pairs have gap 2?
3. What is the approximate average gap? Is it close to $\ln 100 \approx 4.6$?

With a calculator or computer, extend the experiment to 1000 and observe the average moving towards $\ln 1000 \approx 6.9$. If you can, write a short programme of a dozen or so lines: two nested loops imitate circling and crossing out, letting the computer count to a hundred thousand.

Afterwards, pause: what exactly have the table and programme proved? They tell you that up to 100, the average prime gap is about 4.3, the same order as $\ln 100$. This is an observation, a sample of the prime number theorem at $x = 100$. It does not, and cannot, prove that a ratio tends to 1. The same logic applies to the Riemann hypothesis: ten trillion checks remain an experiment. Mathematics has three levels, all represented here: **experiment**, including sieves, tables, and zero calculations, supplies clues; **conjecture**, including twin primes and the Riemann hypothesis, points the way; **proof**, including Euclid, the factorial construction, and the prime number theorem, provides guarantees. Confusing these levels is among number theory's easiest traps and most important lessons.

Mathematical Toolbox

New tools from this chapter:

- **Counting function** $\pi(x)$: Turn the impression of thinning primes into a function to plot and compare.
- **Natural logarithm** $\ln x$: One of the slowest-growing elementary functions, the scale of prime density and also the area under $y = \frac{1}{t}$.
- **Harmonic series and divergence**: Terms tending to zero do not guarantee convergence; the first n terms sum to roughly $\ln n$.
- **Euler product**: $\sum n^{-s} = \prod_p (1-p^{-s})^{-1}$ translates between integer addition and prime multiplication.
- **ζ and analytic continuation**: Extend the series machine to the complex plane; zero locations control prime-distribution error.
- **Distinguishing experiment, conjecture, and proof**: Tables inspire conjectures, conjectures guide proofs, and only proofs guarantee results.

Keep Climbing

For further exploration, remember these signposts: **Chebyshev functions** and elementary proofs of the prime number theorem, given independently by Selberg and Erdos in 1949 without complex analysis; the **logarithmic integral** $\text{Li}(x)$, closer to $\pi(x)$ than $x/\ln x$; **analytic continuation** and complex analysis, the true language of Riemann's paper; modern **sieve methods**, supporting prime-gap research after Yitang Zhang; and **zero computations**, pinning ten trillion zeros to the critical line one by one with fast computers. Any of these could keep you in libraries or online for a whole summer.

Questions to Think About

Observation

1. Record prime gaps up to 100 again and identify every record gap, larger than all preceding gaps. Look at the primes at those positions and guess where the next record might occur. *Hint: a record gap marks a relatively long composite desert. Think about the factorial construction: are deserts likelier near specially structured numbers or just anywhere?*
2. Inspect the table's ratio column from 10^2 to 10^9 . How much must x multiply for each small decrease in the ratio? Get a feel for how slowly it tends to 1. *Hint: multiplying x by 10 moves the ratio by only a fraction of a percentage point. It is not lack of effort; $\ln x$ grows very reservedly.*

Proof

1. Prove that 1000 consecutive positive integers can all be composite. Explain roughly how high your constructed interval lies. *Hint: choose an appropriate factorial for $N = (n + 1)!$ and check why every $N + k$ is composite.*
2. Without consulting a table, prove that a sufficiently long consecutive block of harmonic-series terms, however far out it begins, can have sum greater than 10. *Hint: recall the grouping argument, where each group contributes at least $\frac{1}{2}$. About how many groups exceed 10?*

Challenge

1. Euler's product showed that finitely many primes would give a contradiction as $s \rightarrow 1$. Go further: explain why divergence of prime reciprocals seems almost inevitable from the product formula. *Hint: take logarithms to turn the product into a sum. As $s \rightarrow 1$, the left side blows up; expanding the terms $\ln(1 - p^{-s})$ on the right gives leading terms $\frac{1}{p}$. This requires fluency in turning products into sums with logarithms, so getting stuck is normal.*

The Door to the Next Chapter

We crossed a boundary: continuous, smooth analysis interrogated discrete, individual primes. Look again at the zero diagram. One detail deserves another glance: the Riemann hypothesis says that infinitely many mysterious points all lie on **the same straight line**.

“Lying on a line”, “forming a loop”, “having a hole”: we have used these phrases casually several times. But what is a line? What is a hole? Replace the complex plane by a rubber sheet and allow stretching, bending, and twisting without tearing or joining broken pieces: is the critical line still a line? Under rules permitting deformation but no tearing, why are a coffee cup and a doughnut the same kind of object, yet different from a sphere?

Number theory advances through counting and proof. These questions need a subject devoted to properties that survive deformation. It ignores lengths and angles, asking only about connections, passages, and holes. This is the world of the next chapter: topology.

Part V

Branching Out from Curves and Spaces—Topology, Differential Geometry, Probability, and Foundations

Chapter 16 Topology: What Counts as the Same Shape in Modelling Clay?

16.1 A Map That Does Not Look Like a Map

You have probably travelled by underground and certainly seen its colourful route map: horizontal and vertical lines, evenly spaced dots, and tidy corners. Overlay it on a real city map and almost nothing matches. Stations two centimetres apart on paper may be five kilometres apart; a perfectly straight line on the diagram may twist repeatedly underground. In lengths, angles, and areas, this map tells lies everywhere.

Strangely, passengers do not get lost because of it. When draughtsman Harry Beck drew the first such London Underground map in 1933, the company initially hesitated: would passengers accept something so unrealistic? In practice, it proved so useful that underground systems world-wide copied it. Why? Passengers care about three things: their current station, their destination, and where to change. The tunnels' precise turns, or whether stations are eight hundred or twelve hundred metres apart, do not matter. Beck saw this clearly: he discarded distracting real-world detail and retained only **what connects to what**.

This raises a deep question: a map can be distorted beyond recognition yet still answer certain questions correctly. Which properties of a figure survive arbitrary stretching, bending, and compression? Which vanish at a gentle squeeze? The study of the former is called **topology**, famously nicknamed rubber-sheet geometry. To a topologist, figures are made of unbreakable modelling clay: knead them however you like, but do not tear them or glue separate parts together. The properties that remain unchanged are the “true shape” topology studies.

In this chapter we become clay modellers. We will see how a three-hundred-year-old walking puzzle helped create the subject; why coffee cups and doughnuts look identical to topologists; why a strip of paper loses its front and back after a half-twist; and how these playful ideas serve networks, maps, and robots.

16.2 Seven Bridges and an Impossible Walk

Our story begins in a real city. In eighteenth-century Königsberg, now Kaliningrad in Russia, a river crossed the town, with two islands and seven bridges linking them to its banks. A favourite leisure puzzle reportedly prompted years of tavern debate:

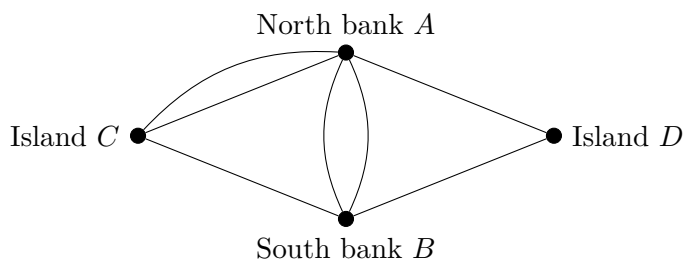
Can you set out from home, cross each of the seven bridges exactly once, and finally return home?

Before reading on, guess: possible or impossible? Many instinctively say yes: enough attempts should find a route. Others say no but cannot explain why. That is the puzzle's charm: it sounds like a riddle, yet nobody can supply a convincing answer. Clever townspeople repeatedly trace routes on maps, always getting stuck at the last bridge or two: either crossing one twice or reaching an island with no unused bridge left.

Trying indefinitely has a fatal weakness: ten thousand failures do not prove the ten-thousand-and-first attempt cannot succeed. Proving impossibility requires a new viewpoint, not more attempts. In 1736, Euler, then living in St Petersburg, heard the puzzle. His first move seemed remarkably unmathematical: he threw away the map.

16.3 What Remains After Scrunching Up the Map?

Euler reasoned that the size of the banks, the islands' shapes, bridge lengths, and road bends were irrelevant. The walker only experiences being on a piece of land and crossing a bridge to another piece. He shrank each of the four land areas—two banks and two islands—to a point, and each bridge to a line connecting points. The result was an ugly little diagram, four points and seven lines, that a primary-school pupil could draw in seconds:



The original question becomes: can we start at a point, **trace every line exactly once**, and return to the start? This is the familiar childhood one-stroke drawing puzzle!

Notice this remarkable step. Before solving the original problem, Euler established that it and the drawing puzzle are **the same problem**. He compressed the map into a little tangle of threads, discarding lengths, areas, and shapes without losing any information the question needed. Keeping only connections is a basic topological skill. The more we discard, the clearer the remaining structure becomes.

Before continuing, we need official names for these points and lines so we can speak precisely.

Definition 16.1 (Graphs, vertices, edges, and degrees). A collection of points, called **vertices**, and connecting lines, called **edges**, forms a **graph**. The number of edges meeting a vertex is its **degree**. Vertices of odd degree are **odd vertices**; those of even degree are **even vertices**. A graph is **connected** if every vertex can be reached from any other by following edges.

Count the degrees in the seven-bridge diagram: A meets 5 edges, B meets 3, C meets 3, and D meets 3. All four vertices are odd. Remember this: it will solve the mystery.

Terminology Card: Topological equivalence (homeomorphism)

Intuition: If two clay figures can be transformed into one another by continuous stretching, bending, and compression, without tearing or gluing, they have the same shape.

Formal definition: Two figures are **homeomorphic** if there is a continuous one-to-one correspondence between them whose inverse is also continuous.

Example: A circle and a square are homeomorphic: stretch the circle like a rubber band into a square frame. A circle and a figure eight are not: the figure eight's central crossing cannot be separated.

16.4 Solving the Mystery: Count the Odd Junctions

Now solve the one-stroke puzzle. Imagine following the graph with a pen. At an intermediate vertex, you must enter along one edge and leave along another, using incident edges in pairs. A vertex that is neither the start nor the finish must therefore have its edges paired as entrances and exits: its degree must be even.

What about the endpoints? If the walk returns to its start, the initial departure and final arrival also pair up, as do intermediate visits, so the starting vertex is even too. There can be no odd vertices anywhere. If return is not required, the start has one extra departure and the finish one extra arrival. These two may be odd, but no others. This is Euler's discovery:

Theorem 16.2 (Euler's one-stroke theorem, 1736). *For a connected graph:*

1. *A route using every edge exactly once and returning to its start exists if and only if there are **no** odd vertices.*
2. *A route using every edge exactly once, without necessarily returning to its start, exists if and only if there are **exactly two** odd vertices, which must be its endpoints.*

Proof. First prove necessity. Each intermediate visit uses two edges, one entering and one leaving, so all vertices other than the endpoints are even. For a closed route, the starting vertex's initial and final edges also pair up, making it even; there are no odd vertices. For an open route, the start and finish each have one unpaired edge and are exactly the two odd vertices.

Now prove sufficiency, beginning with no odd vertices. Start anywhere and follow unused edges arbitrarily. Because edges at every vertex pair up, entering any vertex other than the start always leaves an unused exit. The only place you can get stuck is therefore the start. Continue until stuck: you have a closed route. If unused edges remain, connectivity guarantees that some vertex on this route meets an unused edge. Start another arbitrary walk there; the same reasoning brings it back to that vertex. Insert this new circuit into the original route to make a longer one. Repeat until every edge is used. With exactly two odd vertices, start at one and eventually get stuck at the other, then insert remaining circuits in the same way. $\square$

Return to the bridges: all four vertices are odd, giving 4, neither 0 nor 2. The walk ****does not exist****: not merely undiscovered, but impossible to discover. Without walking a step, Euler counted bridges at each junction and settled years of argument. Reportedly, an eighth bridge later left two odd vertices, finally permitting a walk across every bridge, though not back home. Mathematics predicted possibility and engineering followed: an ironically fitting sequel.

Mathematical Toolbox

Our new tools so far:

- **Compression**: Turn a map into points and lines, retaining only connections.
- **Graphs** (vertices, edges, degrees) and **connectivity**: Minimal language for what connects to what.
- **Parity pairing**: Entrances and exits pair up; parity drives the argument.
- **Homeomorphism**: The criterion for sameness of shape in modelling clay.

16.5 Classifying Letters: What Is a Topological Property?

With homeomorphism as our measuring stick, try a classification game. Imagine printed capital letters as thin clay lines: A, B, C, D, E, and so on. Which have the same shape?

C, I, J, L, M, N, S, U, V, W, and Z form one class: each is a line that can be straightened into a segment. A, D, O, P, Q, and R form another: each has ****one hole****, whether triangular as in A or round as in O. B stands alone with ****two holes****. E, F, T, and Y have no holes but ****one three-way junction****. K and X have a four-way junction. H is special, with two three-way junctions.

What does this game tell us about topology's interests? The classification uses only a few properties:

- **Connectivity**: Is the figure one piece or several? Whether the dot on a lowercase i connects to its stem is a connectivity question.
- **Number of holes**: How many independent loops cannot be bypassed?
- **Junctions and endpoints**: Where do lines branch, and where do they terminate?
- **Boundary**: Does the figure have an edge, and how many boundary components are there?

Lengths, angles, areas, and curvature are absent. Topologists famously joke that they cannot distinguish a coffee cup from a doughnut. Imagine a clay cup with a handle: press in and round its body, leaving the handle as a ring, and it becomes a doughnut. Nothing is torn or glued; the handle's hole becomes exactly the doughnut's hole. A doughnut, however, can never become a ****sphere without a hole****. Every rubber loop on a sphere can shrink to a point, while a loop

around the doughnut can get caught on its ring. Hole count is the first checkpoint separating shapes into species.

Counterexample Alert

“These figures look very different, so they are topologically different” is false: a circle and a square look different but are homeomorphic. Conversely, equal area does not imply topological sameness. A disc and an annulus can have equal areas, yet one has no hole and the other has one; clay deformation cannot create that hole. Homeomorphism concerns continuous deformation, not appearance or size.

16.6 A Number That Counts Holes: The Euler Characteristic

Holes are useful, but counting them can be ambiguous: how many holes does a piece of cheese pierced three times have? Mathematicians need a mechanical, uncontentious measure of how a shape winds around. The method is surprisingly simple: cut the figure into pieces, count three kinds of components, and add and subtract.

Start with convex polyhedra. A cube has 8 vertices, 12 edges, and 6 faces:

$$V - E + F = 8 - 12 + 6 = 2.$$

A regular tetrahedron has $V = 4$, $E = 6$, and $F = 4$, giving $4 - 6 + 4 = 2$. A regular dodecahedron has $V = 20$, $E = 30$, and $F = 12$, again giving $20 - 30 + 12 = 2$. Curious, isn't it? Wildly different component counts always yield 2. Euler—yes, him again, appearing in nearly every corner of mathematics—proved this was no coincidence:

Theorem 16.3 (Euler's polyhedron formula). *Every polyhedron continuously deformable into a sphere, or connected graph drawn on a sphere, satisfies*

$$V - E + F = 2,$$

where V , E , and F count vertices, edges, and faces respectively.

Proof sketch. Imagine the polyhedron as a hollow rubber balloon. Remove one face and flatten the rest onto a table, producing a connected graph of vertices and edges. It divides the plane into regions, 1 fewer than the original face count, since the removed face becomes the unbounded exterior. We need to prove $V - E + F_{\text{regions}} = 1$ for the flattened graph. Perform two demolition operations. If a cycle remains, delete one of its edges: E decreases by 1, and the regions on its two sides merge, decreasing F_{regions} by 1, so $V - E + F_{\text{regions}}$ is unchanged. Once no cycles remain, the graph is a tree. Remove an endpoint and its incident edge: V decreases by 1 and E by 1, again preserving $V - E + F_{\text{regions}}$. Eventually only a lone vertex remains: $V = 1$, $E = 0$, $F_{\text{regions}} = 0$, giving 1. Since neither operation changed the value, the original planar graph also gives 1, and restoring the polyhedron gives 2. $\square$

The number $V - E + F$ is the **Euler characteristic**, usually denoted by the Greek letter χ . Remarkably, however the surface is subdivided—large pieces or small, straight edges or curved—the value of χ remains the same. It is the surface’s fingerprint.

Fingerprints distinguish species. Repeat the game on a doughnut, or torus: draw a connected graph dividing it into regions, and obtain $V - E + F = 0$. A two-holed double torus, like two joined swimming rings, gives -2 . Each added hole lowers χ by 2. Writing g for the hole count, mathematically called the **genus**, we have

$$\chi = 2 - 2g.$$

A sphere has $g = 0$, $\chi = 2$; a doughnut $g = 1$, $\chi = 0$; a double torus $g = 2$, $\chi = -2$. The vague task of counting holes becomes arithmetic with an agreed answer: grid the surface, count, and subtract. A coffee cup can become a doughnut because both have χ equal to 0; neither becomes a sphere because $\chi = 0$ never equals $\chi = 2$.

Terminology Card: Euler characteristic

Intuition: Conduct a surface census. Vertices minus edges plus faces gives an identification number unchanged by continuous deformation.

Formal definition: Subdivide a surface into V vertices, E edges, and F faces; $\chi = V - E + F$, independently of the subdivision.

Example: A cube gives $8 - 12 + 6 = 2$, placing it with the sphere; a torus grid gives 0, smaller by 2, corresponding exactly to its extra hole.

16.7 A Half-Twisted World: The Mobius Strip and Klein Bottle

Our clay game has so far kept an unspoken rule: surfaces have an inside and an outside. A balloon has inner and outer walls; paper has front and back. In 1858, Möbius discovered a surface with only **one side**, absurdly easy to make: take a long paper strip, twist one end halfway around, and join it to the other.

Try it; the experiment below will guide you. Draw a coloured line along the strip’s centre without lifting the pen. Eventually you return to your starting point, having marked “both sides”: there is no other side! Cut along the centre line, and instead of the expected two rings, you obtain one longer ring with a full twist. An ant can crawl over the entire Möbius strip without crossing an edge.

One-Minute Experiment

Cut a scrap-paper strip about 3 centimetres wide and 25 centimetres long.

1. Join the ends directly to make an ordinary ring. Draw around its centre line: only one side is marked. Cut along that line and the ring separates into two rings.
2. Make another strip, twisting one end halfway around before joining it. Draw the centre line: the marks now magically cover the whole strip. Cut along the line and count the twists in the resulting ring.
3. Join another strip after a full twist, 360° . Repeat the drawing and cutting. How does this differ from the previous two?
4. Record the number of resulting rings and their twists after starting with 0, half, one, and one-and-a-half turns. Can you guess a pattern?

A Mobius strip also has one **boundary**: before cutting, its long paper edge surprisingly forms a single connected line. Mathematicians then asked whether a closed surface could have one side and no boundary at all. Yes: the **Klein bottle**. Imagine making a hole in a bottle's base, lengthening its neck, bending it back through the bottle wall, and joining it to the hole. Inside and outside become one surface: walking along the wall takes you from exterior to interior without crossing any edge. An ant can explore the entire Klein bottle.

There is a fact we must admit: a Klein bottle **cannot be built** in our three-dimensional space. The description required passing through the bottle wall. In three dimensions, the neck must penetrate the body to reach the base, violating the clay game's no-puncturing rule. Only in four dimensions can it detour through a fourth direction without touching the wall. Glass "Klein bottles" sold as ornaments all self-intersect: they are three-dimensional projections of a four-dimensional resident, just as a doughnut's shadow on a wall may contain crossings. Mobius strips show that the number of sides is not a given; Klein bottles show that some shapes do not fit our world, yet mathematics describes them precisely.

Connections

Remember turning squares, rotations, and reflections into a symmetry operation table? There we upgraded "looks the same" to "has the same structure". This chapter makes the same upgrade with a different measuring stick. Chapter 6 asked which motions leave a figure unchanged; here we ask which deformations leave properties unchanged. Our graphs speak the same language as Chapter 14's graph theory: the bridges problem is a shared birth certificate for graph theory and topology. Next chapter, see what happens when rulers and protractors replace the modelling clay.

16.8 Another Perspective: Networks, Maps, and Robots

Topology sounds like clay play and paper cutting, but preserving connections while discarding detail is a fundamental principle of the modern world.

Networks. The internet has hundreds of millions of devices, transport networks thousands of junctions, and power grids countless nodes. Engineers often ask first not how long a cable is, but topological questions: is the network connected? Will losing a node split it in two? How large will the pieces be? This is the same thinking as asking how many pieces a figure has. Social networks' six degrees of separation—roughly six friendship links between any two people on average—describe surprisingly tight connections in enormous networks.

Maps. What is the minimum number of colours needed for a world map so neighbouring countries differ? The question depends only on shared borders, not countries' areas, shapes, or coastline curves: it is purely topological. For over a century, people conjectured that four colours suffice, the famous four-colour conjecture. In 1976, mathematicians reduced it to checking over a thousand basic configurations and completed the proof by computer. This became the first major computer-assisted theorem. Some mathematicians still debate whether a proof no individual can check by hand counts as a proof.

Robots. A robotic arm has several joints, each rotating through an angle. Every pose corresponds to a point in a configuration space formed from those angles. Planning motion means finding a continuous collision-free path through that space. Its topology is intriguing: a joint permitting full rotation contributes a circle; two independently rotating joints contribute a torus, our doughnut again! Robot engineers effectively plan routes on doughnuts, spheres, and stranger surfaces. Many obstacles that no turn seems to bypass are fundamentally holes in configuration space.

Note. In these applications, topology answers questions of possibility: can things connect, can a route go around, can four colours suffice? Speed, length, and cost require bringing back lengths and angles and calling on other branches of mathematics. Distinguishing “possible or impossible” from “better or worse” is a first step in problem-solving.

Keep Climbing

For further climbing, remember these keywords: **fundamental group**, making non-shrinkable loops into an algebraic structure and hole counts into rigorous theorems; **homology**, a distant relative of the Euler characteristic that counts holes of any dimension; **knot theory**, studying when knots in three-dimensional space can be undone, from shoelaces to DNA; and the **Poincare conjecture**, the three-dimensional version of “a closed shape without holes must be a sphere”, proved only in 2003 and the first Millennium Prize Problem to be solved.

Questions to Think About

Observation

1. Treat the digits 0 to 9 as thin clay lines, handwritten without crossings, and classify their topological shapes. Which belong together? **Hint:** first count holes and endpoints.
2. Lay an unbuttoned long-sleeved shirt flat. Topologically, it resembles a disc with how many holes? **Hint:** count all the exits once a body is inside, and consider how many holes each exit represents.

Proof

1. Prove that every graph has an even number of odd vertices. **Hint:** when adding every vertex's degree, how many times is each edge counted? Is the total odd or even?
2. Check $V - E + F$ on a sphere with a latitude–longitude grid, say 4 meridians and 2 latitude circles, one being the equator. Count the poles as vertices. Count vertices, edges, and regions to verify Euler's formula. **Hint:** decide how meridian intersections at the poles count and how many pieces each latitude circle is cut into.

Challenge

1. What is the largest number of pairwise adjacent regions on a doughnut, every pair sharing a boundary? On a plane it is at most 4, one source of the four in the four-colour theorem; a doughnut allows more. **Hint:** the answer is 7. First try drawing 5 pairwise adjacent regions, then exploit routes around the ring to fit two more.
2. Join two Möbius strips along their boundaries. Which surface from this chapter results? **Hint:** mentally unfold one strip into a ring-shaped band; consider whether a Klein bottle can be cut into two halves along some line.

The Door to the Next Chapter

We have enjoyed considerable freedom: discard lengths, angles, and areas, and call a figure unchanged as long as its connections and holes remain. In the clay world, a sphere and a potato are the same kind of thing; a teacup and a doughnut are indistinguishable.

Earth's inhabitants will object that Earth is not a potato. Navigation software needs distances; pilots need to know why Beijing–New York flights first head north rather than straight east along a latitude circle. Such questions immediately require what we discarded: length, angle, and curvature. Spherical triangles have angle sums greater than 180° , and straight lines on a torus can wind around themselves. Topology sees none of this, because a squeeze of clay changes it all.

Replace the clay with a firmer curved surface, allowing travel across it but no stretching. The straightest route, shortest distance, and degree of bending then need a new language. How does a surface locally resemble a plane? Can a blindfolded ant tell from underfoot measurements whether it lives on a sphere or a plane? Next we enter differential geometry: distances and directions in curved spaces.

Chapter 17 Differential Geometry:

Distance and Direction in Curved Spaces

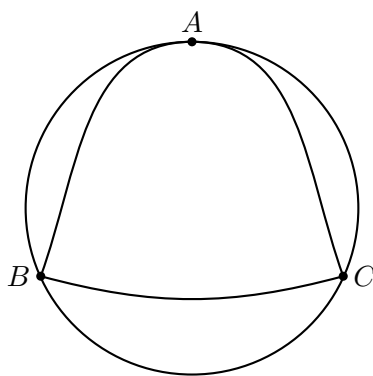
17.1 That Strange Detour on the Map

Open a world map and find Beijing and San Francisco. Imagine you are an airline pilot planning the most fuel-efficient route between them. Intuition says to join the cities with a ruler and fly straight along that line.

Real flights do something else. From Beijing they head northeast, passing over northeastern China and Russia's Far East, near the Aleutian Islands, before turning southeast towards San Francisco. On the map the route bulges conspicuously northward. Airlines burn tens of thousands of tonnes of fuel each year; they would not take a detour without reason.

Consider an older example. Fifteenth-century navigators already knew that sailing due west along one latitude was not the quickest route to a distant port. Clever captains first headed towards higher latitudes, then turned back. Sailors called this "great-circle sailing". They did not know why it worked, only that it was faster.

This leaves us with a question:



A spherical triangle with three great-circle arcs

On a curved surface, what does "straight" mean? How should distance be measured and direction described? These sound like questions for pilots and captains, but they lead to differential geometry, a deep subject studying lengths, angles, and shapes in curved spaces. Early in the twentieth century, it became Einstein's language for gravity. We will travel from a bending flight route to curved spacetime.

17.2 When Intuition Fails: Orange Peel That Will Not Lie Flat

The first natural response to spherical distance is to flatten the sphere. Then a ruler can measure it: easy.

Try this at home. Peel an orange, keeping the peel as intact as possible, and attempt to lay it flat without gaps or overlaps. You cannot: either the edges split or the middle bulges. Compare a flat sheet of paper rolled effortlessly into a cylinder. Unroll it and it returns unchanged, without tears or wrinkles.

This failed kitchen experiment reveals a deep dividing line: cylinders and planes are relatives, deformable into one another without stretching; spheres and planes are not. Every flat map of a sphere must pay a price in distortion.

How large a price? On the familiar Mercator world map, Greenland looks about as large as Africa, though Africa is actually about fourteen times larger. The map is not mistaken; it simply cannot preserve all lengths, angles, and areas. Some maps preserve angles at the expense of areas; others preserve areas at the expense of shapes. Mathematicians have proved that a flat spherical map preserving distance, angle, and area simultaneously cannot exist.

Counterexample Alert

"Flatten the sphere before measuring, and surely we are back in plane geometry?" This plausible plan fails. A sphere cannot be flattened without stretching or tearing, so spherical geometry is not plane geometry in disguise: it has its own laws. Likewise, following a latitude circle is not the shortest route from Beijing to San Francisco. Except for the equator, latitude circles are not great circles, so following one is a detour.

Our first attempt failed but pointed the way. Stop trying to turn surfaces into planes; learn to live inside their geometry, like an ant on a sphere understanding its world through measurements underfoot, without stepping outside.

17.3 Coordinates on a Surface: An Atlas Is Enough

The ant needs no perfect map of the entire sphere, only an atlas. Each page maps a small region, approximately planar. Neighbouring pages overlap, assigning two coordinate systems to the same place, with a conversion rule translating between them.

This idea is important enough to name.

Definition 17.1. If a small surface region corresponds one-to-one and sufficiently smoothly to a disc in the plane, we assign it planar coordinates (u, v) and call it a **coordinate chart**. Charts covering the whole surface, with coordinate conversions on their overlaps, form a **coordinate atlas**.

You have something similar at home. No world map portrays Earth perfectly, but a regional atlas works well: one province per page, with overlaps linking the pages. Likewise, a single chart cannot cover the globe without trouble, particularly at the poles, but two or three suffice. The

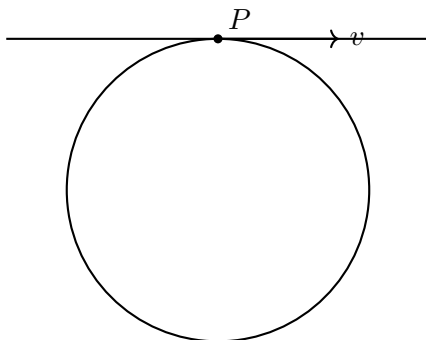
key is that lengths, angles, and areas must be computable in each chart and agree on overlaps. Then the ant has a complete geometry without needing to know that the sphere sits inside three-dimensional space.

Looking only from within a surface is the intrinsic viewpoint. The contrasting external view makes a sphere's curvature immediately visible in three-dimensional space. A turning point in mathematics came with the recognition that intrinsic geometry can stand alone: inhabitants can measure whether their surface curves, and how much, without going outside. A mathematician called the central result the "remarkable theorem": intrinsic curvature belongs to the surface itself. Flattening and rolling that preserve intrinsic curvature cannot fool a measurer living on the surface.

17.4 Tangent Planes and Tangent Vectors: Finding Your Footing

Before an ant can crawl on a sphere, it must ask what direction means.

On a plane, direction is an arrow, a vector. But where does an arrow on a sphere point? To lie on the sphere, it must curve with the surface, no longer resembling a familiar vector. Instead, place an infinitely thin flat board at a point P , touching the sphere only at P and suspended elsewhere. This is the **tangent plane** at P .



The tangent at P : a planar analogue of a tangent plane at P

Definition 17.2. A surface's **tangent plane** at P touches the surface at P and contains all its directions at that point. A vector in this plane starting at P is a **tangent vector** to the surface at P .

Tangent vectors solve a real difficulty: they make taking a small step in a particular direction precise. The step itself curves along the surface, but when infinitesimal it differs negligibly from a straight segment in the tangent plane: the error shrinks faster than the step length. This is the role of "differential": infinitesimally locally, a curved surface and its tangent plane are the same.

You live inside this approximation every day. Earth has radius about 6371 kilometres and the ground beneath you is spherical, yet across a few dozen metres its deviation from a tangent

plane is only millimetres. Building houses and playing football can treat it as flat. Locally planar, globally curved: remember this phrase, which will return in a more important role.

Terminology Card: Tangent plane

Intuition: Gently touch a basketball with a glass plate at one point. The plate lies in the tangent plane. Touch elsewhere and its orientation changes.

Formal definition: The tangent plane at a surface point is spanned by the tangent directions of all smooth curves through that point.

Example: The small patch of ground where you stand approximates Earth's tangent plane beneath your feet; a spirit level measures it.

17.5 Curvature: Measuring How Much Things Bend

With tangent planes, we can quantify bending. The idea is direct: the faster surface points depart from the tangent plane, the more strongly the surface bends there.

Start with a one-dimensional analogy. A circle of radius r bends equally everywhere; its curvature is defined as $1/r$. Smaller radius means sharper bending. As radius tends to infinity, curvature tends to zero and the circle becomes a straight line, whose curvature is zero. For $r = 2$, curvature is $1/2$; for $r = 100$, it is just 0.01 , hardly noticeable over a short walk.

Surfaces are more complicated: at one point, different directions may bend differently. A sphere bends equally in every direction. For radius R , every directional cross-section is a great circle of radius R , with curvature $1/R$. A cylinder differs: across its circular direction the section is a circle of radius R , curvature $1/R$; along a generator, the vertical direction, it is a straight line with zero curvature. Multiplying the curvatures in the two extreme directions gives **Gaussian curvature**. Thus:

$$\text{Sphere's Gaussian curvature} = \frac{1}{R} \cdot \frac{1}{R} = \frac{1}{R^2}, \quad \text{Cylinder's Gaussian curvature} = \frac{1}{R} \cdot 0 = 0.$$

At first this seems wrong: a cylinder visibly bends, so why is its Gaussian curvature zero? Recall the orange peel. A cylinder is made by rolling paper without stretching. An ant living on it makes the same measurements as an ant on a plane: its intrinsic geometry is planar. Gaussian curvature measures intrinsic bending, not external appearance. Conversely, a sphere's everywhere-positive $1/R^2$ quantifies why orange peel cannot lie flat. Earth's radius is about 6371 kilometres, giving surface Gaussian curvature $1/(6.371 \times 10^6)^2 \approx 2.5 \times 10^{-14}$ per square metre. Tiny but positive, it accumulates to prevent any perfect world map.

A third possibility is a saddle. At its centre, the surface bends downwards fore and aft and upwards side to side. Opposite signs multiply to give negative Gaussian curvature. What happens when such a surface is flattened? Unlike the sphere's shortage of material, which causes splitting, it has too much material, producing wrinkles and overlaps like a wood-ear mushroom. Three signs, three fates: positive curvature splits, zero lies flat, negative wrinkles.

17.6 Spherical Triangles: The Angle Sum Rebels

Among our earliest geometry theorems, one is etched into memory: a triangle's angles sum to 180° . Now we will watch it fail on a sphere, in a remarkably orderly way.

First define a spherical triangle. Great circles play the role of straight lines: intersections of the sphere with planes through its centre. The equator and complete meridian circles qualify; other latitude circles do not, since their planes miss the centre.

Definition 17.3. Choose three points on a sphere not lying on one great circle and join them pairwise with three great-circle arcs. The enclosed figure is a **spherical triangle**. Its interior angles are the angles between the tangent lines to the meeting arcs at each vertex.

Consider an extreme, beautiful example. Let A be the North Pole and choose equatorial points B and C with longitude difference 90° . Sides AB and AC follow meridians, while BC follows the equator. At A , the meridians meet at their longitude difference, 90° . At B , meridian and equator are perpendicular, giving 90° ; likewise at C , 90° . All three angles are right angles, summing to 270° , fully 90° more than on a plane!

This excess is not error, but area. Early in the nineteenth century, a mathematician proved one of spherical geometry's loveliest formulas:

Theorem 17.4 (Area of a spherical triangle). *On a sphere of radius R , a spherical triangle with interior angles A , B , C measured in radians and area S satisfies*

$$S = R^2 (A + B + C - \pi).$$

In particular, its angle sum exceeds π , or 180° , and the excess $A + B + C - \pi$ is proportional to its area.

Proof. The proof has three steps, each visible in a diagram.

Step 1: Area of a lune. Two intersecting great circles divide the sphere into four crescent-shaped regions called lunes. A lune of angle A radians occupies the fraction $A/(2\pi)$ of the sphere: imagine slicing a watermelon by longitude through 2π radians and taking A radians. Since the whole sphere has area $4\pi R^2$, the lune of angle A has area

$$4\pi R^2 \cdot \frac{A}{2\pi} = 2R^2 A.$$

Check: at $A = \pi$, the lune is a hemisphere, and the formula gives the correct area $2\pi R^2$.

Step 2: Three pairs of lunes cover the sphere. Let triangle T have angles A , B , and C . The great circles containing its two sides at A enclose a pair of opposite lunes of angle A , each with area $2R^2 A$. Likewise obtain pairs for B and C . These six lunes cover the sphere with overlaps: T is covered 3 times, and the congruent triangle antipodal to T is also covered 3 times, while every other point is covered exactly 1 time.

Step 3: Write the equation. The total area of the six lunes equals the sphere's area plus 4 extra copies of T : 2 extra copies of T and 2 of its antipodal partner:

$$2(2R^2 A + 2R^2 B + 2R^2 C) = 4\pi R^2 + 4S.$$

Divide by $4R^2$ and rearrange:

$$S = R^2(A + B + C - \pi).$$

This completes the proof. $\square$

Check our example: the polar-equatorial triangle has $A = B = C = \pi/2$, sum $3\pi/2$, so $S = R^2(3\pi/2 - \pi) = \pi R^2/2$. Directly, it occupies one-eighth of the globe: meridians differing by 90° select a quarter, then taking the northern half halves that. Its area is $\frac{1}{8} \cdot 4\pi R^2 = \pi R^2/2$, exactly matching.

Corollary 17.5. *No spherical triangle has angle sum exactly 180° . The greater its excess over 180° , the larger the triangle. As a triangle shrinks, the excess tends to zero: locally, spherical geometry approaches plane geometry.*

This locks in our earlier phrase: locally planar, globally curved. Inhabitants measure curvature through the amount by which angle sums betray the planar rule.

Connections

Recall the planar proof that triangle angles sum to 180° : it uses the parallel postulate, giving exactly one parallel through an external point. Any two great circles intersect, so spheres have no parallel lines. The postulate fails and the angle-sum theorem changes. No proof went wrong; the geometric stage changed. Historically, two thousand years of doubt and debate about the parallel postulate led mathematicians towards curved-space geometry. Spherical geometry is one legacy of that internal struggle. Next chapter, another familiar certainty—the precise number from a measurement—will likewise change under randomness.

17.7 Geodesics: Straightest and Shortest

Return to the opening question: what does walking straight on a surface mean? With no compass, an ant understands it as never turning left or right, simply crawling in the direction it faces. Such a route is a **geodesic**.

Terminology Card: Geodesic

Intuition: A route that does not turn on the surface. Stretch a rubber band between two surface points; its resting path is a geodesic.

Formal description: A geodesic is locally shortest: between any sufficiently close pair of its points, no shorter surface route exists. Equivalently, it proceeds straight along the surface with zero sideways acceleration.

Example: On a sphere, geodesics are great circles. The shortest Beijing–San Francisco route is the shorter arc of the great circle through both cities and Earth’s centre.

Why do straightest and shortest coincide? If a route has a turn, cutting the corner shortens it, so a shortest route cannot turn anywhere. Conversely, every small segment of a route that never turns is shortest. On a plane, geodesics are straight lines; on a sphere, great circles. Cut

and flatten a cylinder, and its geodesics become straight segments; roll it back and one may become an ascending helix. A helix can therefore be the straightest route, a startling idea at first hearing.

Calculate the saving. Beijing is roughly 40° north, 116° east; New York roughly 41° north, 74° west. Their latitudes nearly match and longitudes differ by about 170° . Flying along latitude, approximately straight on a flat map, follows the 40° north circle of radius $R \cos 40^\circ$, with arc length

$$\frac{170}{360} \times 2\pi R \cos 40^\circ \approx 0.472 \times 2\pi \times 6371 \times 0.766 \approx 14500 \text{ km.}$$

The actual great-circle route is about 11000 kilometres, saving roughly three thousand five hundred kilometres, more than a Beijing–Urumqi round trip. The curved arc on the map is the sphere’s true straight line.

One-Minute Experiment

Find an orange or table-tennis ball, a marker, soft string, and a protractor, or estimate angles if none is available.

Step 1: Mark three points on the ball. Pull the string taut against the surface between pairs and draw three straight routes along it. These are great-circle arcs, or geodesics.

Step 2: Observe the triangle’s angles, estimate or measure each, and add them. Larger triangles exceed 180° by more; very small ones approach 180° .

Step 3, for comparison: draw a triangle on paper, roll the paper into a cylinder, then unroll it and measure. The angle sum stays unchanged. Why did the spherical triangle change while the paper triangle did not?

17.8 Manifolds: Locally Flat, Globally Full of Possibilities

We have repeatedly used one pattern: locally planar, globally perhaps very different. Mathematicians made that pattern itself an object of study.

Definition 17.6. A space is a **two-dimensional manifold** if a small neighbourhood of every point corresponds smoothly and one-to-one to a planar disc, so it can be mapped by a chart. Replacing the plane by flat n -dimensional space gives an n -dimensional manifold.

Terminology Card: Manifold

Intuition: A manifold is a space coverable by an atlas. Wherever you stand, your surroundings seem flat; only travelling around the world reveals its different global shape.

Formal definition: Every point has a neighbourhood smoothly equivalent to an open disc in flat space, with sufficiently smooth coordinate changes between charts.

Example: A sphere is a two-dimensional manifold, as is a doughnut’s surface, the torus. Two crossing planes are not: no neighbourhood of their intersection can be flattened into a disc.

The torus deserves more discussion. Its outside is sphere-like with positive curvature; its inside saddle-like with negative curvature; globally it is a closed manifold without boundary. An ant drawing triangles finds some with sums above 180° and others below 180° , as curvature changes sign with location. More intriguingly, identifying both pairs of opposite sides of a rectangular paper sheet—first making a cylinder, then joining its ends—also gives a torus, yet this paper torus is intrinsically flat. It supports geometry with triangle angles summing exactly to 180° . One topological shape can carry different geometries. Distinguishing shape from geometry is a starting point of modern geometry.

Surface	Triangle angle sum	Geodesics	Flattens without stretching?
Plane	Exactly 180°	Straight lines	Already flat
Sphere	Above 180° ; greater excess for larger triangles	Great-circle arcs Straight when unrolled;	No: it splits
Cylinder	Exactly 180°	possibly helices	Yes
Saddle	Below 180°	Outward-diverging curves	No: it wrinkles

Why lavish such attention on manifolds? Because we ourselves inhabit one. Looking at the universe, we directly experience only a small, approximately flat neighbourhood. Whether the whole is closed or open, flat or curved cannot be seen at a glance. It must be inferred gradually through internal measurements: triangle angles, convergence or divergence of parallel light beams, and counts of distant galaxies. The ant's method for studying a sphere is the astronomer's method for studying the universe.

17.9 From Local Geometry to the Stars: A Popular-Science Glimpse

In 1854, a mathematician's lecture systematically introduced curved spaces of arbitrary dimension: curvature can vary from point to point, lengths and angles depend on a varying metric, and geodesics replace straight lines. At the time, these tools were pure intellectual play, with no known use.

Sixty-one years later, in 1915, Einstein recognised exactly the language he needed. In popular-science terms, general relativity says that what we call gravity is spacetime curvature. Matter and energy curve spacetime, and freely falling objects follow its geodesics. Planets orbit not because invisible ropes pull them, but because they travel straight through curved spacetime. During the 1919 solar eclipse, astronomers observed distant starlight deflected near the Sun by an amount matching the prediction: light followed spacetime geodesics, and the Sun curved that spacetime.

We must honestly mark the boundary: this is a popular analogy. General relativity treats four-dimensional spacetime, three spatial dimensions and one temporal, with geometry inter-

twined with the passage of time and described by precise equations. A picture of a heavy ball on a rubber sheet cannot capture it. This chapter offers an entrance: curved spaces admit intrinsic measurement, geodesics replace straight lines, and curvature replaces gravity. These are ideas you can now explore directly on a sphere.

Keep Climbing

Further signposts: Gaussian curvature and the remarkable theorem; covariant derivatives and parallel transport, describing vectors carried along curves; the Gauss–Bonnet theorem, linking total curvature to hole count and bridging geometry and topology; Riemannian geometry and metric tensors; and eventually formal courses in differential geometry and general relativity. After multivariable calculus and linear algebra, each of these keywords will become accessible.

Mathematical Toolbox

New tools from this chapter:

- **Charts and atlases:** Cover a surface with planar coordinate patches and convert on overlaps; geometry need not flatten the surface.
- **Tangent planes and vectors:** All directions at a point form its tangent plane; infinitesimally, plane and surface are indistinguishable.
- **Gaussian curvature:** The product of extreme directional curvatures measures intrinsic bending: positive splits when flattened, zero lies flat, negative wrinkles.
- **Spherical triangle area** $S = R^2(A + B + C - \pi)$: Angle excess is area; small triangles approximate planar ones.
- **Geodesics:** Non-turning, locally shortest surface routes; on spheres, great circles.
- **Manifolds:** Spaces locally resembling flat space but globally unrestricted in shape; our own universe very likely is one.

Questions to Think About

Observation

1. On a globe or orange, choose the North Pole and two equatorial points differing in longitude by 120° . Join them into a spherical triangle, inspect its angles, and add them.

Hint: the three angles are equal. First consider the angle between the meridians and the angles between meridians and equator, then check with the area formula.

2. Draw a triangle on the side of a cylindrical cup, then flatten the paper surface, actually or mentally.

Hint: observe what rolling and unrolling preserve: lengths, angles, and angle sum. What does this reveal about intrinsic cylindrical geometry?

Proof

1. Prove that no nondegenerate spherical triangle has angle sum 180° .

Hint: apply $S = R^2(A + B + C - \pi)$ directly, noting that area S is positive.

2. Prove that any two great circles intersect at exactly two antipodal points, the ends of a diameter.

Hint: a great circle is the intersection of a sphere with a plane through its centre; two such planes intersect in a line through the centre.

Challenge

1. Can two spherical triangles have pairwise equal corresponding sides but different angle sums? In the plane, side–side–side guarantees congruence. Does it on a sphere?

Hint: link angle sum and area using the area formula, then consider whether spherical triangle area has an upper bound.

2. Some regions of a doughnut's surface give triangle sums above 180° , others below 180° . Yet a completely flat torus geometry exists by identifying opposite rectangle edges. What different winding patterns can its geodesics have?

Hint: recover the rectangle. Geodesics are straight within it; edge identification means leaving one side and re-entering through its opposite.

The Door to the Next Chapter

We have learnt to measure distances and directions in curved worlds. Notice, though, an assumption quietly passed over: measurements are exact. A taut string gives the arc length; the protractor reading gives the angle.

Real measurements are less obedient. A mountain's height is 8848.86 metres today and 8848.83 with another instrument tomorrow. Navigation satellites place you a few metres differently each time. Even measurements of constants such as light speed fluctuate in their decimal places. Among numbers all plausible yet slightly different, which is the truth? Does truth itself need a new definition?

From differential geometry to general relativity, humanity learnt to describe curved but deterministic spaces. Next comes a more everyday kind of bending: uncertainty itself. Let us enter probability and statistics.

Chapter 18 Probability and Statistics:

Studying Uncertainty Precisely

18.1 A Birthday Bet

At the start of term, several new pupils join a class, bringing its total to thirty. During break, someone suggests a game: everyone announces their birthday, month and day only, to see whether any two match. Class representative Xiaolin confidently says, “There are three hundred and sixty-five days and only thirty of us. A match is far too unlikely. I bet there won’t be one!” His deskmate Xiaoyu smiles: “I’d bet there probably will. A bottle of fizzy drink?”

The pupils start announcing birthdays. Somewhere in the teens, laughter erupts: March 14 has appeared twice. Later, November 2 also repeats. Astonished, Xiaolin pays for the drink, muttering, “Pure luck, surely?”

Was it only luck? Among thirty people, is a shared birthday a rare coincidence or something we should expect? Intuitively, almost everyone sides with Xiaolin: three hundred and sixty-five versus thirty seems an enormous mismatch. This chapter will show that in such problems human intuition nearly always loses and mathematics wins. We can turn luck into a precise number, and show you the calculation.

18.2 Intuition Gets It Spectacularly Wrong

Before seeing the answer, guess the probability that at least two pupils in a class of thirty share a birthday. One per cent? Ten? Most guesses fall below twenty per cent.

Now calculate. Counting at least one match directly is awkward: two may match, three may match, several separate pairs may match. Mathematicians often respond to “at least” by reversing the question: calculate the probability that every birthday differs, then subtract from 1.

Assume 365 days, with each person’s birthday equally likely on any day, ignoring leap years. The first person may occupy any day. The second differs with probability $\frac{364}{365}$; the third differs from both with probability $\frac{363}{365}$; and so on. All thirty birthdays differ with probability

$$\frac{365}{365} \times \frac{364}{365} \times \frac{363}{365} \times \cdots \times \frac{336}{365}.$$

This product is approximately 0.294, so the probability of at least one shared birthday is

$$1 - 0.294 = 0.706,$$

over seventy per cent! With twenty-three people it already exceeds half; with fifty it reaches about ninety-seven per cent. Xiaolin was likely to lose from the outset.

Why does intuition fail so badly? We instinctively compare thirty with three hundred and sixty-five, forgetting that the relevant comparison involves pairs of people, not individuals. How many pairs can thirty people form? The first pairs with twenty-nine others, the second with twenty-eight remaining others, and so on, giving

$$\frac{30 \times 29}{2} = 435$$

pairs. Each of four hundred and thirty-five pairs offers a possible match, quite unlike the impression of thirty versus three hundred and sixty-five. Intuition watches individuals; mathematics watches their relationships. That is why it wins.

This famous phenomenon is called the birthday paradox. It is no logical contradiction, but a theorem sharply conflicting with intuition. In mathematics, the word paradox often warns that our intuition needs upgrading.

18.3 Giving Possibility a Number

We have used probability throughout the calculation, but what is it? Mathematicians struggled with this question for centuries.

Return to a simpler scene: what is the chance a fair coin lands heads? One-half, you immediately answer. Why? There are two symmetric outcomes, with no reason to favour either. A fair die showing six occupies one of six equally likely outcomes, giving $\frac{1}{6}$. This is the classical probability model: with equally likely elementary outcomes,

$$\text{Probability of an event} = \frac{\text{Number of outcomes in the event}}{\text{Total number of outcomes}}.$$

The birthday calculation used this formula, with so many outcomes that counting required the multiplication principle.

Much of life cannot be counted this way. What is the chance of rain tomorrow, or of a particular footballer scoring a penalty? There are no finite collections of equally likely outcomes, and tomorrow cannot be repeated. Try another approach: frequency. Repeat the same experiment many times and count how often the target event appears. Use

$$\text{Frequency} = \frac{\text{Number of occurrences}}{\text{Number of trials}}$$

as an approximation. Ten coin tosses might give seven heads, frequency 0.7; a thousand tosses usually put it near 0.5; a hundred thousand almost certainly very close to 0.5. Experience suggests that increasing trial counts stabilise the frequency near some number: the probability we seek.

Definition 18.1. In a random experiment, the **probability** of event A , denoted $P(A)$, is a number such that the frequency of A stabilises near $P(A)$ when the experiment is repeated independently many times under identical conditions. Probability obeys three basic rules: $0 \leq P(A) \leq 1$ for every event A ; a certain event has probability 1; and mutually exclusive events A and B satisfy $P(A \text{ or } B) = P(A) + P(B)$.

Terminology Card: Frequency and probability

Intuition: Frequency records how often something has happened; probability describes where that frequency settles in the long run. Formal definition: Frequency is a ratio counted from experiments and fluctuates between runs; probability is the number approached as the trial count tends to infinity. Example: In ten thousand fair-coin tosses, 5031 heads give frequency 0.5031, while the probability is 0.5.

The third rule gives a useful corollary: $P(A \text{ does not occur}) = 1 - P(A)$. This reverse calculation spared us complicated casework in the birthday problem. Do not underestimate it: replacing a direct calculation by its complement is one of probability's greatest labour-saving ideas.

Emphasise the words independently repeated. A coin remembers nothing of its previous toss. Unremarkable as that sounds, misunderstanding it can cost real money in casinos.

18.4 After Learning Part of the Story

Tighten the question. A bag contains 3 red balls and 2 white ones. Draw one without replacement, then another. What is the chance the second is red? It depends on the first draw: changing information changes possibilities. Probability given that something has already happened is conditional probability, written $P(A | B)$ and read as the probability of A given B .

Definition 18.2. For $P(B) > 0$, define

$$P(A | B) = \frac{P(A \text{ and } B)}{P(B)}.$$

Why this definition? Repeat the experiment N times, with N large. Event B occurs about $N P(B)$ times. Among those $N P(B)$ cases, the cases where A also occurs are exactly the occurrences of A and B , approximately $N P(A \text{ and } B)$. In the world where B has occurred, the frequency of A is the ratio. The definition simply puts ordinary language into symbols.

If B does not change the chance of A at all, so $P(A | B) = P(A)$, then A and B are independent. Substituting the definition, this is equivalent to

$$P(A \text{ and } B) = P(A) P(B).$$

In two successive coin tosses, the first result does not affect the second. Heads on the first and heads on the second are independent, giving probability $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ for both. Draws without replacement are not independent.

Conditional probability's most striking application is this theorem.

Theorem 18.3 (Bayes' formula). *Let events $B_1, B_2, \dots, B_n$ partition all outcomes without overlap or omission, with every $P(B_i) > 0$. For any event A ,*

$$P(B_i | A) = \frac{P(A | B_i) P(B_i)}{P(A | B_1)P(B_1) + P(A | B_2)P(B_2) + \dots + P(A | B_n)P(B_n)}.$$

Proof. Two steps. First, identify the denominator. Whenever A occurs, it occurs together with some B_j , since the B_j cover all possibilities. The events A and B_j are mutually exclusive across different indices, so

$$P(A) = P(A \text{ and } B_1) + P(A \text{ and } B_2) + \dots + P(A \text{ and } B_n).$$

The definition gives $P(A \text{ and } B_j) = P(A | B_j)P(B_j)$; substitution produces the denominator. Second, examine the numerator. The same definition gives

$$P(A \text{ and } B_i) = P(A | B_i)P(B_i) = P(B_i | A)P(A).$$

Divide the final equality by $P(A)$, then replace $P(A)$ by the first step's expansion. The formula follows. $\square$

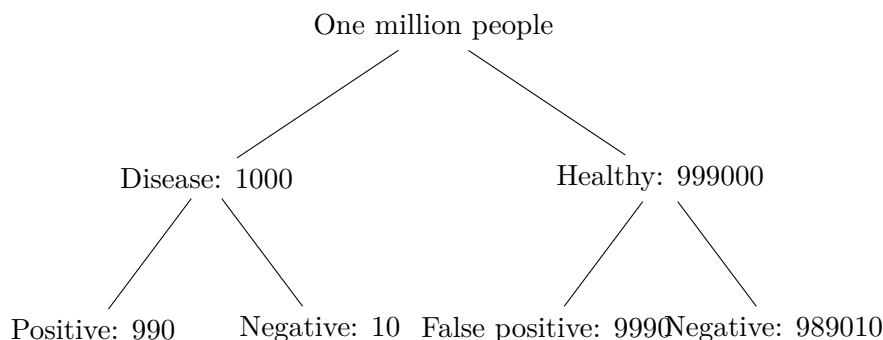
The intimidating formula becomes clear through a realistic example. Suppose a disease affects 0.1% of a population, one in a thousand. A sensitive test returns positive for 99% of affected people and negative for 99% of healthy people, meaning a 1% false-positive rate among the healthy. Someone tests positive. What is the probability they really have the disease?

Intuition shouts ninety-nine per cent: the test is so accurate! Calculate instead. Imagine a city of a million people: about 1000 have the disease, of whom about 990 test positive. About 999000 are healthy; 1%, or 9990, falsely test positive. There are approximately $990 + 9990 = 10980$ positive reports, but only 990 genuinely affected people. Thus

$$P(\text{disease} | \text{positive}) = \frac{990}{10980} \approx 0.09,$$

only about nine per cent! Bayes' formula gives

$$\frac{0.99 \times 0.001}{0.99 \times 0.001 + 0.01 \times 0.999} \approx 0.09.$$



The diagram reveals the secret: the false-positive fraction is small, but the healthy population is so large that false positives outnumber genuine cases ten to one. Intuition sees only 99% test

accuracy and forgets the prior information, prevalence. Bayes' entire spirit fits one sentence: **combine new evidence with the old background**. Prior probability $P(B_i)$ is the background, test result A is new evidence, and $P(B_i | A)$ is the updated judgement. Medical screening, spam filtering, and search-engine spelling correction share this idea.

18.5 What Can an Average Tell Us?

Probability asks whether something happens; next ask what happens on average. A prize box contains 100 tickets: 90 say "Thanks for playing", prize 0 yuan; 9 pay 10 yuan; 1 pays 100 yuan. How much does one draw yield on average?

Average does not mean a single result, which can only be 0, 10, or 100. It means the total winnings across, say, ten thousand draws with replacement, divided by the number of draws. Probabilities predict about 9000 prizes of 0 yuan, 900 of 10, and 100 of 100, totalling roughly $9000 \times 0 + 900 \times 10 + 100 \times 100 = 19000$ yuan, or 1.9 per draw. In symbols,

$$E = 0 \times 0.9 + 10 \times 0.09 + 100 \times 0.01 = 1.9 \text{ (yuan)}.$$

Definition 18.4. If a random quantity X takes values $x_1, x_2, \dots, x_n$ with respective probabilities $p_1, p_2, \dots, p_n$, then

$$E(X) = x_1 p_1 + x_2 p_2 + \dots + x_n p_n$$

is the **mathematical expectation**, or mean, of X .

Expectation is a probability-weighted average. One immediate consequence: if tickets cost 3 yuan but expected prizes are 1.9, long-term players lose an average 1.1 yuan per game. Lotteries and casino games rest on this calculation: the house's expected gain is always positive and players' always negative. Expectation cannot predict the next game, but precisely predicts the long-term result.

The mean alone is insufficient. Two classes average 80 marks; in one, everyone scores between 78 and 82, while in the other half get full marks and half fail. These are very different classes. We need a measure of fluctuation. Take each value's distance from the expectation, square it to prevent positive and negative deviations cancelling, then average with probability weights:

$$\text{Var}(X) = (x_1 - E)^2 p_1 + (x_2 - E)^2 p_2 + \dots + (x_n - E)^2 p_n,$$

called **variance**. Large variance means spread-out, strongly fluctuating results; small variance means results cluster near the mean. The prize draw has large variance, with most leaving empty-handed and a few winning big; a die's outcome has much smaller variance. Investment's phrase "high risk, high return" translates mathematically as large expectation together with large variance.

Counterexample Alert

"The expected prize is 1.9 yuan, so one ticket should give me about two yuan." Wrong: one draw pays only 0, 10, or 100, never 1.9. Expectation describes the long-run average, not the next result. Treating it as a single-trial prediction is a common misreading.

18.6 Enough Throws Turn Disorder into Order

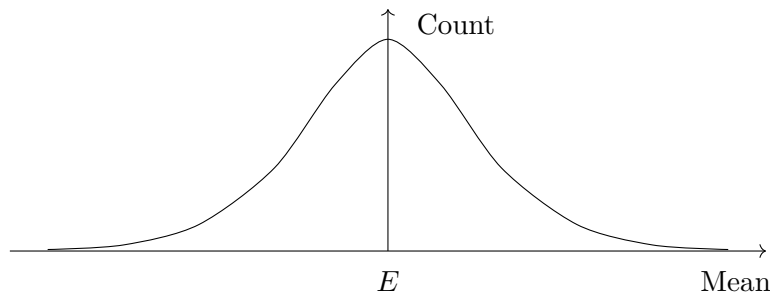
We have repeatedly said that frequency stabilises near probability. Is this a law or merely common experience? Mathematicians made it a theorem.

Theorem 18.5 (Law of large numbers, intuitive version). *Repeat a random experiment independently n times under identical conditions, each result having expectation E . As n grows, the arithmetic mean of the n results approaches E almost surely. With more trials, the probability of deviating from E by more than any prescribed small amount approaches 0.*

A full proof needs more tools, but the idea is tangible: individual fluctuations cancel when averaged. Someone's exceptionally good luck offsets another's bad luck, leaving the expectation. This underpins insurance: nobody knows whether one customer will crash next year, but with a million customers total payouts become nearly deterministic, allowing premiums to be set precisely. Casinos likewise fear not one winning gambler, but too few gamblers. The law of large numbers is on their side.

An even more remarkable fact follows. One die has a flat distribution: 1 through 6 each occurs one-sixth of the time, six equal bars with no bell shape. With two dice, the sum 7 has 6 combinations, while 2 and 12 each have only 1, giving a high centre and low ends. More dice make the sum's distribution approach a smooth symmetric bell. This is not peculiar to dice:

Theorem 18.6 (Central limit theorem, intuitive version). *When many small independent random influences add together, whatever each influence's distribution, their sum or average approaches the same bell-shaped curve, the normal distribution. More influences give a closer approximation.*



This explains why heights, measurement errors, examination marks, and component dimensions all take similar bell shapes: each combines countless small influences. The central limit theorem is called probability's crown because it explains why diverse sources of disorder produce strikingly uniform order. Its proof is beyond this book; controlling precisely how quickly the approximation improves requires Chapter 12's analytic tools.

Having described order, we must confront a stubborn misconception. Roulette produces red seven times in succession; onlookers bet black, declaring it overdue. Wrong: the wheel has no memory, and the next red or black probability is unchanged. This **gambler's fallacy** mistakes long-run balance for short-run compensation. The law of large numbers says proportions approach balance after a hundred thousand trials, not that the next ten repay a debt. In 1913, a Monte

Carlo casino reportedly produced black twenty-six times consecutively. Gamblers betting on overdue red lost piles of francs, all victims of this fallacy.

Counterexample Alert

"After five heads, tails is more likely next." Wrong: independent tosses leave the sixth-toss tail probability at $\frac{1}{2}$. What is rare is the entire six-head sequence, not a particular next step. Confusing a whole history's probability with the next outcome's probability drives the gambler's fallacy.

Connections

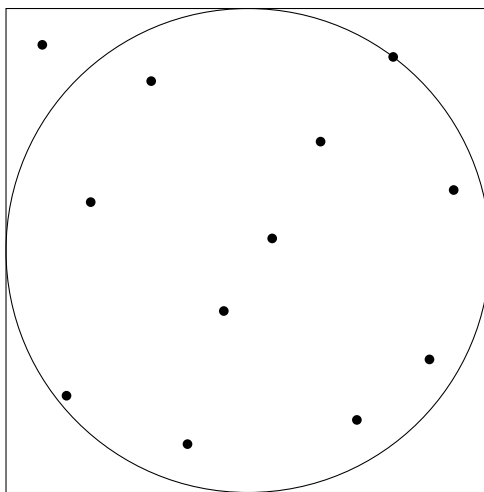
Chapter 9's limits return in new clothes: the law of large numbers says that the sequence of averages converges to the expectation, with a convergence notion allowing randomness. The central limit theorem's bell curve has equation $y = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$, containing both π and e , one from circles and one from growth and compound interest. Circles, growth, and randomness meet in one formula. Mathematics has more hidden passageways than you might expect.

18.7 Finding Pi with Scattered Points

So far, known probabilities predicted frequencies. Reverse that: use frequencies to calculate a deterministic constant seemingly unrelated to randomness. This is the Monte Carlo method, named after the casino city because random numbers fuel it.

Take a square sheet of side 2 and draw an inscribed circle of radius 1. The square's area is 4, the circle's π . Scatter points randomly into the square, perhaps sesame seeds or blind pencil marks. Suppose N points are placed and M fall inside the circle. If every position in the square is equally likely,

$$\frac{M}{N} \approx \frac{\text{Circle area}}{\text{Square area}} = \frac{\pi}{4}, \quad \text{so} \quad \pi \approx \frac{4M}{N}.$$



$$\text{Inside points} / \text{total points} \approx \pi/4$$

The method is charmingly crude: a few hundred points might give $\pi \approx 3.1$; hundreds of thousands, scattered by computer, can reach two or three decimal places. Accuracy improves slowly:

error is roughly inversely proportional to $\sqrt{N}$, so tenfold improvement needs a hundred times as many points. It is not the fastest way to estimate π . Its real strength is measuring bizarrely shaped regions with no area formula, or volumes in high dimensions. Simulating particles passing through matter, pricing complex financial products, and rendering light in computer graphics all share the trick: **probe otherwise inaccessible quantities through random experiments**. Chapter 11's integration returns unexpectedly: Monte Carlo is essentially integrating by throwing dice.

An older version is Buffon's needle. Randomly drop needles of length l onto parallel lines spaced d apart, with $l \leq d$. In the eighteenth century, Buffon proved that the probability of crossing a line is exactly $\frac{2l}{\pi d}$. Repeatedly drop needles and measure the crossing frequency f to infer

$$\pi \approx \frac{2l}{fd}.$$

A needle, paper, and a quiet afternoon let pi emerge from random noise. The proof uses integration: both position and angle are random, and weighting all positions by their chances turns the crossing condition into an integrable region. Details belong to further climbing.

One-Minute Experiment

Estimating π by scattering points at home. Draw a 10×10 square on squared paper. Using one corner as centre and the side length as radius, draw a quarter-circle with compasses. Close your eyes and make random pencil marks, opening them after each to record whether the point lies inside or outside the quarter-circle; redo marks outside the paper. After 100 marks, with M inside, estimate π by $\frac{4M}{100}$. Combine classmates' results: pool five people's 500 points and compare accuracy before and after pooling. Watch the law of large numbers at work: more points give a steadier estimate.

Mathematical Toolbox

New tools from this chapter:

- **Complementary calculation:** For at least one occurrence, calculate none and subtract from 1.
- **Conditional probability and Bayes' formula:** $P(A | B) = \frac{P(A \text{ and } B)}{P(B)}$; combine new evidence with prior background.
- **Checking independence:** Require $P(A \text{ and } B) = P(A)P(B)$, not an intuitive impression.
- **Expectation and variance:** $E(X)$ describes long-term averages; $\text{Var}(X)$ describes fluctuation.
- **Law of large numbers:** With enough trials, averages approach expectations; this concerns the long run, not the next game.
- **Monte Carlo methods:** Approximate areas, constants, and integrals through random-trial frequencies.

18.8 When Statistics Starts to Mislead

Learning the tools also means recognising traps. Beyond birthdays and the gambler's fallacy, several mechanisms particularly catch the half-informed.

First is survivorship bias. During the Second World War, statisticians studied bullet holes in returning bombers. Wings had the most holes, so officers proposed reinforcing them. Mathematicians instead recommended the engines, which showed few holes: aircraft hit there never returned. The visible sample contained only survivors and was inherently incomplete. Arguments that education is useless, citing a few wealthy dropouts, and success courses interviewing only surviving companies make the same mistake.

Second is the illusion of averages. A company's average annual salary of five hundred thousand yuan might combine a boss earning fifty million with everyone else near fifty thousand. Expectations are sensitive to extremes; the median, the middle ranked value, may be more honest. Whenever you see an average, ask which kind, and whose values were averaged.

Third is mistaking correlation for causation. Ice-cream sales and drownings rise together: does ice cream cause drowning? Summer is the common cause. Two rising quantities may have a causal relationship in either direction, a shared third cause, or only coincidence. Statistics detects associations; establishing causality needs controlled experiments or deeper mechanisms. Every citizen in the data age needs this lesson.

Keep Climbing

Further keywords await: **normal distribution**, its exact bell-curve equation and the 3σ rule; **hypothesis testing**, judging quantitatively whether a difference is chance or a genuine effect; **rigorous laws of large numbers**, entered through Chebyshev's inequality; **Markov chains**, random processes where tomorrow depends only on today, powering web ranking; and **information theory**, measuring surprise probabilistically in bits. School courses and competitions also offer elegant probability recurrences, including the famous gambler's ruin problem.

Questions to Think About

Observation

1. Shuffle a deck, turn over cards, and record whether the first two share a suit. Repeat twenty times and estimate the frequency. Then calculate the theoretical probability and compare. Hint: the first card is arbitrary; the second's matching probability depends on how many cards of that suit remain.
2. Collect a week's household electricity usage from differences in meter readings. Calculate the mean, then consider what factor could make one day differ most from it. Is that a source of variance?

Proof

1. Prove that if A and B are independent, then not A and not B are independent too. Hint: use $P(A \text{ does not occur}) = 1 - P(A)$ to expand $P(\text{neither occurs}) = 1 - P(A) - P(B) + P(A)P(B)$, then factorise.
2. Rephrase the birthday problem: how many people make the probability of at least one match exceed 99%? Show your estimation. Hint: exact solution is unnecessary; try all-different probabilities for 40, 50, and 60 people using logarithms or successive products.

Challenge

1. The Monty Hall problem: three doors hide a car and two goats. After you choose a door, the host, who knows the answer, opens another door revealing a goat and offers a switch. What is the probability of winning by switching? Justify by conditional probability or enumeration, and explain why two remaining doors do not mean equal one-half chances. Hint: the host does not open a random door; the action carries information.
2. Design a Monte Carlo experiment estimating $\sqrt{2}$ using only scattered points and area comparisons. Hint: $\sqrt{2}$ is a side or diagonal length of a simple figure; alternatively consider areas near $y = x^2$.

The Door to the Next Chapter

At several crucial moments we said to scatter hundreds of thousands of points, toss a hundred thousand times, or repeat an enormous number of trials. Mathematics supplies formulas, but who performs those dreary repetitions? Hands cannot keep up and minds make mistakes. One machine, however, loves this work: it can throw billions of dice per second without tiring or complaining.

Monte Carlo truly changed the world after electronic computers arrived. Among the earliest tasks for the first general-purpose electronic computer was simulating neutrons travelling through matter with random numbers. New questions immediately followed: if a computer's random numbers come from deterministic formulas, are they truly random? Which problems can it simulate in an instant, and which remain unfinished after ten thousand years? Next chapter opens the door between mathematics and computing.

Chapter 19 Mathematics and Computers: How Far Can Algorithms Go?

19.1 After Ten Million Successes

Start with a game. Write any positive integer and repeatedly apply this rule: if even, divide by 2; if odd, multiply by 3 and add 1. Starting with 6 gives

$$6 \rightarrow 3 \rightarrow 10 \rightarrow 5 \rightarrow 16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1.$$

After reaching 1, the number 1 becomes 4, then 4 becomes 2, and 2 returns to 1, entering the cycle $4 \rightarrow 2 \rightarrow 1$. We stop there.

The rule could scarcely be simpler, yet its question has defeated the world. Mathematicians conjecture that **every** positive starting integer reaches $4 \rightarrow 2 \rightarrow 1$ after finitely many steps. This is the famous “ $3n + 1$ conjecture”, also called the Collatz, Kakutani, or hailstone conjecture: numbers often rise and fall before descending, like hail tumbling in a cloud.

The conjecture remains neither proved nor disproved. But people tried something blunt: let computers test it. An ordinary laptop checks millions of numbers per second; specialist teams using large computers have verified every integer from 1 to about 2^{68} , approximately thirty quadrillion. **Every number** eventually reached $4 \rightarrow 2 \rightarrow 1$, with no exceptions.

Thirty quadrillion successes, no exceptions. Has the conjecture now been proved?

Many instinctively say yes: surely that many tests settle it. Others hesitate: could a counterexample hide much farther out? Starting from that hesitation, we explore mathematics’ close yet subtle relationship with computers: what can machines do for mathematics, and what can they never do?

19.2 Why Is So Much Verification Still Not Proof?

Before taking sides, consider historical stories showing how unreliable repeated success can be.

19.2.1 A Formula That Fooled Us Forty Times

Consider $n^2 - n + 41$. At $n = 1$ it gives 41, prime; $n = 2$ gives 43, prime; $n = 3$ gives 47, again prime. Continue through $n = 4, 5, 6, \dots$ and obtain 43, 47, 53, 61, 71, 83, 97, $\dots$, apparently

all prime. In fact, every value from $n = 1$ through $n = 40$ is prime: forty successes.

A computer might confidently report forty consecutive successes. But at $n = 41$,

$$41^2 - 41 + 41 = 41^2 = 41 \times 41,$$

clearly composite. Failure follows immediately after forty successes.

19.2.2 A Counterexample Beyond Nine Hundred Million

If forty seems too few, what about nine hundred million? In 1919, Polya proposed a conjecture about prime distribution. Its details need not concern us; what matters is how overwhelmingly plausible it seemed. Tests found it true for every number up to nine hundred million. At one number per second, checking that many would take over twenty years without sleep or meals.

Yet in 1958 it was proved false, with the smallest counterexample near 906 150 257, just beyond nine hundred million. All the preceding successes had failed to reach the truth's edge. The first exception hid at a place brute force could barely reach.

19.2.3 Even Great Mathematicians' Conjectures Can Sink

In 1769, Euler conjectured, roughly, that at least n numbers raised to the n th power must be added to obtain another n th power. It stood for nearly two centuries and inspired considerable trust: Euler rarely misjudged such things.

In 1966, two mathematicians used computer search to find a clean counterexample:

$$27^5 + 84^5 + 110^5 + 133^5 = 144^5.$$

Euler's conjecture required at least 5 fifth powers; this uses only 4. One equality overturned nearly two centuries of confidence. Notice that a **computer** helped find it, searching a range impossible by hand. Its role was not prover, but expert fault-finder.

Counterexample Alert

Counterexample alert: declaring success after N tests.

Whether N is a hundred, a million, or thirty quadrillion, the first N successes establish only a **finite range**. Mathematical conjectures usually concern **infinitely many** objects, and infinity cannot be exhausted. Finite tests provide **evidence** and confidence, help **discover patterns**, and may **find a counterexample** that disproves the conjecture. They never replace a proof covering every case. Conversely, one counterexample refutes a conjecture, while proving it must address infinitely many possibilities: the difficulties are inherently asymmetric.

Return to $3n+1$: thirty quadrillion checks build strong confidence, but Polya's story reminds us that an exception might lurk beyond a hundred septillion. Until proof or counterexample arrives, we must honestly call it a **conjecture**, not a theorem.

19.3 Algorithms: Machines of Thought

We keep saying to let a computer verify things. What does it actually do? Without eyes or intuition, its fundamental activity is **executing prewritten instructions step by step**. Such an instruction sequence is an algorithm.

Terminology Card: Algorithm

Intuition: An instruction manual so precise that someone, or something, with no mathematical understanding can mechanically follow it and obtain a result after finitely many steps.

Formal definition: An algorithm is a finite collection of unambiguous instructions that, for every valid kind of input, terminates after finitely many steps and outputs a definite answer. Finite termination matters: a programme that runs forever does not qualify.

Example: Find the greatest common divisor of two positive integers with the two-thousand-year-old Euclidean algorithm. For 1071 and 462, calculate $1071 = 2 \times 462 + 147$, $462 = 3 \times 147 + 21$, and $147 = 7 \times 21 + 0$. Stop at remainder 0; the answer is 21. Every step is explicit and termination is guaranteed: a textbook algorithm.

Why care so much about algorithms as a concept? They draw a precise boundary around mechanical calculation. Once defined, we can ask astonishing questions:

- Can algorithms solve **every** mathematical problem?
- If so, could a machine accept any mathematical statement and automatically declare it true or false?

The second sounds like the ultimate mathematical dream. Proof would become button-pushing and mathematicians could retire together. Early in the twentieth century, many genuinely held this hope.

19.4 Things Algorithms Can Never Do

The dream shattered in 1936. Considering the nature of computation, the British mathematician Turing asked a simple question: **can an algorithm decide whether any given programme will halt?**

What is halting? A programme either finishes after finitely many steps, like the Euclidean algorithm, or loops forever, like instructions to start at 1 and keep adding 1 without stopping. Halting depends on both code and input. Some loops are obvious; others, such as repeatedly applying $3n + 1$ from a large starting number, raise questions nobody knows how to settle.

Definition 19.1. Halting problem: Is there an algorithm H that, given any programme P and any input x , always answers correctly in finite time whether P halts on x ?

Turing's answer was earth-shattering: such an algorithm **does not exist**. Not merely undiscovered, but logically impossible, past, present, and future. This is our central theorem.

Theorem 19.2 (Undecidability of the halting problem). *No general algorithm decides whether an arbitrary programme halts on an arbitrary input.*

Proof. Argue by contradiction. Suppose algorithm H exists. Given programme text and an input, H always returns “halts” or “does not halt” after finitely many steps, and is always correct.

Use H as a component to build a strange programme D . The only input D accepts is programme text Q . The workflow of D is:

1. Call H and ask whether programme Q halts when given **its own text** as input.
2. If H says it halts, D deliberately loops forever.
3. If H says it does not halt, D halts immediately.

In short, D is contrary: **predict halting and I refuse; predict non-halting and I stop**. Constructing D presents no technical obstacle. Given an existing H , wrapping it inside D takes only a few lines of code.

Now the crucial move: feed the text of D into D itself. Does D halt when run on D ?

- Suppose $D(D)$ halts. Then H must say that D halts on itself. But by the design of D , when H says “halts”, D loops forever, so $D(D)$ does not halt: a contradiction.
- Suppose $D(D)$ does not halt. Then H must say “does not halt”. But by the design of D , when H says that, D immediately halts, so $D(D)$ halts: another contradiction.

Both possibilities contradict themselves. The original assumption must be wrong: H does not exist. □

The cleverness lies in self-reference: ask a judging programme to judge one designed to contradict it. This shares a spirit with the liar paradox, “This sentence is false”, and earlier diagonal arguments: **any system powerful enough to discuss itself risks being bitten by self-reference**.

The implications go beyond programmers lacking a universal loop detector. **Some mathematical questions have objectively definite answers—a programme either halts or does not—yet no algorithm can systematically find those answers**. Such problems are called uncomputable or undecidable. Many apparently pure mathematical problems are likewise undecidable, including deciding whether an arbitrary system of polynomial equations has integer solutions, Hilbert’s tenth problem.

Connections**Connection: the halting problem and Godel's incompleteness theorem.**

Earlier we encountered Godel's theorem: every sufficiently strong axiomatic system contains statements true but unprovable within it. This and halting undecidability are two tremors from one earthquake. If a mathematical system were complete enough to decide every statement, we could build a halting decider by submitting the statement that programme P halts. Since halting is undecidable, no complete system can exist. Computation and proof meet at the bottom of the same abyss.

19.5 The Chasm Between Fast and Slow: Computational Complexity

Halting shows that some problems have no algorithm **in principle**. More practically, among problems that **do** have algorithms, are all algorithms equally useful?

Clearly not. Compare two:

Sequential search: In a shuffled stack of n bookmarks, find the one holding a maple leaf by checking one at a time. The worst case needs n checks.

Binary search: If bookmarks are sorted by page number, inspect the middle one. If its page is too high, search the left half; if too low, the right. Each check discards half. Searching n bookmarks takes at most $\log_2 n$ checks: 1000 need only 10, a million only 20.

Different algorithms create enormous workload differences for the same question. Computer science classifies problems by the speed of their best algorithms: **computational complexity**. Its most conspicuous chasm separates two classes:

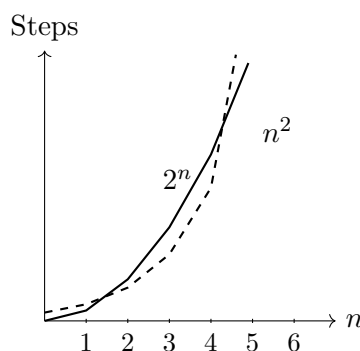
- **Easy-to-solve problems, class P:** An algorithm needs a **polynomial** number of steps in input size, such as n , n^2 , or n^3 . Sorting, greatest common divisors, and shortest paths belong here.
- **Easy-to-check problems, class NP:** An answer may be hard to **find**, but once supplied it can be **verified** in polynomial time. Sudoku is typical: filling the grid may be maddening, but checking a completed solution only requires inspecting each row, column, and box.

Every P problem is naturally NP: if we can solve it, we can check it. What about the converse? **Is every easily checked problem also easily solved?** This million-dollar P versus NP problem remains unsolved. Most experts suspect P differs from NP, meaning some problems really are easy to verify but hard to solve.

Why do polynomial and other growth differ so much? A table makes it clear. Suppose a computer performs a billion steps per second:

Input size n	n^2 steps	2^n steps	Approximate time for 2^n steps
10	100	1024	An instant
20	400	About a million	An instant
30	900	About a billion	About a second
40	1600	About a trillion	About eighteen minutes
60	3600	About a quintillion	About thirty-six years

Polynomials such as n^2 grow moderately: multiply input size a few times and runtime grows dozens of times, still manageable with effort. Exponentials such as 2^n are monsters: adding 1 to input size doubles the workload. Many important problems, such as the shortest tour through dozens of cities, currently have only exponential algorithms. Slightly larger inputs outlast the universe. Unlike the halting problem, these **have** algorithms, but they may be too slow to use. Mathematics distinguishes them from principled impossibility, grading difficulty layer by layer.



Note. This is only a schematic sketch. Both curves accelerate, but exponential growth soon leaves every polynomial far behind, though they may look similar at small n . That is the value of asymptotic thinking: examining the long-term trend.

The chasm is not just academic. Encryption used in shopping and logging in rests on easy verification and difficult inversion: multiplying large primes is easy, but splitting a hundreds-digit product is beyond even supercomputers working together. If someone proved P equals NP and produced a fast algorithm, modern cryptography's foundations would shake.

19.6 Computers Return the Favour to Mathematics

We have emphasised limitations: brute-force checks are not proofs; some problems cannot be computed, others take too long. Concluding that computers are useless to mathematics would be utterly wrong. The real picture is more interesting: they are transforming mathematics in at least three ways.

19.6.1 Mass Computation as Part of Proof: The Four-Colour Theorem

In 1852, someone asked whether four colours suffice for a map with differently coloured neighbours. This troubled mathematicians for over 120 years. In 1976, two mathematicians

announced an unprecedented proof: reduce the problem to **over a thousand** basic configurations and check them all by computer. The theorem held, but over a thousand steps were performed by machine, beyond any person's hand-checking.

The mathematical world erupted: was this a proof? Could anyone trust what no human could independently check? Debate lasted years. The mainstream view now accepts exhaustive machine calculation as part of proof if the programme is carefully audited and its logic explicit. Later versions simplified and reverified the argument. Yet the theorem permanently changed proof's meaning: **reliability shifted from every capable person being able to recheck everything to every logical link being open to scrutiny, even if only a machine can execute one of them.**

19.6.2 Machines as Proofreaders: Formal Proof

A more radical approach avoids debating whether machine-assisted proofs count by writing the **entire** proof in a precise, machine-checkable language. This is formal proof. Assistants such as Lean and Coq require every inference to be unambiguous, then check the logic line by line. Humans supply ideas; machines find mistakes.

The work is painstaking but remarkably rewarding. A famous case is the Kepler conjecture: identical balls pack most densely in the familiar greengrocer's arrangement. Intuitively obvious, it was extraordinarily difficult to prove. In 1998, Hales announced a proof involving substantial computation. After years of checking, referees could only report ninety-nine per cent confidence; the remaining doubt concerned computations too extensive to check. Hales launched a project to formalise the whole proof. In 2014, machines certified every logical step, dispelling the doubt.

Keep Climbing

Further Climbing

Look for these keywords: **Turing machines**, the mathematical model of algorithms underpinning computer science; **P versus NP**, a million-dollar Millennium Prize Problem; **Hilbert's tenth problem**, undecidability of Diophantine solvability; **Lean**, a proof assistant helping mathematicians gradually machine-check modern mathematics; and **SAT solvers**, industrial-strength search engines for verifying chip logic.

19.6.3 Machines as Telescopes: Exploring Geometry, Number Theory, and Analysis

The computer's most romantic role is telescope and microscope: helping humans **see** previously invisible mathematical worlds, while leaving proof to people.

In **geometry**, computers plotted iteration and revealed a whole new world. The Mandelbrot set arises by repeatedly applying $z \mapsto z^2 + c$ to complex numbers and colouring non-escaping points black. Its boundary contains layers of fractal detail, with fresh patterns at every enlargement, endlessly. Before computers, nobody knew such a simple formula concealed such intricate geometry. Fractal geometry now describes coastlines, clouds, and blood vessels.

In **number theory**, computers are enormous sieves. The largest known prime is a Mersenne prime with over twenty million digits, found by worldwide volunteers networking idle computers through GIMPS. Computers also verify Goldbach’s conjecture—every even number greater than 2 is a sum of two primes—over enormous ranges and record fine patterns in prime distribution. Such data inspire conjectures and sometimes supply ammunition to overturn them.

In **analysis**, numerical experiments often reveal patterns before theory. A legendary example is the BBP formula for π : computer searches first uncovered a formula extracting any chosen hexadecimal digit of π directly, with rigorous proof following later. The machine sees the pattern first; humans then understand why. This reversed order is the classic workflow of experimental science.

Mathematical Toolbox

This Chapter’s Mathematical Toolbox

- **Finite verification $\neq$ proof:** However numerous, finite checks supply evidence. One counterexample refutes a conjecture; a proof must cover infinitely many cases.
- **Algorithms:** Finite, explicit, necessarily terminating instructions, giving precision to mechanical solvability.
- **Halting problem:** Some questions are algorithmically undecidable **in principle**; self-reference and contradiction expose such barriers.
- **Computational complexity:** Among algorithmic problems, distinguish fast from slow. Polynomial growth is manageable, exponential growth monstrous; P versus NP remains a great mystery.
- **Human-machine teamwork:** Machines perform massive calculations, rigorous checks, and visual exploration; humans ask questions, invent concepts, and supply insight-dependent reasoning.

19.7 A Paper Computer Lab: Hands-On Experiments

One-Minute Experiment

Experiment 1: Run $3n + 1$ by hand.

Start with 27 and apply the rule on paper: divide evens by 2, multiply odds by 3 and add 1. Patiently count the steps to 1, around a hundred, and record the largest value. This small 27 rises as high as 9232 before falling. Try 31 or 97 too. Feel why verification is not effortless: trajectories rise and fall unpredictably, yet **so far** every tested one lands safely.

Experiment 2: Feel binary search with books.

Arrange 16 books by spine height or page number. Ask a classmate to choose one mentally. You may ask only whether it is a selected book or lies to its left or right. Always start in the middle and record the maximum questions needed. Then search from the first book sequentially and record the worst case. For 16 books, binary search needs at most 4 checks, sequential search 16. Repeat with 32: binary search adds just 1 check, while sequential search doubles. Handle logarithmic and linear growth yourself and you will not forget the difference.

Experiment 3: Sketch the lying programme.

Work only on paper. Suppose your deskmate claims a universal programme H detects whether any programme loops forever. Following the proof, describe the contrary programme D and ask whether D halts on D . Whatever the answer, identify the contradiction. This game gives tangible understanding of contradiction and self-reference.

Questions to Think About

Observation

1. Evaluate $n^2 - n + 41$ for $n = 1, 2, \dots, 8$, check primality, then calculate directly at $n = 41$.

Hint: at $n = 41$, the first two terms $41^2 - 41$ share a factor 41, so the whole expression has factor 41.

2. In the $3n + 1$ process, what must the final few values of any trajectory reaching 1 be?

Hint: which number becomes 1 after division by 2? Then work backwards once more.

Proof

3. Imitate the halting proof to show no algorithm G can accept any programme P such that G always decides correctly whether P prints the two-character Chinese greeting ni hao, meaning hello. Hint: assume G exists and build a programme that asks G whether it will print hello, refuses if told yes, and prints it if told no; then have it judge itself.

4. Prove that binary search finds any one of n ordered books within at most $\lceil \log_2 n \rceil + 1$ checks. Hint: each check at least halves the search range. How many halvings reduce n to 1?

Challenge

5. Someone announces a proof that P equals NP and a polynomial-time Sudoku algorithm. Assuming this is correct, name three aspects of everyday life that would change radically, with reasons. Hint: recall cryptography, then think further: is checking a mathematical proof difficult?

If easily verified things are easily found, what would automatic theorem proving mean?

6. The $3n + 1$ conjecture is unproved. Someone suggests that, since halting is undecidable, perhaps the $3n + 1$ conjecture is inherently undecidable too. Which parts of that reasoning are defensible, and which confuse concepts? Hint: no universal algorithm deciding **all** programmes differs from **one** particular statement being unprovable. Nonetheless, mathematicians have constructed specific number-theoretic problems equivalent to halting.

The Door to the Next Chapter

We have seen a strange picture. Computers are humanity's most powerful calculating tools, capable of checking thirty quadrillion numbers in a blink, auditing proofs longer than a lifetime's checking, and drawing unimaginable fractals. Yet insurmountable walls remain: some problems have no algorithm, some algorithms take too long, and a hundred million verifications still do not mean universally true.

Machines' power and boundaries reflect mathematics' own. Godel makes incompleteness unavoidable for sufficiently strong axiomatic systems; Turing makes undecidability an unavoidable reef for computation. Unsolved questions such as P versus NP remind us that much beneath our proud mathematical building remains unexplored.

What are those foundations? How did nineteenth-century mathematicians shore up limits, infinity, and continuity, concepts that make school mathematics feel less secure? How do deep theorems, from the fundamental theorem of calculus to Galois theory, stitch apparently unrelated branches into one map? Next we stand at mathematics' edge and look back at the whole: where are its boundaries, and where does its map lead?

Chapter 20 Mathematics' Boundaries and a Unified Map

20.1 A Map Hanging on the School Wall

Imagine your school preparing for an anniversary. The administration commissions a huge campus map for the playground wall, marking every tree and bench. The art teacher strives for perfection, then has a thought: once the map hangs on the wall, it too belongs to the campus. Should a complete campus map therefore include the map hanging on that wall?

Including it creates trouble: the little map inside the map is part of the campus too, so it needs a smaller map inside, which needs an even smaller one, endlessly. The teacher's perfectionism has led into a dead end.

This apparent brain-teaser touches twentieth-century mathematics' deepest wound: **what happens when something starts talking about itself?** A campus map also describes itself. A sentence can discuss another sentence, or itself: is "This sentence is false" true or false? A set contains things; can it contain itself?

We will follow this crack down to discoveries made over a century ago: infinity comes in different sizes, unrestricted sets can explode into contradiction, and even the hope that mathematics can prove every truth is provably impossible. Then we turn around for an equally important task: mapping this book's journeys and seeing how six lenses—number, shape, change, structure, randomness, and proof—colour the same world differently. This final chapter is both a map and a door.

20.2 Does Infinity Come in Different Sizes?

Begin with a simpler question: **is there a largest infinity?**

Most people instinctively shake their heads. Infinity is infinity; how can endlessly many things be compared? Natural numbers $1, 2, 3, \dots$ never end, neither do even numbers $2, 4, 6, \dots$, nor decimals between 0 and 1. Are they not all just infinitely many?

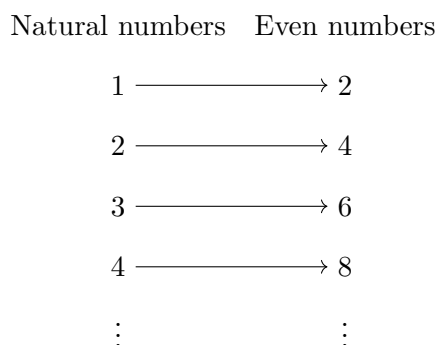
Here comes intuition's first failure. Guess which collection is larger, even numbers or natural numbers. Many immediately say natural numbers: evens are only half of them! Apparently self-evident, this is wrong because it transfers finite-set experience directly to infinity. With two piles of sweets, count each and compare the totals. When neither count ends, the word number

itself loses its familiar meaning.

Comparing infinities requires a new method.

Definition 20.1 (Equal cardinality). Sets A and B have **equal cardinality** if their elements admit a one-to-one correspondence: every element of A pairs with exactly one element of B , and every element of B is paired exactly once, with neither omissions nor repetition.

One-to-one correspondence is ancient: shepherds could tie one knot per departing sheep without knowing numerals, then use the pairing to check for losses. This definition avoids waiting until counting is finished: **it requires no completed count**.



Pair each natural number n with $2n$, as shown. Every even number is reached and every natural number has a partner. Thus the two collections have **the same size**, although evens are only part of the naturals! A part can equal the whole: infinity's first strange rule. Galileo noticed it, but it troubled mathematicians for over two centuries. Cantor finally clarified it in the late nineteenth century. Fiercely attacked by colleagues, some calling his theory a disease, he nevertheless made one-to-one correspondence the measuring stick for infinity.

He measured familiar sets. Integers, including zero and negatives, match natural numbers by listing $0, -1, 1, -2, 2, \dots$. More surprisingly, **rational numbers also match natural numbers**. Rationals densely fill the line, infinitely many in every interval: how can they be listed? Group positive fractions by numerator-plus-denominator: sum 2 gives $\frac{1}{1}$; sum 3 gives $\frac{1}{2}, \frac{2}{1}$; sum 4 gives $\frac{1}{3}, \frac{2}{2}, \frac{3}{1}$; and so on. Every fraction eventually appears and every rational receives a queue number. A set matching the naturals is **countable**: the count never ends, but everyone can queue for a number.

So far infinity seems to have one size, merely playing hide-and-seek. The real thunder comes next.

Theorem 20.2 (Cantor's diagonal theorem). *The real and natural numbers do not have equal cardinality: real numbers **cannot be listed** in a single sequence. Infinity therefore has at least two sizes.*

Proof. Suppose, for contradiction, that someone claims to list every real number between 0 and

1:

Number 1: 0.**3**1415 ...Number 2: 0.2**7**182 ...Number 3: 0.14**1**42 ...Number 4: 0.577**2**1 ...

⋮

Construct a new decimal r . Its first digit differs from the first listed number's first digit, say changing 3 to 4; its second differs from the second number's second digit, changing 7 to 8; its third changes the diagonal 1 to 2; its fourth changes 2 to 3; continue along the diagonal. The result, $r = 0.4823\dots$, is a genuine real number between 0 and 1.

Is it listed? Not first, because the first digit differs; not second, because the second differs; not n th, because its n th digit differs from that of the n th number. Every entry mismatches in a chosen position, so **the supposedly complete list omitted it**. No proposed list escapes this construction: give any list and diagonalisation produces a missing number. The assumption is false. Reals cannot be listed; there are more of them than naturals. $\square$

Terminology Card: Cardinality

Intuition: A set's size, measured by pairing rather than counting. Collections admitting a one-to-one correspondence have equal cardinality.

Formal description: The naturals' cardinality is $\aleph_0$, read "aleph-null". The larger cardinality of the reals is called the cardinality of the continuum. It can be proved that $\aleph_0$ is the smallest infinite cardinal.

Example: Evens, integers, and rationals have cardinality $\aleph_0$; reals, decimals between 0 and 1, and points on the number line have continuum cardinality.

The diagonal number r is not merely an imagined hypothetical object; its digits can be constructed. Try producing a missing number yourself.

One-Minute Experiment

Write five decimals between 0 and 1, each with five decimal places, in five rows. Choose random digits or borrow fragments of pi. For each number take its i th digit, meaning the digit at position i in row i , and add 1, wrapping 9 to 0. Assemble the five new digits into a decimal. Check that it differs from row 1 in position 1, row 2 in position 2, and so on: it is absent from the list. Repeat with new numbers. You have performed Cantor's diagonal argument, except that his list was infinite. The logic of a deliberately mismatching position is identical.

The story continues. Cantor proved that for any set A , the set of all its subsets, its power set, is strictly larger than A . There is therefore no largest infinity, only larger ones: beyond $\aleph_0$ lies the continuum, beyond that still larger sizes, without end. Is the continuum the very next

infinity after $\aleph_0$? This natural question, the continuum hypothesis, has an answer stranger than the question. We return to it later.

20.3 The Barber and the Impossible Catalogue

Different sizes of infinity are startling enough, but a deeper crack appeared in sets themselves.

Early twentieth-century mathematicians hoped sets could underpin all mathematics: numbers could be defined as sets, and figures, functions, and equations reduced to them. Secure foundations would support an eternal mathematical building. The central assumption sounded harmless: **any clearly stated condition defines the set of all things satisfying it**. Being even defines the even numbers; being red defines red things.

In 1901, Russell found a fatal condition: **not being an element of oneself**. Most sets satisfy it: the set of all apples is not itself an apple and therefore does not contain itself. Collect every such set into R :

$$R = \{ x \mid x \notin x \},$$

Does R contain itself? If $R \in R$, then R meets the condition of not containing itself, so $R \notin R$. If $R \notin R$, then R qualifies for membership in R , so $R \in R$. Both directions contradict themselves: **this set cannot even exist**.

The popular version is a village barber who shaves exactly those villagers who do not shave themselves. Does he shave himself? If yes, he belongs among self-shavers and must not; if no, he belongs among non-self-shavers and must. Such a barber cannot exist, not through poor skill but through a contradictory job description.

Counterexample Alert

"Every clearly stated condition defines a set" sounds unassailable, but Russell's paradox shows it can destroy the system. Another tempting mistake says that all infinities are alike and therefore offer nothing to study. Cantor's diagonal proof shows not only differences, but endlessly many levels of difference. **Intuition often stops working before infinity; that is why rigorous definitions are essential.**

Russell's paradox forced rules for sets' birth: specify exactly which sets may be constructed and how. These rules are **axioms**. The most widely used system, ZF or ZFC with the axiom of choice, roughly permits the empty set, unions of existing sets, power sets, and subsets selected from infinite sets by definite rules, among other constructions. Notice that subset selection permits choosing qualifying elements from a known set A , but **not** collecting all qualifying things out of nowhere. Russell's R requires precisely that unrestricted collection, so it is politely excluded and the paradox dissolves.

Terminology Card: Axiom

Intuition: The rules of a game. Instead of asking why the rules hold, ask whether following them creates contradictions and what play they allow.

Formal description: An axiom is a starting statement accepted without proof; a theorem is a conclusion derived from axioms through rules of inference.

Example: Through an external point there is exactly one line parallel to a given line: an axiom of Euclidean geometry. Change it and obtain different geometries, as in our earlier parallel-postulate story.

This changed mathematicians' view of axioms. Ancient Greeks treated them as self-evident truths; twentieth-century mathematicians treated them as game rules. The key questions became **consistency**, avoiding contradiction, and **fruitfulness**, yielding rich theorems. Euclidean geometry, set theory, and probability all establish rules before exploring their consequences.

20.4 Gödel: Mathematics Cannot Contain Every Truth

Axiomatisation inspired a grand hope: could a sufficiently strong formal system of axioms and inference rules be **consistent**, never deriving contradictions; **complete**, proving every true natural-number statement; and able to prove its own consistency mathematically? Hilbert's programme sought a permanent certificate of mathematical safety.

In 1931, the twenty-five-year-old Gödel proved two theorems refuting central parts of this plan. In very informal language:

Theorem 20.3 (Gödel's incompleteness theorems, informal version). *Every consistent axiomatic system that is sufficiently strong, at least expressing natural-number arithmetic, and has mechanically checkable rules contains statements **true but unprovable within that system**. Moreover, the system **cannot prove its own consistency**.*

We must explain the simplification and its **limits of applicability**. These are precise theorems about formal systems, with technical assumptions. Mechanically checkable rules, for instance, mean a computer can check the axiom list item by item. The conclusion is **not** that mathematics is unreliable, **not** that nothing can be trusted, and certainly not permission to believe anything unproved. It says precisely that **provability** and **truth** cannot fully coincide in any fixed sufficiently strong rule system. Some truths require stepping outside it or adding axioms.

Gödel's construction is astonishing. His encoding, now called Gödel numbering, translates formulas and proofs into natural numbers. Thus the statement that proposition P is provable becomes an arithmetic statement: the system discusses its own proofs. Echoing the liar paradox, he constructs an arithmetic statement G meaning " G is not provable in this system". If the system proves G , it proves a falsehood, contradicting consistency. Hence, assuming consistency, G is indeed unprovable; but that is exactly what G says, so G is true. That produces truth

without proof. Remember our map inside a map? Taming self-reference creates mathematics' sharpest breakthroughs.

Connections

Godel's idea of encoding symbol strings as numbers so systems can discuss themselves changed the world in another form fifty years later. Programmes are fundamentally numbers too, since code is a symbol string; programmes analyse other programmes and themselves. Turing's proof that no universal programme decides all halting questions uses the same diagonal self-reference. The continuum hypothesis belongs to this family too: mathematicians proved it **neither provable nor refutable** in ZFC. Godel's inaccessible truths are not an abstract threat: concrete important questions enter the blind spot. Infinity, self-reference, and computability, apparently unrelated at our opening, are three knots in one deep net.

What does this mean for us? Honestly, mathematics is not a final-answer stamping machine, but a territory with ever-new frontiers. Each solved question exposes new blank spaces at the map's edge. This is not failure, but evidence that mathematics remains alive.

Mathematical Toolbox

New tools from this chapter:

- **One-to-one correspondence:** Compare set sizes, even infinite ones, without counting to the end.
- **Diagonalisation:** Alter the n th feature of the n th object to create something outside a list, proving uncountability and powering self-reference.
- **Axiomatic thinking:** View theories as rules plus deduction, valuing consistency and richness rather than self-evidence.
- **A meta-level viewpoint:** Step outside a system to study it. Questions of what can be proved, computed, or completely mapped often become clear only one level up.

20.5 Six Lenses: One Map of the Whole Book

After twenty chapters, it is time to map our travels. Six overlapping, mutually illuminating lenses offer views of mathematics:

Lens	Central questions	An unresolved conjecture
Number	Counting, measurement, exact representation	Riemann hypothesis: fine prime-distribution patterns
Shape	Position, shape, distance, symmetry	Optimal sphere packing in most higher dimensions
Change	Rates, accumulation, limits	Smoothness of Navier–Stokes solutions
Structure	Relations, operations, patterns themselves	P versus NP : is checking always easier than solving?
Randomness	Regularity within uncertainty	Many questions between random processes and chaos
Proof	Grounds for belief	How far machine proof can go: an open question

Number was our starting point. From tally knots to fractions, negatives, irrationals, and imaginary numbers, every surprised "That counts as a number too?" expanded the territory. Now even infinities receive classifications and numbers.

Shape led from Pythagoras to symmetry, transformations, and non-Euclidean geometry. It is not merely drawing, but turning positional relationships into computable objects.

Change leaps from static quantities to processes: speed, slope, and area transform into one another, differentiation and integration become inverse operations, and mathematics captures motion on paper.

Structure is the most abstract and modern lens. Addition, rotation, permutation, and matrices look unrelated yet share operational patterns. Groups, graphs, and vector spaces study relations between relations.

Randomness accepts unavoidable chance, then asks what necessity hides inside it. Large numbers, expectations, and statistical inference support definite decisions under uncertainty.

Proof binds the other five lenses and runs through the book: from apparently true to why true, from inductive guesses to deductive chains. Here we pushed that thread to its end. Proof itself can be studied, and it has boundaries.

These lenses are not islands. The next question shows them meeting in one landscape.

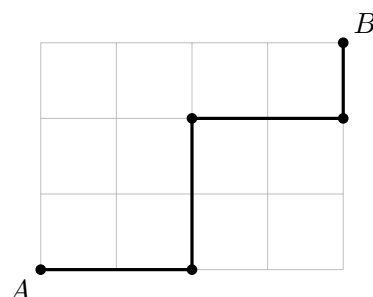
20.6 One Question, Three Paths: The Shortest Route

Consider an old puzzle. An ant must crawl from vertex A to the opposite vertex B of a cube-shaped cardboard box, staying on its surface. How long is the shortest route?

First approach, through shape: unfolding. Cut along cube edges and flatten the surface. Surface routes become planar lines, where a straight segment is shortest. For cube side 1, unfold two adjacent faces into a 2×1 rectangle. The distance from A to B is $\sqrt{2^2 + 1^2} = \sqrt{5}$. A transformation translates a difficult surface question into an easy planar one. Geometry often gains power not by harder calculation but by **changing the space we look in**.

Second approach, through structure: graphs and greed. Modify the question: a city has junctions and known road lengths; find the shortest home-to-school route. Draw junctions

as vertices and roads as weighted edges, forming a **graph**. An elegant greedy strategy keeps the currently known distances from home, repeatedly **fixing** the nearest junction, since detouring through another cannot reach it sooner, then updating its neighbours. When school is fixed, its shortest distance is known. Dijkstra's algorithm is one heart of navigation software, relying not on geometric intuition but on extracting distance structure step by step.



Grid shortest paths: several routes, equal length

Third approach, through change: let nature calculate. Modify it again. Light bends entering water from air, choosing an angle that **minimises total travel time**: Fermat's least-time principle. Deriving refraction needs no account of what light thinks; write travel time as a function of path and ask when small changes cease to reduce it. Calculus and the calculus of variations optimise among **infinitely many possible paths**, not a finite shortlist. Many physical laws, including least action, share this form.

One shortest-path idea, three spirits: geometry transforms, algorithms exploit structure, and analysis optimises among infinitely many candidates. They translate and inspire one another: navigation contains discretised geometry; variational problems contain continuous versions of greedy reasoning. A unified mathematical map does not merge every subject into one course. It recognises that **every ascent of the same mountain meets at its summit**. Prime distribution likewise belongs to number through sieves and factors, to shape through strange diagonal patterns in prime spirals, to randomness through lottery-like large-scale behaviour, and to change through the logarithmic density of the prime number theorem. Deep questions never belong to just one lens.

Keep Climbing

Further keywords: **set theory**, arithmetic of infinity; **mathematical logic**, mathematics of proof; **halting problem**, computation's boundaries; **graph theory**, shortest paths and network flows; **calculus of variations**, optimisation among infinite choices; and **Riemann hypothesis**, the million-dollar prime mystery. Each opens another book.

Questions to Think About

Observation

1. Following the correspondence between evens and naturals, construct one between odd numbers and all integers, including negatives. *Hint: list integers as $0, 1, -1, 2, -2, \dots$, then*

match them with the odd-number queue.

2. The cube can be unfolded in different ways. Draw another net, calculate its route length, and compare with $\sqrt{5}$. *Hint: locate A and B on the unfolded grid. Some unfoldings give $\sqrt{5}$, others longer distances; the shortest wins.*

Proof

3. Prove that the reals between 0 and 1 are as numerous as all reals. *Hint: find a one-to-one correspondence stretching a finite interval onto the line. Start with the graph of $y = \tan x$ and explain its one-to-one property.*

4. Use diagonal reasoning to show that there is no set of all sets. *Hint: if such a set U existed, the power set of U should be no larger than U , but Cantor proved the power set strictly larger.*

Challenge

5. The barber paradox is resolved by saying no such barber exists. Someone changes the rule: "I shave everyone who does not shave themselves, except me." Does the paradox vanish? Are there other difficulties? *Hint: the new rule is consistent, but who is omitted among non-self-shavers, and who tends the barber's beard? Compare ZF's restriction to selecting subsets from an existing set.*

6. Gödel's statement G says it is unprovable. Why not add G as an axiom and make the system complete? Why is this not a permanent solution? *Hint: the new system remains sufficiently strong and consistent if the original was, so the theorem applies again.*

The Door to the Next Chapter

This book has no next chapter. Yours is just beginning.

If the book is a map, the edges marked "Here be dragons" point towards your real destinations. Here are signposts for travellers with different tastes.

If you enjoy **competitions**, they squeeze every drop from school mathematics: combinatorics, number theory, geometry, and inequalities teach ingenious moves with minimal resources. Their essence is depth, not speed. Finding three solutions to one problem takes you farther than grinding through three hundred problems.

If you enjoy **programming**, you already hold the book's handiest laboratory. Test a hundred million Goldbach examples, simulate the law of large numbers, or implement Dijkstra in a maze solver. Mathematics gives vision and code gives brute force; together they almost feel like cheating.

If you enjoy **modelling**, the real world has been waiting: why epidemic curves bend as they do, how delivery riders should route themselves, or why a song sounds pleasing. Modelling translates tangled reality into one of our six lenses. Wrong translations are nothing to fear; errors teach better ones.

If you head towards **university mathematics**, you will meet the full versions of these stories. Calculus becomes analysis, geometry becomes topology and manifolds, randomness becomes

measure theory, and axioms, diagonals, and incompleteness become mathematical logic. Behind every informal explanation here, you will discover a genuine proof.

If you continue from **pure curiosity**, congratulations: you have the best reason. Many great advances began with a useless "What if?" Nobody knew Cantor's questions about infinity would help underpin computer science; prime-number players never imagined their games guarding the world's banking passwords.

How did the art teacher resolve the campus map? Reportedly, she drew a tiny map in its corner, placed a nearly invisible dot inside that, and stopped. Beside it she wrote: "For anything smaller, go into the campus and look for yourself."

Mathematics is like that. No map, however detailed, replaces walking. We have lit the spark; the road ahead is yours to illuminate.

Appendix A The Basic Grammar of Mathematical Proof

Learning a language begins with its grammar; reading mathematical proofs is no different. A proof is not a juggling act with symbols. It follows rules for making things clear: what makes a statement true, what refutes it, and what counts as a reason. This appendix gathers the grammar used throughout the book, with examples at middle school level. Whenever the main text leaves you stuck, return here for help.

A.1 Propositions: Statements That Can Be True or False

Mathematics discusses a special kind of sentence: a **proposition**, a statement that can be judged true or false. “The interior angles of a triangle add to 180° ” is a proposition, and it is true. “All prime numbers are odd” is also a proposition, but false: 2 is a counterexample. “Mathematics is fascinating” is not a proposition because it has no definite truth value. On its own, “ $x > 3$ ” is not one either, since we cannot decide its truth without specifying x . Quantifiers address this issue, as we will see below.

Many propositions have the form “if ... then ...”. For example: “If an integer is divisible by 4, then it is divisible by 2.” The part after “if” is the hypothesis; the part after “then” is the conclusion. Exchanging them produces the **converse**. Here the converse is: “If an integer is divisible by 2, then it is divisible by 4.” Notice that the original is true while its converse is false: 6 is divisible by 2 but not by 4.

Counterexample Alert

“If a proposition holds, so does its converse” is a common misunderstanding. “Vertically opposite angles are equal” is true, but its converse, “Equal angles are vertically opposite,” is plainly false. Corresponding angles formed by parallel lines are equal without being vertically opposite. A proposition and its converse require separate judgments.

A.2 Sufficient and Necessary Conditions: Two Different Hats

When “if p , then q ” holds, we say p is a **sufficient condition** for q : having p is enough to guarantee q . At the same time, q is a **necessary condition** for p : for p to hold, q must hold too.

Example A.1. Being a rectangle guarantees that a quadrilateral is a parallelogram, so “is a rectangle” is sufficient for “is a parallelogram.” Conversely, being a parallelogram is only necessary for being a rectangle. A rectangle must be a parallelogram, but parallel sides alone are not enough: right angles are required too.

An everyday analogy: electricity is necessary for an electric bulb to light—without it, the bulb cannot glow—but not sufficient, since a broken filament prevents it from lighting even with power. Sufficient need not mean necessary, and necessary need not mean sufficient. When both hold, p is a **necessary and sufficient condition** for q , written $p \Leftrightarrow q$. The two then stand or fall together: for example, “a triangle is equilateral” and “all three of its interior angles are 60° .”

A.3 Quantifiers: Being Clear About “All” and “There Exists”

Two of the most common expressions in propositions are “all” or “for every,” written $\forall$, and “there exists” or “at least one,” written $\exists$. They behave very differently:

- For “Every even number is divisible by 2” to be true, **every** even number must pass the test.
- For “There is an integer x such that $x^2 = 4$ ” to be true, **finding** one is enough: $x = 2$ works.
- One counterexample refutes an “all” statement. Refuting an “exists” statement requires showing that none can be found.

Negation has a clear rule too. The negation of “All ravens are black” is not “All ravens are nonblack,” but “There is a raven that is not black.” Negating a universal statement produces an existential one, and vice versa. This exchange is used constantly in proof by contradiction.

A.4 Counterexamples: One Is Enough

A counterexample is a specific case satisfying the hypothesis but not the conclusion. It is mathematics’ most economical weapon: one counterexample overturns even the most beautiful conjecture. Consider “For all integers a, b , if $a^2 > b^2$, then $a > b$.” Plausible enough—until we take $a = -3$, $b = 1$. Then $a^2 = 9 > 1 = b^2$, yet $-3 < 1$. One counterexample ends the conjecture. Conversely, checking a million examples is still not a proof. Examples show only that a claim has not yet been defeated, not that it never will be.

A.5 Mathematical Induction: Toppling a Row of Dominoes

Suppose we want to prove a proposition $P(n)$ for all positive integers n , such as

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

Checking each case would never end. Mathematical induction requires just two tasks: (1) the **base case**: verify $P(1)$; (2) the **inductive step**: prove that whenever $P(k)$ holds, $P(k+1)$ follows. Think of a row of dominoes: if the first falls and each falling domino knocks over the next, the entire row falls.

Proof. (Base case) When $n = 1$, the left side is 1 and the right side is $1^2 = 1$, so the claim holds. (Inductive step) Assume $1 + 3 + \cdots + (2k-1) = k^2$ for $n = k$. For $n = k+1$,

$$1 + 3 + \cdots + (2k-1) + (2k+1) = k^2 + (2k+1) = (k+1)^2,$$

which is exactly the required identity. By induction, the equality holds for every positive integer n . $\square$

Neither step can be omitted. A base case alone leaves open a possible break farther along; an inductive step alone may describe a row that was never pushed.

A.6 Proof by Contradiction: Assume the Opposing Claim Is Right

Some propositions resist a direct attack. Try a different approach: **assume the conclusion is false and reason onward until you reach a contradiction**. The contradiction shows that the starting assumption was wrong, so the conclusion must hold. A classic example is that there are infinitely many primes.

Proof. Suppose there are only finitely many primes, and list them all: $p_1, p_2, \dots, p_n$. Construct $N = p_1 p_2 \cdots p_n + 1$. Dividing N by any p_i leaves remainder 1, so none of the listed primes divides N . But every integer greater than 1 is either prime or divisible by a prime. Therefore a prime outside the list must exist, contradicting the claim that the list contains every prime. Thus there cannot be only finitely many primes. $\square$

This argument already appeared in Euclid's *Elements* over two thousand years ago. Proof by contradiction lets you borrow a false proposition as a ladder, then dismantle it after climbing.

Mathematical Toolbox

A quick reference to the grammar of proof:

- A proposition is a statement with a truth value. Its converse exchanges hypothesis and conclusion and must be judged independently.
- $p \Rightarrow q$: p is sufficient and q is necessary. When each is necessary and sufficient for the other, write $p \Leftrightarrow q$.
- Negating $\forall$ gives $\exists$ with “not”; negating $\exists$ gives $\forall$ with “not.”
- One counterexample refutes “all.” Induction can prove statements for all positive integers, using a base case and an inductive step.
- When a direct proof is difficult, try contradiction: assume the conclusion is false and derive a contradiction.

Reflection Questions

Observation

1. Write the converse of “A square has four equal sides” and determine whether it is true. Hint: find a shape with four equal sides that is not a square.

Proof

1. Prove $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$ by induction. Hint: for the inductive step, add $k+1$ to both sides, then combine the fractions and simplify.

Challenge

1. Prove by contradiction that $\sqrt{2}$ cannot be written as a ratio of two integers. Hint: suppose $\sqrt{2} = \frac{p}{q}$ in lowest terms, square both sides, and examine the parity of p .

Appendix B An Elementary Mathematics Toolbox

This appendix is a checklist before setting out. The main text assumes you have encountered these tools, but does not demand mastery. If something seems unfamiliar, ten minutes here should refresh it. For each tool we answer just three questions: what it is, what difficulty it resolves, and what a typical example looks like. The list roughly follows frequency of use in the main text, with the most common tools first. Consult it when needed; you need not read it all beforehand. But do work through each example. A tool truly belongs in your toolbox only once you can check how it works.

B.1 Fractions and Ratios: Parts of a Whole

The fraction $\frac{a}{b}$ means dividing the unit “1” into b equal parts and taking a of them. It addresses the shortage of integers: share two pancakes among three people and each receives $\frac{2}{3}$ of a pancake, which no integer expresses. Two main rules govern fractions. For addition and subtraction, first use a common denominator: $\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$. Multiplication and division are direct: $\frac{2}{3} \times \frac{3}{5} = \frac{2}{5}$, and dividing by a fraction means multiplying by its reciprocal.

A ratio describes the relative sizes of two quantities. A map scale of 1 : 50000 means 1 centimetre on the map corresponds to 50000 centimetres, or 500 metres, on the ground. A measured 3.6 centimetres represents $3.6 \times 500 = 1800$ metres. Percentages express ratios with denominator 100: a twenty-percent discount means paying 80% of the original price. Direct proportion, $y = kx$, means doubling one quantity doubles the other. Inverse proportion, $y = \frac{k}{x}$, means doubling one halves the other: double the speed and the time for the same distance halves.

When describing change with percentages, always identify the reference whole. Raising a price by 10% and then lowering it by 10% does not restore the original price. The increase uses a smaller base than the decrease, leaving 99% of the original price. Many everyday numerical traps hide in this step.

B.2 Equations: Working with an Unknown as Though It Were Known

The central idea of equations fits in one sentence: **give the unknown a name, then calculate with it as though it were known.** “Three times a number plus 5 equals 20”: call the number x and write $3x + 5 = 20$. Solving uses the balance principle. Adding, subtracting, multiplying, or dividing by the same number on both sides preserves equality, provided the divisor is not 0. Thus $3x = 15$ and $x = 5$.

Here is a complete example. “A father is 40 and his son is 12. In how many years will the father be twice the son’s age?” Let that be x years from now. Write $40 + x = 2(12 + x)$, expand to $40 + x = 24 + 2x$, and rearrange to get $x = 16$. Check by substitution: after 16 years, their ages will be 56 and 28, exactly a ratio of two. Name the unknown, set up the equation, solve it, and substitute to check: all four steps matter, though the last is the easiest to skip.

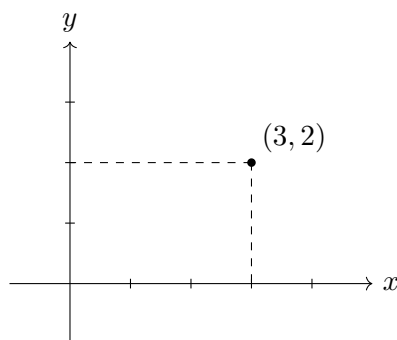
A system of two linear equations uses two balances together. For example,

$$\begin{cases} x + y = 10, \\ x - y = 4, \end{cases}$$

adding eliminates y : $2x = 14$, so $x = 7$, and substitution gives $y = 3$. A quadratic equation such as $x^2 - 5x + 6 = 0$ can often be factored as $(x - 2)(x - 3) = 0$, giving $x = 2$ or $x = 3$. When factoring is difficult, the quadratic formula is available: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

B.3 Coordinates: An Address for Every Point in the Plane

In a Cartesian coordinate system, each point corresponds to an ordered pair of real numbers (x, y) . Move x units east, or west if negative, then y units north, or south if negative. This translates shapes into numbers, allowing algebraic calculation of geometric problems and pictures of algebraic expressions.



The distance formula between (x_1, y_1) and (x_2, y_2) , $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$, is simply the Pythagorean theorem in coordinate clothing: the horizontal and vertical differences are the perpendicular legs. The midpoint formula $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$ says to average the two addresses component by component. The midpoint of $(1, 2)$ and $(3, 6)$ is $(2, 4)$. The graph of $y = 2x + 1$ is a straight line whose slope 2 means that each increase of 1 in x raises y by 2.

B.4 Triangles: The Building Blocks of Plane Geometry

A triangle is the most rigid of polygons: once its three side lengths are fixed, so is its shape. That is why roof supports and bicycle frames use triangles. Its three most frequently used properties are:

- Angle sum: the three interior angles add to 180° .
- Pythagorean theorem: in a right triangle with legs a, b and hypotenuse c , $a^2 + b^2 = c^2$; for example, $3^2 + 4^2 = 5^2$.
- Triangle inequality: the sum of any two sides exceeds the third. Thus 2, 3, 6 cannot form a triangle, since $2 + 3 < 6$.

A triangle's area is half its base times its height. This formula will return in integration as cutting into narrow strips and adding them. Two triangles are congruent if they coincide exactly, as guaranteed by criteria such as side-side-side. They are similar if their shapes agree, even if their sizes differ, with proportional corresponding sides. A useful consequence of similarity is that a side-length ratio of k gives an area ratio of k^2 . Double the sides and the area quadruples, explaining why larger pizzas can be disproportionately good value. Measuring a flagpole's height from its shadow uses similarity: at the same moment, its height-to-shadow ratio equals yours.

B.5 Exponents and Logarithms: Accelerating Multiplication and Reversing It

Exponents abbreviate repeated multiplication: 2^{10} means multiplying ten copies of 2, giving 1024. The law $a^m \cdot a^n = a^{m+n}$ adds the numbers of factors. In particular, $a^0 = 1$ is the empty product and $a^{-1} = \frac{1}{a}$ is the reciprocal. Scientific notation applies exponents in everyday work: light travels at approximately 3×10^8 metres per second, and a hydrogen atom has a mass of approximately 1.67×10^{-27} kilograms. Exponents make enormous and tiny numbers manageable to write.

A logarithm is an exponent in reverse. $\log_a b$ asks: "To what power must a be raised to give b ?" Since $2^{10} = 1024$, $\log_2 1024 = 10$. Logarithms address the burden of excessive multiplication. The rule $\log_a(xy) = \log_a x + \log_a y$ turns multiplication into addition. Before calculators, astronomers used logarithm tables to reduce months of multiplication to days of addition.

Terminology Card: Logarithm

Intuitive explanation: the power of the base needed to obtain a number. Formal definition: if $a^x = b$, with $a > 0$, $a \neq 1$, and $b > 0$, then $x = \log_a b$. Example: $\log_{10} 1000 = 3$, because $10^3 = 1000$.

Mathematical Toolbox

Six tools to carry with you: fractions and ratios (parts and relative sizes), equations (calculating with unknowns), coordinates (translating between shapes and numbers), triangles (angle sums, the Pythagorean theorem, and the triangle inequality), exponents (abbreviating repeated multiplication), and logarithms (asking the inverse exponential question). Return to this appendix whenever the main text uses them.

Reflection Questions

Observation

1. A glass of sugar water contains 10% sugar. After you drink half and refill it with water, what is the concentration? Hint: track the sugar remaining, not the water added.

Proof

1. Prove that a square of side length 1 has diagonal length $\sqrt{2}$. Hint: what kind of triangles does the diagonal produce?

Challenge

1. Without solving the equation, determine whether $x^2 + x + 1 = 0$ has real solutions. Hint: examine $b^2 - 4ac$ under the square root in the quadratic formula.

Appendix C Symbols and Glossary

Mathematical symbols are abbreviations, not secret codes. Behind each lies an intuitive idea that would take longer to say in words. This appendix groups the book’s principal symbols and terms by field. “Intuition” gives the picture to bring to mind; “Example” offers a minimal check. When you encounter an unfamiliar symbol, look it up here, then return to the main text to see it at work. You can consult the relevant section for an unfamiliar term or scan a table before a chapter to meet its cast of characters. These tables are maps, not memorization assignments. Travel the routes often enough and you will remember them naturally.

C.1 Numbers

The world of numbers has expanded floor by floor: negative numbers for subtraction, fractions for division, and irrational numbers to express lengths such as the diagonal of a square with side length 1. The symbols below are the nameplates for those floors.

Symbol/term	Name	Intuition	Example
$\mathbb{N}$	Naturals	Counting numbers, starting at 0 or 1	$0, 1, 2, 3, \dots$
$\mathbb{Z}$	Integers	Natural numbers and their negative counterparts	$-3, 0, 7$
$\mathbb{Q}$	Rationals	Numbers expressible as ratios of two integers	$\frac{2}{3}, 0.25$
$\mathbb{R}$	Reals	Every point on the number line, with no gaps	$\sqrt{2}, \pi$
$a \mid b$	a divides b	b divides exactly by a , with no remainder	$3 \mid 12$, but $3 \nmid 14$
$\gcd(a, b)$	GCD	Greatest common divisor: the largest integer dividing both a and b	$\gcd(12, 18) = 6$
$ x $	Absolute value	Distance from x to the origin, regardless of direction	$ -5 = 5$
$\approx$	Approximately	Equal within an acceptable error	$\pi \approx 3.14$

C.2 Algebra

Algebra’s central move is representing numbers with symbols, allowing us to study calculation rules independently of specific values. The symbol x is not a mysterious number but a placeholder: it holds a space so a pattern can emerge.

Symbol/term	Intuition	Example
a^n	A product of n copies of a : abbreviated multiplication	$2^5 = 32$
$\sqrt{a}$	The nonnegative number whose square is a	$\sqrt{9} = 3$
$\log_a b$	The answer to “what power of a equals b ?”	$\log_2 8 = 3$
$\sum$	Summation: add a sequence of numbers according to a rule	$\sum_{k=1}^4 k = 1 + 2 + 3 + 4 = 10$
$f(x)$	A function: a machine giving one output for input x	If $f(x) = 2x$, then $f(3) = 6$
Polynomial	An expression using only ad- dition, subtraction, and mul- tiplication of unknowns	$x^2 - 3x + 2$
Factorization	Expressing a polynomial as a product	$x^2 - 1 = (x + 1)(x - 1)$

C.3 Geometry

Most geometric symbols grow out of the shapes themselves. Seeing $\triangle$ suggests a triangle; seeing $\perp$ suggests a right-angled corner. Sketching a symbol’s meaning while learning it is more effective than repeating its name ten times.

Symbol/term	Intuition	Example
$\angle ABC$	The angle opening at vertex B	$\angle ABC = 60^\circ$
$\triangle ABC$	The triangle with vertices A, B, C	Interior angles sum to 180°
$\cong$	Congruent: able to coincide exactly	Side-side-side establishes congruence
$\sim$	Similar: same shape, possibly different size	An enlarged photograph is similar to the original
$\perp$	Perpendicular: meeting at exactly 90°	The sides of a right-angled corner
$\parallel$	Parallel: coplanar lines that never meet	The two rails of a railway track
π	Circumference divided by diameter, the same for all circles	Circumference $= 2\pi r$

C.4 Analysis

Analysis, the rigorous development of calculus, studies approaching without limit. Its symbols almost all address one question: how close are we to the target? The difficult process of making “close” precise is the story of Chapters 9 through 12.

Symbol/term	Intuition	Example
$\lim$	Limit: a target approached arbitrarily closely, not necessarily reached	$\frac{1}{n} \rightarrow 0$ (larger n brings it closer to 0)
∞	Infinity: a direction beyond every finite number, not a number	Counting $1, 2, 3, \dots$ never ends
ε	An error allowance as small as you choose	Accuracy within $\varepsilon = 0.001$
Δx	Change: a small step in x	From 3 to 3.1, $\Delta x = 0.1$
$f'(x)$	Derivative: the instantaneous slope of a curve	The derivative of displacement is velocity
$\int$	Integral: combining infinitely many small strips into a total	Integrating speed over time gives distance

C.5 Probability

Probability theory gives uncertainty rules. Likelihood is represented by a number from 0 to 1, with 0 for impossibility and 1 for certainty. The hardest part is often translating a real problem into the correct events, rather than calculating.

Symbol/term	Intuition	Example
$P(A)$	The likelihood of event A , between 0 and 1	Coin toss: $P(\text{heads}) = \frac{1}{2}$
$P(A B)$	The probability of A given that B occurs	Probability of carrying an umbrella given rain
$E(X)$	Expectation: the average outcome in the long run	Expected result of a die roll: 3.5
Random variable	A measurement whose result involves randomness	The number showing on a rolled die
Variance	The size of fluctuations around the expectation	More consistent scores have smaller variance

C.6 Discrete Mathematics

Discrete mathematics studies separate objects: integers, sets, and graphs. It is computer science's native language: the basic objects in a computer are discrete, with no "half" of one. This section also includes the quantifiers from Appendix A, among the symbols most frequently encountered in proofs throughout the book.

Symbol/term	Intuition	Example
$\in$	Membership: an object belongs to a set	$2 \in \{1, 2, 3\}$
$\forall$	For every: each member must satisfy the claim	Every even number is divisible by 2
$\exists$	There exists: at least one can be found	There is an even prime
$a \equiv b \pmod{m}$	Congruence: a and b leave the same remainder modulo m	$17 \equiv 2 \pmod{5}$
$n!$	Factorial: the number of orders for n distinct objects	$3! = 6$
$\binom{n}{k}$	Binomial coefficient: unordered choices of k objects from n	$\binom{4}{2} = 6$
Graph	A network of points and connections between them	A transport network with cities as vertices and roads as edges

A general tip for reading symbols: when one is unfamiliar, read it aloud and ask three questions. What objects does it act on? What question does it answer? What is its inverse operation? For instance, $\log$ acts on positive numbers, asks “what power?”, and is reversed by exponentiation. Once you can answer all three, the symbol becomes less intimidating.

Connections

Symbols travel between fields. In algebra, $\sum$ adds finitely many terms; in analysis, it appears in infinite series. In set theory, $\in$ records membership; in probability, it can say that an event contains an outcome. If a symbol leaves you puzzled in any chapter, first locate it here, then see what new task its new setting gives it.

One final reminder: a symbol’s meaning comes from its definition, not its appearance. The same symbol may play different roles in different chapters. In an equation, x is an unknown; in a function, an independent variable; in geometry, perhaps a horizontal coordinate. Asking “who is it this time?” at each encounter is a basic mathematical reading skill.

Appendix D Mathematical Experiments and Programming Resources

Mathematicians have laboratories too, but their equipment is much cheaper: a pen, paper, a spreadsheet, or a computer on which to write a few lines of code. This appendix introduces three accessible experimental tools, each with activities you can try immediately. First, one ground rule: **experiments provide observations; proofs establish conclusions**. Tools help you see more and make better guesses, but no quantity of observations replaces proof.

D.1 Spreadsheets: An Underrated Mathematical Instrument

You need no specialized software. A spreadsheet on a computer or phone supports serious exploration. Enter a formula, fill it down, and patterns emerge.

Activity one: the power of compound interest. In the first column, enter years 0 through 30. Start the second column with a principal of 1000, then make each following cell equal to the previous one times 1.05, representing an annual interest rate of 5%. Fill down: after 30 years, the amount is approximately 4322, more than 4 times the principal. Better still, it doubles roughly every 14 years. This is exponential growth, explaining why interest on interest can sound mild yet become striking over time. A useful trick is to keep the interest rate in its own cell and reference that cell in the formula. Change it once and the whole curve changes. Separating an adjustable quantity this way is the mathematical idea of a parameter.

Activity two: sifting out primes. Enter 1 through 200 in the first column. In the second, determine whether each number is prime by trying division by earlier primes. Count the primes in each hundred: there are 25 from 1 to 100, but only 21 from 101 to 200. You will see primes becoming sparser, an observation pointing toward a major theorem in number theory.

D.2 GeoGebra: Geometry That Moves

GeoGebra is free dynamic geometry software that you can use in a browser. Its greatest strength is dragging: move a vertex after drawing a figure and every dependent quantity changes in real time.

Activity one: an angle sum you cannot drag apart. Draw any triangle and ask the software to measure its three interior angles and display their sum. Drag the vertices vigorously,

making the triangle broad, flat, or sharp. The sum stays at 180° . This is no proof, but it makes you feel the fact deserves one.

Activity two: approach π yourself. Draw a circle and an inscribed regular polygon. Increase the side count from 6 to 12, 24, and 48, watching the ratio of the polygon's perimeter to the circle's diameter approach $3.14159\dots$. As you move the side-count slider, notice how the gap between polygon and circle shrinks without ever disappearing completely. This is an honest picture of approximation. In minutes on a screen, you can reenact the polygon method mathematicians used to estimate π over two thousand years ago.

One-Minute Experiment

You can experiment without a computer. Wrap string around a circular cup rim, measure the diameter, and compare the string's length with it. The circumference is always a little more than 3 times the diameter. Try a bottle cap, a bowl, and a plate. The ratio barely changes: you have just observed π .

D.3 Python: Letting a Computer Count Ten Thousand Times

Python is a programming language with syntax close to English. Free online environments let you run it without installing anything. For mathematics, its value is tirelessness: it never minds trying a million examples.

Activity one: test a conjecture. Write a small loop checking whether $n^2 + n + 41$ is prime for n from 1 to 1000. Astonishingly, the first several dozen values are prime. This famous prime-generating formula has a run of 40 successes. But keep going: at $n = 40$, the result is 41×41 , and the pattern collapses. This experiment makes “many examples are still not a proof” a lesson you can feel.

Activity two: estimate π by scattering points. Have a program place one hundred thousand random points in a square of side length 1 and count how many lie inside its inscribed circle, closer to the centre than the radius. The circle-to-square area ratio is $\frac{\pi}{4}$, so dividing the inside count by the total and multiplying by 4 estimates π . More points make the estimate steadier. This illustrates relative frequency approaching probability.

Activity three: approximate $\sqrt{2}$. Begin with $[1, 2]$. Take its midpoint, compare the midpoint's square with 2, and discard the appropriate half, retaining $\sqrt{2}$ in the remaining interval. After 30 repetitions, the interval is less than a billionth wide: $\sqrt{2}$ is trapped in a tiny gap. This is bisection, an algorithmic form of the limit idea. You never seize the target exactly, but can enclose it in an arbitrarily small cage.

Computational experiments have two basic habits: change parameters and watch what changes. Replace 1000 in activity one with 100000, or change the formula to $n^2 - n + 41$. Predict the outcome, then run it to check. Wrong guesses often teach more than right ones.

Counterexample Alert

“The computer verified a million examples, so the proposition is true.” That conclusion does not follow. The prime formula succeeds 40 times and fails on the 41st; mathematical history has examples that fail only after hundreds of millions of successes. A program can refute a false conjecture by finding a counterexample or strengthen our confidence. To **establish** a claim about infinitely many objects requires proof. An experiment is a telescope, not a verdict.

D.4 Where to Find These Tools

Any common office suite provides a spreadsheet. Search for GeoGebra to find its official site and open it in your browser. Search for an online Python interpreter to find an environment requiring no installation. You do not need a hefty tutorial to begin: try this appendix’s activities and, when stuck, search for something like “how to write a spreadsheet formula” or “Python for loop.” Learning while doing is the quickest route. Do not panic at error messages. They tell you what the computer did not understand, which is useful training in logic in itself.

Keep Climbing

Keywords for further exploration: numerical computation (having computers solve equations), Monte Carlo methods (using random sampling to calculate areas and probabilities), and computer-assisted proof. The famous four-colour theorem used a computer to verify many cases, prompting years of debate about what counts as proof. University courses in computational mathematics and scientific computing explore these topics.

Reflection Questions**Observation**

1. Change the annual interest rate in activity one to 10%. Roughly how many years does doubling take? Hint: watch how many rows separate doublings in the second column and compare with 14 years at 5%; exact calculation is unnecessary.

Proof

1. Prove that each halving in activity three leaves $\sqrt{2}$ in the retained interval. Hint: split into two cases according to whether the midpoint’s square is greater or less than 2.

Challenge

1. Design an experiment to estimate the probability that a needle randomly dropped on paper marked with parallel lines crosses a line: Buffon’s needle problem. Hint: first identify what is random when the needle lands. Position is one quantity; what else?

Appendix E Reading Routes by Level and Interest

You need not work through all twenty chapters from the first page to the last. Mathematics is less a one-way street than a metro map, with several main lines and many interchanges. This appendix offers a default route and four routes by interest. Choose your own pace and preferences, and change lines whenever you like.

E.1 The Default Route: Four Steps

If you have no particular preference, these four steps provide a steady path, each preparing the next.

Step one: Chapters 1–4, foundations. Start with algebraic symbols, equations, and functions to practise symbolic reasoning. These chapters supply the book’s shared language, needed on any later route. Read them in order and consult Appendix B whenever elementary material is unfamiliar.

Step two: Chapters 5–8, going deeper. Polynomials, transformations, and vectors put the first step’s tools to deeper use. Here you first appreciate how the same object, such as a polynomial, can have several algebraic and geometric faces. Changing the face through which you view a problem is one of a mathematician’s handiest moves.

Step three: Chapters 9–12, the language of higher mathematics. Move from change to infinity, meeting intuitive pictures of limits, derivatives, and integrals. The keyword is approximation. Allow yourself time to become comfortable with rigorous statements about infinity. First grasp the graphs and numerical examples; rigor will follow. Letting intuition fall behind is the greater difficulty.

Step four: Chapter 13 onward, branching by interest. From here, chapters no longer form a strict sequence. Number theory, topology, geometry, probability, and mathematical foundations develop their own lines. After step three, you can jump to whatever looks intriguing. The following routes are designed for that freedom.

E.2 Four Routes by Interest

Route one: problem solving—Chapters 1–6, 13, and 14. If you enjoy the relief of solving a problem that initially seemed baffling, this route leads to mathematics’ oldest hunting

ground: integers. The first six chapters sharpen algebraic tools; Chapters 13 and 14 enter number theory, where a question may take one sentence to understand, such as a conjecture about primes, yet stump mathematicians for centuries. Keep scrap paper nearby and work each example yourself before reading on. The unit of problem-solving ability is not pages read, but sheets of working.

Route two: shapes and spaces—Chapters 3, 4, 6–12, 16, and 17. If pictures feel friendlier than formulas, begin with coordinates and transformations, continue through function graphs, rates of change, and curves, and arrive at topology and differential geometry: geometry on rubber sheets and shortest paths in curved spaces. Draw often, even badly. Geometric intuition comes from making drawings, not merely looking at them.

Route three: infinity and change—Chapters 8–12, 15, and 17. Infinity is one of mathematics' most captivating cliffs. What is an infinitesimal? How can infinitely many positive numbers have a finite sum? This route begins with rates of change, approaches limits and infinite series directly, then extends into related geometric and analytic ideas. Bring patience. For every conclusion about infinity, ask how it is approached step by step.

Route four: codes, computation, and applications—Chapters 2, 5, 7, 13–15, 18, and 19. If your question is “what is mathematics useful for?”, begin with equations and algebraic structures, pass through number theory—a foundation of modern cryptography, with prime-number properties used in encrypted phone payments—and continue to probability and real-world modeling. Combine this route with Appendix D's experiments. Many ideas along it become immediately playable in a spreadsheet or a few lines of code.

Connections

The routes' connections around Chapter 13 are no accident. Number theory uses elementary tools such as divisibility and remainders, yet connects to cryptography, geometry, and even analysis. Mathematics has many hidden passages between branches. The book's Connections boxes keep pointing them out. Wandering from one route onto another is a normal and rewarding part of reading, not something to correct.

E.3 Three Suggestions for Reading

First, **getting stuck is normal**. The reflection questions provide hints only. Returning after several days of being stuck often teaches more than immediately reading an answer. Mathematicians themselves can spend years stuck on a problem. Second, **you need not understand everything at once**. Follow the main story on a first reading and save proofs for a second. Recognizing where the difficulty lies is progress. Third, **give hands-on work priority**. Stop for the One-Minute Experiments. Five minutes of doing often beats half an hour of staring.

The other four appendices are tools to keep beside you. Appendix A is a grammar of proof, useful when proofs first seem bewildering. Appendix B refreshes elementary tools such as fractions, equations, and coordinates. Appendix C is a symbol dictionary, and Appendix D provides experimental equipment. Together with this appendix, they form the book's support

team, ready wherever your current chapter takes you.

Keep Climbing

To keep climbing after this book, choose the next mountain by route. Problem solving leads to competition mathematics and elementary number theory texts. Shapes lead to *What Is Mathematics?* and introductory books on topology. Infinity leads to first courses in calculus and mathematical analysis. Applications lead to accessible books and open courses on cryptography and probability. With keywords in hand, libraries and the internet offer many paths.

Reflection Questions

Observation

1. Browse the contents, choose the three chapters you most want to read, and identify their routes. Hint: a chapter can belong to several routes, and intersections are often especially interesting.

Proof

1. This appendix has no theorem to prove, but try arguing to a classmate why Chapters 1–4 precede almost every route. Hint: open any later chapter and count the tools it uses from those early chapters.

Challenge

1. Design a route absent from this appendix, perhaps for someone interested in mathematical history and mathematicians' lives. List the chapter stops and explain why each deserves a visit. Hint: first decide what you want to see at the destination, then work backward to the preparation needed along the way.